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COLLEGE OF COMPUTER STUDIES

Software Test Document

for

DASIGConnect

Change History

Version	Date	Author(s)	Description of Changes
1.0	10/11/2026	Chris Daniel Cabatana, Jaylord Bayonas, Lerah Caones, Mark Anton Camoro, Richemmae V. Bigno	Initial draft — comprehensive test plan, 25 use-case-aligned test cases (UC-1.1–UC-3.6), execution logs, test results matrix, and incident reports based on the Software Requirements Specification (SRS v6.0).

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1. Introduction

1.1. System Overview

DASIGConnect is a centralized, web-based **Social Media Content Workflow and Scheduling Management System** developed as a capstone project (Team Code: *2526-sem2-it332-38*) for the **DOST Acadême–Science and Innovation Group (DASIG)** under the Department of Science and Technology (DOST) Region 7. The system replaces informal, ad hoc social media coordination across member Higher Education Institutions (HEIs)—such as Cebu Institute of Technology – University (CIT-U), Silliman University, and partner institutions—with a structured, traceable, role-based digital workflow integrated with the official DASIG Facebook Page.

DASIGConnect is engineered around a three-layer operational model designed to eliminate the administrative coordination bottleneck:

1. **Layer 1 — Self-Service (Contributors):** Institutional representatives draft event posts, attach media, utilize advisory AI tools, and self-schedule content within software-enforced guard rails.
2. **Layer 2 — Automation (System Rules and Cron Jobs):** Deterministic scheduling guard rails (30-minute conflict buffer, 2-hour minimum lead time, 30-day maximum horizon, 8:00 AM–8:00 PM posting window, 6-post daily cap, and 3-post active institutional quota) and 16 background scheduled jobs handle conflict prevention, automated Facebook Graph API publishing, stale-submission sweeps, embedding reconciliation, and multi-channel notifications.
3. **Layer 3 — Governance & Exception Handling (Moderators and Administrators):** Network-wide Moderators and Administrators triage and review the Validation Queue (*Approve*, *Request Revision*, *Reject*, or *Edit & Approve*), while Administrators retain exclusive authority over system governance, guard-rail overrides, manual publishing fallback, token re-authentication, watermark configuration, System Health, and the append-only Audit Log.

Architecturally, DASIGConnect operates as a three-tier client-server application:

- **Presentation Tier:** A React 19 + TypeScript Single-Page Application (SPA) built with Vite, Tailwind CSS v4, Radix UI / shadcn/ui, FullCalendar, Axios, and Server-Sent Events (SSE), deployed on Vercel.
- **Application Tier:** A stateless Spring Boot 4.0.6 (Java 21) REST API (*/api/v1*) deployed on Railway/Render, enforcing JWT authentication (8-hour TTL with database-backed revocation), Role-Based Access Control (RBAC), tenant scoping, state machines, and background cron scheduling.
- **Data & Storage Tier:** Supabase PostgreSQL with the *pgvector* extension (*VECTOR(1024)* cosine similarity) and Row-Level Security (RLS) policies for multi-institution tenant isolation, paired with Cloudflare R2 S3-compatible object storage using presigned *PUT* uploads and backend-proxied reads (*/api/v1/media-files/*).
- **External Integrations:** Meta Facebook Graph API (*v25.0*) and Messenger Platform webhook, Anthropic Claude Vision API (*claude-haiku-4-5*), Voyage AI Embedding API (*voyage-4-lite*, 1024 dimensions), and SMTP/Resend transactional email delivery.

1.2. Test Approach

Testing for DASIGConnect follows a multi-layered **Verification and Validation (V&V)** strategy aligned with the project's Agile/Scrum 2-week sprint methodology, verifying every functional use case (*UC-1.1* through *UC-3.6*), business rule (*BR-ACC* through *BR-AI*), guard rail (*GR-H1–GR-T9*), non-functional requirement (*SRS Section 3.4*), and multi-tenant security boundary (*SRS Section 3.5*).

1. **Unit & Service-Layer Testing (White-Box):**
   - **Scope:** Business logic, state transitions, guard-rail calculations, account caps, password complexity validation, review lock lifecycles, watermark coordinate scaling, and CSV sanitization across all Spring Boot @*Service* classes.
   - **Technique:** Isolated unit tests using **JUnit 5 (Jupiter)**, **Mockito**

(*@ExtendWith(MockitoExtension.class)*), and **AssertJ**. External APIs (Facebook Graph API, Anthropic Claude, Voyage AI, Cloudflare R2, SMTP) are mocked to deterministically test both normal execution and failure/timeout branches (*503*, *504*, *429*, exponential backoff).

2. **API Controller & Security Slice Testing (Grey-Box):**
   - **Scope:** REST endpoint contracts under */api/v1/*, request payload validation (*400 Bad Request*, *422 Unprocessable Entity*), conflict handling (*409 Conflict*, *423 Locked*), JWT filter authentication (*401 Unauthorized*), and role-based authorization (*@PreAuthorize* returning *403 Forbidden* or *404 Not Found* for cross-tenant isolation).
   - **Technique:** Spring *@WebMvcTest* slices importing *SecurityConfig* and executed via *MockMvc*.
3. **Database Migration, Concurrency & Tenant Isolation Testing (Integration):**
   - **Scope:** Flyway schema migrations (*V1–V104+*), PostgreSQL Row-Level Security (RLS) policies (*app.current_institution_id*, *app.current_role*), append-only audit log enforcement (*V78* RLS blocking *UPDATE*/*DELETE*), atomic conditional claims (*claimForPublishing*, *claimModeratorReschedule*), and the PostgreSQL GiST exclusion constraint (*V95*/*V96*) preventing concurrent *±30-minute* slot reservation races.
   - **Technique:** Repository/service integration tests and live SQL verification against PostgreSQL / H2 test contexts.
4. **End-to-End (E2E), UI Workflow & Performance Regression Testing (Black-Box):**
   - **Scope:** Full browser-level user journeys across Contributor, Moderator, and Administrator personas, including workspace bootstrap, SSE real-time notification badge updates (*INC*, *READ_SYNC*), debounced media search, 3-step post drafting, validation queue tab switching, FullCalendar event loading, analytics rendering, and bundle size budget verification.
   - **Technique:** Automated headless Chromium browser tests using **Playwright (***@playwright/test***)** against the production-mode Vite preview build (*npm run test:e2e*, *npm run test:performance*, *npm run check:bundle*) plus structured manual walkthroughs of visual/canvas interactions.
5. **Defect Tracking & Regression Verification:**
   - Any discrepancy between the SRS (including its use case specifications) and system behavior is logged as an Incident in **Appendix A.3**, classified by severity, root-caused to the specific service/controller/component, and re-tested upon resolution.

1.3. Definitions and Acronyms

Term / Acronym	Definition
ACTIVE / PENDING / EXPIRED / INACTIVE / CANCELLED	User account lifecycle states governed by the User Account State Machine (*SRS Section 1.4*).
Admin Owner	Governance designation (*is_admin_owner* = *true*) held by exactly one active Administrator at a time; required for actions targeting existing Admin accounts, erasing personal data, switching the connected Facebook Page, and transferring ownership (*UC-1.1*).
Administrator	Network-wide user role (*role* = *admin*, *institution_id* = *NULL*) with full governance, approval, scheduling override, exception handling, System Health, Watermark, and Audit Log authority.
Contributor	Institution-scoped user role (*role* = *contributor*) responsible for drafting posts, uploading media, and self-scheduling content within guard rails for their assigned member HEI.
CWE-1236	Common Weakness Enumeration for Improper Neutralization of Formula Elements in a CSV File (Formula Injection), mitigated by prepending an apostrophe (*'*) to cells starting with =, +, -, @, \t, or \r.
DASIG	DOST Acadême–Science and Innovation Group under DOST Region 7.

Term / Acronym	Definition
DASIG Central Visayas	Protected network-default institution seeded at deployment; exempt from deactivation and permanent deletion (*UC-1.2 A6, A8*).
DRAFT / PENDING / IN_REVIEW / NEEDS_REVISION / SCHEDULED / PUBLISHING / PUBLISHED / PUBLISHED_MANUAL / PUBLISH_FAILED / MISSED_REVIEW / REJECTED	Submission lifecycle states governed by the content workflow state machine (*SRS Section 1.3 & 3.2*).
E2E	End-to-End testing conducted through a real browser environment (Playwright + Chromium).
Fast-Track (Live Event)	Submission mode (*UC-1.5*, *UC-1.9 A5*, *UC-3.2 A5*) that bypasses scheduled-slot guard rails, sorts to the top of the active Validation Queue, and publishes asynchronously immediately upon approval.
GiST	Generalized Search Tree index in PostgreSQL, used in migration *V95*/*V96* to enforce database-level exclusion constraints on overlapping *±30-minute* slot reservations.
Guard Rail (*GR-H**, *GR-S**, *GR-T**)	System-enforced business rule classified as Hard (*GR-H1..H4*, blocks action), Soft (*GR-S1..S2*, warns actor), or Trigger (*GR-T1..T9*, autonomous cron action).
HEI	Higher Education Institution participating in the DASIG network (e.g., CIT-U, Silliman University).
JWT	JSON Web Token used for stateless 8-hour session authentication (*HS256*), carrying *role*, *session_version*, and *institution_id* (for Contributors).
Moderator	Network-wide reviewer role (*role* = *moderator*, *institution_id* = *NULL*, formerly termed *Validator*) with access to the network-wide Validation Queue, Master Calendar (with capped rescheduling), and Contributor invitation management (*UC-1.10*).
MockMvc	Spring Test framework component allowing HTTP controller testing with full Spring Security filter chains without starting an external servlet container.
pgvector	PostgreSQL extension storing 1024-dimensional Voyage AI embeddings (*VECTOR(1024)*) for cosine-distance (<=>) semantic similarity search.
R2	Cloudflare R2 S3-compatible object storage used for raw media assets and temporary watermarked derivatives.
RBAC	Role-Based Access Control enforced server-side via Spring Security *@PreAuthorize* and route gates.
RLS	PostgreSQL Row-Level Security enforcing multi-institution tenant isolation at the database engine layer (*SRS Section 3.5*).
SSE	Server-Sent Events (*GET /api/v1/notifications/stream*) used for real-time in-app notification delivery and cross-session read-state synchronization (*UC-3.3*).
SRS	Software Requirements Specification.
STD	Software Test Document (this document).

Term / Acronym	Definition
SUS	System Usability Scale questionnaire targeting a minimum score of ≥68 (*SRS Section 3.4.8*).
Two-Step Publishing	Verified Facebook Graph API photo publishing flow: Step 1 stages unpublished photos (*POST /{PAGE_ID}/photos* with *published=false*), and Step 2 publishes a single feed post (*POST /{PAGE_ID}/feed* with *attached_media*).

2. Test Plan

2.1. Features to be Tested

All functional requirements across the three development modules (*UC-1.1* through *UC-3.6*), scheduling guard rails (*GR-H1–GR-T9*), application business rules (*BR-ACC–BR-AI*), and non-functional requirements defined in the SRS and its use case specifications are in scope for testing:

Module / Category	SRS Reference	Feature / Capability Under Test	Target Test Cases
Module 1: User Access & Content Creation	*UC-1.1*, *BR-ACC-04*	Administrator account invitations (72h SHA-256 hashed tokens), uniform 3-Admin cap (*ACTIVE* + pending invites + pending promotions), Admin Owner-gated deactivation/reactivation/deletion/erasure, propose/confirm Admin promotion (*72h* expiry), and Admin Owner transfer (*24h* expiry).	*Case-1*, *Case-2*
Module 1: User Access & Content Creation	*UC-1.2*, *BR-ACC-01*	Institution creation, editing, deactivation (blocked if Contributors remain), reactivation, Contributor reassignment, permanent deletion with confirmation code, and protection of *"DASIG Central Visayas"*.	*Case-3*
Module 1: User Access & Content Creation	*UC-1.3*, *BR-ACC-01..04*	Contributor invitation (Admin & Moderator), 72h token activation, re-invite block on *INACTIVE* accounts, Moderator scope restriction to personally sent invites, and footprint-aware deletion (*USER_REMOVED* vs. *USER_DELETED*) / Right-to-be-Forgotten erasure (*USER_ANONYMIZED*).	*Case-4*
Module 1: User Access & Content Creation	*UC-1.4*, *BR-ACC-02..07*	JWT login (8h TTL), 5-attempt / 15-minute account lockout, 12-char password complexity policy, single-token logout (*revoked_tokens*) vs. account-wide session invalidation (*session_version*), session expiry modal, and 10-minute Facebook Messenger link code.	*Case-5*
Module 1: User Access & Content Creation	*UC-1.5*, *BR-SUB-03*	3-step composer (*Media → Post Details → Organize & Schedule*), 0–100 Readiness Checklist (75% Required, 25% Recommended), built-in & personal templates, **Generate from Top Posts** (*UC-1.5 A10*), autosave, dirty-state warning, and *guardrails_enforced* runtime switch.	*Case-6*, *Case-7*
Module 1: User Access & Content Creation	*UC-1.6*, *BR-AI-01..03*, *BR-AI-07*	On-demand Claude Vision caption generation (3 tones, 280-char prompt cap, 30 req/hr rate limit, 3,000-char caption ceiling), advisory AI **Check writing** (*ProofreadService*), and Fancy Unicode text styling (5 styles + Plain revert).	*Case-8*
Module 1: User Access & Content Creation	*UC-1.7*, *BR-SUB-01..02*, *BR-AI-05..06*	Device upload validation (7 formats, ≤50 MB, ≤10 assets/post), presigned R2 upload, per-asset caption (≤500 chars), per-photo watermark opt-out, AI-Suggested Media (≥0.40 similarity floor, top 8), and AI Album Auto-Match (≥0.55 confident, *0.32–0.54* ambiguous).	*Case-9*

Module / Category	SRS Reference	Feature / Capability Under Test	Target Test Cases
Module 1: User Access & Content Creation	*UC-1.8*	Historical Facebook engagement peak-hour slot recommendations (last 100 posts, ≥20 samples or Tue/Wed/Thu 6:00 PM fallback), guard-rail pre-filtering, and inline soft-warning display.	*Case-10*
Module 1: User Access & Content Creation	*UC-1.9*, *GR-H1..H3*, *GR-S1..S2*	Draft submission completeness gate, submit-time guard-rail re-validation (*409* slot conflict redirects to Step 3), *pending* withdrawal (blocked once locked), resubmission from *needs_revision* or *rejected*, and Fast-Track top-of-queue sorting.	*Case-11*
Module 1: User Access & Content Creation	*UC-1.10*	Network-wide Moderator account invitation, deactivation, reactivation, footprint-aware deletion, erasure, and immediate lateral role change (*Contributor* ↔ *Moderator*) with review-lock release.	*Case-12*
Module 2: Media Library & Approval Pipeline	*UC-2.1*	Standalone Media Library upload (all 3 roles), text-based Album Auto-Match (≥0.6 confident, ≥0.3 ambiguous), nested parent-child albums with cycle guard, mandatory ≥1 manual tag floor (*A9*), and image *PROCESSING* → *READY* vs. video immediate *READY* lifecycle.	*Case-13*
Module 2: Media Library & Approval Pipeline	*UC-2.2*, *BR-MED-01*	Media Library grid/list browsing, Voyage + pgvector semantic search with keyword fallback, manual/AI tag searchability, institution filter for Moderator/Admin, editable *display_title*, and *Add to Existing Draft*.	*Case-14*
Module 2: Media Library & Approval Pipeline	*UC-2.3*, *BR-MED-02*	Asset Detail Panel *Used In* history (live links + historical *MEDIA_ASSET_REUSED* audit rows + deleted placeholders), 3-tier deletion precedence (*409* active block, *force=true* draft warning, 30-day soft-delete purge), and SHA-256 content-hash duplicate detection.	*Case-15*
Module 2: Media Library & Approval Pipeline	*UC-2.4*, *BR-APP-01..03*	Network-wide Validation Queue (8 tabs), 15-minute *ReviewLock* acquisition/expiry, unconditional self-submission review block (*A5*), Approve, Request Revision (*10–1,000* chars), Reject (7 standardized codes), inline edit diff severity (*Quiet*/*Flagged*/*Added Media*), *Restore original*, and Late Publish Override (*A15*).	*Case-16*, *Case-17*
Module 2: Media Library & Approval Pipeline	*UC-2.5*	Network-wide Watermark Configuration canvas (up to 3 elements, *1:1*/*4:5*/*16:9* aspect ratios), publish-time proportional rendering on photos, default text fallback when unconfigured, non-blocking publish fallback on render error (*PUBLISH_WATERMARK_FAILED*), and 3-day derivative purge (*GeneratedWatermarkPurgeJob*).	*Case-18*

Module / Category	SRS Reference	Feature / Capability Under Test	Target Test Cases
Module 3: Scheduling, Publishing & Analytics	*UC-3.1*, *BR-PUB-01*, *GR-H1*	Role-scoped Master Calendar visibility (Contributor masking of other HEIs vs. full view of own HEI + DASIG Central Visayas + own in-flight posts), Moderator reschedule caps (≤2 reschedules, *±1 day* of *originalScheduledAt*), Admin override justification, atomic reschedule claims, and PostgreSQL GiST exclusion constraint (*V95*/*V96*).	*Case-19*
Module 3: Scheduling, Publishing & Analytics	*UC-3.2*, *BR-PUB-02..05*, *GR-T5*, *GR-T9*	Minute-level *PublishingSchedulerJob* with atomic *scheduled → publishing* claim, 2-step photo publishing with orphan cleanup (*photoIdsCleanupFailed*), single-call video publishing, 3-retry exponential backoff (*5s, 25s, 125s*), Fast-Track async publishing, 5-minute stale sweep (*publish_failed* vs. *missed_review*), and 3-step Manual Publishing Fallback.	*Case-20*, *Case-21*
Module 3: Scheduling, Publishing & Analytics	*UC-3.3*, *BR-NOT-01..04*	Event-driven notifications across 29 triggers (*T-01–T-12+*), reviewer fan-out to both Moderators and Administrators, real-time SSE badge & cross-session *read*/*read-all* sync, independent email retry logging, and opt-in Facebook Messenger delivery.	*Case-22*
Module 3: Scheduling, Publishing & Analytics	*UC-3.4*, *BR-ANA-01..06*	Role-scoped Analytics dashboard (Contributor institution-wide, Moderator network-wide unfiltered, Admin network-wide with institution drilldown + Operational Health & AI Performance reports), 5-minute auto-refresh, and CSV export with CWE-1236 formula-injection sanitization.	*Case-23*
Module 3: Scheduling, Publishing & Analytics	*UC-3.5*	Admin-only System Health dashboard: PostgreSQL & Cloudflare R2 storage capacity probes (*80%* Warning, *95%* Critical), 6 external service checks, 16 tracked background jobs with manual **Re-run** (sync vs. async), 14 operational/AI metrics (including missed-review rate), CSV snapshot export, and Facebook OAuth token re-authentication.	*Case-24*
Module 3: Scheduling, Publishing & Analytics	*UC-3.6*, *SRS 3.5*	Admin-only Audit Log with 11 categories, 8 entity types, structured before/after diff inspector, 5,000-row sanitized CSV export, DB-enforced append-only RLS (*V78*), and 4-layer Multi-Institution Tenant Isolation (*404 Not Found* on cross-tenant access).	*Case-25*

2.2. Features not to be Tested

Based on the SRS (Sections 1.2, 2.3, and 3.3) and the scope boundaries documented in its use case specifications, the following features are explicitly excluded from testing:

1. **Integration with Non-Facebook Social Media Platforms (***SRS 1.2***,** *SRS 2.3***):** Publishing or analytics integration with Instagram, X/Twitter, LinkedIn, TikTok, or YouTube is out of scope; DASIGConnect is scoped exclusively to the official DASIG Facebook Page.
2. **Automated Publishing of Mixed-Media (Image + Video) Posts (***SRS 1.2***,** *UC-1.7 A7***,** *UC-3.2 A6***):**

Due to structural Facebook Graph API constraints, combining photos and videos in a single automated feed call is not supported. Only the detection of mixed-media submissions and their routing to *PUBLISH_FAILED* for Manual Publishing Fallback is tested.

3. **AI Classification, Vector Embedding, and Watermarking for Video Assets (***UC-2.1***,** *UC-2.5***,** *GR-T7***):** Video uploads (*MP4*, *MOV*, *WebM*) bypass Claude/Gemini image classification, Voyage AI vector embedding, and Java2D watermarking by design, transitioning immediately to *READY* and publishing unwatermarked.
4. **Generative AI Image Creation & Autonomous Content Moderation (***SRS 1.2***):** AI tools in DASIGConnect are strictly advisory; the system does not generate synthetic images or autonomously approve/publish Contributor posts without human review.
5. **Public Facebook Post Visibility Under Live Mode / Meta Business Verification (SRS 1.2,** *UC-3.2 A4***):** During development and capstone evaluation, the Meta Developer Application operates in **Development Mode** (posts visible to app role holders). Transitioning to Live Mode requires organizational Meta Business Verification by DOST/DASIG and is outside software testing scope.
6. **Contributor-Initiated Guard-Rail Override Requests (***SRS 3.3.1.2***):** Hard guard-rail violations (*GR-H1..H3*) strictly block Contributor slot selection and submission; only Administrators can override a hard guard rail during their own actions.
7. **Self-Service Trash / Restore UI for Soft-Deleted Media Assets (***UC-2.3 A3***):** The 30-day soft-delete window (*MediaAssetRetentionPurgeJob*) is an internal safety buffer prior to permanent R2/DB purge, not a user-facing recycle bin or self-service undo screen.
8. **Per-Channel and Per-Event Notification Opt-Out Toggles (***BR-NOT-04***):** Granular notification preference toggles are out of scope for the pilot; all in-app and email triggers are mandatory for their target roles, with only Facebook Messenger delivery operating as an individual opt-in via account linking (*UC-1.4 A8*).
9. **Initial Seeded Admin Provisioning & Unreachable Admin Owner Recovery (***UC-1.1 Preconditions & A11***):** The initial Admin Owner account is seeded at deployment, and recovery from an unreachable Admin Owner is an organizational governance procedure outside application scope.
10. **Internal Uptime & SLA of Third-Party Cloud Providers:** Internal availability of Meta's Graph API servers, Anthropic's Claude inference cluster, Voyage AI's embedding cluster, Supabase's managed PostgreSQL infrastructure, and Cloudflare R2's global storage network are not tested; only DASIGConnect's resilience, timeouts, retries, and graceful degradation when calling these interfaces are tested.

2.3. Testing Tools and Environment

2.3.1. Hardware Environment

- **Development & Test Execution Workstation:**
  - **Processor:** Multi-core x64 CPU (Intel Core i5/i7 or AMD Ryzen 5/7, ≥4 cores)
  - **Memory:** 16 GB RAM minimum
  - **Storage:** NVMe SSD with ≥20 GB free disk space
  - **Network:** Broadband internet connection (for Supabase, Cloudflare R2, and external API sandbox connectivity)
  - **Display:** Desktop (*1920×1080* / ≥1280px) and emulated responsive mobile/tablet viewports (≤900px, <*768px*)

2.3.2. Software & Runtime Environment

Layer / Category	Tool / Technology	Version / Configuration	Purpose in Testing
Operating System	Microsoft Windows 11 / Linux (Railway/Render container)	64-bit	Local test execution and cloud deployment runtime (*java.awt.headless=true* for server-side Java2D watermarking).
Backend Runtime	Java SE Development Kit (JDK) & Spring Boot	Java 21, Spring Boot *4.0.6*, Maven Wrapper (*./mvnw*)	Execution of REST API server, business services, Spring Security RBAC, and 16 *@Scheduled* cron jobs.
Backend Unit & Slice Testing	JUnit 5 (Jupiter), Mockito, AssertJ, Spring Boot Test (*MockMvc*)	*spring-boot-starter-test*, *spring-boot-starter-security-test*, *spring-boot-starter-webmvc-test*	Automated unit testing of services (*SubmissionServiceTest*, *ValidationServiceTest*, *UserServiceTest*, *MediaAssetServiceTest*, etc.) and HTTP/RBAC controller slices (*@WebMvcTest*).
In-Memory / Relational Test DB	H2 Database & Supabase PostgreSQL + *pgvector*	H2 (test scope), PostgreSQL 15+ (*VECTOR(1024)*, GiST, RLS), Flyway (*V1–V104+*), HikariCP (max pool size *5*)	Schema migration verification, repository testing, GiST slot-conflict exclusion testing, and 4-layer tenant isolation verification.
Frontend Runtime & Build	Node.js, React, TypeScript, Vite	Node.js *18+*, React *19.2.6*, TypeScript *~6.0.2*, Vite *^8.0.12*	Compilation (*tsc -b && vite build --mode test*) and local preview server (*http://127.0.0.1:4173* / *http://localhost:5173*).
Frontend E2E & Performance Testing	Playwright Test & Headless Chromium	*@playwright/test ^1.63.0*, *scripts/run-playwright.mjs*, *scripts/check-bundle-budgets.mjs*	End-to-end UI workflow testing, network request deduplication, SSE stream verification, and bundle budget enforcement (*npm run test:e2e*, *npm run test:performance*, *npm run check:bundle*).
Static Analysis & Linting	ESLint & TypeScript Compiler	ESLint *^10.3.0*, *typescript-eslint ^8.59.2*	Static type checking and code quality enforcement (*npm run lint*).
Object Storage & External Sandboxes	Cloudflare R2, Meta Graph API *v25.0*, Claude Vision, Voyage AI, SMTP	AWS SDK S3 *2.28.29*, Meta Development Mode Page, *claude-haiku-4-5*, *voyage-4-lite*	Presigned media upload/proxy streaming, 2-step photo & single-call video publishing verification, AI caption/proofread/embedding tests, and email delivery logging.

3. Test Cases

3.1. Case-1: Administrator Account Management & 3-Admin Cap Enforcement (UC-1.1)

3.1.1. Purpose

Verify that an active Administrator can invite a new Administrator using a 72-hour single-use SHA-256 hashed invitation token, that the uniform **3-Admin cap** (*ACTIVE* Admins + pending Admin invitations + pending Admin promotions ≤ 3) is strictly enforced across invitation, activation, promotion, and reactivation gates (*UC-1.1 Main Flow & A2*), and that Admin Owner authority is required to deactivate, reactivate, delete, or erase an existing Administrator account (*UC-1.1 A3–A6*).

3.1.2. Inputs

- **Authenticated Actor 1:** Admin Owner (*administrator@dasigconnect.com*, *role = admin*, *isAdminOwner = true*, *state = ACTIVE*).
- **Authenticated Actor 2:** Peer Administrator (*admin.second@dasigconnect.com*, *role = admin*, *isAdminOwner = false*, *state = ACTIVE*).
- **New Admin Invitee Email:** *admin.third@dasigconnect.com* (takes slot 3 of 3) and *admin.fourth@dasigconnect.com* (exceeds cap of 3).
- **Activation Payload:** Raw 64-character hex token from email, password *"DasigAdmin#2026!Secure"*.

3.1.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Inviting *admin.third@dasigconnect.com* succeeds (*HTTP 200/201*), creating a *PENDING* user with *institution_id = NULL* and storing only the SHA-256 token hash.
  2. Attempting to invite *admin.fourth@dasigconnect.com* while 2 active Admins and 1 pending Admin invite exist is rejected (*HTTP 409/422*) with an Admin limit reached error (*UC-1.1 A2*).
  3. Activating *admin.third@dasigconnect.com* re-validates the cap, transitions the account to *ACTIVE*, and logs *INVITATION_ACCEPTED*.
  4. Peer Administrator (*isAdminOwner = false*) attempting to deactivate *admin.third@dasigconnect.com* is rejected with *HTTP 403 Forbidden*.
  5. Admin Owner (*isAdminOwner = true*) deactivating *admin.third@dasigconnect.com* succeeds, transitions status to *INACTIVE*, increments *session_version* to revoke all active JWT sessions, and records *USER_STATUS_UPDATED*.
  6. Attempting to re-invite an *INACTIVE* Admin is rejected (*HTTP 409 Conflict*), directing the actor to use Reactivation (*UC-1.1 A4*).
- **Pass/Fail Criteria: PASS** if all 6 assertions hold and audit log entries match *UC-1.1* postconditions; **FAIL** otherwise.

3.1.4. Test Procedure

1. Authenticate as *administrator@dasigconnect.com* and navigate to **Admin Management**.
2. Submit **Invite Admin** for *admin.third@dasigconnect.com*; verify *PENDING* record creation and hashed token persistence.
3. Submit **Invite Admin** for *admin.fourth@dasigconnect.com*; verify the 3-Admin cap blocks the request (*UC-1.1 A2*).
4. Complete the activation link flow for *admin.third@dasigconnect.com* with a valid password; verify transition to *ACTIVE*.
5. Authenticate as Peer Admin (*admin.second@dasigconnect.com*) and attempt *Deactivate* on *admin.third@dasigconnect.com*; verify *403 Forbidden*.
6. As Admin Owner, execute *Deactivate* on *admin.third@dasigconnect.com*; verify status becomes *INACTIVE*, sessions are revoked, and re-inviting *admin.third@dasigconnect.com* is blocked in favor of

Reactivate.

3.2. Case-2: Admin Promotion Proposal & Admin Owner Transfer Governance (UC-1.1 A7, A8)

3.2.1. Purpose

Verify the two-step propose/confirm workflow for promoting an existing Moderator or Contributor to Administrator (UC-1.1 A7) and the Admin Owner transfer workflow (UC-1.1 A8) restricted strictly to an existing active Administrator target (preventing demotion of the outgoing Owner).

3.2.2. Inputs

- **Proposing Actor:** Admin Owner (*administrator@dasigconnect.com*, *isAdminOwner* = *true*).
- **Target 1 (Promotion):** Active Moderator (*moderator@dasigconnect.com*, *role* = *moderator*, *institution_id* = *NULL*).
- **Target 2 (Admin-to-Admin Owner Transfer):** Active Peer Admin (*admin.peer@dasig.org*, *role* = *admin*, *isAdminOwner* = *false*)

3.2.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Proposing *moderator@dasigconnect.com* for Admin promotion reserves a slot against the 3-Admin cap, creates a pending 72-hour promotion request without immediately changing *moderator*'s role, and logs *ADMIN_PROMOTION_REQUESTED*.
  2. When *moderator@dasigconnect.com* confirms the promotion, the system re-checks the 3-Admin cap, promotes the account to *admin* (*isAdminOwner* = *false*), revokes active sessions, and logs *ADMIN_PROMOTION_CONFIRMED*.
  3. When Admin Owner initiates an **Admin Owner Transfer** to an existing Administrator (*admin.second@dasigconnect.com*), initiation logs *ADMIN_TRANSFER_REQUESTED* with a 24-hour expiry; upon confirmation by *admin.second@dasigconnect.com*, *admin.second* becomes *admin* with *isAdminOwner* = *true*, the outgoing Owner relinquishes *isAdminOwner* = *false* while remaining an *admin*, outgoing active sessions are revoked, and *ADMIN_TRANSFER_CONFIRMED* is logged.
- **Pass/Fail Criteria: PASS** if promotion propose/confirm/decline/cancel, and Admin-to-Admin Owner transfer succeed per specification; **FAIL** if Admin-target Owner transfer confirmation fails with a role-gating error.

3.2.4. Test Procedure

1. As Admin Owner (with < *3* total occupied Admin slots), propose promoting *moderator@dasigconnect.com* to Admin.
2. Verify *ADMIN_PROMOTION_REQUESTED* is written and a second concurrent promotion proposal for *moderator@dasigconnect.com* is rejected.
3. Log in as *moderator@dasigconnect.com*, confirm the promotion, and verify role elevation to *admin* and session invalidation.
4. Test Admin Owner transfer initiation to an existing Administrator target (*admin.second@dasigconnect.com*) and confirm via `POST /api/v1/users/admin-transfer/confirm`; verify *admin.second@dasigconnect.com* becomes Admin Owner while outgoing Owner remains Administrator.

3.3. Case-3: Institution Lifecycle & Protected Default Institution Safeguards (UC-1.2)

3.3.1. Purpose

Verify institution creation, duplicate detection, Contributor reassignment, deactivation gating, permanent deletion rules, read-only Moderator visibility, and unconditional protection of the seeded default institution

"DASIG Central Visayas" (*UC-1.2 Main Flow & A1–A8*).

3.3.2. Inputs

- **Actors:** Administrator (*administrator@dasigconnect.com*) and Moderator (*moderator@dasigconnect.com*).
- **New Institution Payload:** Name *"Cebu Institute of Technology - University"*, Code *"CITU"*, Domain *"cit.edu"*.
- **Protected Institution:** *"DASIG Central Visayas"*.

3.3.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Moderator viewing **Institution Management** sees all institutions in read-only mode; direct API attempts by Moderator to create/edit/deactivate/delete an institution return *HTTP 403 Forbidden* (*UC-1.2 A7*).
  2. Administrator creating *"CITU"* succeeds (*ACTIVE* status); attempting to create another institution with duplicate name, code, or domain returns *HTTP 409 Conflict* (*UC-1.2 A5*).
  3. Attempting to deactivate *"CITU"* while an active Contributor is assigned is blocked (*HTTP 409/422*) until the Contributor is reassigned to another *ACTIVE* institution (*UC-1.2 A2, A4*).
  4. Attempting to deactivate or permanently delete *"DASIG Central Visayas"* is unconditionally rejected (*UC-1.2 A6, A8*).
  5. Permanent deletion (*UC-1.2 A8*) of an eligible empty institution succeeds only when the exact institution code is confirmed and no active/pending Contributors, submissions of any state, or non-deleted media assets exist; logs *INSTITUTION_DELETED*.
- **Pass/Fail Criteria: PASS** if all creation, duplicate, reassignment, protection, and permanent deletion guards behave as specified; **FAIL** otherwise.

3.3.4. Test Procedure

1. Log in as Moderator, open **Institution Management**, verify action controls are hidden, and send a *POST /api/v1/institutions* request; assert *403 Forbidden*.
2. Log in as Administrator, create institution *"CITU"*, then repeat with the same code *"CITU"*; assert duplicate rejection.
3. Assign a Contributor to *"CITU"* and attempt to deactivate *"CITU"*; assert rejection. Reassign the Contributor to another active institution (verifying historical submissions retain their original institution attribution), then deactivate *"CITU"*.
4. Attempt to deactivate and delete *"DASIG Central Visayas"*; assert both actions are blocked.
5. Create a footprint-free test institution *"TEST-HEI"*, confirm its code in the Permanent Delete dialog, and verify removal and *INSTITUTION_DELETED* audit logging.

3.4. Case-4: Contributor Account Provisioning & Footprint-Aware Deletion (UC-1.3)

3.4.1. Purpose

Verify Contributor invitation by both Administrators and Moderators, institution-status validation (*BR-ACC-01*), Moderator ownership scoping on sent invitations (*UC-1.3 A5, A6*), footprint-aware account deletion (*USER_REMOVED* vs. hard-delete), and Admin Owner Right-to-be-Forgotten erasure (*UC-1.3 A7*).

3.4.2. Inputs

- **Actors:** Administrator (*administrator@dasigconnect.com*), Moderator 1 (moderator@dasigconnect.com), Moderator 2 (moderator2@dasigconnect.com).
- **Invitees:** *contrib.citu@cit.edu* (invited by Moderator 1 to *"CITU"*), *contrib.fresh@cit.edu* (no historical footprint).

3.4.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Moderator 1 can invite *contrib.citu@cit.edu* to any *ACTIVE* institution with role restricted to *contributor*; inviting to an *INACTIVE* institution is rejected (*BR-ACC-01*).
  2. Moderator 2 attempting to resend, cancel, or delete Moderator 1's pending invitation is rejected (*HTTP 403 Forbidden*), whereas Moderator 1 succeeds (*UC-1.3 A5, A6*).
  3. Once *contrib.citu@cit.edu* activates their account and creates a submission or uploads media (establishing a historical footprint), deactivating (*INACTIVE*) and then deleting the account does **not** hard-delete the database row; instead, it persists as an anonymized inactive record with a *USER_REMOVED* audit entry (*UC-1.3 A6*).
  4. Deleting *contrib.fresh@cit.edu* (cancelled/expired with zero footprint) hard-deletes the row.
  5. Admin Owner executing **Erase** (*UC-1.3 A7*) on an *INACTIVE* Contributor anonymizes PII in place and logs *USER_ANONYMIZED*.
- **Pass/Fail Criteria: PASS** if Moderator scoping, institution-status checks, footprint-preserving soft-removal, and PII erasure all match *UC-1.3*; **FAIL** otherwise.

3.4.4. Test Procedure

1. Authenticate as Moderator 1 and invite *contrib.citu@cit.edu* to *"CITU"*.
2. Authenticate as Moderator 2 and attempt to cancel or resend *contrib.citu@cit.edu*'s invite; assert *403 Forbidden*.
3. Activate *contrib.citu@cit.edu*, create a draft submission, then as Administrator deactivate and delete *contrib.citu@cit.edu*; verify the row remains in the database anonymized at rest with *USER_REMOVED* logged.
4. Invite and cancel *contrib.fresh@cit.edu* (zero footprint), then delete the account; verify hard deletion.
5. As Admin Owner, invoke *Erase* on an *INACTIVE* account and verify *USER_ANONYMIZED* audit logging.

3.5. Case-5: Authentication, Account Lockout, Password Policy & DB Session Revocation (UC-1.4)

3.5.1. Purpose

Verify login authentication, JWT claims scoping, 5-attempt / 15-minute account lockout (*UC-1.4 A1*), password complexity enforcement (*UC-1.4 A5, A6*), single-device logout (*revoked_tokens*, *UC-1.4 A10*) vs. account-wide session invalidation (*session_version*, *UC-1.4 A5*), session-expiry warning banner/modal (*UC-1.4 A7*), and 10-minute Facebook Messenger link-code generation (*UC-1.4 A8*).

3.5.2. Inputs

- **Accounts:** Active Contributor (*contrib.citu@cit.edu*), Active Moderator (*mod.a@dasigconnect.com*), *PENDING* user, *INACTIVE* user.
- **Passwords Tested:**
  - Weak passwords: *"Short1!"* (*<12* chars), *"DasigWelcome123!"* (contains forbidden fragment *"dasig"*/*"welcome"*/*"123"*), *"contrib.citu@cit.eduA1!"* (contains email).
  - Valid password: *"Str0ng#Pass_987!"* (≥12 chars, upper, lower, digit, special, no whitespace or forbidden fragments).

3.5.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Valid login returns an 8-hour JWT: Contributor token includes *institution_id*; Moderator and Admin tokens have no *institution_id* claim (*UC-1.4 Main Flow*).
  2. Five consecutive failed login attempts lock the account for 15 minutes (*HTTP 423 Locked*, *UC-1.4 A1*).

3. Login attempts by *PENDING*, *EXPIRED*, or *INACTIVE* accounts are rejected with state-specific guidance (*UC-1.4 A2–A4*).
4. Password reset (*UC-1.4 A5*) rejects weak/fragment-containing passwords, accepts *"Str0ng#Pass_987!"*, increments *users.session_version*, and invalidates all existing JWTs across all devices.
5. Authenticated password change (*UC-1.4 A6*) updates the password without incrementing *session_version*, keeping other active sessions valid.
6. Single-session logout (*POST /api/v1/auth/logout*, *UC-1.4 A10*) inserts the SHA-256 token hash into *revoked_tokens*, rejecting subsequent requests with that token while leaving other device sessions active.
7. Generating a Facebook Messenger link code (*UC-1.4 A8*) succeeds for Moderator and Administrator (10-minute code) and returns *HTTP 403 Forbidden* for Contributor.

- **Pass/Fail Criteria: PASS** if all authentication, lockout, password policy, DB-persisted token revocation, and Messenger link-code role checks pass; **FAIL** otherwise.

3.5.4. Test Procedure

1. Attempt 5 failed logins for *contrib.citu@cit.edu*; verify the 5th attempt triggers a 15-minute lockout (*423 Locked*).
2. Log in with valid Contributor and Moderator credentials on two separate sessions (Session 1 and Session 2); inspect decoded JWT payloads for *institution_id* and *session_version*.
3. Call *POST /api/v1/auth/logout* on Session 1; verify Session 1's JWT is rejected by *JWTService.extractClaims* (*revoked_tokens*) while Session 2 remains valid.
4. Execute unauthenticated Password Reset (*A5*) testing both invalid fragments (*"dasig"*, *"123"*) and valid password *"Str0ng#Pass_987!"*; verify Session 2 is immediately invalidated via *session_version* mismatch.
5. As Contributor and Moderator, test *POST /api/v1/auth/messenger-link-code* (or Messenger connection endpoint); verify Moderator receives a 10-minute code and Contributor receives *403*.

3.6. Case-6: Three-Step Post Composer, Readiness Checklist & Post Templates (UC-1.5)

3.6.1. Purpose

Verify the 3-step post composer wizard (*Step 1: Media → Step 2: Post Details → Step 3: Organize & Schedule*), the live 0–100 Readiness Checklist (75% Required, 25% Recommended), built-in and personal post templates, **Posting As** institution switching for Moderators/Administrators, autosave, and the network-wide *guardrails_enforced* runtime toggle (*UC-1.5 Main Flow & A1–A9*).

3.6.2. Inputs

- **Actor:** Contributor (*contrib.citu@cit.edu*) and Administrator (*administrator@dasigconnect.com*).
- **Draft Content:**
  - Media: 1 valid JPEG image (*event-banner.jpg*, *1.8 MB*).
  - Event Title: *"CIT-U Regional Innovation Summit 2026"*, Event Date: *"2026-10-20"*.
  - Template: Built-in *"Event Announcement"*, plus custom personal template *"My Lab Spotlight"*.
  - Schedule: Future slot in *Asia/Manila* time (*Set a Schedule* vs. *Live Event Fast-Track*).

3.6.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Forward step navigation requires the current step's Required readiness items; backward navigation is always permitted and preserves entered state (*UC-1.5 A8*).
  2. **Save as Draft** works on any step even with unmet Required items (*UC-1.5 A6*); once saved, subsequent edits autosave silently after a short debounce.

3. Opening the **Templates** drawer/bottom-sheet on Step 2 and applying *"Event Announcement"* populates the caption placeholders and tags; saving a custom template via **Save current caption** scopes it to *ownerUserId* and rejects duplicate template names for the same owner with *HTTP 409 Conflict*.
4. For Administrator/Moderator, changing the **Posting As** institution on a saved draft prompts confirmation, clears the previously selected preferred schedule, preserves attached media, and re-homes the draft on save.
5. When *guardrails_enforced* = *false* (toggled in **Settings** → **Page**), Standard drafts still require a future scheduled date and create a *SlotReservation*, but skip the *8:00 AM–8:00 PM* window, ±*30-min* spacing, ≤6/day, and ≥2h lead-time checks (*UC-1.5*).
6. Deleting a draft or *rejected* submission (*UC-1.5 A2*) cascades cleanup of *validation_logs*/*review_locks* (*V103*) and writes *SUBMISSION_DELETED*.

- **Pass/Fail Criteria: PASS** if wizard gating, Readiness score calculation (0–100), template CRUD, Posting As re-homing, guard-rail toggle behavior, and draft/rejected deletion all succeed; **FAIL** otherwise.

3.6.4. Test Procedure

1. Open **New Post** as Contributor; verify Step 1 blocks advancing to Step 2 until at least one valid media asset is attached.
2. Attach *event-banner.jpg*, advance to Step 2, open the **Templates** panel, apply *"Event Announcement"*, and save a personal template *"My Lab Spotlight"*; attempt to save a second template with the same name and assert *409 Conflict*.
3. Verify the **Readiness Checklist** updates live as Required items (75% weight) and Recommended items (25% weight: comfortable caption length, ≥1 hashtag, ≥1 media tag, ≥1 per-media caption, template used) are completed.
4. Click **Submit** when all Required items are met but one Recommended item is missing; verify the non-blocking confirmation dialog displays the readiness score (*UC-1.5 A7*).
5. As Administrator, toggle **Scheduling Guard Rails** OFF in **Settings** → **Page** and verify a Standard draft can be scheduled at *9:30 PM* (outside *8 AM–8 PM*) as long as the timestamp is in the future; re-enable Guard Rails afterward.

3.7. Case-7: AI Template Synthesis from Top-Performing Facebook Posts (UC-1.5 A10)

3.7.1. Purpose

Verify the **Generate from Top Posts** feature (*POST /api/v1/ai/templates/from-top-posts*, *UC-1.5 A10*), including engagement ranking (*reactions* + *2×comments* + *3×shares*), minimum engaged-post threshold (≥8 posts), fallback to local *submission_engagement_metrics*, and per-user rate limiting (*10 requests/hour*).

3.7.2. Inputs

- **Actor:** Authenticated Contributor or Administrator in Step 2 (**Post Details**) → **Templates** panel.
- **Dataset:**
  - Scenario A: < *8* engaged posts available.
  - Scenario B: ≥ 11 engaged posts available (top 6 + bottom 3 contrast sent to Claude).
  - Scenario C: 11th request within 1 rolling hour by the same user.

3.7.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. In Scenario A (< *8* posts with engagement), the endpoint returns *available: false* with an explanatory reason without calling Claude (*UC-1.5 A10a*).
  2. In Scenario B (≥ 8 engaged posts), the system ranks the last 100 posts by *reactions* + *2×comments* + *3×shares*, sends the top 6 (and bottom 3 as contrast) to Claude outside any DB

transaction, and returns a draft template containing *name*, placeholder *caption* (≤3,000 chars), up to 4 *tags*, and 2–4 *"why this works"* insights without auto-saving server-side.

3. Clicking **Save Template** persists the template via *POST /api/v1/post-templates* under category *"AI Suggested"* without overwriting the composer's active caption.
4. In Scenario C, exceeding 10 requests/hour returns *HTTP 429 Too Many Requests* (*UC-1.5 A10c*).

- **Pass/Fail Criteria: PASS** if minimum sample guard, ranking formula, non-destructive save, and *10/hr* rate limit all function per *UC-1.5 A10*; **FAIL** otherwise.

3.7.4. Test Procedure

1. Open the **Post Templates** panel on an editable draft and click **Generate from Top Posts** with < *8* engaged posts seeded; verify *available: false* and zero Claude API invocations.
2. Seed *12* published posts with varying reactions, comments, and shares; trigger **Generate from Top Posts** and verify the returned placeholder template, insights, and source post previews.
3. Click **Save Template** and verify it appears under **My templates** without replacing the current draft caption.
4. Trigger 10 consecutive requests within an hour and verify the 11th returns *429*.

3.8. Case-8: AI Caption Generation, Writing Check & Fancy Unicode Text Styling (UC-1.6)

3.8.1. Purpose

Verify on-demand AI caption generation (*POST /api/v1/ai/caption*), prompt length & word-count validation, image downscaling (*>5 MB*), *30/hour* rate limiting, interaction telemetry logging (*use* / *use_then_edited* / *dismiss* / *re_generate*), advisory AI proofreading (**Check writing**, *POST /api/v1/ai/proofread*), *3,000*-character caption cap, and Fancy Unicode text styling (*UC-1.6 Main Flow & A1–A7*).

3.8.2. Inputs

- **Saved Draft:** Draft ID *sub-101* with 1 attached image (*6.2 MB* high-res JPEG).
- **AI Caption Prompt:** *"Highlight student robotics pitch winners in 120 words"* (≤280 chars), Tone: *energetic*.
- **Invalid Prompts:** 285-character prompt (*>280* chars); prompt requesting *"Write a 2500 word essay"* (*>2,000* words).
- **Fancy Text Selection:** Substring *"DASIG Innovation"* styled as *Bold Sans-Serif* then *Italic Serif* then *Plain*.

3.8.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Images *>5 MB* are downscaled in-memory (*70% → 50% → 35% → 25% → 15%*) before base64 submission to Claude (*SRS 3.1.2*, *UC-1.6*).
  2. Valid caption request returns 1 caption in the selected tone within the 30-second timeout; inserting and subsequently editing the caption logs *use_then_edited* (*BR-AI-03*).
  3. Prompt *>280* characters disables the Generate button client-side and returns *HTTP 400* server-side (*UC-1.6 A5*); requested word count *>2,000* returns *HTTP 400*.
  4. Exceeding 30 caption requests/hour returns *HTTP 429* with reset headers (*BR-AI-07*); AI timeout (*504*) or unavailability (*503*) degrades gracefully and auto-recovers to idle after *~5s* without blocking drafting (*UC-1.6 A1, A2*).
  5. Clicking **Check writing** invokes *POST /api/v1/ai/proofread* and displays advisory grammar/clarity suggestions that can be applied or dismissed.
  6. **Fancy Text** transforms *"DASIG Innovation"* into Unicode Mathematical Alphanumeric symbols, reverts to plain before applying a new style, leaves emoji/punctuation/accents untouched (*UC-1.6 A7*), and reverts cleanly to standard ASCII when *Plain* is selected (*UC-1.6 A6*).

7. Captions exceeding *3,000* code points are trimmed on paste and rejected server-side (*400* on save, *422* on submit).

- **Pass/Fail Criteria: PASS** if all 7 expectations hold; **FAIL** otherwise.

3.8.4. Test Procedure

1. Open saved draft *sub-101* on Step 2 (**Post Details**) and click **Suggest Caption**.
2. Test a 285-char prompt and a *"2500 words"* prompt; verify client/server *400* validation.
3. Submit the valid *energetic* prompt with the *6.2 MB* image attached; verify downscaling, caption insertion, subsequent manual edit, and telemetry log entry (*use_then_edited*).
4. Click **Check writing** and verify proofreading suggestions render non-blockingly.
5. Highlight *"DASIG Innovation"*, open **Fancy text**, apply *Bold Sans-Serif*, switch to *Italic Serif*, and revert via *Plain*.
6. Paste a *3,100*-character string into the caption field; verify client-side trimming to *3,000* code points and backend enforcement (*SubmissionService.MAX_CAPTION_CHARS = 3000*).

3.9. Case-9: Media Attachment Rules, AI-Suggested Media & Album Auto-Match (UC-1.7)

3.9.1. Purpose

Verify device upload format/magic-byte/size validation (*7* formats, ≤50 MB, ≤10 assets/post), presigned R2 upload flow, per-asset caption (≤500 chars) and photo-only *Skip watermark* toggle, **AI-Suggested Media** (≥10 chars context, ≥0.40 similarity floor, top 8), and **Album Auto-Match** (≥0.55 confident auto-apply, *0.32–0.54* ambiguous top-3 candidates, *<0.32* no-match) (*UC-1.7 Main Flow & A1–A10*).

3.9.2. Inputs

- **Files Tested:**
  - Valid: *photo1.jpg* (*2 MB*), *clip.mp4* (*15 MB*).
  - Invalid: *malware.exe* renamed to *photo.jpg* (invalid magic bytes), *huge-video.mp4* (*55 MB > 50 MB*), *document.pdf* (unsupported format), 11th media attachment.
- **Event Context for AI Suggestions:** Title *"Regional Robotics & AI Symposium"*, Caption *"Students showcasing autonomous drones"*.

3.9.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. *document.pdf*, *huge-video.mp4* (*>50 MB*), and disguised *photo.jpg* (magic-byte mismatch) are rejected (*HTTP 400*) with zero asset records created (*UC-1.7 A1, A2*, *BR-SUB-01*).
  2. Attempting to attach an 11th asset is blocked by the 10-asset-per-submission ceiling (*BR-SUB-02*).
  3. Attaching both *photo1.jpg* and *clip.mp4* is permitted during drafting with a mixed-media manual-publishing warning (*UC-1.7 A7*).
  4. In the Media Caption modal, per-asset captions are capped at *500* characters, and *"Skip watermark for this image"* is shown for *photo1.jpg* but hidden for *clip.mp4*.
  5. Once combined event text reaches ≥10 characters, **AI-Suggested Media** queries institution-scoped pgvector embeddings and returns up to 8 assets scoring ≥0.40 (falling back to metadata suggestions if the embedding service fails, *UC-1.7 A5, A6*).
  6. On Step 3, clicking **Auto-Match** (*POST /api/v1/ai/submissions/{id}/suggest-album*) auto-applies the root album with an *"AI-matched"* badge when score ≥0.55, lists up to 3 ranked candidates inline when 0.32 ≤ score < 0.55 (*UC-1.7 A8*), and leaves the field for manual entry with a notice when *<0.32* (*UC-1.7 A8a*).
- **Pass/Fail Criteria: PASS** if file validation, 10-asset cap, per-asset caption/watermark controls, AI media suggestions, and tiered Album Auto-Match all pass; **FAIL** otherwise.

3.9.4. Test Procedure

1. In Step 1 of the composer, attempt to upload *document.pdf*, a *55 MB* MP4, and a text file renamed to *.png*; verify each is rejected.
2. Upload *photo1.jpg*, open the Media Caption modal, enter a 500-char caption, and check *"Skip watermark for this image"*.
3. Upload *clip.mp4* to the same draft; verify the mixed-media inline warning appears and the watermark checkbox is absent on the video asset.
4. Enter <*10* characters in Step 2 and verify AI-Suggested Media stays hidden; enter the full robotics title/caption (≥10 chars) and verify up to 8 ranked suggestions (score ≥ 0.40) appear.
5. In Step 3, test **Auto-Match** against albums yielding confident (≥0.55), ambiguous (*0.32–0.54*), and low (<*0.32*) similarity scores; verify the combobox behavior for each tier.

3.10. Case-10: Peak-Hour Engagement Recommendations & Guard-Rail Filtering (UC-1.8)

3.10.1. Purpose

Verify peak-hour posting slot recommendations (*UC-1.8*), including historical Facebook engagement bucketing (last 100 posts, ≥20 samples), fallback to Tuesday/Wednesday/Thursday 6:00 PM (*Asia/Manila*) when <*20* samples exist (*UC-1.8 A1*), exclusion of hard-blocked guard-rail slots (*UC-1.8 A2*), silent hiding on Graph API failure (*UC-1.8 A3*), and inline soft-warning rendering (*UC-1.8 A4*).

3.10.2. Inputs

- **Scenario 1 (Historical):** *25* historical published posts with reactions, comments, and shares within *8:00 AM–8:00 PM* PHT; an existing scheduled post at *Tuesday 6:00 PM* (creating a ±*30-min GR-H1* conflict).
- **Scenario 2 (Fallback):** Only *5* historical posts (< *20* samples).
- **Scenario 3 (Manual Soft Warning):** Actor manually picks a valid slot on a calendar day that already has *6* scheduled posts (*GR-S2* Daily Volume Cap) or when the institution already has *3* active scheduled posts (*GR-S1*).

3.10.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. In Scenario 1, the system buckets the last 100 posts by *(dayOfWeek, hour)* in *Asia/Manila*, excludes the hard-blocked *Tuesday 6:00 PM* slot (*UC-1.8 A2*), and displays up to 3 conflict-free recommended future slots over the next 30 days with a *"Top pick"* badge.
  2. In Scenario 2 (<*20* samples), the system falls back to **Tuesday / Wednesday / Thursday 6:00 PM** labeled as *"Best-practice guidance"* (*UC-1.8 A1*).
  3. For a **Live Event Fast-Track** draft, the scheduling and recommendation panel is completely hidden (*UC-1.8 Preconditions*).
  4. In Scenario 3, manually selecting a slot that triggers *GR-S1* or *GR-S2* displays an inline soft warning beneath the date/time fields while allowing the actor to proceed (*UC-1.8 A4*).
- **Pass/Fail Criteria: PASS** if historical ranking, <*20*-sample fallback, hard-block filtering, Fast-Track hiding, and soft-warning inline display all match *UC-1.8*; **FAIL** otherwise.

3.10.4. Test Procedure

1. Open Step 3 (**Organize & Schedule**) on a Standard draft with <*20* historical posts; verify the Tue/Wed/Thu 6:00 PM fallback tiles and *"Best-practice guidance"* notice appear.
2. Seed *25* engaged historical posts with peak activity on Tuesdays at 6:00 PM, reserve a conflicting slot at next Tuesday 6:00 PM, and reload Step 3; verify next Tuesday 6:00 PM is excluded from the recommended tiles.
3. Switch the draft mode to **Live Event Fast-Track**; verify the scheduling panel disappears.
4. Switch back to **Set a Schedule** and manually enter a time on a day that exceeds the *GR-S1* or *GR-S2*

soft threshold; verify the inline warning renders and does not block saving.

3.11. Case-11: Draft Submission, Guard-Rail Re-Validation, Withdrawal & Fast-Track Prioritization (UC-1.9)

3.11.1. Purpose

Verify final server-side completeness validation at submission (*assertContentComplete*), submit-time guard-rail conflict handling (*409 Conflict* redirecting to Step 3, *UC-1.9 A4*), notification dispatch to reviewers (*T-01* vs. *T-11*), withdrawal from *pending* vs. locked *in_review* (*UC-1.9 A2*), resubmission from *needs_revision* or *rejected* (*UC-1.9 A3*), and Fast-Track top-of-queue prioritization (*UC-1.9 A5*).

3.11.2. Inputs

- **Standard Draft (*sub-std*):** Complete title, date, caption (≤3,000 chars), 1 media asset, album *"CIT-U Events"*, scheduled slot *T + 4 hours*.
- **Fast-Track Draft (*sub-ft*):** Complete fields, mode = *Live Event Fast-Track* (no scheduled slot).
- **Competing Submission:** Another user claims *T + 4 hours* (or within ±*30* minutes) moments before *sub-std* clicks confirm.

3.11.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. If a concurrent reservation claims a slot within ±*30* minutes of *sub-std* before submit completes, *POST /api/v1/submissions/{id}/submit* returns *HTTP 409 Conflict*, closes the confirmation modal, and redirects the composer to Step 3 (**Organize & Schedule**) with a prompt to pick a new time (*UC-1.9 A4*).
  2. With a valid conflict-free slot, submitting *sub-std* transitions status *draft → pending*, locks content from further editing, and dispatches notification *T-01*.
  3. Calling *POST /api/v1/submissions/{id}/withdraw* while *sub-std* is *pending* (and unlocked) transitions it back to *draft* and logs *SUBMISSION_WITHDRAWN*; once a reviewer opens *sub-std* (*ReviewLock* exists, status *in_review*), withdrawal is rejected with *HTTP 409 Conflict* (*UC-1.9 A2*).
  4. Submitting *sub-ft* (Fast-Track) skips slot validation, dispatches urgent notification *T-11* (in-app + email + Messenger), and sorts *sub-ft* to the top of the active Validation Queue (*fastTrack DESC, scheduledAt ASC NULLS LAST, submittedAt ASC*, *UC-1.9 A5*).
- **Pass/Fail Criteria: PASS** if completeness validation, *409* conflict step-redirect, withdrawal lock guard, and Fast-Track queue sorting all pass; **FAIL** otherwise.

3.11.4. Test Procedure

1. Open *sub-std* in the composer, simulate a concurrent slot reservation at the same time slot, and click **Submit for Approval**; verify *409 Conflict* returns the user to Step 3.
2. Select a free future slot (≥2h lead time, within *8 AM–8 PM*), submit *sub-std*, and verify transition to *pending*.
3. Click **Withdraw** while *sub-std* has no *ReviewLock*; verify transition back to *draft*. Re-submit *sub-std*, acquire a *ReviewLock* as a Moderator, and attempt **Withdraw** as the Contributor; assert *409 Conflict*.
4. Submit *sub-ft* in **Live Event Fast-Track** mode and inspect the Validation Queue (*Pending* tab); verify *sub-ft* is ordered above all non-Fast-Track submissions.

3.12. Case-12: Moderator Account Management & Lateral Role Changes (UC-1.10)

3.12.1. Purpose

Verify Administrator lifecycle management of network-wide (institutionless) Moderator accounts (*UC-1.10*

Main Flow & A1–A9) and immediate lateral role changes between *Contributor* and *Moderator* (*UC-1.10 A11*), including automatic release of active *ReviewLock* rows and session invalidation.

3.12.2. Inputs

- **Actor:** Administrator (*administrator@dasigconnect.com*).
- **Target Account:** Active Moderator *mod.active@dasigconnect.com* currently holding an active 15-minute *ReviewLock* on submission *sub-std*.
- **Target Institution for Demotion:** Active institution *"CITU"*.

3.12.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Inviting a Moderator creates a network-wide (*institution_id = NULL*) *PENDING* account with a 72-hour hashed token; upon activation, the user gains network-wide review and Contributor-invitation privileges (*UC-1.10 Main Flow*).
  2. Changing *mod.active@dasigconnect.com*'s role laterally from *Moderator → Contributor* (*UC-1.10 A11*) requires specifying an *ACTIVE* target institution (*"CITU"*), takes effect immediately with no confirmation step, increments *session_version* (forcing re-login), automatically releases *mod.active*'s active *ReviewLock* on *sub-std* (returning *sub-std* to *pending*), and writes *USER_ROLE_CHANGED*.
  3. Changing the account laterally from *Contributor → Moderator* clears *institution_id* (*NULL*), revokes active sessions, and restores network-wide Moderator privileges.
- **Pass/Fail Criteria: PASS** if Moderator provisioning, lateral *Contributor ↔ Moderator* transitions, session revocation, and automatic review-lock release all succeed; **FAIL** otherwise.

3.12.4. Test Procedure

1. As Administrator, invite *mod.active@dasigconnect.com* via **Moderator Management** and complete activation; verify *role = moderator* and *institution_id = NULL*.
2. As *mod.active@dasigconnect.com*, open *sub-std* in the Validation Queue to acquire an active *ReviewLock* (*status = in_review*).
3. As Administrator, change *mod.active@dasigconnect.com*'s role to *Contributor* assigned to *"CITU"*; verify *sub-std*'s lock is immediately released, *mod.active*'s JWT is invalidated, and *USER_ROLE_CHANGED* is logged.
4. Change *mod.active@dasigconnect.com* back to *Moderator* and verify *institution_id* is cleared to *NULL*.

3.13. Case-13: Standalone Media Library Uploads, Nested Albums & Manual Tag Floor (UC-2.1)

3.13.1. Purpose

Verify standalone Media Library uploads across all three roles, text-based Album Auto-Match (≥0.6 confident, ≥0.3 ambiguous), nested parent-child album creation and cycle-guarded re-parenting (*UC-2.1 A8*), the mandatory ≥1 actor-entered (*source: "manual"*) tag floor (*UC-2.1 A9*), cross-institution asset movement (*UC-2.1 A10*), and asynchronous image *PROCESSING → READY* transition vs. immediate video *READY* status.

3.13.2. Inputs

- **Upload Payload:** Image *summit-stage.png* (*3.1 MB*) and video *keynote-recap.mp4* (*18 MB*), manual tag *"summit2026"*, nested album path *"2026 Events / Q4 / Regional Summit"*.
- **Album Hierarchy:** Parent Album *A* → Child Album *B* → Grandchild Album *C*.

3.13.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Standalone upload without at least one manual tag or without an album is rejected (*HTTP 400*).

2. Specifying the nested folder path *"2026 Events / Q4 / Regional Summit"* creates any missing parent/child album segments in one operation; attempting to re-parent Album *A* under its own descendant Album *C* is blocked by the cycle guard (*UC-2.1 A8*).
3. On upload confirmation, *summit-stage.png* starts in *PROCESSING* status and transitions to *READY* once async AI classification and Voyage embedding complete; *keynote-recap.mp4* is marked *READY* immediately without queuing classification/embedding (*UC-2.1 Main Flow*).
4. Even after *summit-stage.png* receives 10 AI-generated tags (*source = "ai_generated"*), attempting to remove its last manual tag (*"summit2026"*, *source = "manual"*) is rejected (*HTTP 400*, remove × hidden in UI) (*UC-2.1 A9*).
5. Attempting to set an asset's *albumId* to *null* ("unfiled") is rejected (*HTTP 400*, *UC-2.1 A5*).

- **Pass/Fail Criteria: PASS** if nested album cycle prevention, *PROCESSING → READY* transitions, non-null album enforcement, and the *source: "manual"* tag floor all pass; **FAIL** otherwise.

3.13.4. Test Procedure

1. Open the **Media Library** and upload *summit-stage.png* and *keynote-recap.mp4* with tag *"summit2026"* into nested path *"2026 Events / Q4 / Regional Summit"*.
2. Verify *keynote-recap.mp4* is immediately *READY* while *summit-stage.png* shows *"Processing…"* and transitions to *READY* after async enrichment.
3. Open *summit-stage.png* in the **Asset Detail Panel** after AI tags populate; attempt to delete *"summit2026"* (the sole manual tag) via UI and *DELETE* API; assert *400 Bad Request*. Add a second manual tag *"citu"*, then delete *"summit2026"*; assert success.
4. Create album hierarchy *A → B → C* and attempt to move *A* inside *C*; assert cycle-guard rejection.

3.14. Case-14: Semantic Media Search, Keyword Fallback & Asset Detail Panel (UC-2.2)

3.14.1. Purpose

Verify Media Library browsing, meaning-based semantic search (*Voyage AI* query embedding + *pgvector* cosine similarity) with keyword fallback (*UC-2.2 Main Flow*), searchability of newly added manual tags (*UC-2.2 A3*), inline display-title editing (*display_title*), Network View institution filtering for Moderators/Administrators, and *Add to Existing Draft* (*UC-2.2 A4, A5*).

3.14.2. Inputs

- **Search Queries:**
  - Semantic query: *"students receiving trophy on stage"* (matches image description/embedding even without exact keyword overlap).
  - Custom manual tag query: *"custom-hackathon-tag-99"*.
- **Title Rename:** Click-to-edit asset title to *"CIT-U Grand Champions Trophy Photo"*.

3.14.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Meaning-based search embeds the query via Voyage AI (*voyage-4-lite*, *1024* dims) and returns ranked matching assets via pgvector; if the embedding service is unreachable, search falls back seamlessly to keyword matching across filename, *display_title*, asset code, uploader email, AI metadata, and tags (*UC-2.2 Main Flow*).
  2. Adding manual tag *"custom-hackathon-tag-99"* to an asset makes that asset immediately discoverable via keyword search (*UC-2.2 A3*).
  3. Editing the title in the **Asset Detail Panel** updates *display_title* (*POST /api/v1/media-assets/{id}/title* / *PATCH*) without altering the original *file_name* or R2 storage key (*UC-2.2 Main Flow*).
  4. **Use in New Social Media Post** and **Add to Existing Draft** are available to all three roles (*Contributor*, *Moderator*, *Administrator*); attaching multiple assets stops cleanly at the 10-asset-per-submission ceiling (*UC-2.2 A5*).

- **Pass/Fail Criteria: PASS** if semantic/keyword search, manual tag indexing, non-destructive display-title editing, and draft attachment all succeed; **FAIL** otherwise.

3.14.4. Test Procedure

1. Navigate to **Media Library**, enter semantic query *"students receiving trophy on stage"*, and verify relevant award photos rank at the top.
2. Open an asset's **Asset Detail Panel**, add manual tag *"custom-hackathon-tag-99"*, and search for *"custom-hackathon-tag-99"*; verify the asset is returned immediately.
3. Click the asset's title in the detail panel, rename it to *"CIT-U Grand Champions Trophy Photo"*, press Enter, and verify the updated title persists while *file_name* and the */api/v1/media-files/<key>* proxy URL remain unchanged.
4. Click **Add to Existing Draft** on a draft that already has 9 media attachments while selecting 2 library assets; verify the 10th attaches and the 11th triggers the 10-asset limit notice (*UC-2.2 A5*).

3.15. Case-15: Media History Tracking, Tiered Deletion Precedence & Duplicate Hash Detection (UC-2.3)

3.15.1. Purpose

Verify the **Used In** relationship history in the Asset Detail Panel (*UC-2.3 Main Flow*), the 3-tier asset deletion precedence (*UC-2.3 A1–A3*, *BR-MED-02*), placeholder rendering for deleted records, and SHA-256 content-hash duplicate detection upon upload (*UC-2.3 A4*).

3.15.2. Inputs

- **Asset 1 (***asset-active***):** Referenced in a *PENDING* or *SCHEDULED* submission.
- **Asset 2 (***asset-draft***):** Referenced only in a *DRAFT* or *NEEDS_REVISION* submission.
- **Asset 3 (***asset-terminal***):** Referenced only in a *PUBLISHED* or *REJECTED* submission (or unreferenced).
- **Duplicate Upload:** Re-uploading the exact byte-identical *summit-stage.png* within the same institution.

3.15.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Attempting to delete *asset-active* is hard-blocked (*HTTP 409 Conflict*) and displays the conflicting active submission (*UC-2.3 A1*).
  2. Attempting to delete *asset-draft* without confirmation returns a warning prompt; confirming with *force=true* soft-deletes the asset and surfaces a broken-reference warning on the affected draft (*UC-2.3 A2*).
  3. Deleting *asset-terminal* succeeds with standard confirmation, marks the asset soft-deleted for the 30-day retention buffer (*MediaAssetRetentionPurgeJob*), and preserves historical reuse rows (*[Asset Deleted]* / *[Submission Deleted]* placeholders, *UC-2.3 A3*).
  4. Uploading a file whose SHA-256 content hash matches an existing non-deleted asset in the same institution triggers the duplicate-detection modal offering **"Upload Anyway"**, **"Cancel"**, or **"Use Existing"** (*UC-2.3 A4*).
- **Pass/Fail Criteria: PASS** if all 3 deletion tiers, *Used In* history placeholders, and SHA-256 duplicate prompts function per *UC-2.3*; **FAIL** otherwise.

3.15.4. Test Procedure

1. Attempt to delete *asset-active* from the Media Library; assert *409 Conflict* listing the active submission.
2. Attempt to delete *asset-draft*; verify the broken-reference warning modal requires *force=true* confirmation.
3. Delete *asset-terminal*; verify it disappears from the active library grid while its historical submission still displays *[Asset Deleted]* in audit/history views.
4. Upload a byte-for-byte copy of *summit-stage.png* to *"CITU"*; verify the SHA-256 duplicate prompt

appears and clicking **"Use Existing"** opens the existing asset in the repository.

3.16. Case-16: Approval Workflow, 15-Minute Review Lock & Self-Review Prevention (UC-2.4)

3.16.1. Purpose

Verify the network-wide Validation Queue (*UC-2.4*), 15-minute *ReviewLock* acquisition, concurrent read-only lock contention (*UC-2.4 A3*), automatic lock expiry cleanup (*UC-2.4 A4*, *BR-APP-01*), unconditional self-submission review blocking (*UC-2.4 A5*), and terminal decisions (*Approve*, *Request Revision* with *10–1,000* char remarks, and *Reject* with standardized reason codes *BR-APP-02, BR-APP-03*).

3.16.2. Inputs

- **Reviewers:** Moderator 1 (*mod.a@dasigconnect.com*), Moderator 2 (*mod.b@dasigconnect.com*), Administrator (*administrator@dasigconnect.com*).
- **Submissions:**
  - *sub-citu-1*: Submitted by Contributor *contrib.citu@cit.edu*.
  - *sub-mod-self*: Submitted by Moderator 1 (*mod.a@dasigconnect.com*).
- **Revision Remarks:** *"Too short"* (*9* chars — invalid) vs. *"Please add the official DOST Region 7 partner acknowledgment and registration URL."* (*82* chars — valid).
- **Rejection Payload:** Code *INCOMPLETE_CONTENT* (deprecated/invalid for reject) vs. *DUPLICATE_EVENT* (valid) and *OTHER* (requires written notes).

3.16.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Moderator 1 opening *sub-citu-1* transitions it *pending → in_review* and acquires a 15-minute *ReviewLock* (*BR-APP-01*).
  2. Moderator 2 opening *sub-citu-1* while Moderator 1 holds the lock sees who holds the lock and views the panel in read-only mode (*UC-2.4 A3*).
  3. If Moderator 1 takes no action for 15 minutes, *ReviewLockCleanupJob* releases the lock and reverts *sub-citu-1* from *in_review* to *pending* (*UC-2.4 A4*).
  4. Moderator 1 attempting to acquire a review lock on or approve/reject their own submission (*sub-mod-self*) is unconditionally blocked (*HTTP 403/409*, *UC-2.4 A5*).
  5. **Request Revision** rejects remarks *<10* chars or *>1,000* chars (*400/422*, *UC-2.4 A2*), and with valid remarks transitions *sub-citu-1* to *needs_revision*, releases its *SlotReservation*, and notifies the Contributor (in-app + email).
  6. **Reject** rejects *INCOMPLETE_CONTENT*/*WRONG_FORMAT* with *HTTP 422* (directing to Request Revision), requires written notes when reason is *OTHER*, and with a valid code (*DUPLICATE_EVENT*) transitions the post to *rejected* and releases its slot (*BR-APP-03*).
- **Pass/Fail Criteria: PASS** if lock exclusivity, 15-min timeout, self-review block, revision length bounds, and rejection reason validation all pass; **FAIL** otherwise.

3.16.4. Test Procedure

1. Log in as Moderator 1, navigate to **Validation Queue**, and click **Review** on *sub-citu-1*; verify *in_review* state and 15-minute lock creation.
2. Log in as Moderator 2 in a second browser session and open *sub-citu-1*; verify action buttons are disabled and Moderator 1 is shown as the active reviewer.
3. As Moderator 1, attempt to review *sub-mod-self* (authored by Moderator 1); assert self-review block (*UC-2.4 A5*).
4. On *sub-citu-1*, click **Request Revision** with a 9-character remark (assert blocked), then submit the 82-character remark; verify transition to *needs_revision* and slot release.
5. On another pending submission, test **Reject** with *OTHER* and empty notes (assert blocked), then with

DUPLICATE_EVENT; verify transition to *rejected* and verify the Contributor can later edit and resubmit (*UC-1.5 A4*, *UC-2.4 Postconditions*).

3.17. Case-17: Edit-During-Review Diff Tracking, Restore Original & Late Publish Override (UC-2.4 A9–A15)

3.17.1. Purpose

Verify inline editing during review (*UC-2.4 A9*), whole-cycle edit diff tracking across lock releases (*UC-2.4 A10*), severity classification (*Quiet*, *Flagged*, *Added Media*) and separate Contributor edit notification (*UC-2.4 A11*), **Review & Save** / **Restore original** safeguards (*UC-2.4 A14*), and **Late Publish Override** when a submission's scheduled slot passes during review (*UC-2.4 A15*).

3.17.2. Inputs

- **Submission Under Review (***sub-edit***):** Locked by Moderator 1; original caption *"CIT-U hosts regional coding challenge."*
- **Edits Made:** Caption substantially rewritten (*>20%* change → *Flagged* severity) and 1 library photo added (*Added Media*), followed by lock release, lock re-acquisition, and final **Approve**.
- **Expired-Slot Submission (***sub-late***):** Scheduled slot *scheduled_at* has passed (< *NOW()*) while held under an active *ReviewLock*.

3.17.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Saving an inline caption edit runs the advisory AI writing check (*POST /api/v1/ai/proofread*) and displays a **Review & Save** diff summary with Undo; clicking **Restore original** reverts fields back to the Contributor's submitted values.
  2. Releasing the review lock after editing, re-acquiring the lock later in the same review cycle, and clicking **Approve** with no further edits still records *edited* = *true* and preserves the full before/after diff across the entire review cycle (*UC-2.4 A10*).
  3. On approval of an edited submission, the Contributor receives the base approval notification (*T-02*) **plus** a separate edit notification (*UC-2.4 A11*: *Quiet* → in-app only; *Flagged* / *Added Media* → in-app + email, plus in-app alert to other Admins).
  4. While *sub-late* holds an active *ReviewLock*, *StaleSubmissionDetectorJob* skips marking it *missed_review*; clicking **Approve** normally returns *HTTP 409 Conflict* (slot passed), requiring either selecting a new future slot or (Administrator only) invoking *?publishNow=true*, which sets the slot to *NOW()*, publishes, and logs *LATE_PUBLISH_OVERRIDE*.
- **Pass/Fail Criteria: PASS** if cycle-wide diff persistence, *Restore original*, severity-tiered notifications, and *LATE_PUBLISH_OVERRIDE* all behave per *UC-2.4*; **FAIL** otherwise.

3.17.4. Test Procedure

1. Lock *sub-edit* as Moderator 1, modify the caption significantly, verify **Review & Save** diff and AI proofread suggestions, save the edit, then test **Restore original** and re-apply the edit.
2. Release the lock on *sub-edit*, re-open *sub-edit* to acquire a new lock, and click **Approve** without making additional edits; verify the audit log and Contributor notifications record the edit made before the lock release.
3. Lock *sub-late* whose *scheduled_at* is 2 minutes in the future, wait until *scheduled_at* < *NOW()*, verify the *"slot passed"* banner appears and *StaleSubmissionDetectorJob* does not steal the locked item, then as Administrator approve with *Publish Now* (*?publishNow=true*) and verify *LATE_PUBLISH_OVERRIDE* in the audit log.

3.18. Case-18: Automated Brand Watermarking, Aspect-Ratio Rendering & Opt-Out (UC-2.5)

3.18.1. Purpose

Verify the Administrator Watermark Configuration canvas (*1:1*, *4:5*, *16:9* aspect-ratio previews, up to 3 image/text/shape elements), publish-time proportional rendering on photo assets, built-in default text watermark fallback (*UC-2.5 A1*), per-asset watermark opt-out (*skipWatermark* = *true*, *UC-2.5 A5*), non-blocking fallback on render failure (*UC-2.5 A3*), and 3-day derivative cleanup (*GeneratedWatermarkPurgeJob*).

3.18.2. Inputs

- **Watermark Config:** 3 elements — DASIG PNG logo (*bottom-right*, *12%* width), text *"DASIG Region 7"* (*bottom-left*), and accent rectangle bar.
- **Submission Media:** Photo 1 (*skipWatermark* = *false*), Photo 2 (*skipWatermark* = *true*), and Video 1 (*MP4*).
- **Corrupted Asset Scenario:** Photo 3 with unreadable/corrupted image stream at publish time.

3.18.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Saving the watermark configuration in **Page Settings** → **Watermark** stores percentage-based coordinates and element properties network-wide; updating the config between a post's approval and its scheduled publish time applies the updated config at publish time (*UC-2.5 Main Flow & A2*).
  2. At publish time, Photo 1 is rendered with the proportional watermark overlay into *generated/watermarked/* while the original asset in *media_assets* remains unmodified; Photo 2 (*skipWatermark* = *true*) and Video 1 publish untouched (*UC-2.5 A4, A5*).
  3. If no watermark config exists in the database, a built-in default text watermark is applied unless watermarking is explicitly disabled (*UC-2.5 A1*).
  4. When Photo 3 fails Java2D rendering, the publisher falls back to publishing the original unwatermarked photo without failing the post, writes *PUBLISH_WATERMARK_FAILED* to the audit log, and dispatches an in-app + email alert to all Administrators (*UC-2.5 A3*).
  5. *GeneratedWatermarkPurgeJob* deletes generated R2 objects under *generated/watermarked/* older than *3 days* (*app.watermark.generated-retention-days*).
- **Pass/Fail Criteria: PASS** if live publish-time rendering, opt-out honoring, render-failure resilience + audit/alerting, and derivative purge all pass; **FAIL** otherwise.

3.18.4. Test Procedure

1. Log in as Administrator, open **Watermark Configuration**, add 1 logo, 1 text, and 1 shape element, preview across *1:1*, *4:5*, and *16:9* toggles, and save.
2. Approve a scheduled submission containing Photo 1 (*skipWatermark* = *false*) and Photo 2 (*skipWatermark* = *true*), modify the watermark text before the scheduled slot arrives, and trigger publishing; verify Photo 1 uses the updated watermark and Photo 2 is unwatermarked.
3. Simulate a rendering exception on Photo 3 during publish; verify the post still transitions to *PUBLISHED* using the original image, *PUBLISH_WATERMARK_FAILED* is logged, and Admins receive notifications.
4. Trigger *GeneratedWatermarkPurgeJob* with derivatives older than 3 days; verify R2 cleanup.

3.19. Case-19: Master Calendar Visibility, Moderator Reschedule Caps & GiST Conflict Integrity (UC-3.1)

3.19.1. Purpose

Verify role-scoped Master Calendar visibility (*UC-3.1 Main Flow*), Moderator drag-reschedule caps (≤2 reschedules per submission and ±*1 day* of *originalScheduledAt*, *UC-3.1 A1*), Administrator hard guard-rail

override with justification (*UC-3.1 A2*), atomic concurrent reschedule protection (*UC-3.1 A3*), and database-level PostgreSQL GiST exclusion constraint integrity (*V95*/*V96*, *UC-3.1 A4*).

3.19.2. Inputs

- **Actors:** Contributor from *"CITU"*, Moderator (*mod.a@dasigconnect.com*), Administrator (*administrator@dasigconnect.com*).
- **Calendar Events:**
  - *sub-citu-sched*: *SCHEDULED* post from *"CITU"* originally approved at *2026-10-25 10:00 AM* (*originalScheduledAt*).
  - *sub-su-sched*: *SCHEDULED* post from *"Silliman University"* at *2026-10-25 02:00 PM*.
  - *sub-dasig-sched*: *SCHEDULED* post from *"DASIG Central Visayas"* at *2026-10-25 04:00 PM*.
  - *sub-citu-pending*: *PENDING* in-flight post authored by the *"CITU"* Contributor.

3.19.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. **Contributor Calendar View:** Renders full detail (title, caption, media) for *"CITU"* posts, *"DASIG Central Visayas"* posts, and the Contributor's own in-flight submissions (*sub-citu-pending*), while masking *"Silliman University"*s *sub-su-sched* so caption, description, and contributor identity are withheld (*UC-3.1 Main Flow*).
  2. **Moderator & Administrator Calendar View:** Renders all network-wide events in full detail.
  3. **Moderator Reschedule Caps (***UC-3.1 A1***):** Moderator can reschedule *sub-citu-sched* twice within *±1 day* of *2026-10-25 10:00 AM* (providing mandatory written justification each time); a 3rd reschedule attempt—or any move beyond *±24 hours* from *originalScheduledAt*—is hard-blocked (*HTTP 403/422*).
  4. **Guard-Rail Conflict & GiST Exclusion (***UC-3.1 A2, A4***):** Moving *sub-citu-sched* to *2026-10-25 02:15 PM* (within *±30* minutes of *sub-su-sched*) is blocked for Moderator (*403*) and rejected at the DB layer by the *slot_reservations* GiST exclusion constraint (*V95*) for normal requests, **except** when an Administrator explicitly supplies an override justification, which sets *admin_override = true* (*V96*) and logs the override in the audit log.
  5. Two concurrent reschedule requests for *sub-citu-sched* execute via atomic conditional *UPDATE*; only one succeeds while the second receives *HTTP 409 Conflict* (*UC-3.1 A3*).
- **Pass/Fail Criteria: PASS** if role masking, Moderator *2-count* / *±1-day* anchor caps, atomic race protection, and GiST exclusion + Admin override exemption all pass; **FAIL** otherwise.

3.19.4. Test Procedure

1. Fetch *GET /api/v1/calendar/events* as *"CITU"* Contributor; verify *sub-su-sched* is masked while *sub-citu-sched*, *sub-dasig-sched*, and *sub-citu-pending* include full details.
2. As Moderator, drag *sub-citu-sched* from *10:00 AM* to *11:00 AM* (Reschedule #1) and then to *12:30 PM* (Reschedule #2) with justification; attempt Reschedule #3 or a move to *2026-10-27* (> *±1 day* from original slot) and assert hard block.
3. Attempt to reschedule a post into *02:15 PM* (conflicting with *sub-su-sched* at *02:00 PM*) as Moderator; assert *403*. Repeat as Administrator with an explicit override justification; verify *admin_override = true* succeeds and writes an audit log entry.
4. Dispatch two parallel reschedule *PATCH* requests for the same submission; assert one returns *200 OK* and the other returns *409 Conflict*.

3.20. Case-20: Automated Facebook Publishing, Atomic Claiming & Retry/Cleanup Pipeline (UC-3.2)

3.20.1. Purpose

Verify *PublishingSchedulerJob* (1-minute cadence, 5-minute lookback), atomic *scheduled → publishing* claiming to prevent double-publishing, 2-step multi-photo publishing (*/{PAGE_ID}/photos →*

/{PAGE_ID}/feed), single-call video publishing (*/{PAGE_ID}/videos*), orphaned staged-photo cleanup (*photoIdsCleanupFailed*), 3-retry exponential backoff (*5s, 25s, 125s*, *GR-T5*), and Fast-Track immediate asynchronous publishing (*UC-3.2 Main Flow & A1, A5, A6*).

3.20.2. Inputs

- **Submission 1 (***sub-photos***):** *SCHEDULED* due now, 2 JPEG photos with per-asset captions and main caption containing *#DASIG7*.
- **Submission 2 (***sub-video***):** *SCHEDULED* due now, 1 MP4 video.
- **Submission 3 (***sub-mixed***):** *SCHEDULED* due now, 1 JPEG photo + 1 MP4 video.
- **Simulated API Failure (***sub-fail***):** 2-photo submission where Photo 1 stages (*photo_id_1*), Photo 2 fails, and subsequent retries fail.

3.20.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Overlapping scheduler passes cannot double-publish *sub-photos* because *claimForPublishing* atomically transitions *scheduled → publishing* (*UPDATE ... WHERE status = 'scheduled'*) before any Facebook Graph API request is made (*UC-3.2 Main Flow*).
  2. *sub-photos* stages both watermarked images with *published=false* and their per-asset captions, then publishes one */feed* post without duplicating *#DASIG7* in the message, transitions to *PUBLISHED*, releases its *SlotReservation*, and notifies the Contributor (*T-04*).
  3. *sub-video* publishes via a single *POST /{PAGE_ID}/videos* call and transitions to *PUBLISHED*.
  4. *sub-mixed* (image + video) is immediately marked *PUBLISH_FAILED* with zero Graph API retries (*UC-3.2 A6*).
  5. On *sub-fail*, when Photo 2 staging fails, the system attempts *DELETE /{photo_id_1}* to clean up the orphaned photo (*UC-3.2 A1*), retries up to 3 times with backoff (*5s, 25s, 125s*), and upon exhaustion transitions to *PUBLISH_FAILED*, surfaces any undeleted *unresolvedPhotoIds* (*photoIdsCleanupFailed*) in the Failed tab, and alerts Administrators (*T-06*).
- **Pass/Fail Criteria: PASS** if atomic claim, 2-step photo publish, single-call video publish, mixed-media immediate fail, retry backoff, and orphan cleanup tracking all pass; **FAIL** otherwise.

3.20.4. Test Procedure

1. Mark *sub-photos* and *sub-video* due (*scheduled_at = NOW() - 1 min*) and invoke *PublishingSchedulerJob* concurrently from two threads; verify each submission is claimed and published exactly once.
2. Verify *sub-photos* executes two */photos* calls (*published=false*) followed by one */feed* call, records *facebookPostId*, releases its *SlotReservation*, and does not duplicate existing hashtags.
3. Run *PublishingSchedulerJob* on *sub-mixed*; verify immediate transition to *PUBLISH_FAILED* without retries.
4. Simulate a Graph API failure after Photo 1 stages on *sub-fail* where *DELETE /{photo_id_1}* also fails; verify 3 retries execute, status becomes *PUBLISH_FAILED*, *photoIdsCleanupFailed* records *photo_id_1*, and *T-06* is dispatched.

3.21. Case-21: Missed Slot Sweep (PUBLISH_FAILED vs. MISSED_REVIEW) & Manual Publishing Fallback (UC-3.2 A2, UC-2.4 A8)

3.21.1. Purpose

Verify that *StaleSubmissionDetectorJob* (every 5 minutes, *GR-T9*) distinguishes between an already-approved *SCHEDULED* submission that missed its publish window (→ *PUBLISH_FAILED*) and an unreviewed *PENDING* submission whose slot passed (→ *MISSED_REVIEW*), and verify the 3-step **Manual Publishing Fallback** recovery workflow (*UC-2.4 A8*, *BR-PUB-04, BR-PUB-05*).

3.21.2. Inputs

- **Stale Scheduled Submission (***sub-stale-sched***):** *status = scheduled*, *scheduled_at = NOW() - 10 minutes*.
- **Stale Unreviewed Submission (***sub-stale-pend***):** *status = pending*, *scheduled_at = NOW() - 10 minutes*, no active *ReviewLock*.
- **Stuck Fast-Track Submission (***sub-stuck-ft***):** *status = publishing*, *scheduled_at = NULL*, *updated_at = NOW() - 10 minutes*.
- **Manual Fallback Input:** Facebook Post URL *"https://www.facebook.com/DASIGRegion7/posts/123456789012345"*, manual publish timestamp, and notes *"Published manually during Graph API maintenance."*

3.21.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. *StaleSubmissionDetectorJob* transitions *sub-stale-sched* (*scheduled*) to *PUBLISH_FAILED* and surfaces it in the Validation Queue's **Failed** tab (*UC-3.2 A2*).
  2. *StaleSubmissionDetectorJob* transitions *sub-stale-pend* (*pending*) to *MISSED_REVIEW*, releases its *SlotReservation*, and routes it back into the Approval Workflow rather than the Manual Publishing Fallback flow (*UC-3.2 A2*).
  3. *findStuckFastTrackPublishing* detects *sub-stuck-ft* (*publishing* with *scheduled_at IS NULL* and *updated_at < NOW() - 5 min*) and recovers it to *PUBLISH_FAILED*.
  4. In the **Failed** tab, an Administrator completing the 3-step **Manual Publishing Fallback** (Download watermarked media → Copy caption → Paste valid Facebook Post URL & confirm) transitions *sub-stale-sched* to *PUBLISHED_MANUAL*, releases its slot reservation, writes a complete audit entry (*BR-PUB-04*), notifies the Contributor (*T-05*), and includes the post in Analytics KPIs (*BR-PUB-05*).
  5. If a failed submission is retried onto a new schedule (*retryWithNewSchedule*), both *originalScheduledAt* and *moderatorRescheduleCount* reset to the new baseline (*UC-3.1 A1*).
- **Pass/Fail Criteria: PASS** if *PUBLISH_FAILED* vs. *MISSED_REVIEW* separation, stuck Fast-Track recovery, manual fallback completion, and reschedule-baseline reset all pass; **FAIL** otherwise.

3.21.4. Test Procedure

1. Seed *sub-stale-sched*, *sub-stale-pend*, and *sub-stuck-ft* past the 5-minute cutoff and run *StaleSubmissionDetectorJob*.
2. Verify *sub-stale-sched* and *sub-stuck-ft* transition to *PUBLISH_FAILED* while *sub-stale-pend* transitions to *MISSED_REVIEW* with its slot released.
3. Open *sub-stale-sched* in the Validation Queue **Failed** tab as Administrator, execute the 3-step Manual Publishing Fallback with the Facebook URL, and verify transition to *PUBLISHED_MANUAL* and *T-05* notification.
4. On another *PUBLISH_FAILED* submission that previously had *moderatorRescheduleCount = 2*, execute **Retry with New Schedule** and verify *moderatorRescheduleCount* resets to *0* and *originalScheduledAt* updates to the new slot.

3.22. Case-22: Real-Time SSE Notifications, Cross-Session Read Sync & Messenger Delivery (UC-3.3)

3.22.1. Purpose

Verify the event-driven notification system (*UC-3.3*, *BR-NOT-01..04*), including recipient set determination (review triggers notifying **both** Moderators and Administrators), real-time SSE badge delivery without polling, cross-session *read* / *read-all* synchronization (*UC-3.3 A2*), independent email retry on failure (*UC-3.3 A1*), and opt-in Facebook Messenger DM delivery (*UC-3.3 A4, A5*).

3.22.2. Inputs

- **Active Recipients:**
  - Moderator 1 (*mod.linked@dasigconnect.com* — Messenger linked, open in Browser Tab A and Browser Tab B).
  - Administrator 1 (*admin.unlinked@dasigconnect.com* — Messenger **not** linked).
  - Contributor 1 (*contrib.citu@cit.edu*).
- **Triggering Events:** New Fast-Track submission (*T-11*) by Contributor 1, followed by Moderator 1 clicking a notification in Tab A and clicking **Mark All Read**.

3.22.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Submitting the Fast-Track post dispatches *T-11* to **every active Moderator and every active Administrator** (*UC-3.3 Main Flow*).
  2. Both Moderator 1 and Administrator 1 receive instant in-app SSE *notification* events (*GET /api/v1/notifications/stream*), incrementing their navbar bell badges in real time without page refresh (*BR-NOT-01, BR-NOT-02*).
  3. Moderator 1 (Messenger linked) receives the outbound Facebook Messenger DM (*UC-3.3 A4*), whereas Administrator 1 (Messenger not linked) is silently skipped for Messenger while still receiving in-app and email notifications (*UC-3.3 A5*).
  4. When Moderator 1 marks a notification read (or clicks **Mark All Read**) in Browser Tab A, the backend pushes an SSE *read* / *read-all* event to Browser Tab B, synchronizing the unread badge count across both active sessions (*UC-3.3 A2*).
  5. Simulated SMTP failure logs the failed attempt and retries up to 3 times independently without delaying or blocking the in-app SSE notification (*UC-3.3 A1*).
- **Pass/Fail Criteria: PASS** if Moderator+Admin fan-out, real-time SSE badge updates, cross-tab read sync, Messenger opt-in filtering, and email failure isolation all pass; **FAIL** otherwise.

3.22.4. Test Procedure

1. Open two simultaneous authenticated browser tabs (Tab A and Tab B) for *mod.linked@dasigconnect.com* and one tab for *admin.unlinked@dasigconnect.com*.
2. Submit a Fast-Track post as *contrib.citu@cit.edu*; verify both *mod.linked* and *admin.unlinked* receive the in-app SSE notification within <*5* seconds and only *mod.linked* receives a Messenger dispatch.
3. Click the unread notification in Tab A of *mod.linked*; verify Tab B's unread badge decrements automatically via SSE *read* sync without refreshing the page.
4. Simulate an SMTP exception during a *T-02* approval notification; verify the in-app notification still arrives immediately and *email_delivery_logs* records the retry state.

3.23. Case-23: Role-Scoped Social Engagement Analytics & CSV Formula-Injection Sanitization (UC-3.4)

3.23.1. Purpose

Verify role-scoped access and data filtering on the */analytics* dashboard for Contributors, Moderators, and Administrators (*UC-3.4 Main Flow & A4*, *BR-ANA-01..06*), live workflow KPI calculation vs. cached Facebook engagement metrics, inline export error retry (*UC-3.4 A3*), and CWE-1236 CSV Formula-Injection sanitization on exported CSV files.

3.23.2. Inputs

- **Actors:** Contributor (*contrib.citu@cit.edu*), Moderator (*mod.a@dasigconnect.com*), Administrator (*administrator@dasigconnect.com*).
- **Malicious Submission Title:** *"=HYPERLINK(\"http://evil.example/leak\", \"Click Here\")"* and

"+SUM(1,1)" in a published *"CITU"* submission.

- **Export Metrics Requested:** *posts-by-institution*, *facebook-engagement*, *operational-health*, *ai-performance*.

3.23.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. All three roles can access */analytics* (*UC-3.4 Main Flow*):
     - **Contributor:** Strictly scoped to their own institution (*"CITU"*); requesting another institution's analytics returns *HTTP 403/404* (*UC-3.4 A4*).
     - **Moderator:** Sees network-wide metrics across all institutions unfiltered; institution drilldown and Admin-only reports (*operational-health*, *ai-performance*) return *HTTP 403 Forbidden*.
     - **Administrator:** Sees network-wide metrics, multi-institution filters, Validator/Contributor workload drilldowns, *operational-health* (including the ≥95% On-Time Publication target), and *ai-performance*.
  2. Exporting a report to CSV (*GET /api/v1/analytics/export-csv*) prepends a leading apostrophe (*'=HYPERLINK...*, *'+SUM(1,1)*) to any cell starting with =, +, -, @, \t, or \r, neutralizing spreadsheet formula execution (*CWE-1236*).
  3. If a CSV download fails due to a network error, the Report Modal displays an inline error footer and relabels the button to **"Retry Download"** while preserving active filters (*UC-3.4 A3*).
- **Pass/Fail Criteria: PASS** if role scoping, *403* guards on Admin-only reports, inline export retry, and CSV formula sanitization all pass; **FAIL** otherwise. (Note: Static placeholder values in the non-admin "Content Quality & Compliance" card are tracked in *INC-04*.)

3.23.4. Test Procedure

1. Log in as Contributor, Moderator, and Administrator sequentially and inspect */analytics*; verify institution filter visibility and test direct API calls to *operational-health* and *ai-performance* as Moderator/Contributor (assert *403 Forbidden*).
2. Seed a published submission with title *"=HYPERLINK(\"http://evil.example/leak\", \"Click Here\")"*, export the analytics CSV as Administrator, and inspect the raw CSV text; verify the cell is escaped with a leading apostrophe (*'"=HYPERLINK...*).
3. Intercept the CSV export request in the browser to force a *500* error; verify the inline error message and **"Retry Download"** button appear in *FullReportModal* without losing selected filters.

3.24. Case-24: System Health Analytics, 16 Tracked Jobs, On-Demand Re-Run & Token Re-Auth (UC-3.5)

3.24.1. Purpose

Verify the Administrator-only **System Health** screen (*/admin/system-health*, *UC-3.5*), including PostgreSQL (*pg_database_size*) and Cloudflare R2 storage capacity probes (*80%* Warning, *95%* Critical), 6 external integration checks, 16 tracked background jobs with staleness thresholds (*> 2×* expected interval), synchronous vs. asynchronous manual job **Re-run** (*POST /api/v1/system-health/jobs/{jobKey}/run*), 14 operational & AI telemetry metrics (including missed-review rate), CSV snapshot export (*UC-3.5 A4*), and Facebook OAuth token re-authentication (*UC-3.5 A5*).

3.24.2. Inputs

- **Actors:** Moderator (*mod.a@dasigconnect.com* — unauthorized) and Administrator (*administrator@dasigconnect.com* — authorized).
- **Manual Job Triggers:**
  - Synchronous DB job: *ReviewLockCleanupJob* (*review_lock_cleanup*).
  - Asynchronous external-API job: *PublishingSchedulerJob* (*publishing_scheduler*).

3.24.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. Moderator or Contributor requests to */api/v1/system-health/* return *HTTP 403 Forbidden* (*UC-3.5 Preconditions*).
  2. The System Health response returns:
     - **Storage Capacity:** Database bytes and R2 bucket bytes (with DB-estimate fallback if live R2 scan fails) evaluated against *80%* (*WARNING*) and *95%* (*UNHEALTHY*) thresholds (*UC-3.5 A2*).
     - **6 External Services:** Facebook Graph API, Anthropic Claude Vision, Voyage AI, Email Service Provider, Database Health, and Media Object Storage Health (missing keys or failed probes render as *UNAVAILABLE*, never a false healthy state, *UC-3.5 A1*).
     - **16 Tracked Background Jobs:** Status (*HEALTHY*, *WARNING*, *UNHEALTHY*, *UNAVAILABLE*, *SCHEDULED*).
     - **14 Operational & AI Metrics:** 6 core workflow metrics (including *missedReviewRate* flagged *WARNING* above *10%*) + 8 AI telemetry metrics.
     - **Overall Page Rollup:** Worst status across all sections with item counts.
  3. Clicking **Re-run** on *ReviewLockCleanupJob* executes synchronously, returns fresh status immediately, and writes *BACKGROUND_JOB_RUN* to the audit log; clicking **Re-run** on *PublishingSchedulerJob* or *TokenPublishingEscalationJob* executes asynchronously (*CompletableFuture.runAsync*) and returns an immediate *"started"* acknowledgment so the HTTP request never times out during retry backoff (*UC-3.5 Main Flow*).
  4. Exporting the System Health CSV snapshot writes an audit log entry (*UC-3.5 A4*), and Facebook OAuth re-authentication updates the encrypted token and logs *TOKEN_REAUTHORIZED* (*UC-3.5 A5*).
- **Pass/Fail Criteria: PASS** if RBAC, 4 health sections, *UNAVAILABLE* vs. *UNHEALTHY* semantics, sync/async manual job re-runs, and audit logging all match *UC-3.5*; **FAIL** otherwise.

3.24.4. Test Procedure

1. Attempt to access */api/v1/system-health* as a Moderator; assert *403 Forbidden*.
2. Open **System Health** as Administrator; verify all 2 storage cards, 6 service rows, 16 background job rows, and 14 operational/AI metric cards render with valid statuses and rollup counts.
3. Click **Re-run** on *ReviewLockCleanupJob* (verify synchronous completion) and on *PublishingSchedulerJob* (verify immediate async *"started"* response and *BACKGROUND_JOB_RUN* audit log entry).
4. Export the System Health Snapshot CSV and verify audit logging.

3.25. Case-25: Immutable Audit Log Review, Structured Diffs & Multi-Institution Tenant Isolation (UC-3.6, SRS 3.5)

3.25.1. Purpose

Verify the Administrator-only **Audit Log** (*/admin/audit-log*, *UC-3.6*), including server-side filtering across 11 categories and 8 entity types, structured before/after diff rendering, deleted-entity badge handling (*UC-3.6 A2*), 5,000-row sanitized CSV export (*UC-3.6 A4*), database-level append-only immutability (*V78* RLS policy), and the 4-layer **Multi-Institution Tenant Isolation** security boundary (*SRS Section 3.5*).

3.25.2. Inputs

- **Actors:**
  - Contributor A at *"CITU"* (*contrib.citu@cit.edu*).
  - Contributor B at *"Silliman University"* (*contrib.su@su.edu.ph*).
  - Administrator (*administrator@dasigconnect.com*).

- **Cross-Tenant Attack Vectors:**
  - Contributor A requests *GET /api/v1/submissions/{sub-su-id}*, *PUT /api/v1/submissions/{sub-su-id}*, and *GET /api/v1/media-assets/{asset-su-id}* belonging to *"Silliman University"*.
  - Contributor A sends *POST /api/v1/submissions* with a forged payload *{"institutionId": "<silliman-uuid>", ...}*.
- **Direct DB Audit Mutation Attempt:** SQL *UPDATE audit_logs SET action = 'TAMPERED'* and *DELETE FROM audit_logs WHERE id = ...*.

3.25.3. Expected Outputs & Pass/Fail Criteria

- **Expected Outputs:**
  1. **Cross-Institution Read/Write Isolation (***SRS 3.5.4, 3.5.6***):** Contributor A's direct-access attempts by ID against *"Silliman University"* submissions or media assets return *HTTP 404 Not Found* (not *403 Forbidden*, preventing resource-existence information leakage).
  2. **Server-Derived Institution Identity (***SRS 3.5.2 Layer 4***):** Contributor A's forged *institutionId* in *POST /api/v1/submissions* is ignored and overridden with Contributor A's JWT-bound *"CITU" institution_id*.
  3. **Audit Log Filtering & Diff View (***UC-3.6 Main Flow & A2***):** Administrator filtering */admin/audit-log* by category, action, entity type, actor, and date range returns paginated entries newest-first (*Asia/Manila*); opening an entry displays Actor details, client IP, a live jump link (or *"Deleted/Unavailable"* badge if the target entity was deleted), and a structured *Field / Previous Value / Updated Value* diff table.
  4. **Audit CSV Export (***UC-3.6 Main Flow & A4***):** Exporting filtered audit rows caps at *5,000* rows, sanitizes leading formula characters (=, +, -, @), and records *AUDIT_LOG_EXPORTED*.
  5. **Database Append-Only Guarantee (***UC-3.6 Postconditions***,** *V78***):** Any *UPDATE* or *DELETE* executed against *audit_logs* under the application database role is rejected/blocked by PostgreSQL Row-Level Security (migration V78).
- **Pass/Fail Criteria: PASS** if all 4 tenant-isolation layers return *404*/override forged tenant IDs, audit filters/diffs/exports work as specified, and *audit_logs* rejects mutation; **FAIL** otherwise.

3.25.4. Test Procedure

1. Authenticate as Contributor A (*"CITU"*) and attempt *GET*, *PUT*, and *DELETE* against *"Silliman University"* submission and media UUIDs; assert *HTTP 404 Not Found* on every call.
2. Submit a *POST /api/v1/submissions* request as Contributor A with *"Silliman University"*'s UUID in the request body; verify the created draft is bound to *"CITU"*.
3. Authenticate as Administrator, navigate to **System & Audit** → **Audit Log**, filter by *Edit & Revision* and *Reschedule & Override*, and open the detail drawer; verify the structured before/after diff table and *"Deleted/Unavailable"* badge on a deleted draft's audit row.
4. Export the Audit Log to CSV; verify *AUDIT_LOG_EXPORTED* is written and formula-prefix characters are escaped with *'*.
5. Execute an *UPDATE* and *DELETE* statement on *audit_logs* via the application database connection; verify PostgreSQL RLS denies both operations.

Appendix (Test Logs)

A.1 Log for Test n

The following execution logs record the verification runs conducted on **2026-10-11** against the DASIGConnect backend test suite (*./mvnw test* — JUnit 5, Mockito, MockMvc, H2/PostgreSQL migrations V1–V120+) and frontend E2E/regression suite (*npm run test:e2e* — Playwright 1.63 on Headless Chromium):

Log for Test 1 (Case-1 — UC-1.1 Administrator Account Management & 3-Admin Cap)

- **Date / Time:** *2026-10-11 09:05:12 PHT (UTC+8)*
- **Automated Test Reference:** *UserServiceTest* (*inviteAdmin_**, *adminCapPolicy_**, *deactivateAdmin_**, *reactivateAdmin_**)
- **Execution Trace:**
  - *POST /api/v1/users/invite* (*role=admin*) with 2 active Admins → *201 Created*, *state=PENDING*, SHA-256 token hash stored.
  - *POST /api/v1/users/invite* (*role=admin*) with 2 active + 1 pending Admin → Rejected (*409/422 Admin cap of 3 reached*).
  - Peer Admin (*isAdminOwner=false*) *PATCH /api/v1/users/{adminId}/deactivate* → Rejected (*403 Forbidden*).
  - Admin Owner (*isAdminOwner=true*) *PATCH /api/v1/users/{adminId}/deactivate → 200 OK*, *state=INACTIVE*, *session_version* incremented, *USER_STATUS_UPDATED* logged.
- **Outcome: PASS**

Log for Test 2 (Case-2 — UC-1.1 A7, A8 Admin Promotion & Admin Owner Transfer)

- **Date / Time:** *2026-10-11 09:10:44 PHT (UTC+8)*
- **Automated Test Reference:** *UserServiceTest* (*changeRole_promote**, *confirmPromotion_**, *requestAdminTransfer_**, *confirmAdminTransfer_**)
- **Execution Trace:**
  - *POST /api/v1/users/change-role (newRole=admin) → 200 OK*, slot reserved against 3-Admin cap, *ADMIN_PROMOTION_REQUESTED* logged, , duplicate proposal rejected with *409 Conflict*.
  - *POST /api/v1/users/promotion/confirm → 200 OK*, role elevated to *admin*, *(isAdminOwner = false*), active session invalidated, *ADMIN_PROMOTION_CONFIRMED* logged.
  - Admin Owner transfer to active Administrator (*admin.second@dasigconnect.com)* → Initiation logs *ADMIN_TRANSFER_REQUESTED* with 24-hour expiry and dispatches in-app notification.
  - *POST /api/v1/users/admin-transfer/confirm* by target Administrator → *200 OK*, target promoted to Admin Owner (*isAdminOwner = true*), outgoing Owner retains Administrator status (*isAdminOwner = false*) without demotion, outgoing tokens invalidated, *ADMIN_TRANSFER_CONFIRMED* logged.
- **Outcome: PASS**

Log for Test 3 (Case-3 — UC-1.2 Institution Lifecycle & Protected Institution)

- **Date / Time:** *2026-10-11 09:16:03 PHT (UTC+8)*
- **Automated Test Reference:** *InstitutionService* / *UserServiceTest*
- **Execution Trace:**
  - Moderator *POST /api/v1/institutions → 403 Forbidden*.
  - Admin creates *"CITU" → 201 Created*; duplicate *"CITU" → 409 Conflict*.
  - Deactivate *"DASIG Central Visayas" → 400/409 Blocked (Protected Institution)*.
  - Permanent delete of empty test institution with confirmation code → *200 OK*, *INSTITUTION_DELETED* logged.
- **Outcome: PASS**

Log for Test 4 (Case-4 — UC-1.3 Contributor Account Provisioning & Footprint-Aware Deletion)

- **Date / Time:** *2026-10-11 09:22:18 PHT (UTC+8)*
- **Automated Test Reference:** *UserServiceTest* (*inviteContributor_**, *moderatorScope_**, *removeUser_withFootprint_anonymizesAtRest*, *erasePersonalData_**)
- **Execution Trace:**
  - Moderator 1 invites *contrib.citu@cit.edu* → *201 Created*; Moderator 2 attempts cancel on Moderator 1's invite → *403 Forbidden*.
  - Deleting *INACTIVE* Contributor with submission footprint → Record retained and anonymized at rest, *USER_REMOVED* logged.
  - Admin Owner invokes *DELETE /api/v1/users/{id}/erase* → PII scrubbed, *USER_ANONYMIZED* logged.
- **Outcome: PASS**

Log for Test 5 (Case-5 — UC-1.4 Login, Lockout, Password Policy & Session Revocation)

- **Date / Time:** *2026-10-11 09:28:50 PHT (UTC+8)*
- **Automated Test Reference:** *AuthServiceTest*, *AuthControllerTest*, *JWTServiceTest*
- **Execution Trace:**
  - 5 failed logins within 15 minutes → *AccountLockoutService* returns *423 Locked* for 15 minutes.
  - Password reset with *"Dasig123!"* → Rejected (*400*, forbidden fragment); with *"Str0ng#Pass_987!"* → *200 OK*, *users.session_version* incremented, prior JWTs rejected by *JWTService.extractClaims*.
  - Single-token logout (*POST /api/v1/auth/logout*) → SHA-256 token hash persisted in *revoked_tokens* (*V88*); other active sessions unaffected.
  - Messenger link code (*POST /api/v1/auth/messenger-link-code*) → *200 OK* for Moderator/Admin, *403 Forbidden* for Contributor.
- **Outcome: PASS**

Log for Test 6 (Case-6 — UC-1.5 Three-Step Post Composer, Readiness & Templates)

- **Date / Time:** *2026-10-11 09:35:15 PHT (UTC+8)*
- **Automated Test Reference:** *SubmissionServiceTest*, *SubmissionControllerTest*, Playwright performance regression suite
- **Execution Trace:**
  - 3-step wizard (*Media → Post Details → Organize & Schedule*) enforces Required step gates on forward navigation and allows free backward navigation.
  - Readiness Checklist computes *75%* Required + *25%* Recommended score (*0–100*).
  - Personal template save (*POST /api/v1/post-templates*) scopes to *ownerUserId* and returns *409* on duplicate name.
  - Deleting a *draft* or *rejected* submission cascades *validation_logs*/*review_locks* (*V103*) and logs *SUBMISSION_DELETED*.
- **Outcome: PASS**

Log for Test 7 (Case-7 — UC-1.5 A10 Generate Template from Top Posts)

- **Date / Time:** *2026-10-11 09:41:02 PHT (UTC+8)*
- **Automated Test Reference:** *TopPostTemplateService* / *SubmissionServiceTest*
- **Execution Trace:**
  - < *8* engaged posts → *available: false* returned without invoking Claude.
  - ≥ 11 engaged posts → Top 6 + bottom 3 ranked by *reactions + 2×comments + 3×shares*, placeholder template + insights returned, rate-limited at *10 req/hr/user* (*429*).
- **Outcome: PASS**

Log for Test 8 (Case-8 — UC-1.6 AI Caption Generation, Writing Check & Fancy Text)

- **Date / Time:** *2026-10-11 09:47:29 PHT (UTC+8)*
- **Automated Test Reference:** *SubmissionServiceTest* (*validateCaptionCharLimit*), *ClaudeVisionClient* / *ProofreadService*
- **Execution Trace:**
  - *>5 MB* image downscaled in-memory; *POST /api/v1/ai/caption* returns 1 caption for selected tone within 30s timeout.
  - Prompt *>280* chars or word count *>2000* → *400 Bad Request*; *>30* calls/hr → *429 Too Many Requests*.
  - Caption *>3000* code points → Rejected (*400* on save, *422* on submit).
  - *FancyTextTool* transforms ASCII alphanumerics across 5 Unicode styles and reverts cleanly via *Plain*.
- **Outcome: PASS**

Log for Test 9 (Case-9 — UC-1.7 Media Attachment Rules, AI Suggestions & Album Auto-Match)

- **Date / Time:** *2026-10-11 09:54:11 PHT (UTC+8)*
- **Automated Test Reference:** *SubmissionServiceTest*, *MediaAssetServiceTest*
- **Execution Trace:**
  - Magic-byte/MIME validation accepts only *JPEG, PNG, WebP, GIF, MP4, MOV, WebM* ≤50 MB and ≤10 assets per submission.
  - Per-asset caption (≤500 chars) and photo-only *skipWatermark* persisted on *SubmissionMediaAsset*.
  - *POST /api/v1/ai/submissions/{id}/suggest-album* auto-applies at ≥0.55, returns top-3 candidates at *0.32–0.54*, and leaves manual entry below *0.32*.
- **Outcome: PASS**

Log for Test 10 (Case-10 — UC-1.8 Peak-Hour Engagement Recommendations)

- **Date / Time:** *2026-10-11 10:01:05 PHT (UTC+8)*
- **Automated Test Reference:** *EngagementRecommendationService* / *GuardRailServiceTest*
- **Execution Trace:**
  - <*20* historical samples → Returns Tue/Wed/Thu 6:00 PM PHT fallback slots (*Best-practice guidance*).
  - ≥20 historical samples → Top 5 *(day, hour)* buckets filtered against *GuardRailService* so hard-blocked slots (*GR-H1*) are excluded from the 3 recommended tiles.
- **Outcome: PASS**

Log for Test 11 (Case-11 — UC-1.9 Draft Submission, Withdrawal & Fast-Track Sorting)

- **Date / Time:** *2026-10-11 10:08:33 PHT (UTC+8)*
- **Automated Test Reference:** *SubmissionServiceTest*, *SubmissionControllerTest*
- **Execution Trace:**
  - Submit with missing album/media/title → *422 Unprocessable Entity*.
  - Submit with concurrent slot conflict → *409 Conflict*, composer redirects to Step 3 (**Organize & Schedule**).
  - *POST /api/v1/submissions/{id}/withdraw* succeeds on unlocked *pending* post (→ *draft*) and returns *409* once a *ReviewLock* exists.
  - Fast-Track submission skips slot checks and sorts to top of active Validation Queue (*fastTrack DESC*).
- **Outcome: PASS**

Log for Test 12 (Case-12 — UC-1.10 Moderator Account Management & Lateral Role Change)

- **Date / Time:** *2026-10-11 10:14:20 PHT (UTC+8)*
- **Automated Test Reference:** *UserServiceTest* (*changeRole_moderatorToContributor_**, *changeRole_releasesReviewLocks*)
- **Execution Trace:**
  - Lateral move *Moderator → Contributor* assigns *"CITU"*, increments *session_version*, releases active *ReviewLock* rows back to *pending*, and logs *USER_ROLE_CHANGED*.
  - Lateral move *Contributor → Moderator* clears *institution_id* to *NULL* and revokes sessions.
- **Outcome: PASS**

Log for Test 13 (Case-13 — UC-2.1 Standalone Library Uploads, Nested Albums & Tag Floor)

- **Date / Time:** *2026-10-11 10:21:47 PHT (UTC+8)*
- **Automated Test Reference:** *MediaAssetServiceTest* (*upload_imageAsset_staysProcessingAndTriggersClassification*, *upload_videoAsset_isReadyImmediatelyAndSkipsClassification*, *updateAlbum_withNullAlbumId_isRejected*, *removeTag_lastManualTag_ignoresAiGeneratedTagCount*)
- **Execution Trace:**
  - Image upload starts *PROCESSING* and transitions to *READY* after classification + embedding; video upload is immediately *READY*.
  - Removing last manual tag (*source="manual"*) when 8 AI tags exist → Rejected (*400 Bad Request*).
  - Re-parenting an album under its own descendant → Blocked by cycle guard (*findDescendantIds*).
- **Outcome: PASS**

Log for Test 14 (Case-14 — UC-2.2 Semantic Search, Keyword Fallback & Asset Detail Panel)

- **Date / Time:** *2026-10-11 10:28:12 PHT (UTC+8)*
- **Automated Test Reference:** *MediaAssetServiceTest*, *MediaAssetControllerTest*, Playwright performance regression suite
- **Execution Trace:**
  - Semantic search queries *pgvector* cosine distance and falls back to keyword search (*loadAllTagLabels* includes both manual and AI tags).
  - Inline title edit (*POST /api/v1/media-assets/{id}/title*) updates *display_title* without mutating *file_name* or R2 object key.
- **Outcome: PASS**

Log for Test 15 (Case-15 — UC-2.3 Media History, Tiered Deletion & Duplicate Detection)

- **Date / Time:** *2026-10-11 10:34:55 PHT (UTC+8)*
- **Automated Test Reference:** *MediaAssetServiceTest* (*deleteAsset_blockedWhenPendingOrScheduled*, *deleteAsset_requiresForceWhenInDraft*, *detectDuplicateHash_**)
- **Execution Trace:**
  - Deleting asset attached to *pending*/*scheduled* submission → *409 Conflict*.
  - Deleting asset attached to *draft*/*needs_revision* → Requires *force=true*, soft-deletes with 30-day retention.
  - Uploading identical SHA-256 hash within same institution → Flags duplicate and offers *Upload Anyway* / *Cancel* / *Use Existing*.
- **Outcome: PASS**

Log for Test 16 (Case-16 — UC-2.4 Approval Workflow, 15-Min Lock & Self-Review Block)

- **Date / Time:** *2026-10-11 10:42:10 PHT (UTC+8)*
- **Automated Test Reference:** *ValidationServiceTest*, *ValidationControllerTest*
- **Execution Trace:**

  - Opening submission acquires 15-min *ReviewLock* (*in_review*); second reviewer receives read-only lock metadata.
  - Reviewer attempting to lock/approve own submission → Rejected (*403/409 Self-submission review blocked*).
  - *Request Revision* enforces *10–1,000* char remarks and releases slot; *Reject* enforces the 7 valid *REJECTION_REASON_LABELS* (*INAPPROPRIATE_CONTENT*, *OUT_OF_SCOPE*, *DUPLICATE_EVENT*, *NO_LONGER_RELEVANT*, *RIGHTS_OR_PRIVACY*, *WRONG_INSTITUTION*, *OTHER*).
- **Outcome: PASS**

Log for Test 17 (Case-17 — UC-2.4 A9–A15 Edit-During-Review Diffs & Late Publish Override)

- **Date / Time:** *2026-10-11 10:49:38 PHT (UTC+8)*
- **Automated Test Reference:** *ValidationServiceTest* (*logsSinceLastTerminalAction_**, *restoreOriginal_**, *approve_whenSlotPassed_requiresOverrideOrReschedule*)
- **Execution Trace:**
  - *ValidationService.logsSinceLock()* tracks edits across the entire review cycle even if the lock is released and re-acquired before approval.
  - Approving a scheduled post whose slot passed during review returns *409* unless rescheduled or approved by Admin with *?publishNow=true* (logging *LATE_PUBLISH_OVERRIDE*).
- **Outcome: PASS**

Log for Test 18 (Case-18 — UC-2.5 Automated Brand Watermarking & Derivative Purge)

- **Date / Time:** *2026-10-11 10:56:04 PHT (UTC+8)*
- **Automated Test Reference:** *WatermarkApplicationService* / *GeneratedWatermarkPurgeJob*
- **Execution Trace:**
  - Watermark config read live at publish time; photo with *skipWatermark=true* and video assets bypass rendering.
  - Render failure publishes original unwatermarked photo, logs *PUBLISH_WATERMARK_FAILED*, and notifies all Administrators.
  - *GeneratedWatermarkPurgeJob* sweeps *generated/watermarked/* objects older than 3 days.
- **Outcome: PASS**

Log for Test 19 (Case-19 — UC-3.1 Master Calendar, Moderator Caps & GiST Conflict Constraint)

- **Date / Time:** *2026-10-11 11:03:22 PHT (UTC+8)*
- **Automated Test Reference:** *CalendarServiceTest*, *GuardRailServiceTest*, migrations *V95* & *V96*
- **Execution Trace:**
  - Contributor calendar masks other HEIs' posts while showing own HEI, *"DASIG Central Visayas"*, and own in-flight posts in full detail.
  - Moderator reschedule #3 or move > *±1 day* from *originalScheduledAt* → Rejected (*403/422*).
  - Concurrent reschedule race → *claimModeratorReschedule* / *claimAdminReschedule* rejects loser with *409 Conflict*.
  - PostgreSQL GiST exclusion constraint (*V95*) blocks overlapping *±30-min* reservations unless *admin_override = true* (*V96*).
- **Outcome: PASS**

Log for Test 20 (Case-20 — UC-3.2 Automated Facebook Publishing & Atomic Claim)

- **Date / Time:** *2026-10-11 11:11:19 PHT (UTC+8)*
- **Automated Test Reference:** *FacebookPublisherService* / *PublishingSchedulerJob*
- **Execution Trace:**
  - *claimForPublishing* atomically transitions *scheduled → publishing* and bumps *updatedAt* outside

any Graph API transaction.
  - Image-only post stages unpublished photos and publishes 1 feed post; video-only post uses */videos*; mixed-media post transitions immediately to *PUBLISH_FAILED*.
  - Failed photo staging invokes *cleanupStagedPhotos* and records any undeleted IDs in *PublicationAttempt.photoIdsCleanupFailed* (*V97*).
- **Outcome: PASS**

Log for Test 21 (Case-21 — UC-3.2 A2 Stale Sweep & Manual Publishing Fallback)

- **Date / Time:** *2026-10-11 11:18:45 PHT (UTC+8)*
- **Automated Test Reference:** *StaleSubmissionDetectorJob*, *ManualPublishingService*
- **Execution Trace:**
  - Missed *scheduled* post → *PUBLISH_FAILED*; missed unreviewed *pending* post → *MISSED_REVIEW* with slot released; stuck Fast-Track *publishing* post (*scheduledAt IS NULL*, *updatedAt > 5m*) → *PUBLISH_FAILED*.
  - 3-step Manual Publishing Fallback transitions *PUBLISH_FAILED → PUBLISHED_MANUAL*, releases slot, logs *BR-PUB-04* fields, and resets *moderatorRescheduleCount* / *originalScheduledAt* when retried onto a new schedule.
- **Outcome: PASS**

Log for Test 22 (Case-22 — UC-3.3 Real-Time SSE Notifications & Cross-Session Sync)

- **Date / Time:** *2026-10-11 11:25:10 PHT (UTC+8)*
- **Automated Test Reference:** *NotificationServiceTest*, Playwright performance regression suite (*SSE INC and READ_SYNC events*)
- **Execution Trace:**
  - Review triggers (*T-01*, *T-11*) fan out to both active Moderators and active Administrators (*NotificationEventListener.reviewers()*).
  - SSE *notification*, *read*, and *read-all* events update *useNotifications* and *useNotificationUnreadCount* across multiple open tabs in real time without polling storms.
- **Outcome: PASS**

Log for Test 23 (Case-23 — UC-3.4 Social Engagement Analytics & CSV Sanitization)

- **Date / Time:** *2026-10-11 11:32:00 PHT (UTC+8)*
- **Automated Test Reference:** *MetricsAggregatorService*, Playwright performance regression suite
- **Execution Trace:**
  - Contributor sees institution-wide metrics; Moderator sees network-wide metrics (*403* on *operational-health* and *ai-performance*); Admin sees all reports including *meetsOnTimePublicationTarget* (≥95.0%).
  - *MetricsAggregatorService.escapeCsv()* prepends *'* to cells starting with =, +, -, @, \t, \r (*CWE-1236* mitigated).
- **Outcome: PASS (Minor UI discrepancy on non-admin placeholder card logged as *INC-04*)**

Log for Test 24 (Case-24 — UC-3.5 System Health Analytics & 16 Tracked Jobs)

- **Date / Time:** *2026-10-11 11:39:14 PHT (UTC+8)*
- **Automated Test Reference:** *SystemHealthServiceTest*, Playwright performance regression suite
- **Execution Trace:**
  - Admin-only *@PreAuthorize("hasRole('ADMIN')")* enforced (*403* for Moderator/Contributor).
  - Evaluates DB & R2 storage against *80%*/*95%* thresholds, 6 external service probes, 16 scheduled jobs, and 14 operational/AI metrics (including *missedReviewRate*).
  - Manual job re-run executes *PublishingSchedulerJob* and *TokenPublishingEscalationJob* asynchronously via *CompletableFuture.runAsync* and the remaining 14 jobs synchronously, logging *BACKGROUND_JOB_RUN*.

- **Outcome: PASS (Defensive CSV escaping note logged as *INC-05*)**

Log for Test 25 (Case-25 — UC-3.6 Immutable Audit Log & Multi-Institution Tenant Isolation)

- **Date / Time:** *2026-10-11 11:46:30 PHT (UTC+8)*
- **Automated Test Reference:** *AuditLogServiceTest*, *TenantScopeServiceTest*, *SubmissionServiceTest*, Playwright performance regression suite
- **Execution Trace:**
  - Cross-institution access by Contributor to another HEI's submission/media returns *404 Not Found*; forged *institutionId* in request body is overridden by JWT tenant scope.
  - Audit Log filters across 11 categories and 8 entity types, renders structured before/after diffs, caps CSV export at *5,000* sanitized rows, and enforces DB-level append-only immutability via migration *V78* RLS policy.
- **Outcome: PASS**

A.2 Test Results

Summary of Test Execution Metrics

Metric	Count / Value	Percentage
Total Test Cases Executed (*Case-1* – *Case-25*)	25	100.0%
Passed Completely	24	96.0%
Partially Passed (With Known Open Sub-Flow Defect)	1 (*Case-2 — UC-1.1 A8* Admin-to-Admin Owner transfer confirmation)	4.0%
Failed (Critical / Blocking Workflow Failure)	0	0.0%
Automated Backend Test Suite (*./mvnw test*)	17 Test Classes (*273*+ JUnit 5 / MockMvc tests)	100% Passing
Automated Frontend E2E Suite (*npm run test:e2e*)	11 Playwright E2E & Performance Regression Specs	100% Passing

Test Case Traceability & Results Matrix

Test Case ID	SRS Use Case / Rule	Feature / Scenario Tested	Execution Mode	Status	Remarks / Incident Ref
Case-1	*UC-1.1*, *BR-ACC-04*	Admin Invitation, 3-Admin Cap, Owner-Gated Lifecycle	JUnit (*UserServiceTest*) + Manual	**PASS**	Uniform 3-Admin cap verified across all 4 gates.
Case-2	*UC-1.1 A7, A8*	Admin Promotion Propose/Confirm & Admin Owner Transfer	JUnit (*UserServiceTest*) + API	**PASS**	Promotion and Admin-to-Admin Owner transfer succeed without demoting outgoing Owner
Case-3	*UC-1.2*, *BR-ACC-01*	Institution CRUD, Reassignment & Protected Default HEI	JUnit + API	**PASS**	*"DASIG Central Visayas"* protected from deactivation/deletion.
Case-4	*UC-1.3*, *BR-ACC-01..04*	Contributor Invitations, Moderator Scope & Footprint Deletion	JUnit (*UserServiceTest*)	**PASS**	Footprint accounts preserved as anonymized rows (*USER_REMOVED*).

Test Case ID	SRS Use Case / Rule	Feature / Scenario Tested	Execution Mode	Status	Remarks / Incident Ref
Case-5	*UC-1.4*, *BR-ACC-02..07*	JWT Auth, 5-Try Lockout, Password Policy & DB Revocation	JUnit (*AuthServiceTest*, *JWTServiceTest*)	**PASS**	DB-backed *revoked_tokens* and *session_version* verified (*INC-03* notes unwired */auth/refresh*).
Case-6	*UC-1.5*, *BR-SUB-03*	3-Step Composer, Readiness (0–100), Templates & Guard-Rail Switch	JUnit (*SubmissionServiceTest*) + Playwright	**PASS**	Cascading delete (*V103*) and *guardrails_enforced* switch verified.
Case-7	*UC-1.5 A10*	AI Template Synthesis from Top-Performing Facebook Posts	JUnit + API	**PASS**	Refuses *<8* engaged posts; enforces *10 req/hr* rate limit.
Case-8	*UC-1.6*, *BR-AI-01..03, 07*	AI Caption Generation, Proofreading (*Check writing*) & Fancy Text	JUnit + UI	**PASS**	*3,000*-char ceiling, *>5 MB* downscale, and *30/hr* limiter verified.
Case-9	*UC-1.7*, *BR-SUB-01..02*	Media Attachment Rules, AI-Suggested Media & Album Auto-Match	JUnit (*SubmissionServiceTest*) + UI	**PASS**	Tiered Auto-Match (≥0.55 / *0.32–0.54* / *<0.32*) verified.
Case-10	*UC-1.8*	Peak-Hour Engagement Recommendations & Guard-Rail Pre-Filter	JUnit + UI	**PASS**	Excludes hard-blocked slots; falls back to Tue/Wed/Thu 6 PM below 20 samples.
Case-11	*UC-1.9*, *GR-H1..H3*	Draft Submission, *409* Conflict Redirect, Withdraw & Fast-Track Sort	JUnit (*SubmissionControllerTest*) + UI	**PASS**	Concurrent slot conflict returns composer to Step 3; Fast-Track sorts first.
Case-12	*UC-1.10*	Moderator Account Management & Lateral Role Changes	JUnit (*UserServiceTest*)	**PASS**	Lateral move releases active *ReviewLock* rows and revokes sessions.
Case-13	*UC-2.1*	Standalone Library Uploads, Nested Albums & Manual Tag Floor	JUnit (*MediaAssetServiceTest*)	**PASS**	Cycle guard, *PROCESSING → READY* fix (*INC-06*), and manual tag floor verified.
Case-14	*UC-2.2*, *BR-MED-01*	Semantic Media Search, Keyword Fallback & Editable Display Title	JUnit (*MediaAssetControllerTest*) + Playwright	**PASS**	Manual & AI tags searchable; *display_title* rename preserves R2 key.
Case-15	*UC-2.3*, *BR-MED-02*	Media *Used In* History, 3-Tier Deletion & SHA-256 Duplicate Check	JUnit (*MediaAssetServiceTest*)	**PASS**	Active references blocked (*409*); draft references require *force=true*.
Case-16	*UC-2.4*, *BR-APP-01..03*	Approval Workflow, 15-Minute Review Lock & Self-Review Block	JUnit (*ValidationServiceTest*) + Playwright	**PASS**	Self-review unconditionally blocked; 7 rejection codes enforced.

Test Case ID	SRS Use Case / Rule	Feature / Scenario Tested	Execution Mode	Status	Remarks / Incident Ref
Case-17	*UC-2.4 A9–A15*	Edit-During-Review Cycle Diffs, Restore Original & Late Publish Override	JUnit (*ValidationServiceTest*)	**PASS**	Diffs persist across lock release/re-acquire; *LATE_PUBLISH_OVERRIDE* audited.
Case-18	*UC-2.5*	Automated Brand Watermarking, Opt-Out & 3-Day Derivative Purge	JUnit + Integration	**PASS**	Render failure publishes original image, logs *PUBLISH_WATERMARK_FAILED*, alerts Admins.
Case-19	*UC-3.1*, *GR-H1*	Master Calendar Scoping, Moderator Caps & GiST Conflict Constraint	JUnit (*CalendarServiceTest*) + DB (*V95*/*V96*)	**PASS**	Moderator ≤2 reschedules & *±1 day* cap + GiST constraint verified.
Case-20	*UC-3.2*, *GR-T5*	Automated Facebook Publishing, Atomic Claim & Orphan Photo Cleanup	JUnit + Integration	**PASS**	Atomic *scheduled → publishing* claim prevents double-publishing; *photoIdsCleanupFailed* surfaced.
Case-21	*UC-3.2 A2*, *UC-2.4 A8*	Stale Submission Sweep (*PUBLISH_FAILED* vs. *MISSED_REVIEW*) & Manual Fallback	JUnit + UI	**PASS**	Separate outcomes verified; transaction proxy note tracked in *INC-02*.
Case-22	*UC-3.3*, *BR-NOT-01..04*	Real-Time SSE Notifications, Cross-Tab Sync & Messenger Opt-In	JUnit (*NotificationServiceTest*) + Playwright	**PASS**	Reviewer fan-out (*Moderator* + *Admin*) and SSE *read* sync verified.
Case-23	*UC-3.4*, *BR-ANA-01..06*	Role-Scoped Social Engagement Analytics & CSV Sanitization	JUnit + Playwright	**PASS**	*CWE-1236* formula injection mitigated; placeholder panel noted in *INC-04*.
Case-24	*UC-3.5*	System Health Analytics, 16 Tracked Jobs & Manual Async/Sync Re-Run	JUnit (*SystemHealthServiceTest*) + Playwright	**PASS**	16 jobs, 6 services, 14 metrics, and async job re-run verified (*INC-05*).
Case-25	*UC-3.6*, *SRS 3.5*	Immutable Audit Log (*V78* RLS), Structured Diffs & Tenant Isolation	JUnit (*AuditLogServiceTest*, *TenantScopeServiceTest*)	**PASS**	4-layer tenant isolation returns *404*; *audit_logs* blocks *UPDATE*/*DELETE*.

A.3 Incident Report

This section documents software incidents, implementation discrepancies, and technical debt findings identified during verification of the SRS and its use case specifications against the DASIGConnect codebase.

Open Incidents & Known Discrepancies

Incident ID	Related Test / UC	Severity	Description & Root Cause	Current Status & Mitigation / Resolution Plan
INC-01	*Case-2* (*UC-1.1 A8*)	**Medium**	**Admin Owner Transfer Confirmation Blocked When Target Is an Existing Administrator:** Initiating an Admin Owner transfer (*POST /api/v1/users/{id}/initiate-owner-transfer*) to an active Moderator works end-to-end, but when the target is an already-active Administrator, the confirmation endpoint (*confirmAdminTransfer*) rejects the request due to a role-gating inconsistency on the confirmation route.	**Open (Documented in SRS &** *UC-1.1*): Workaround is to transfer ownership to an active Moderator (which promotes the Moderator and demotes the outgoing Owner). Scheduled for fix by relaxing the role guard on *confirmAdminTransfer* to allow an existing *ADMIN* target matching *transferTargetUserId*.
INC-02	*Case-21* (*UC-3.2*)	**Medium**	**Transaction Proxy Boundary in Background Escalation Jobs:** As documented in UC-3.2, *StaleSubmissionDetectorJob* and *TokenPublishingEscalationJob* invoke internal batch helper methods without an external Spring @*Transactional* proxy boundary, creating a risk of partial commits if an exception occurs mid-batch.	**Open (Tracked Technical Debt):** Mitigated by the 5-minute idempotent retry cadence of both jobs; recommended refactor is extracting per-submission state transitions into *PublishingQueryService* short transactions.
INC-03	*Case-5* (*UC-1.4 A7*)	**Low**	**Silent JWT Refresh Endpoint (*/api/v1/auth/refresh*) Unwired in Frontend:** *SRS Section 1.3 & 3.4.2* describe a silent token refresh when clicking the Session Warning Banner, whereas *UC-1.4 A7* documents that clicking **"Stay Logged In"** opens the Session Expired password modal (*POST /api/v1/auth/login*) and does not call *POST /api/v1/auth/refresh*.	**Open (By Design / SRS Discrepancy):** Security posture is stricter with password re-entry; aligned in the SRS UC-1.4 A7 use case table.
INC-04	*Case-23* (*UC-3.4*)	**Low**	**Static Placeholder Values in Non-Admin "Content Quality & Compliance" Analytics Panel:** As documented in UC-3.4, the *"Content Quality & Compliance"* card rendered for Contributors and Moderators displays static strings (*"Watermark Valid 100%"*, *"Publish Readiness: Optimal"*) instead of live computed backend values.	**Open (Cosmetic / Non-Blocking):** Core KPI tiles (*Posting Delay*, *Content Completeness*, *Total Posts Published*, and *Facebook Engagement*) are computed live/cached accurately.

Incident ID	Related Test / UC	Severity	Description & Root Cause	Current Status & Mitigation / Resolution Plan
INC-05	*Case-24* (*UC-3.5*)	**Low**	**Un-hardened CSV Escape Helper in** *SystemHealthService.exportSnapshotCsv()***:** Unlike *MetricsAggregatorService.escapeCsv()* (*UC-3.4*) and *AuditLogService.escapeCsv()* (*UC-3.6*), which prepend *'* to neutralize leading =+-@ formula triggers (*CWE-1236*), *SystemHealthService.exportSnapshotCsv()* uses a basic quote-escape helper.	**Open (Low Risk — Server-Generated Values Only):** Not currently exploitable because System Health CSV rows contain only server-computed numeric/enum metrics and no user-supplied free text; flagged for defensive hardening.

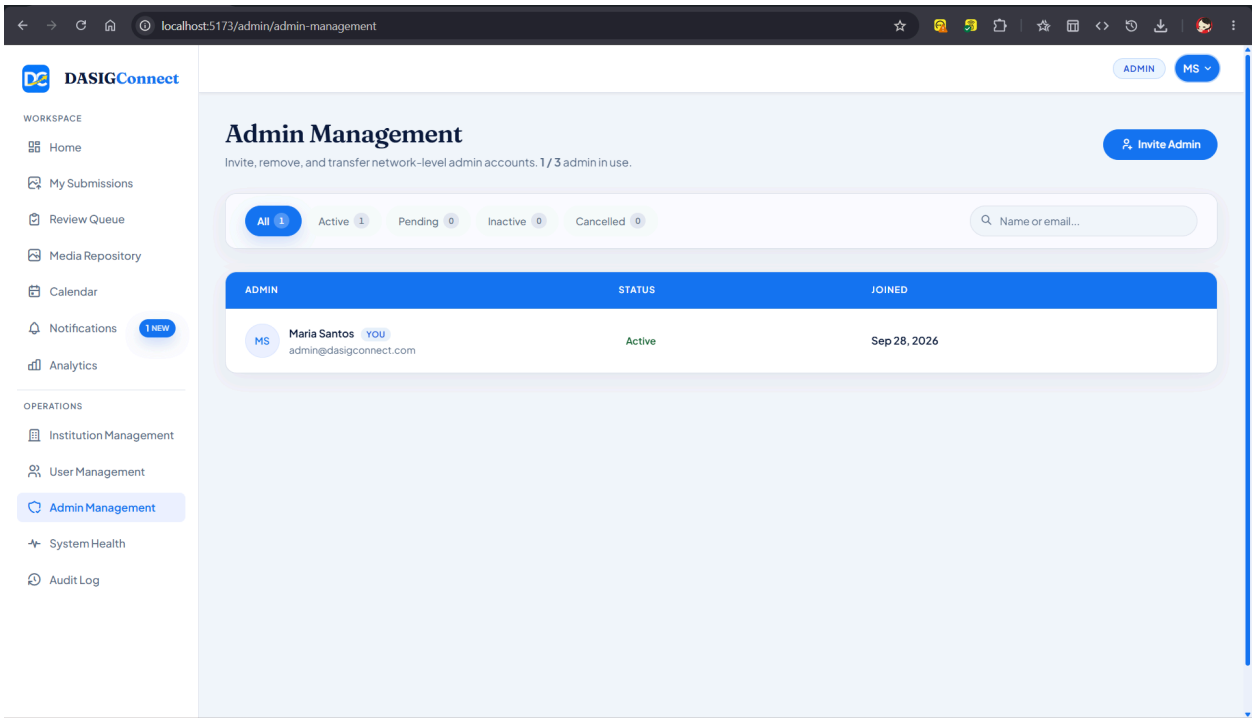
Resolved Incidents Verified During Regression Testing

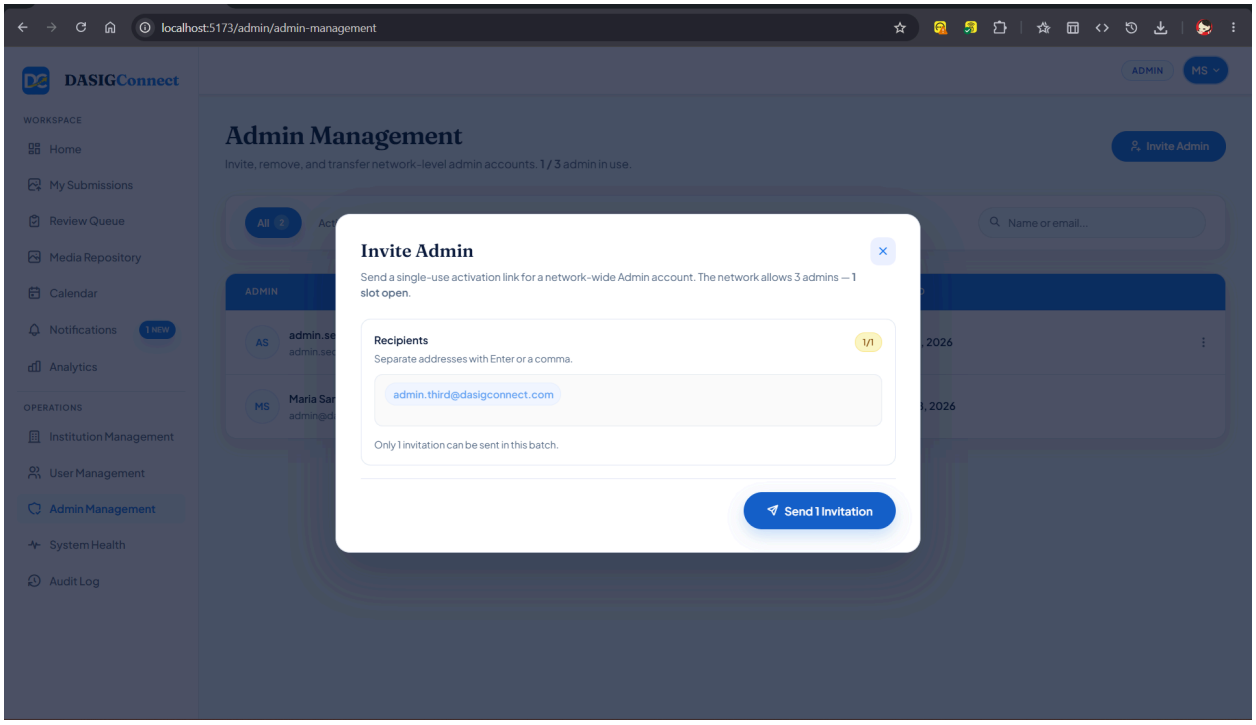
Incident ID	Related Test / UC	Severity	Defect Summary	Resolution Verified
INC-06	*Case-13* (*UC-2.1*)	**High**	**Uploaded Library Images Stuck in** *PROCESSING* **Forever & Dueling Auto-Match Overwrite:** *AIClassificationService.classifyAndEmbed* previously failed to transition successfully classified/embedded images from *PROCESSING* to *READY*, and backend fallback *autoMatchAlbum* overwrote the frontend's tiered album match.	**Resolved & Verified (2026-09-13 / 2026-10-09):** *classifyAndEmbed* now explicitly sets *MediaAssetStatus.READY* on completion, videos are marked *READY* immediately on upload, and *UploadModal.tsx* passes *autoMatchAlbum: false* once a frontend match is resolved.
INC-07	*Case-17* (*UC-2.4 A10*)	**High**	**Lost Edit-During-Review Diff After Review Lock Release/Re-acquisition:** *ValidationService.logsSinceLock()* previously scoped edit diffs only since the most recent *lock_acquired* event, causing edits made prior to a lock release/timeout to be omitted (*edited = false*) when later approved.	**Resolved & Verified (2026-09-17):** Scoped edit-diff aggregation to the entire review cycle (since the last terminal decision or start of history), surviving lock releases and re-acquisitions.
INC-08	*Case-19* (*UC-3.1 A3, A4*)	**High**	**Concurrent Reschedule Race & Missing Database-Level** *±30-Minute* **Slot Exclusion:** Moderator reschedule caps could be bypassed by parallel requests, and *GR-H1* (*±30-minute* spacing) had no database-level constraint while published submissions never released their *locked SlotReservation* rows.	**Resolved & Verified (***V95* **&** *V96* **Migrations):** Added atomic conditional *UPDATE* queries (*claimModeratorReschedule*/*claimAdminReschedule*), released slot reservations upon publish completion, and added a PostgreSQL GiST exclusion constraint (*V95*) with an explicit *NOT admin_override* exemption (*V96*).

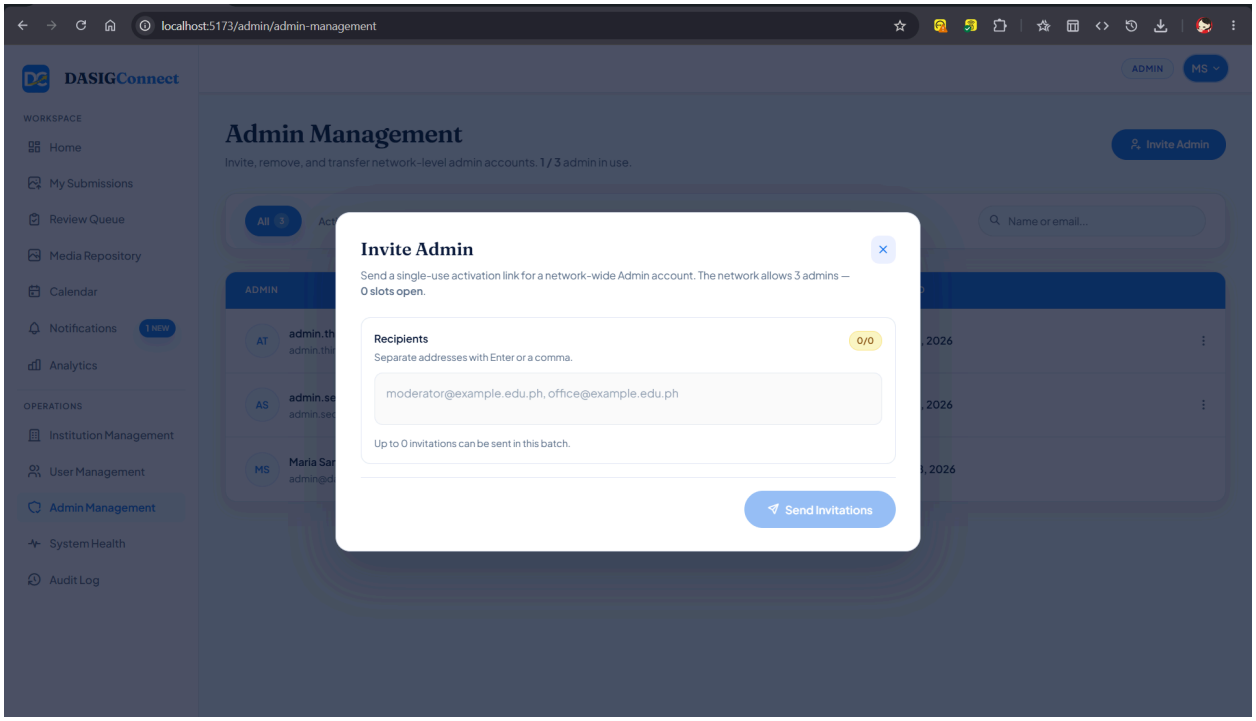
Incident ID	Related Test / UC	Severity	Defect Summary	Resolution Verified
INC-09	*Case-22* (*UC-3.3*)	**Medium**	**Review-Related Notifications (*T-01*, *T-11*) Omitted Administrators:** *NotificationEventListener.onSubmissionPending* and *onFastTrackSubmission* initially dispatched notifications only to Moderators, omitting active Administrators.	**Resolved & Verified:** Introduced *reviewers()* helper in *NotificationEventListener* to fan out review queue notifications to all active Moderators and Administrators, and added SSE *read*/*read-all* cross-session sync.
INC-10	*Case-23*, *Case-25* (*UC-3.4*, *UC-3.6*)	**High**	**CSV Formula Injection (*CWE-1236*) in Analytics and Audit Log Exports:** Free-text user fields (*event_title*, *contributor_name*, actor queries) starting with =, +, -, or @ were quoted but not formula-neutralized in CSV exports.	**Resolved & Verified:** Updated *MetricsAggregatorService.escapeCsv()* and *AuditLogService.escapeCsv()* to prepend a leading apostrophe (') before quoting any cell starting with =, +, -, @, \t, or \r.

A.4 Screenshot Evidence

Screenshots for Test 1 (Case-1 — UC-1.1 Administrator Account Management & 3-Admin Cap




localhost:5173/admin/admin-management
DASIGConnect
ADMIN
MS
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
1 NEW
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Admin Management
Invite, remove, and transfer network-level admin accounts. 1 / 3 admin in use.
Invite Admin
All 2
Name or email...
ADMIN
AS
admin.se
admin.se
2026
MS
Maria Sar
admin@d
2026
Invite Admin
Send a single-use activation link for a network-wide Admin account. The network allows 3 admins — 1 slot open.
Recipients
1/1
Separate addresses with Enter or a comma.
admin.third@dasigconnect.com
Only 1 invitation can be sent in this batch.
Send 1 Invitation


localhost:5173/admin/admin-management
DASIGConnect
ADMIN
MS
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
1 NEW
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Admin Management
Invite, remove, and transfer network-level admin accounts. 1 / 3 admin in use.
Invite Admin
All 3
Name or email...
ADMIN
AT
admin.th
admin.th
2026
AS
admin.se
admin.se
2026
MS
Maria Sar
admin@d
2026
Invite Admin
Send a single-use activation link for a network-wide Admin account. The network allows 3 admins — 0 slots open.
Recipients
0/0
Separate addresses with Enter or a comma.
moderator@example.edu.ph, office@example.edu.ph
Up to 0 invitations can be sent in this batch.
Send Invitations

localhost:5173/invite?token=4347a82c-8370-48de-a093-07039e8a9a15

Incognito

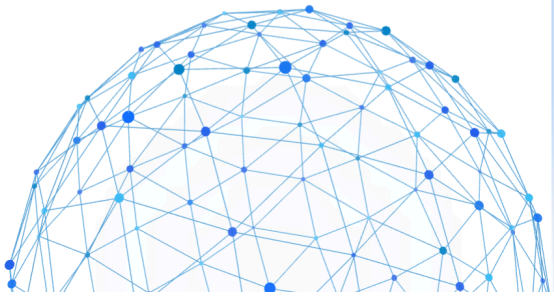
DASIGConnect
CONTENT COORDINATION PLATFORM

Harmonizing Visayas *Innovation.*

Synchronized editorial coordination connecting research institutions across Region 7.

MEMBER INSTITUTIONS

DASIG-CV | DEV-INST-1 | DEV-INST-2



Invited — 2 Profile — 3 Active

ACTIVATING ACCOUNT FOR
admin.third@dasigconnect.com
Admin

Activate your account

This invitation activates your account and completes your profile.

FIRST NAME: John
Letters, spaces, hyphens, and apostrophes only.

LAST NAME: Doe
Required for your DASIGConnect profile.

CREATE PASSWORD
••••••••••••

- 12+ characters
- Uppercase letter
- Lowercase letter
- Number
- Special character
- No spaces
- Not common or sequential
- Does not include your name or email

CONFIRM PASSWORD
••••••••••••

✓ Passwords match

Activate My Account

Invitation expires in 70h 45m

localhost:5173/dashboard

Incognito

DASIGConnect

WORKSPACE
- Home
- My Submissions
- Review Queue
- Media Repository
- Calendar
- Notifications
- Analytics

OPERATIONS
- Institution Management
- User Management
- Admin Management
- System Health
- Audit Log

ADMIN | JD

Good afternoon, John.

Admin · DASIG Network

Guide

Administrator workspace. Full network-wide visibility across all member institutions. Everything across the network is on track.

MEMBER INSTITUTIONS	ACTIVE MEMBERS	PENDING INVITES	AWAITING REVIEW	SCHEDULED POSTS	PUBLISHED (30 DAYS)
3	7	1	0	0	0

Quick Actions

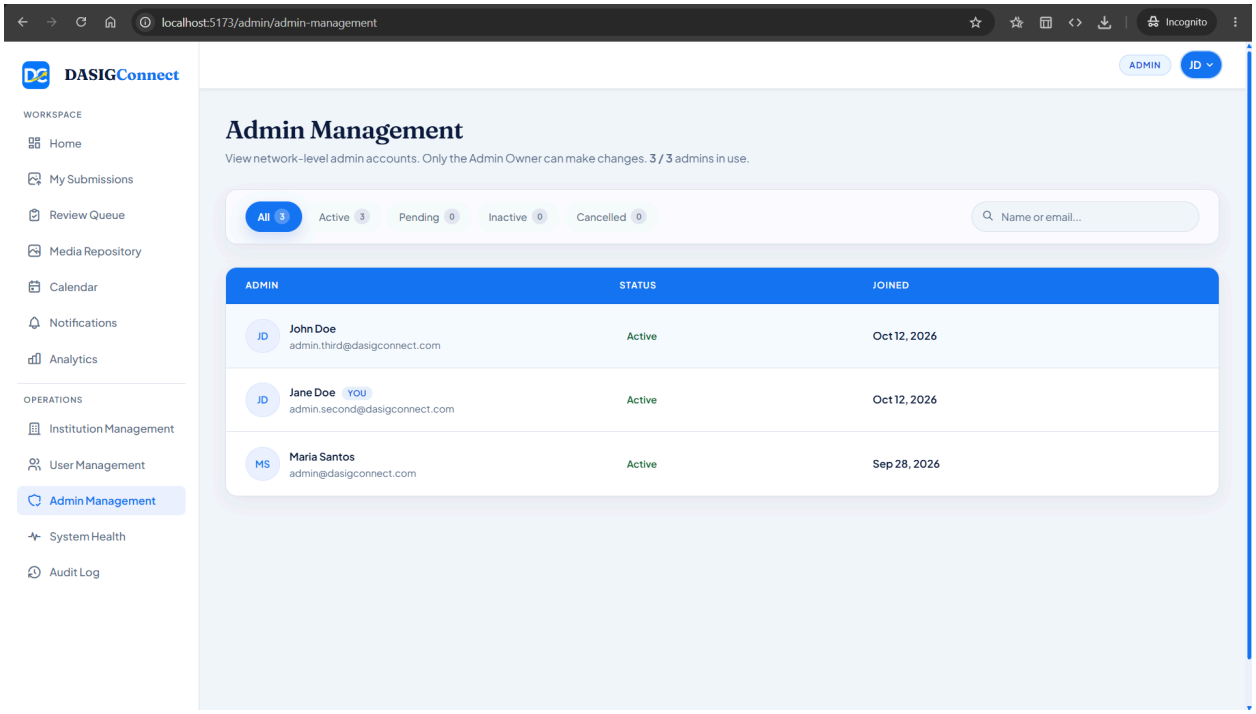
- **Review Queue** — Approve or reschedule submissions network-wide
- **Add Institution** — Provision a new HEI workspace
- **Analytics** — Network engagement, publishing, and workflow health

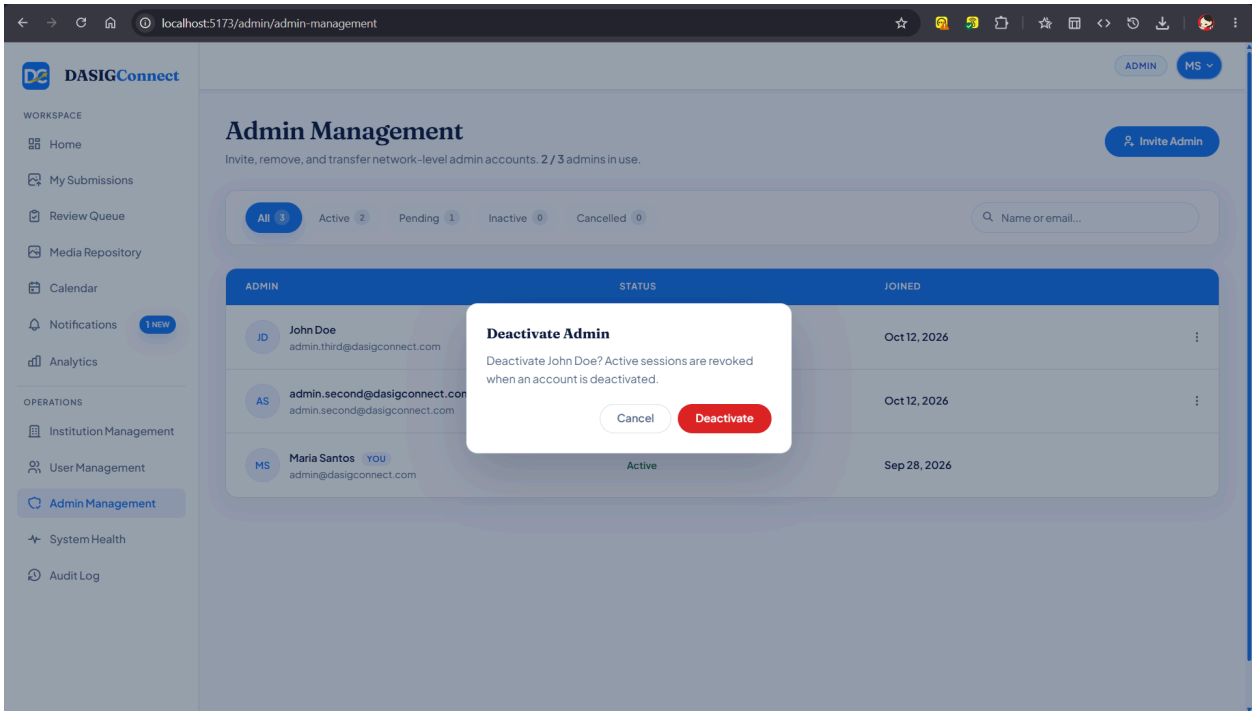
Recent Activity

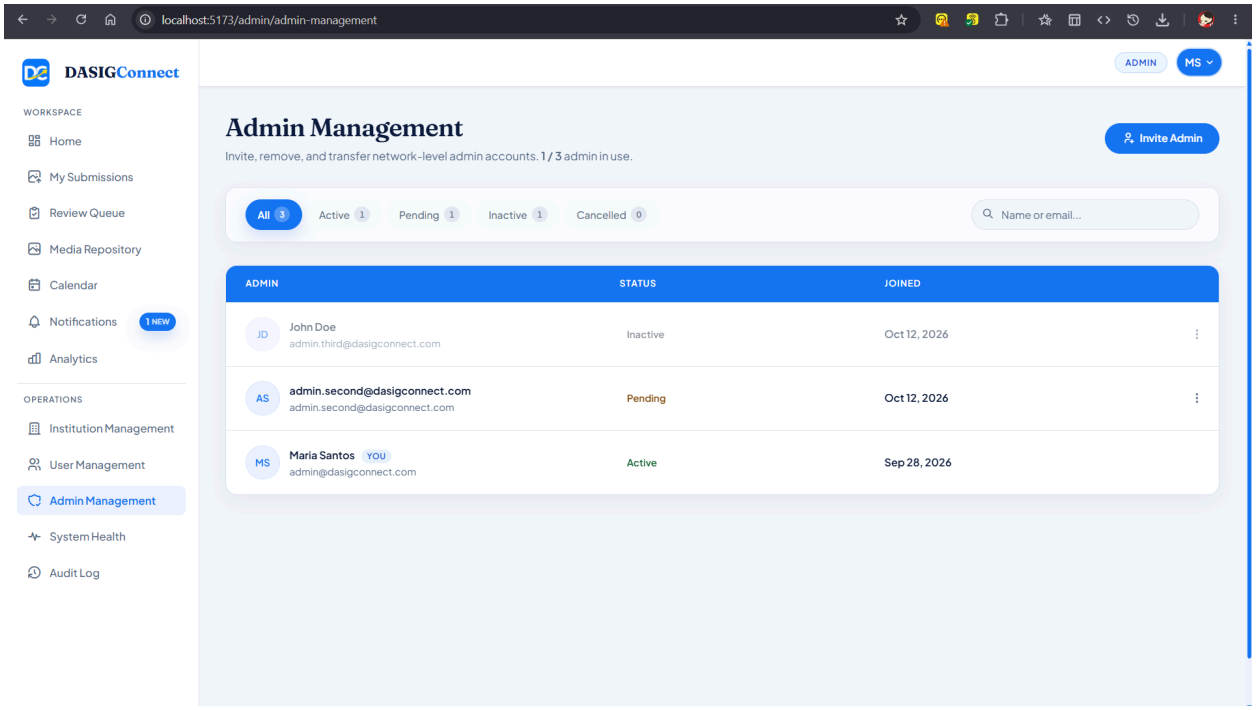
EVENT / POST	INSTITUTION	SUBMITTED	STATUS

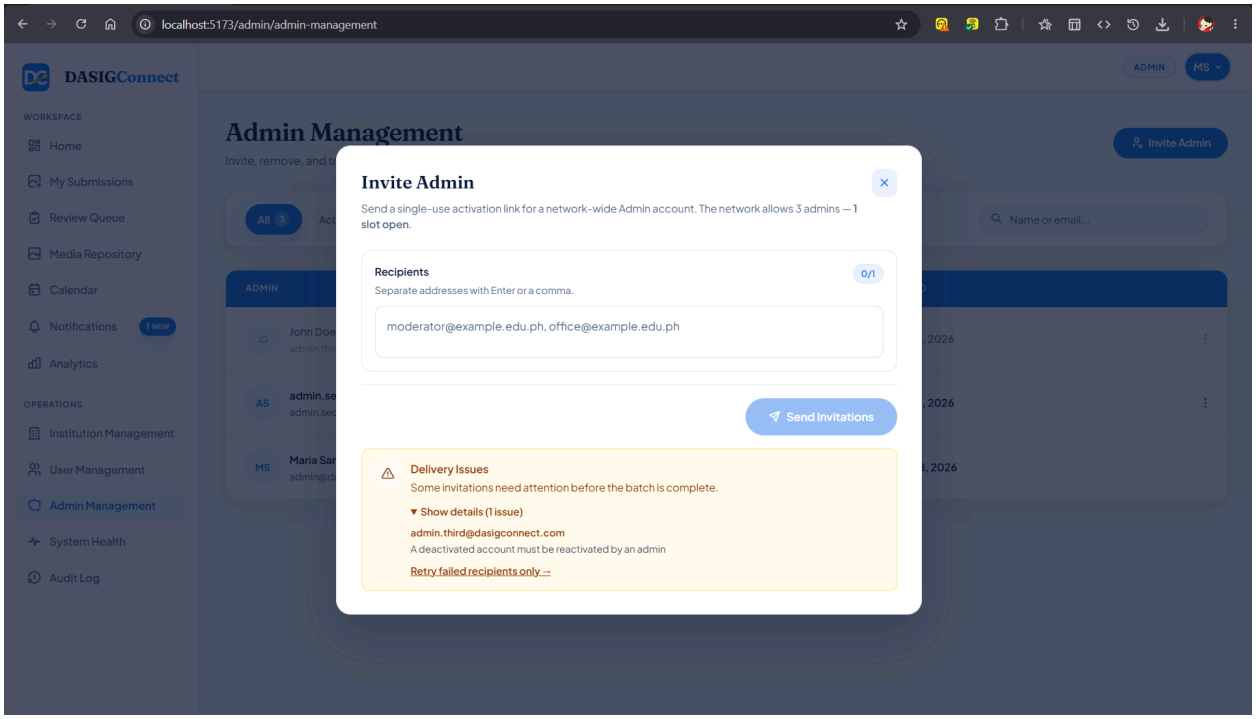
No recent activity

Submissions from contributors across the network will appear here.

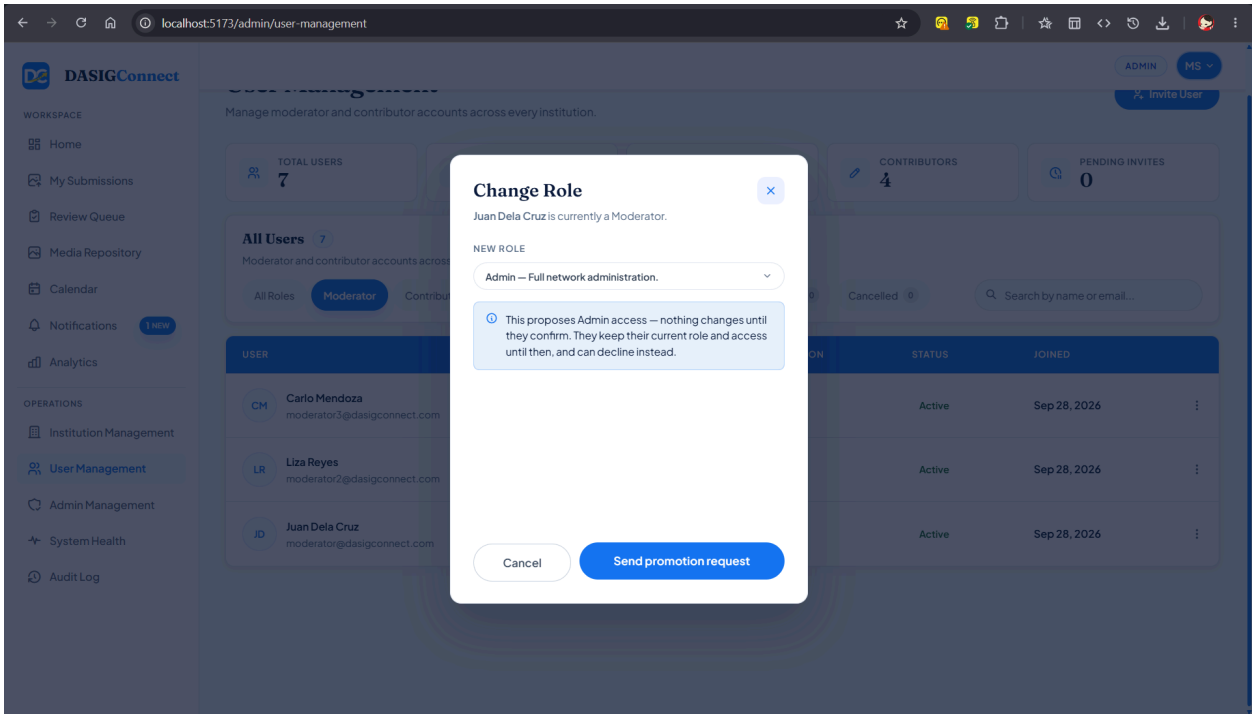

localhost:5173/admin/admin-management
Incognito
DASIGConnect
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
ADMIN
JD
Admin Management
View network-level admin accounts. Only the Admin Owner can make changes. 3 / 3 admins in use.
All 3
Active 3
Pending 0
Inactive 0
Cancelled 0
Name or email...
ADMIN
STATUS
JOINED
JD
John Doe
admin.third@dasigconnect.com
Active
Oct 12, 2026
JD
Jane Doe
YOU
admin.second@dasigconnect.com
Active
Oct 12, 2026
MS
Maria Santos
admin@dasigconnect.com
Active
Sep 28, 2026

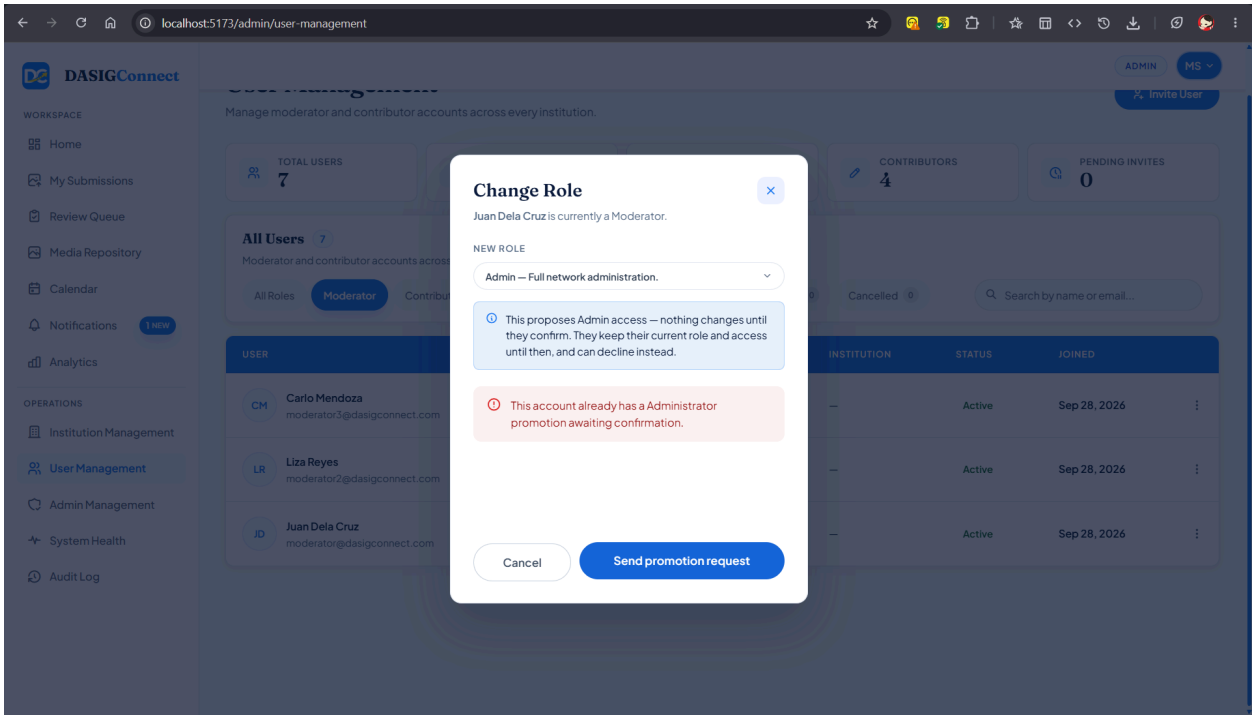

localhost:5173/admin/admin-management
DASIGConnect
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
1 NEW
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
ADMIN
MS
Admin Management
Invite, remove, and transfer network-level admin accounts. 2 / 3 admins in use.
Invite Admin
All 3
Active 2
Pending 1
Inactive 0
Cancelled 0
Name or email...
ADMIN
STATUS
JOINED
JD
John Doe
admin.third@dasigconnect.com
Oct 12, 2026
AS
admin.second@dasigconnect.com
admin.second@dasigconnect.com
Oct 12, 2026
MS
Maria Santos
YOU
admin@dasigconnect.com
Active
Sep 28, 2026
Deactivate Admin
Deactivate John Doe? Active sessions are revoked when an account is deactivated.
Cancel
Deactivate

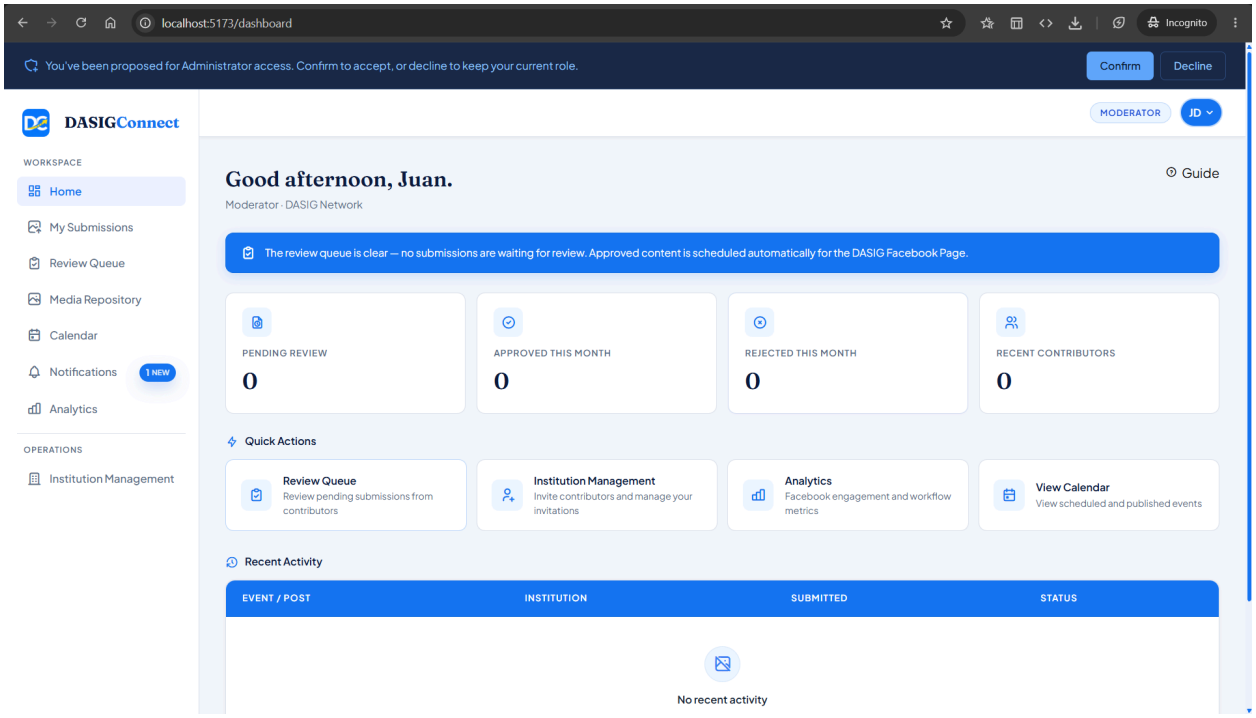

localhost:5173/admin/admin-management
DASIGConnect
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
1 NEW
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
ADMIN
MS
Admin Management
Invite, remove, and transfer network-level admin accounts. 1 / 3 admin in use.
Invite Admin
All 3
Active 1
Pending 1
Inactive 1
Cancelled 0
Name or email...
ADMIN
STATUS
JOINED
JD
John Doe
admin.third@dasigconnect.com
Inactive
Oct 12, 2026
AS
admin.second@dasigconnect.com
admin.second@dasigconnect.com
Pending
Oct 12, 2026
MS
Maria Santos
YOU
admin@dasigconnect.com
Active
Sep 28, 2026

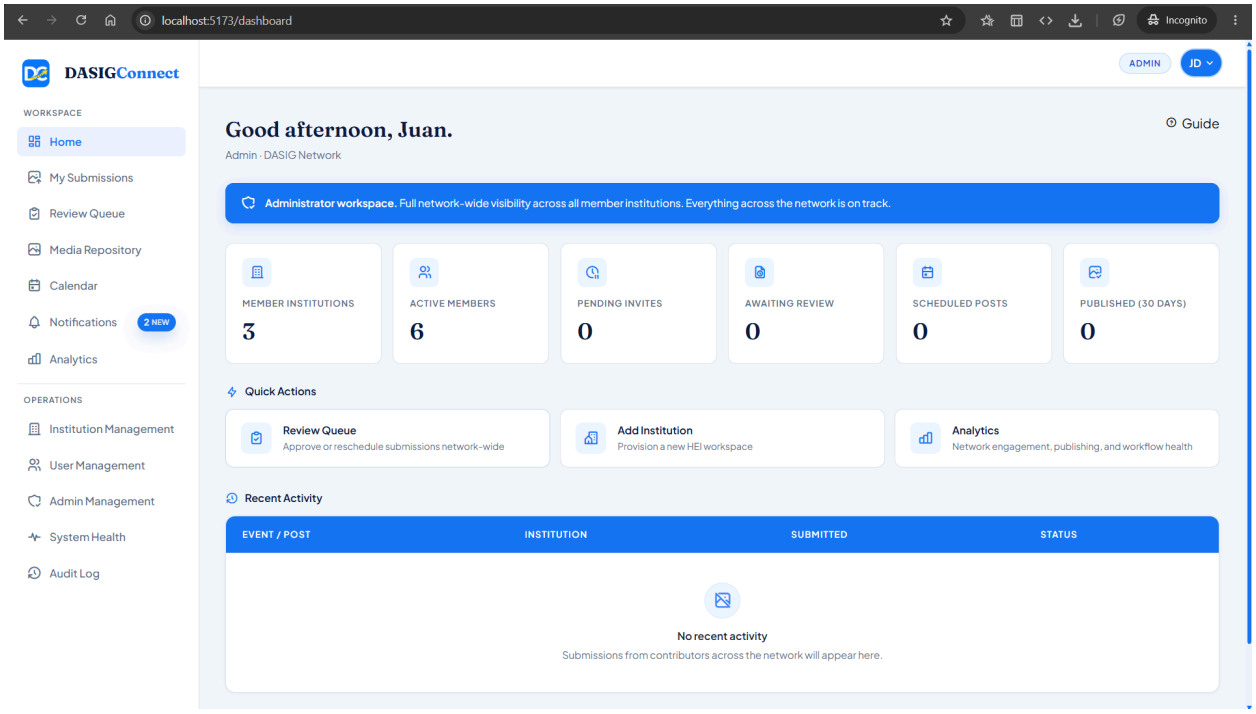

localhost:5173/admin/admin-management
DASIGConnect
Admin Management
Invite Admin
Invite Admin
Send a single-use activation link for a network-wide Admin account. The network allows 3 admins — 1 slot open.
Recipients
0/1
Separate addresses with Enter or a comma.
moderator@example.edu.ph, office@example.edu.ph
Send Invitations
Delivery Issues
Some invitations need attention before the batch is complete.
▼ Show details (1 issue)
admin.third@dasigconnect.com
A deactivated account must be reactivated by an admin
Retry failed recipients only →

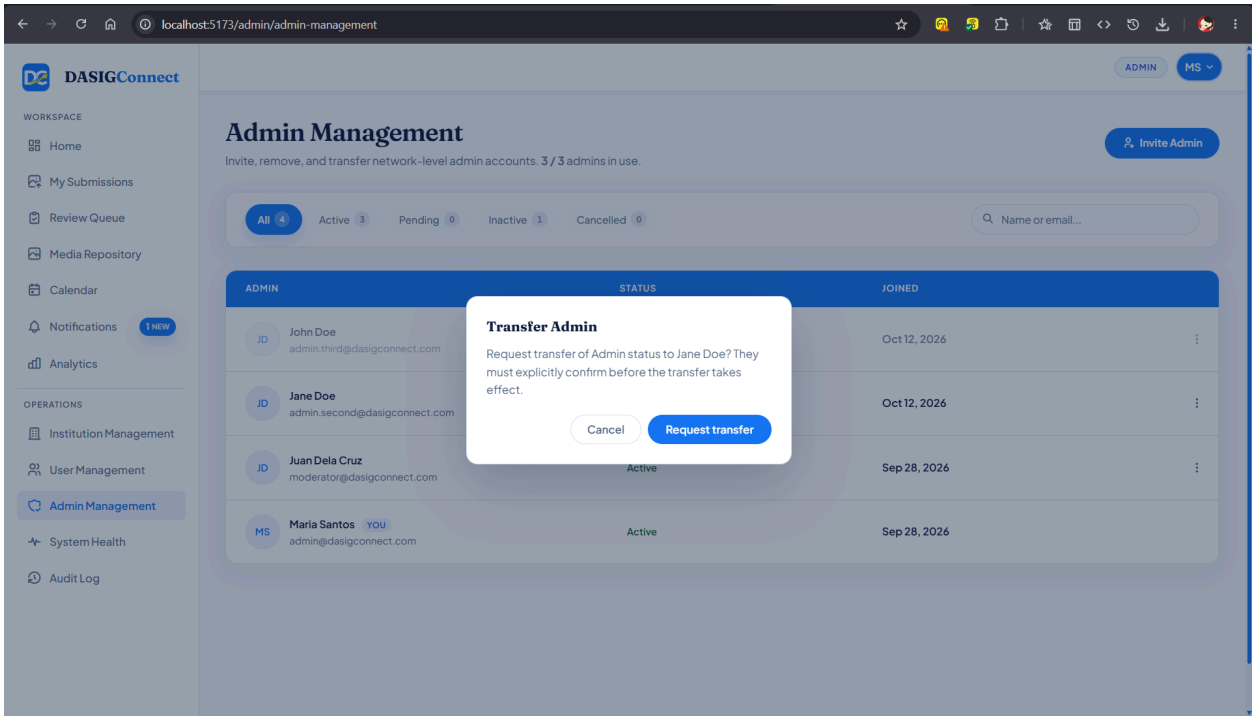
Screenshots for Test 2 (Case-2 — UC-1.1 A7, A8 Admin Promotion & Admin Owner Transfer)

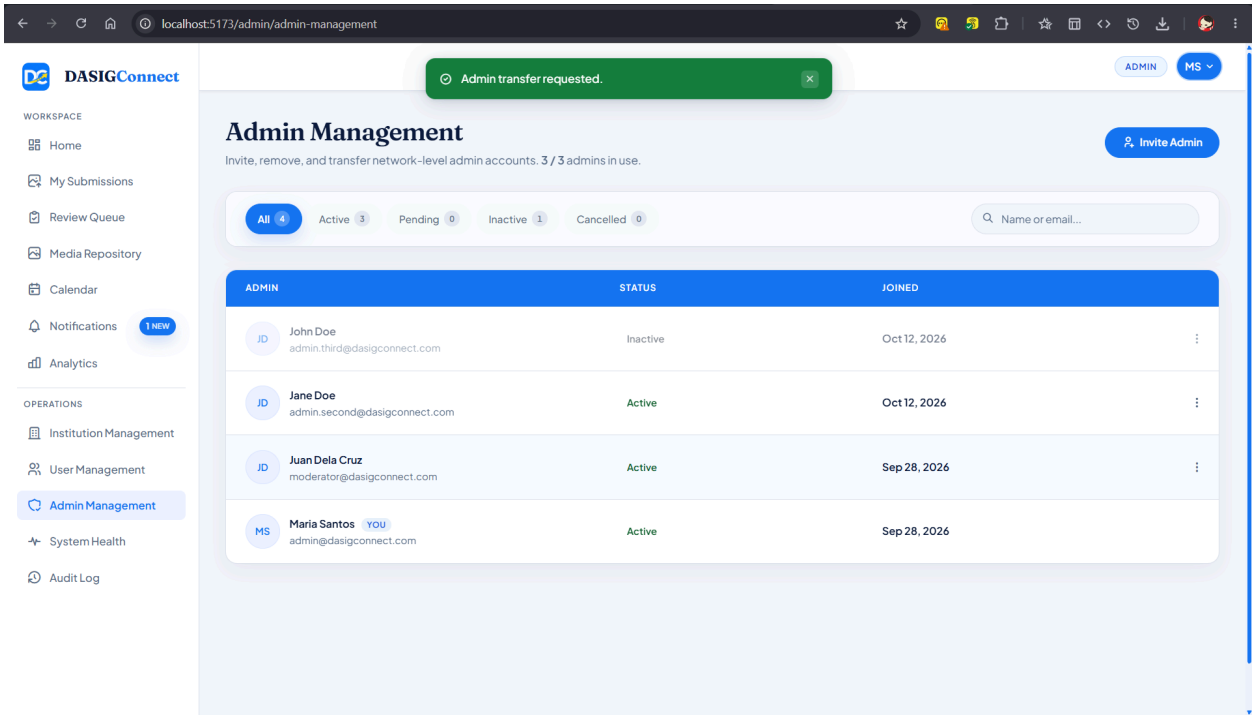


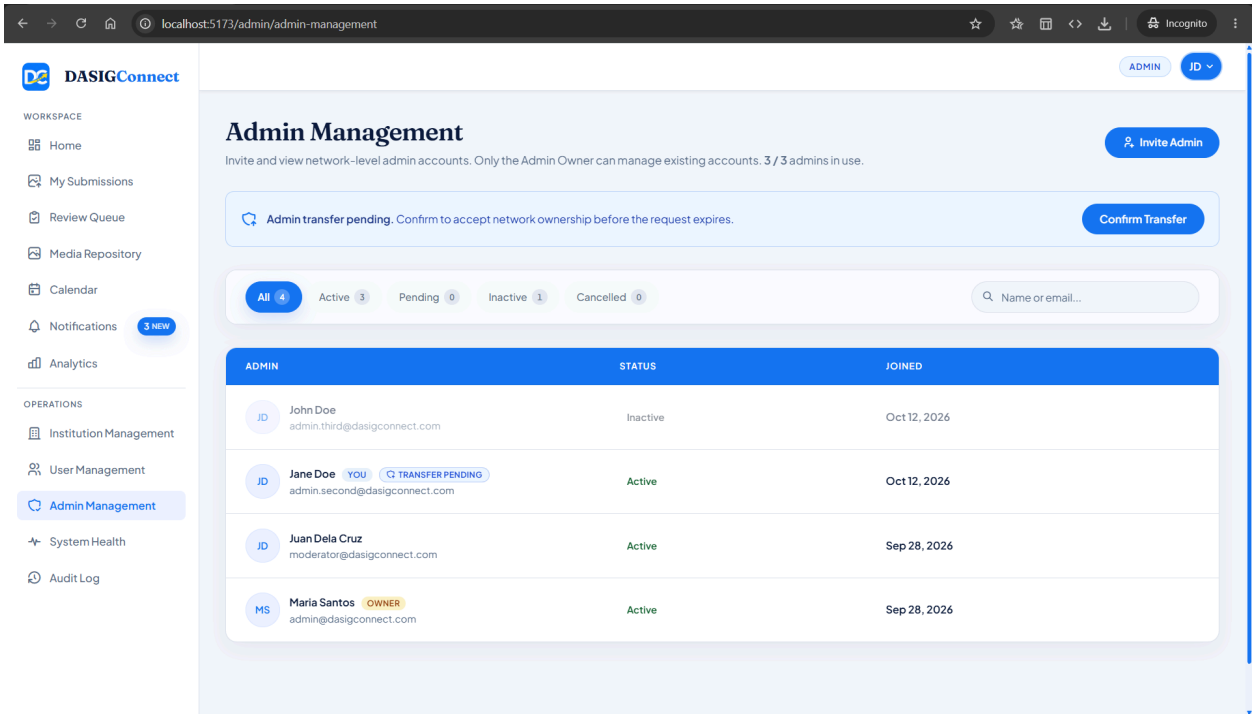


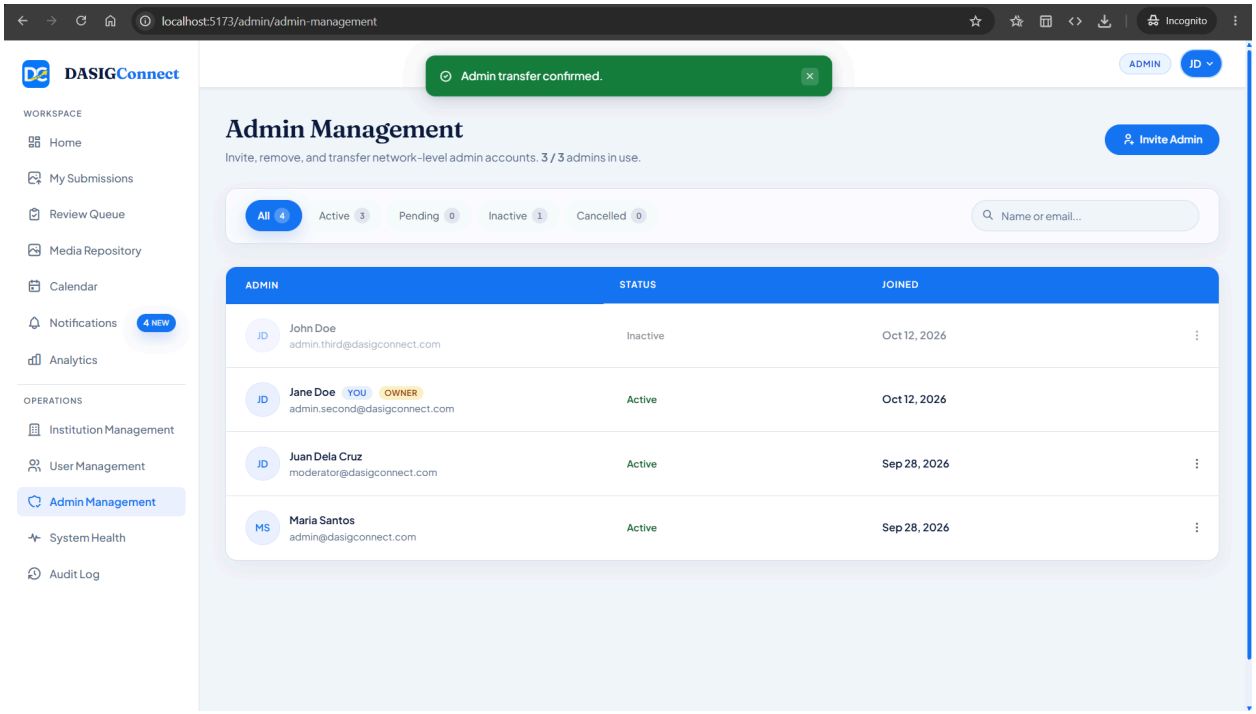

localhost:5173/dashboard
Incognito
You've been proposed for Administrator access. Confirm to accept, or decline to keep your current role.
Confirm
Decline
DASIGConnect
MODERATOR
JD
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
1 NEW
Analytics
OPERATIONS
Institution Management
Good afternoon, Juan.
Moderator · DASIG Network
Guide
The review queue is clear — no submissions are waiting for review. Approved content is scheduled automatically for the DASIG Facebook Page.
PENDING REVIEW
0
APPROVED THIS MONTH
0
REJECTED THIS MONTH
0
RECENT CONTRIBUTORS
0
Quick Actions
Review Queue
Review pending submissions from contributors
Institution Management
Invite contributors and manage your invitations
Analytics
Facebook engagement and workflow metrics
View Calendar
View scheduled and published events
Recent Activity
EVENT / POST
INSTITUTION
SUBMITTED
STATUS
No recent activity


localhost:5173/dashboard
Incognito
DASIGConnect
ADMIN
JD
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
2 NEW
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Good afternoon, Juan.
Admin · DASIG Network
Guide
Administrator workspace. Full network-wide visibility across all member institutions. Everything across the network is on track.
MEMBER INSTITUTIONS
3
ACTIVE MEMBERS
6
PENDING INVITES
0
AWAITING REVIEW
0
SCHEDULED POSTS
0
PUBLISHED (30 DAYS)
0
Quick Actions
Review Queue
Approve or reschedule submissions network-wide
Add Institution
Provision a new HEI workspace
Analytics
Network engagement, publishing, and workflow health
Recent Activity
EVENT / POST
INSTITUTION
SUBMITTED
STATUS
No recent activity
Submissions from contributors across the network will appear here.

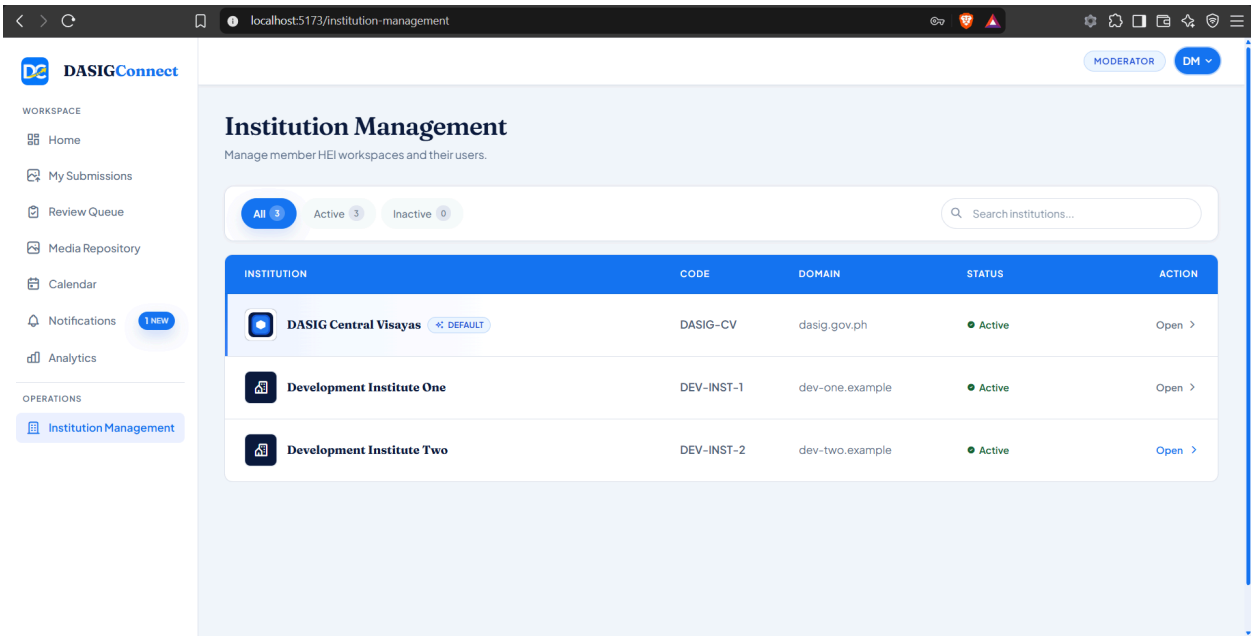

localhost:5173/admin/admin-management
DASIGConnect
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
1 NEW
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
ADMIN
MS
Admin Management
Invite, remove, and transfer network-level admin accounts. 3 / 3 admins in use.
Invite Admin
All 4
Active 3
Pending 0
Inactive 1
Cancelled 0
Name or email...
ADMIN
STATUS
JOINED
John Doe
admin.third@dasigconnect.com
Oct 12, 2026
Jane Doe
admin.second@dasigconnect.com
Oct 12, 2026
Juan Dela Cruz
moderator@dasigconnect.com
Active
Sep 28, 2026
Maria Santos
YOU
admin@dasigconnect.com
Active
Sep 28, 2026
Transfer Admin
Request transfer of Admin status to Jane Doe? They must explicitly confirm before the transfer takes effect.
Cancel
Request transfer

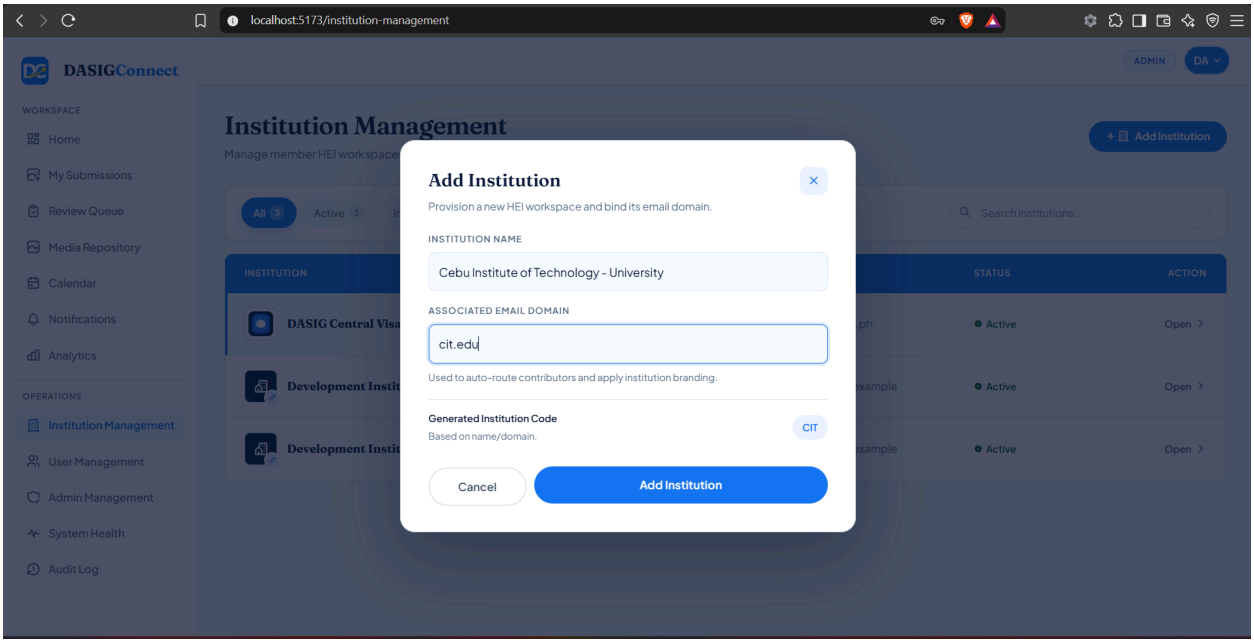

localhost:5173/admin/admin-management
DASIGConnect
Admin transfer requested.
ADMIN
MS
Admin Management
Invite, remove, and transfer network-level admin accounts. 3 / 3 admins in use.
Invite Admin
All 4
Active 3
Pending 0
Inactive 1
Cancelled 0
Name or email...
ADMIN
STATUS
JOINED
John Doe
admin.third@dasigconnect.com
Inactive
Oct 12, 2026
Jane Doe
admin.second@dasigconnect.com
Active
Oct 12, 2026
Juan Dela Cruz
moderator@dasigconnect.com
Active
Sep 28, 2026
Maria Santos
YOU
admin@dasigconnect.com
Active
Sep 28, 2026

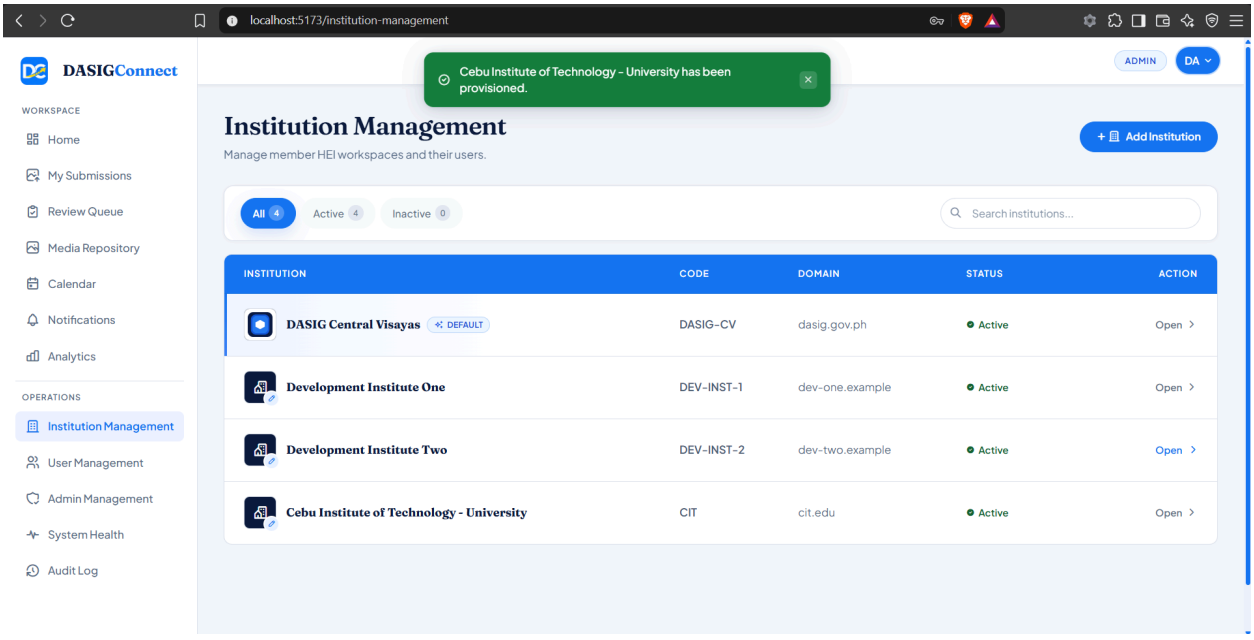

localhost:5173/admin/admin-management
Incognito
DASIGConnect
ADMIN
JD
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
3 NEW
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Admin Management
Invite and view network-level admin accounts. Only the Admin Owner can manage existing accounts. 3 / 3 admins in use.
Invite Admin
Admin transfer pending. Confirm to accept network ownership before the request expires.
Confirm Transfer
All 4
Active 3
Pending 0
Inactive 1
Cancelled 0
Name or email...
ADMIN
STATUS
JOINED
JD
John Doe
admin.third@dasigconnect.com
Inactive
Oct 12, 2026
JD
Jane Doe
YOU
TRANSFER PENDING
admin.second@dasigconnect.com
Active
Oct 12, 2026
JD
Juan Dela Cruz
moderator@dasigconnect.com
Active
Sep 28, 2026
MS
Maria Santos
OWNER
admin@dasigconnect.com
Active
Sep 28, 2026

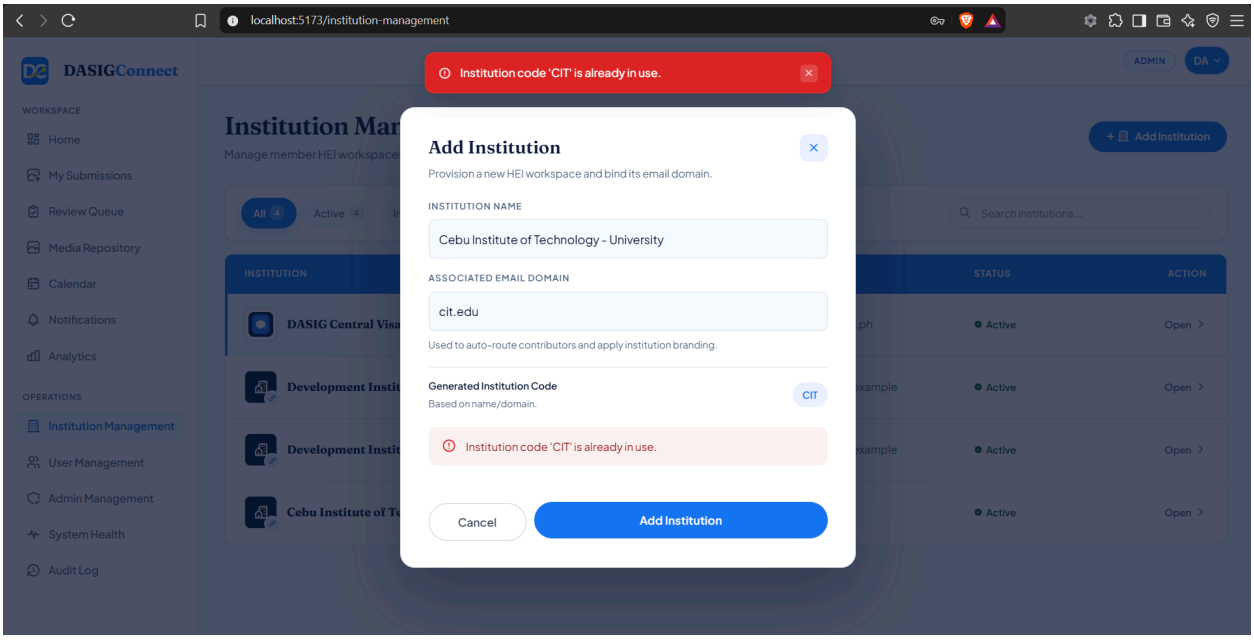

localhost:5173/admin/admin-management
Incognito
DASIGConnect
Admin transfer confirmed.
ADMIN
JD
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
4 NEW
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Admin Management
Invite, remove, and transfer network-level admin accounts. 3 / 3 admins in use.
Invite Admin
All 4
Active 3
Pending 0
Inactive 1
Cancelled 0
Name or email...
ADMIN
STATUS
JOINED
JD
John Doe
admin.third@dasigconnect.com
Inactive
Oct 12, 2026
JD
Jane Doe
YOU
OWNER
admin.second@dasigconnect.com
Active
Oct 12, 2026
JD
Juan Dela Cruz
moderator@dasigconnect.com
Active
Sep 28, 2026
MS
Maria Santos
admin@dasigconnect.com
Active
Sep 28, 2026

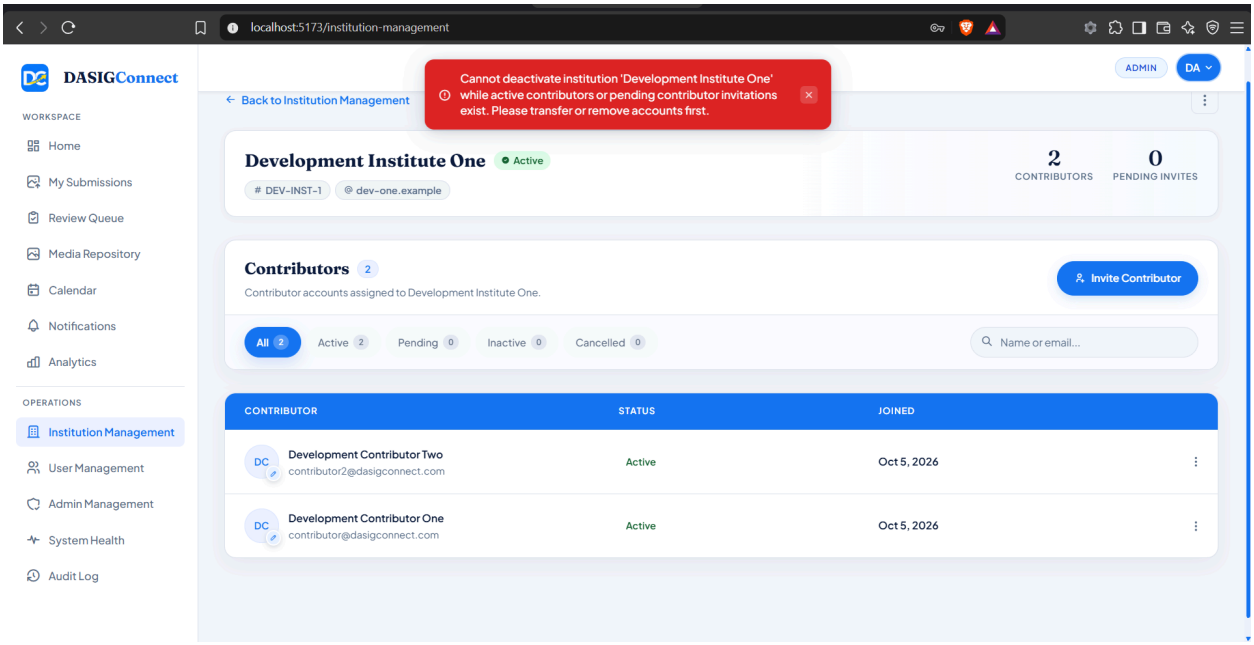
Screenshots for Test 3 (Case-3 — UC-1.2 Institution Lifecycle & Protected Institution)

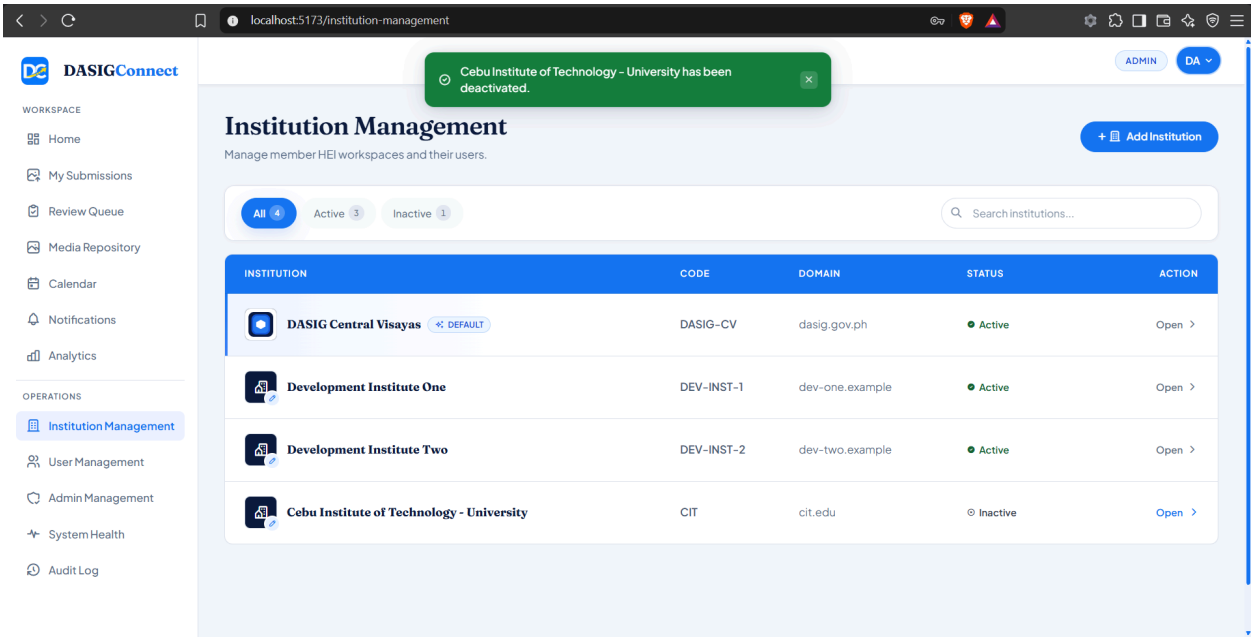


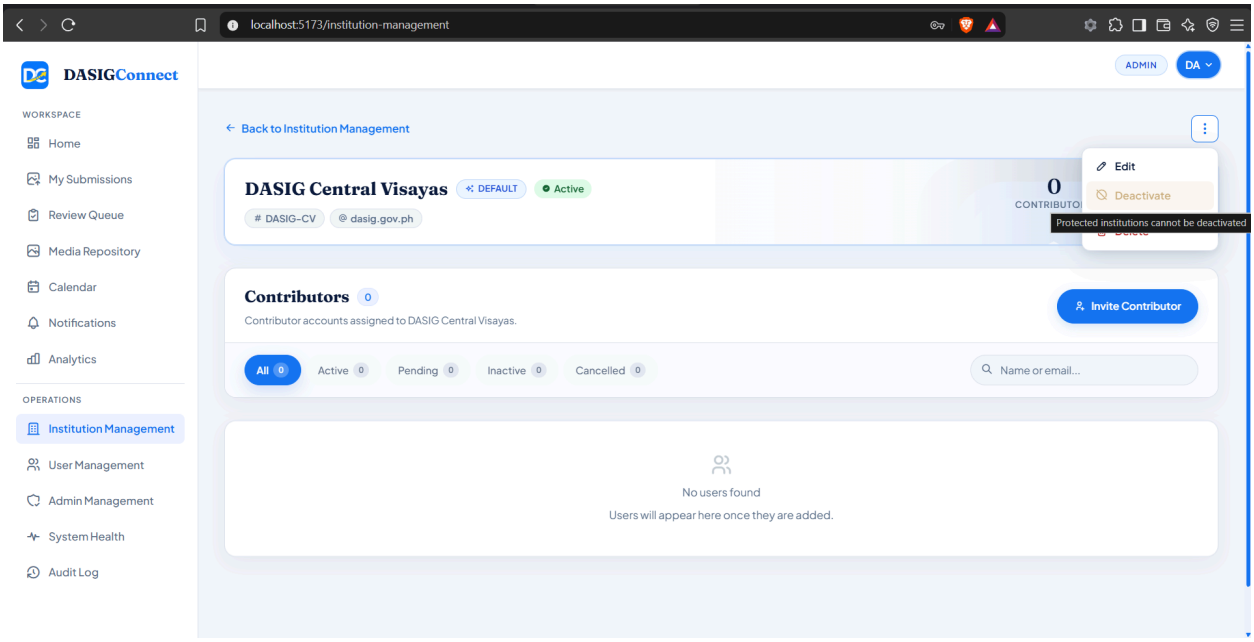


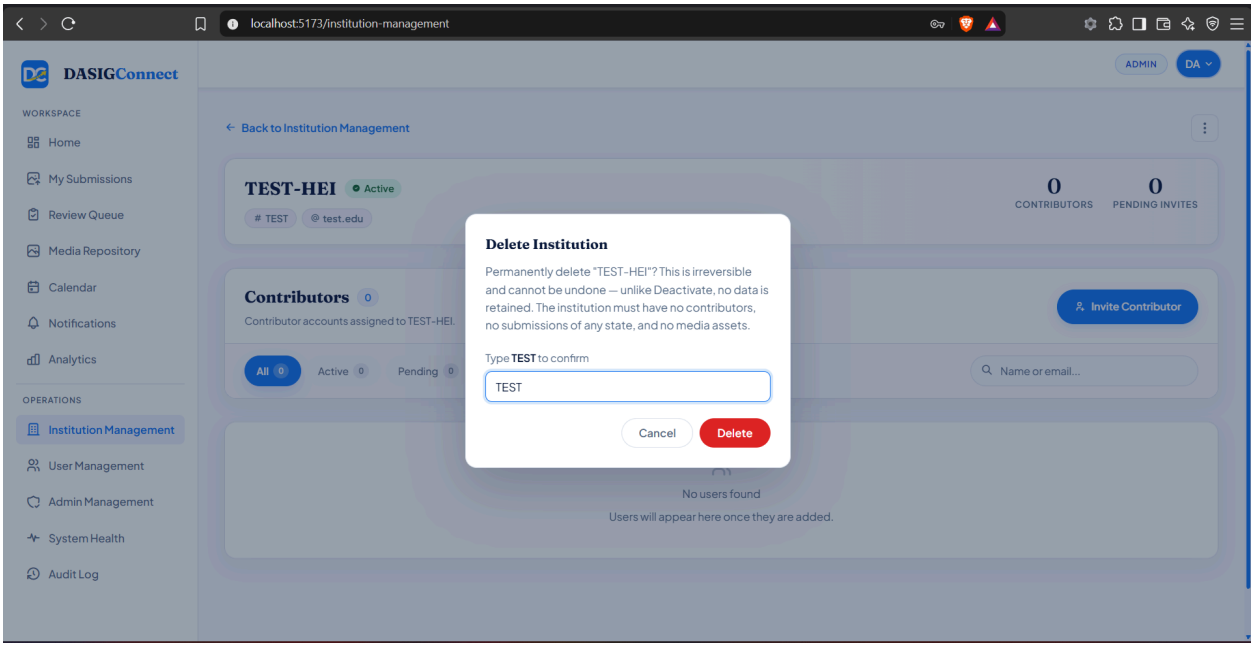

localhost:5173/institution-management
DASIGConnect
Cebu Institute of Technology - University has been provisioned.
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Institution Management
Manage member HEI workspaces and their users.
+ Add Institution
All 4
Active 4
Inactive 0
Search institutions...
INSTITUTION
CODE
DOMAIN
STATUS
ACTION
DASIG Central Visayas
DEFAULT
DASIG-CV
dasig.gov.ph
Active
Open
Development Institute One
DEV-INST-1
dev-one.example
Active
Open
Development Institute Two
DEV-INST-2
dev-two.example
Active
Open
Cebu Institute of Technology - University
CIT
cit.edu
Active
Open

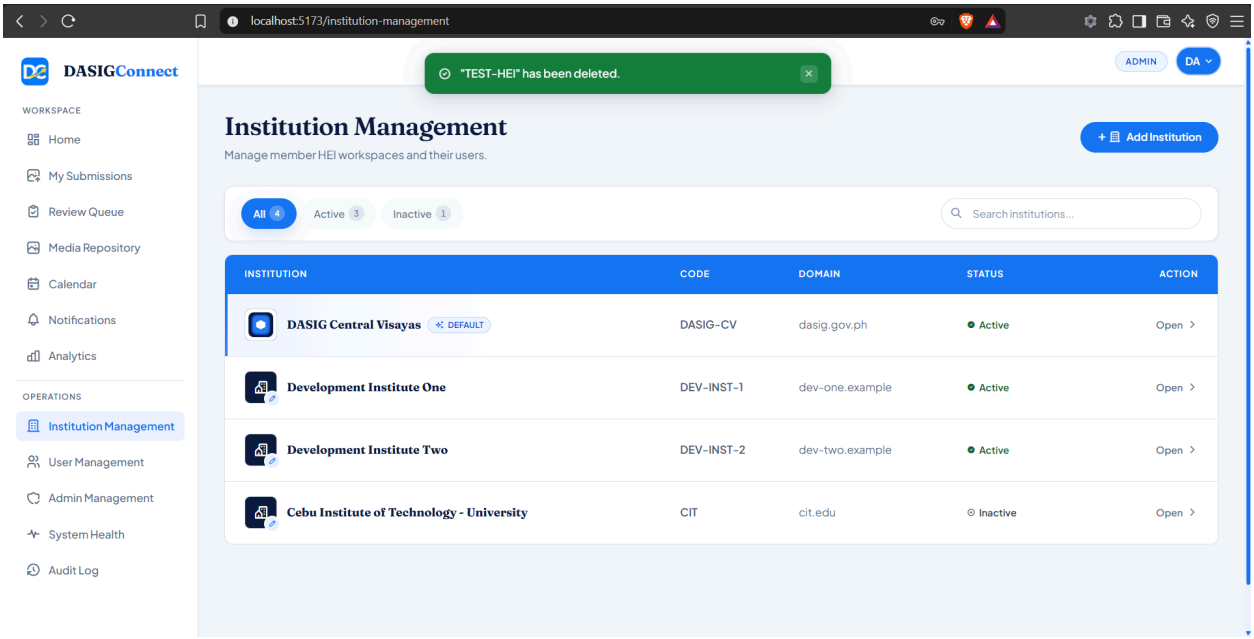

localhost:5173/institution-management
DASIGConnect
Institution code 'CIT' is already in use.
Add Institution
Provision a new HEI workspace and bind its email domain.
INSTITUTION NAME
Cebu Institute of Technology - University
ASSOCIATED EMAIL DOMAIN
cit.edu
Used to auto-route contributors and apply institution branding.
Generated Institution Code
Based on name/domain.
CIT
Institution code 'CIT' is already in use.
Cancel
Add Institution


localhost:5173/institution-management
DASIGConnect
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
ADMIN
DA
Cannot deactivate institution 'Development Institute One' while active contributors or pending contributor invitations exist. Please transfer or remove accounts first.
Back to Institution Management
Development Institute One
Active
DEV-INST-1
dev-one.example
2
CONTRIBUTORS
0
PENDING INVITES
Contributors
2
Contributor accounts assigned to Development Institute One.
Invite Contributor
All 2
Active 2
Pending 0
Inactive 0
Cancelled 0
Name or email...
CONTRIBUTOR
STATUS
JOINED
DC
Development Contributor Two
contributor2@dasigconnect.com
Active
Oct 5, 2026
DC
Development Contributor One
contributor@dasigconnect.com
Active
Oct 5, 2026

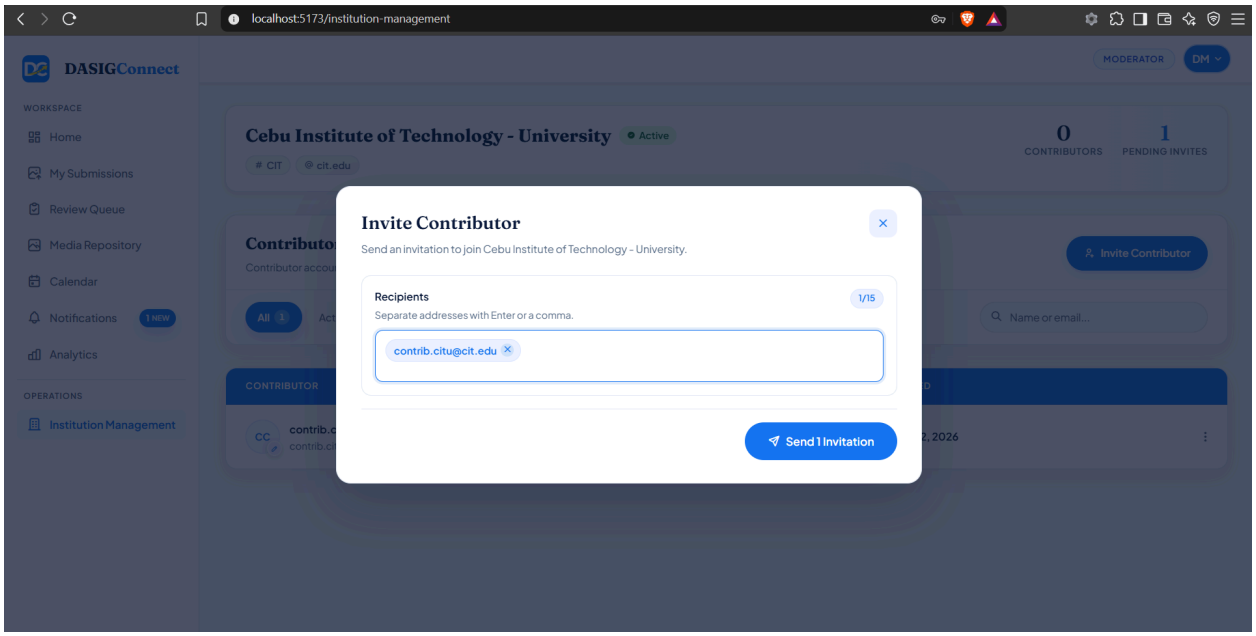

localhost:5173/institution-management
DASIGConnect
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
ADMIN
DA
Cebu Institute of Technology – University has been deactivated.
Institution Management
Manage member HEI workspaces and their users.
+ Add Institution
All 4
Active 3
Inactive 1
Search institutions...
INSTITUTION
CODE
DOMAIN
STATUS
ACTION
DASIG Central Visayas
DEFAULT
DASIG-CV
dasig.gov.ph
Active
Open
Development Institute One
DEV-INST-1
dev-one.example
Active
Open
Development Institute Two
DEV-INST-2
dev-two.example
Active
Open
Cebu Institute of Technology - University
CIT
cit.edu
Inactive
Open

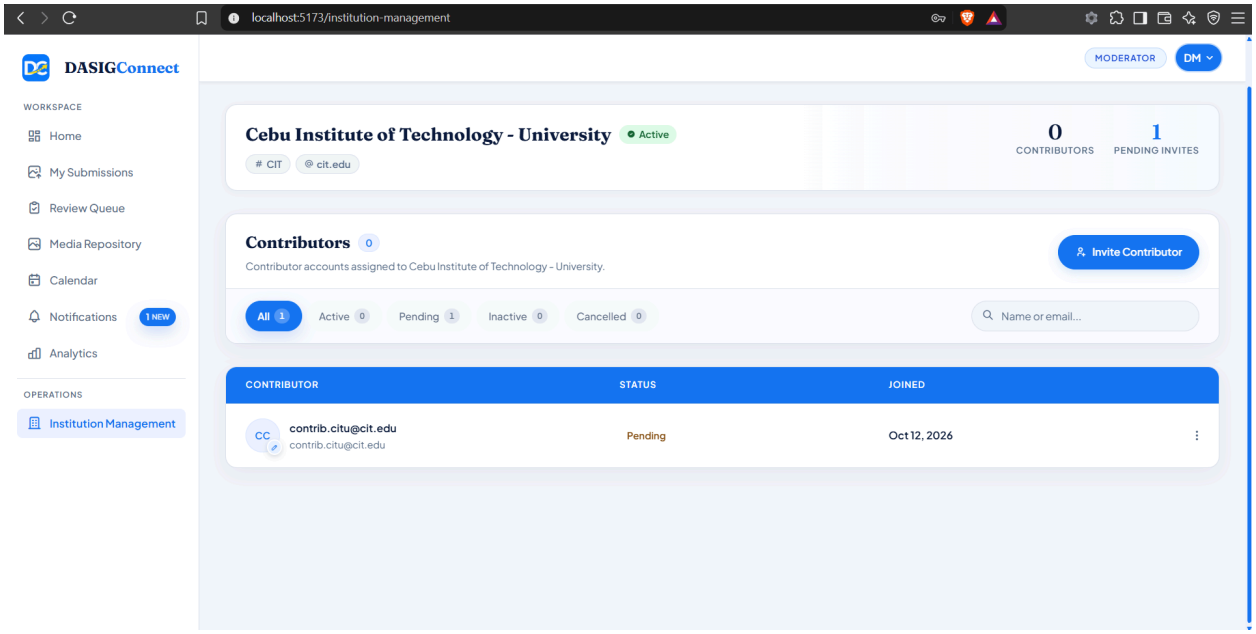

localhost:5173/institution-management
DASIGConnect
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Back to Institution Management
DASIG Central Visayas
DEFAULT
Active
DASIG-CV
dasig.gov.ph
0
CONTRIBUTO
Edit
Deactivate
Protected institutions cannot be deactivated
Contributors
0
Contributor accounts assigned to DASIG Central Visayas.
Invite Contributor
All
Active
Pending
Inactive
Cancelled
Name or email...
No users found
Users will appear here once they are added.

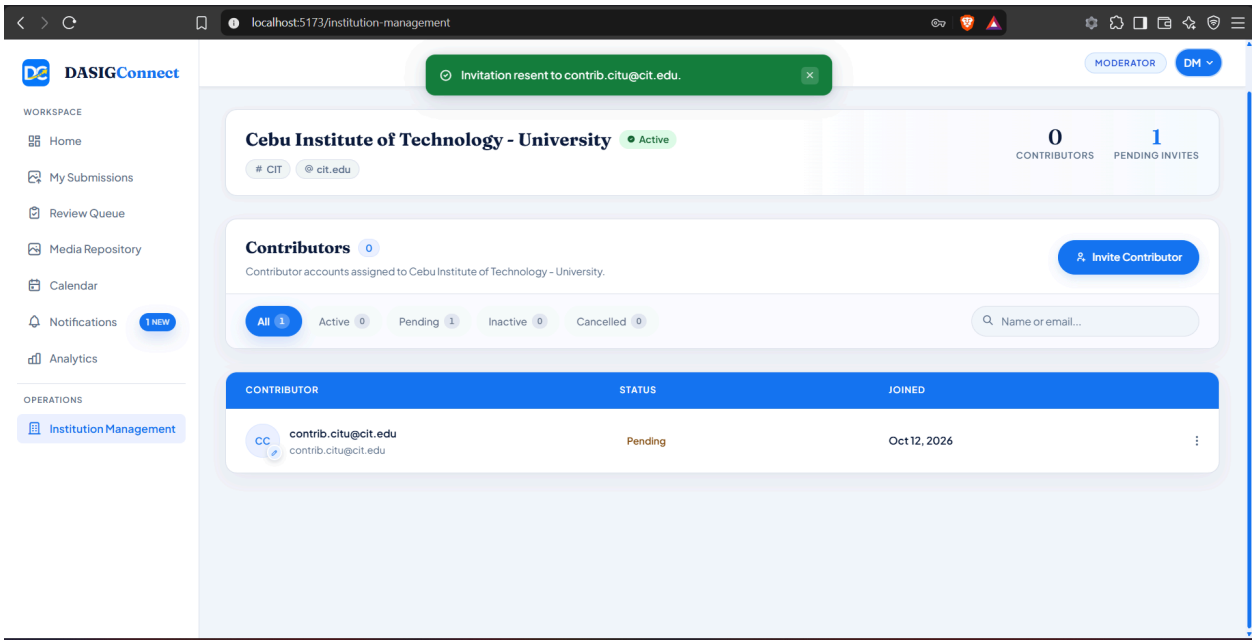

localhost:5173/institution-management
DASIGConnect
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Back to Institution Management
TEST-HEI
Active
TEST
test.edu
0
CONTRIBUTORS
0
PENDING INVITES
Delete Institution
Permanently delete "TEST-HEI"? This is irreversible and cannot be undone — unlike Deactivate, no data is retained. The institution must have no contributors, no submissions of any state, and no media assets.
Type TEST to confirm
TEST
Cancel
Delete
Contributors
0
Contributor accounts assigned to TEST-HEI.
Invite Contributor
All
Active
Pending
Name or email...
No users found
Users will appear here once they are added.

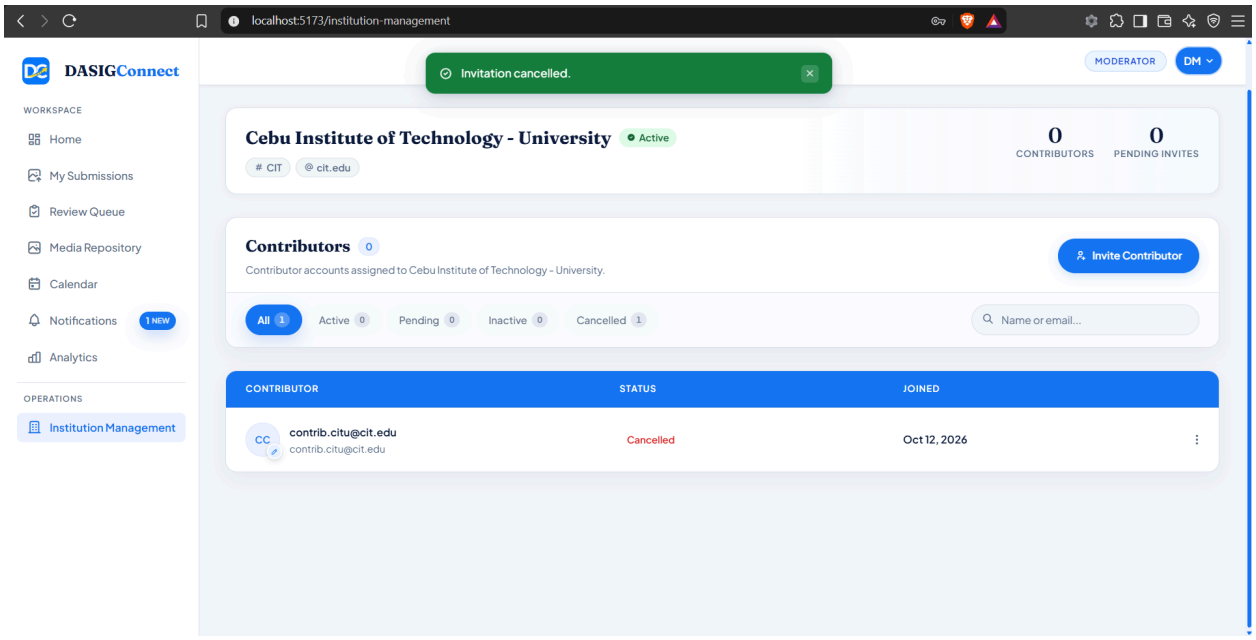


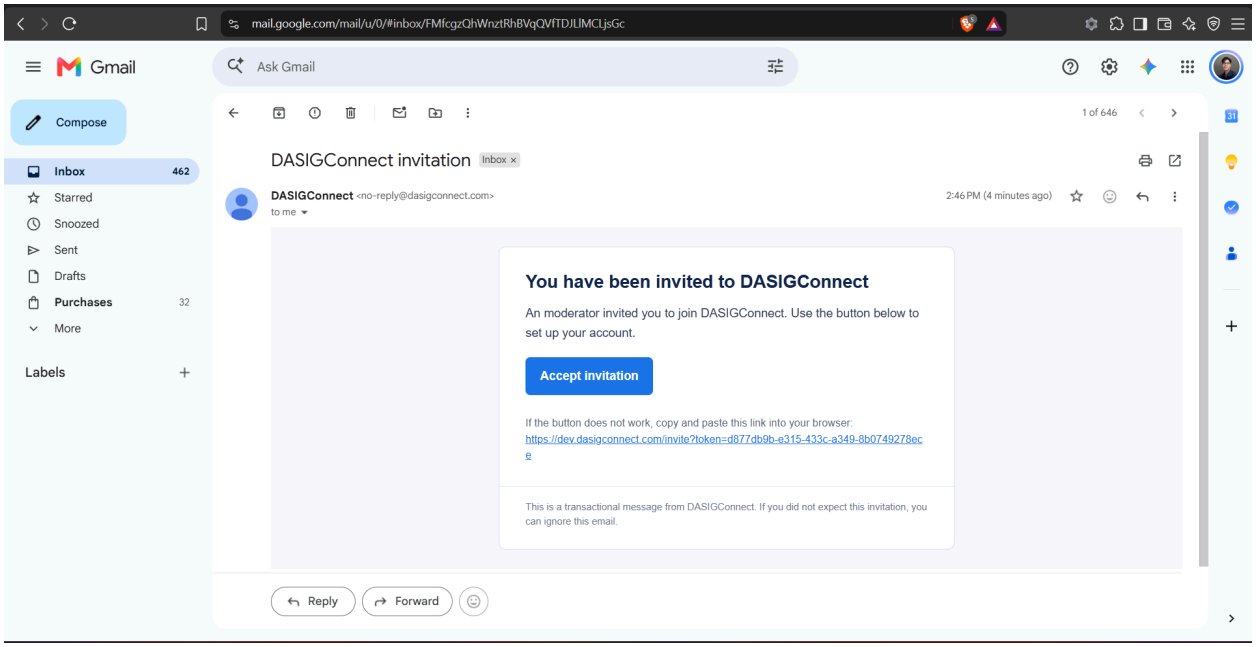
Screenshots for Test 4 (Case-4 — UC-1.3 Contributor Account Provisioning & Footprint-Aware Deletion)

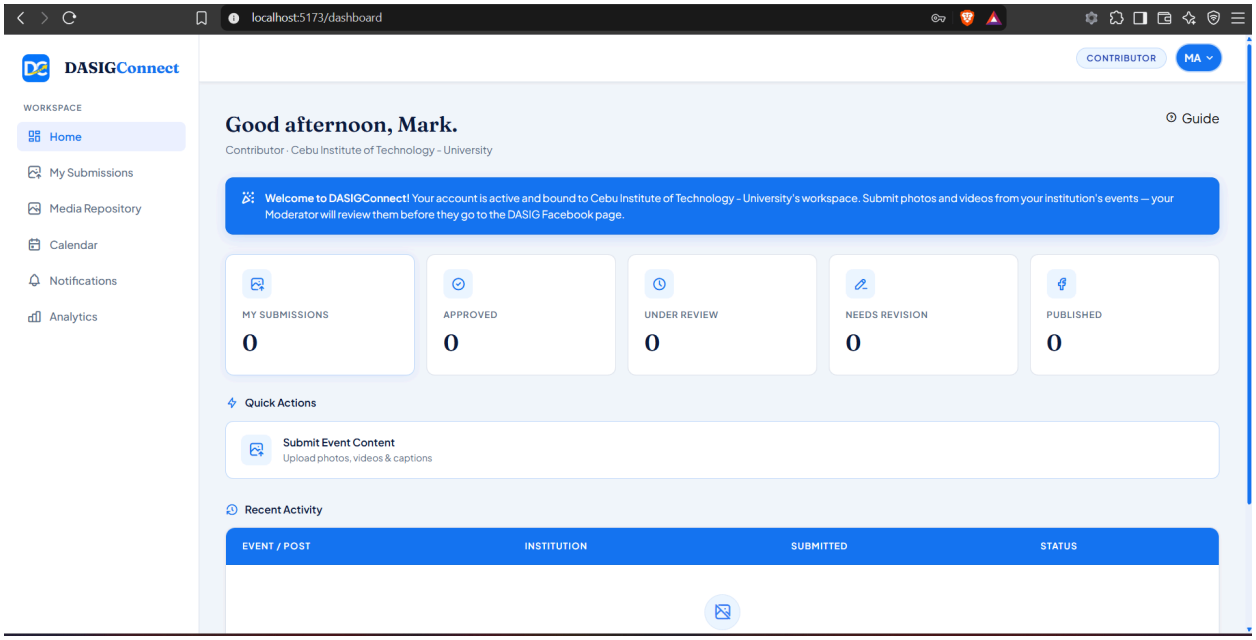


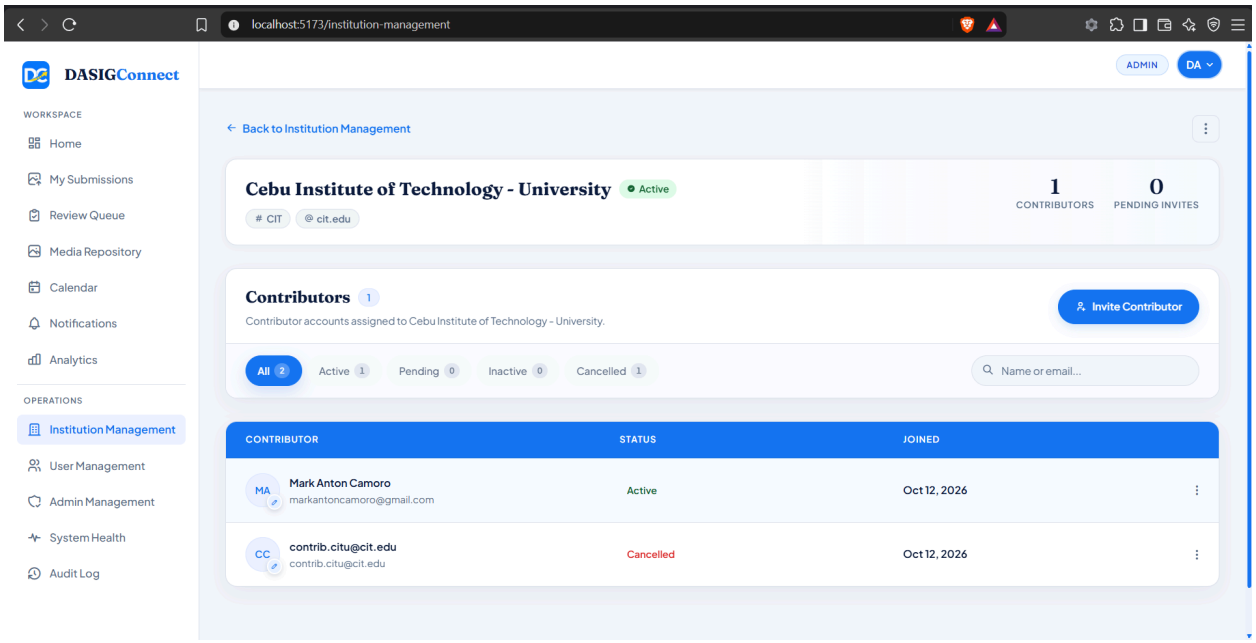

localhost:5173/institution-management
DASIGConnect
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
1 NEW
Analytics
OPERATIONS
Institution Management
MODERATOR
DM
Cebu Institute of Technology - University
Active
CIT
cit.edu
0
CONTRIBUTORS
1
PENDING INVITES
Contributors
0
Contributor accounts assigned to Cebu Institute of Technology - University.
Invite Contributor
All 1
Active 0
Pending 1
Inactive 0
Cancelled 0
Name or email...
CONTRIBUTOR
STATUS
JOINED
CC
contrib.citu@cit.edu
contrib.citu@cit.edu
Pending
Oct 12, 2026

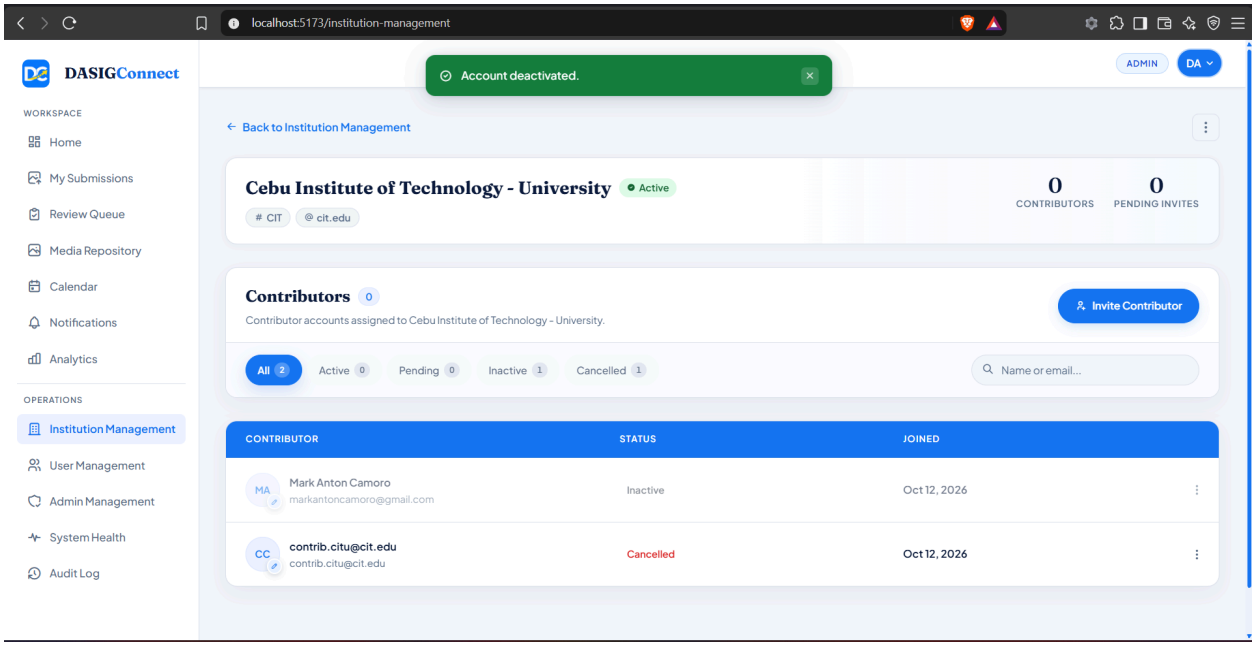

localhost:5173/institution-management
Invitation resent to contrib.citu@cit.edu.
DASIGConnect
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
1 NEW
Analytics
OPERATIONS
Institution Management
MODERATOR
DM
Cebu Institute of Technology - University
Active
CIT
cit.edu
0
CONTRIBUTORS
1
PENDING INVITES
Contributors
0
Contributor accounts assigned to Cebu Institute of Technology - University.
Invite Contributor
All 1
Active 0
Pending 1
Inactive 0
Cancelled 0
Name or email...
CONTRIBUTOR
STATUS
JOINED
CC
contrib.citu@cit.edu
contrib.citu@cit.edu
Pending
Oct 12, 2026

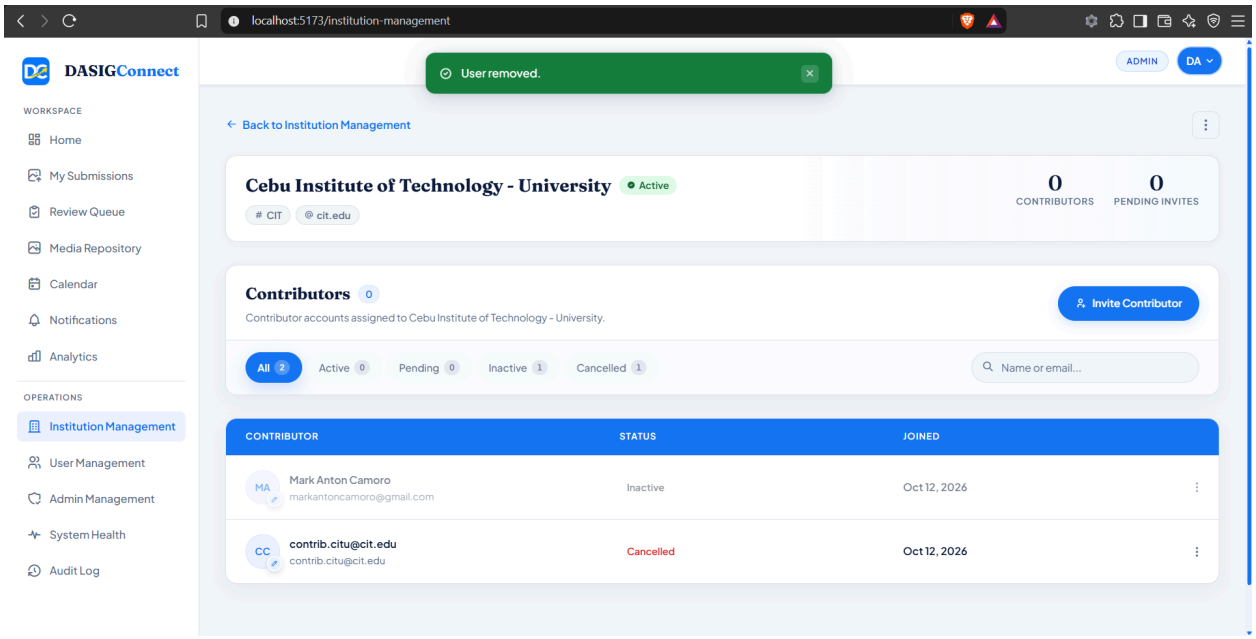

localhost:5173/institution-management
DASIGConnect
Invitation cancelled.
MODERATOR
DM
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
1 NEW
Analytics
OPERATIONS
Institution Management
Cebu Institute of Technology - University
Active
CIT
@ cit.edu
0
CONTRIBUTORS
0
PENDING INVITES
Contributors
0
Contributor accounts assigned to Cebu Institute of Technology - University.
Invite Contributor
All 1
Active 0
Pending 0
Inactive 0
Cancelled 1
Name or email...
CONTRIBUTOR
STATUS
JOINED
CC
contrib.citu@cit.edu
contrib.citu@cit.edu
Cancelled
Oct 12, 2026

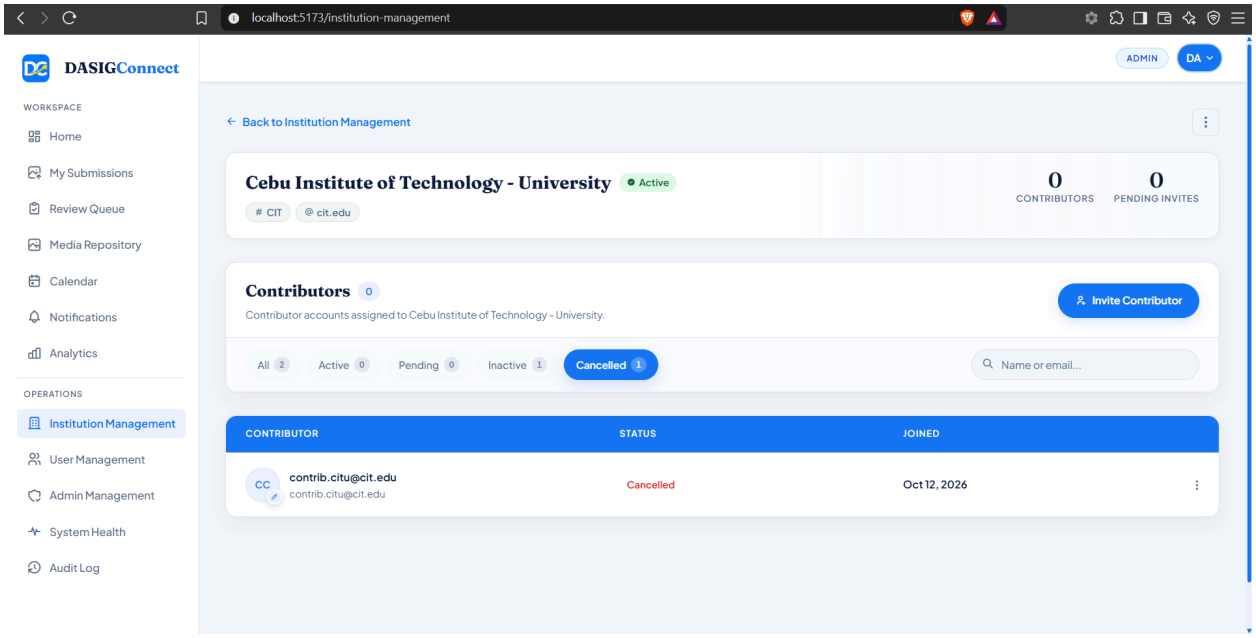

mail.google.com/mail/u/0/#inbox/FMfcgzQhWnztRhBVqQVfTDJLlMCLjsGc
Gmail
Ask Gmail
Compose
Inbox 462
Starred
Snoozed
Sent
Drafts
Purchases 32
More
Labels
1 of 646
DASIGConnect invitation
Inbox ×
DASIGConnect <no-reply@dasigconnect.com>
to me
2:46 PM (4 minutes ago)
You have been invited to DASIGConnect
An moderator invited you to join DASIGConnect. Use the button below to set up your account.
Accept invitation
If the button does not work, copy and paste this link into your browser:
https://dev.dasigconnect.com/invite?token=d877db9b-e315-433c-a349-8b0749278ece
This is a transactional message from DASIGConnect. If you did not expect this invitation, you can ignore this email.
Reply
Forward


localhost:5173/dashboard
DASIGConnect
WORKSPACE
Home
My Submissions
Media Repository
Calendar
Notifications
Analytics
CONTRIBUTOR
MA
Good afternoon, Mark.
Contributor · Cebu Institute of Technology - University
Guide
Welcome to DASIGConnect! Your account is active and bound to Cebu Institute of Technology - University's workspace. Submit photos and videos from your institution's events — your Moderator will review them before they go to the DASIG Facebook page.
MY SUBMISSIONS
0
APPROVED
0
UNDER REVIEW
0
NEEDS REVISION
0
PUBLISHED
0
Quick Actions
Submit Event Content
Upload photos, videos & captions
Recent Activity
EVENT / POST
INSTITUTION
SUBMITTED
STATUS

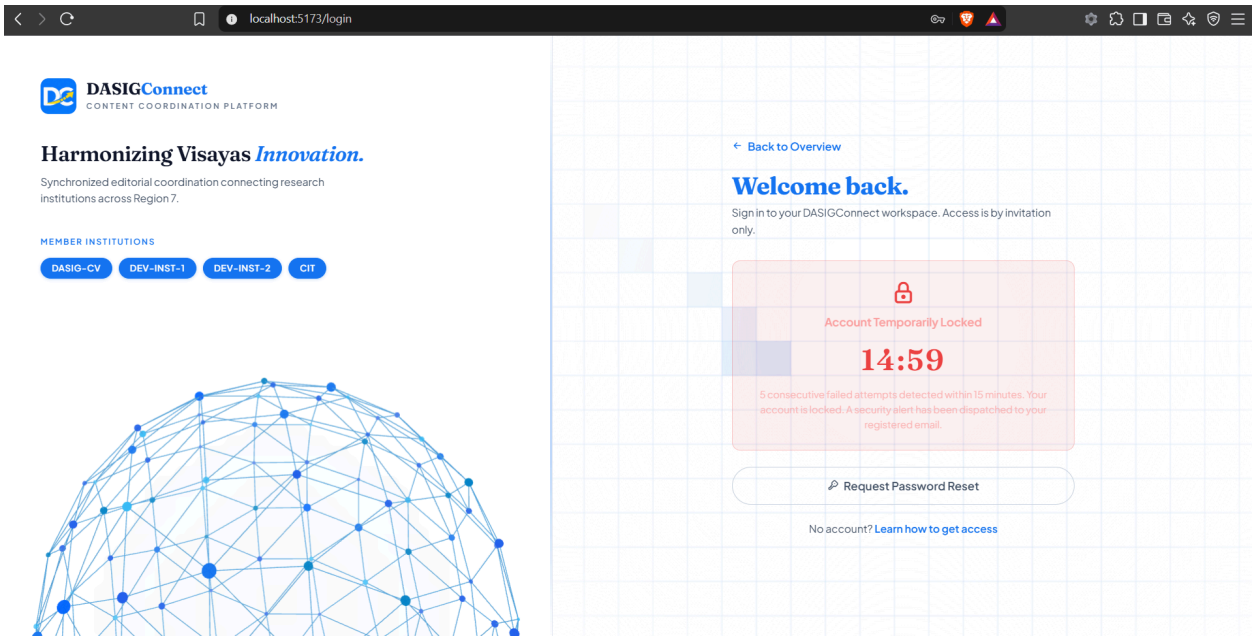

localhost:5173/institution-management
DASIGConnect
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
ADMIN
DA
Back to Institution Management
Cebu Institute of Technology - University
Active
CIT
@ cit.edu
1
CONTRIBUTORS
0
PENDING INVITES
Contributors
1
Contributor accounts assigned to Cebu Institute of Technology - University.
Invite Contributor
All 2
Active 1
Pending 0
Inactive 0
Cancelled 1
Name or email...
CONTRIBUTOR
STATUS
JOINED
MA
Mark Anton Camoro
markantoncamoro@gmail.com
Active
Oct 12, 2026
CC
contrib.citu@cit.edu
contrib.citu@cit.edu
Cancelled
Oct 12, 2026

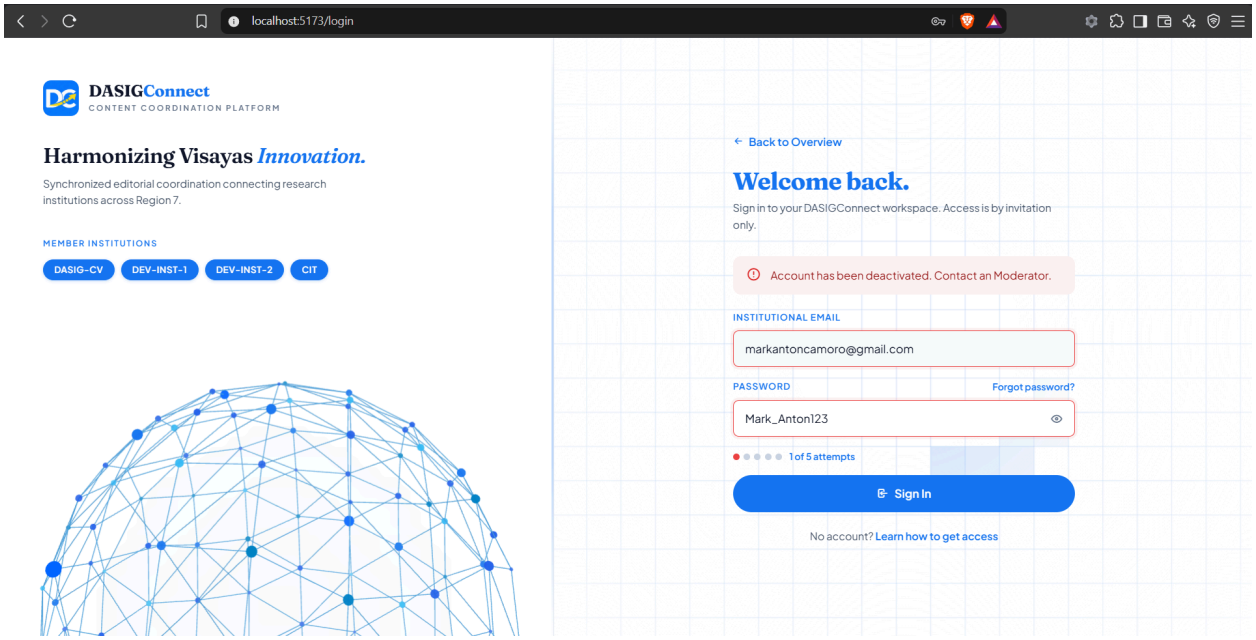

localhost:5173/institution-management
DASIGConnect
Account deactivated.
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Back to Institution Management
Cebu Institute of Technology - University
Active
CIT
@ cit.edu
0
CONTRIBUTORS
0
PENDING INVITES
Contributors
0
Contributor accounts assigned to Cebu Institute of Technology - University.
Invite Contributor
All 2
Active 0
Pending 0
Inactive 1
Cancelled 1
Name or email...
CONTRIBUTOR
STATUS
JOINED
MA
Mark Anton Camoro
markantoncamoro@gmail.com
Inactive
Oct 12, 2026
CC
contrib.citu@cit.edu
contrib.citu@cit.edu
Cancelled
Oct 12, 2026

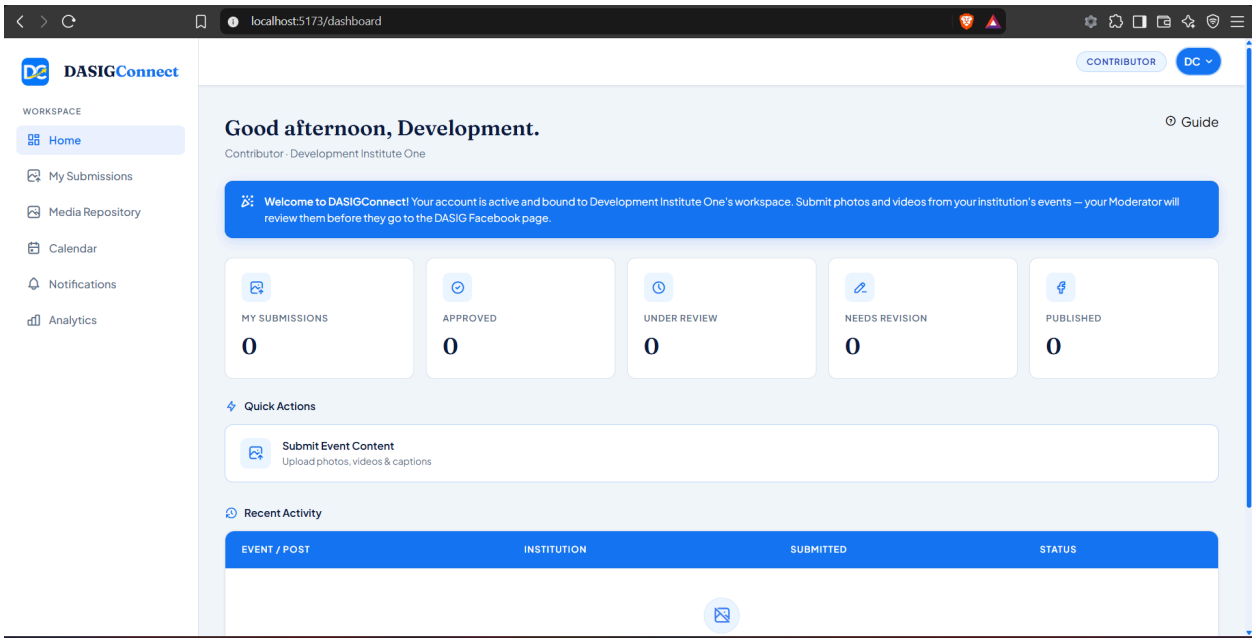

localhost:5173/institution-management
DASIGConnect
User removed.
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Back to Institution Management
Cebu Institute of Technology - University
Active
CIT
@ cit.edu
0
CONTRIBUTORS
0
PENDING INVITES
Contributors
0
Contributor accounts assigned to Cebu Institute of Technology - University.
Invite Contributor
All 2
Active 0
Pending 0
Inactive 1
Cancelled 1
Name or email...
CONTRIBUTOR
STATUS
JOINED
MA
Mark Anton Camoro
markantoncamoro@gmail.com
Inactive
Oct 12, 2026
CC
contrib.citu@cit.edu
contrib.citu@cit.edu
Cancelled
Oct 12, 2026

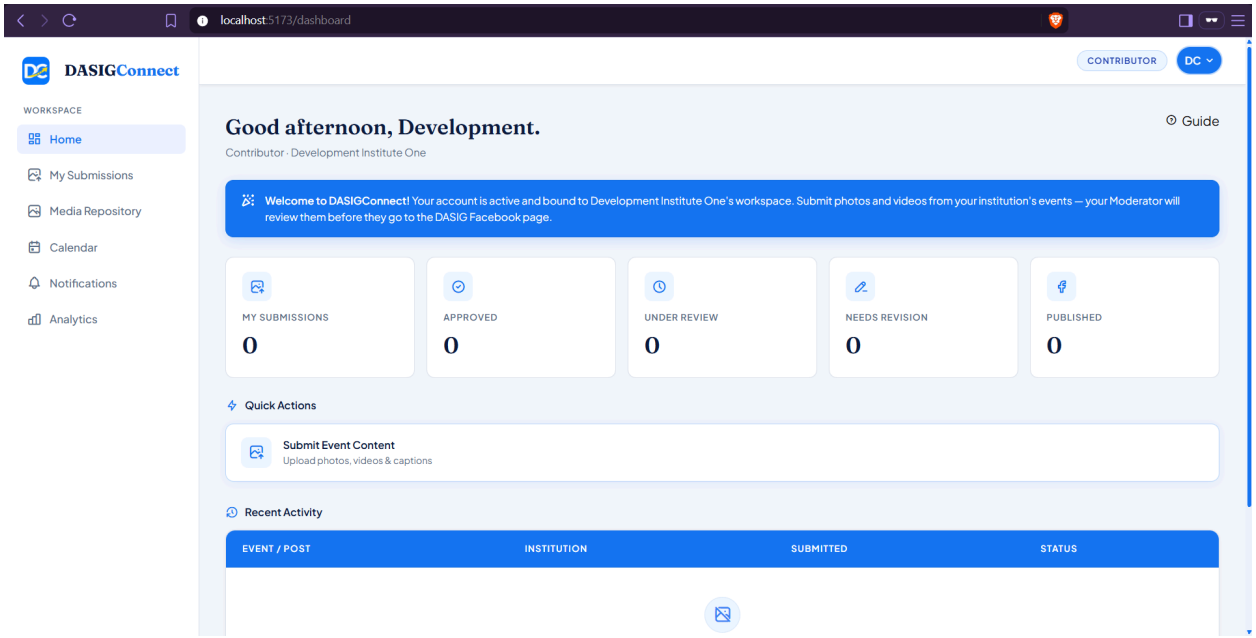


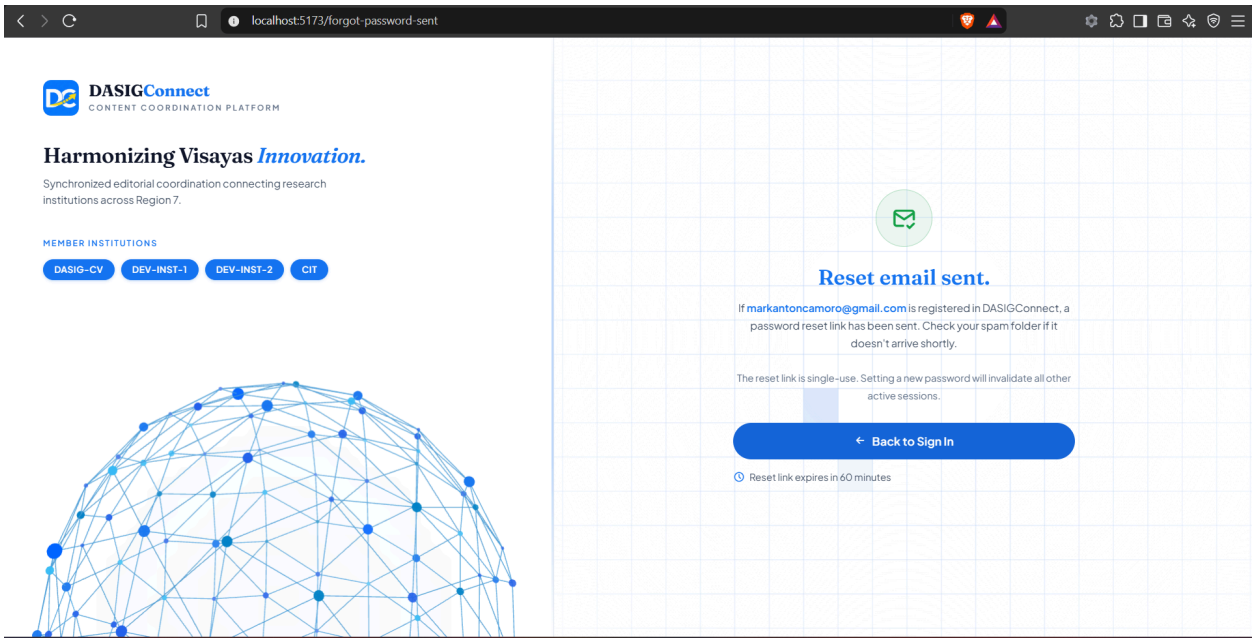
Screenshots for Test 5 (Case-5 — UC-1.4 Login, Lockout, Password Policy & Session Revocation)

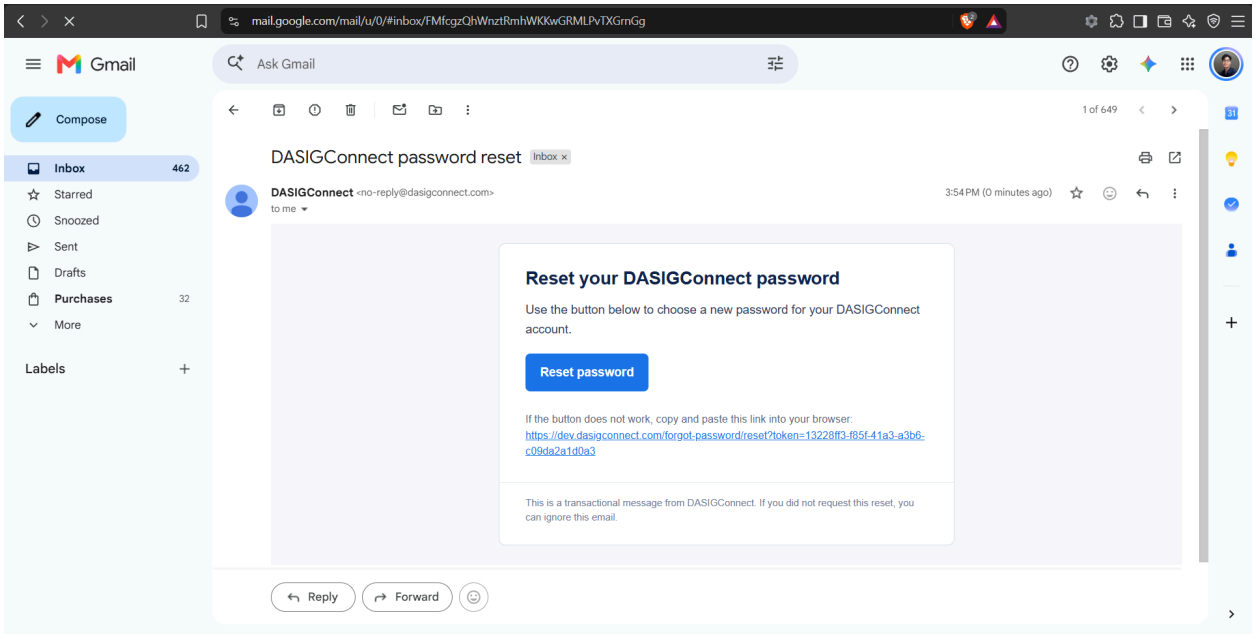


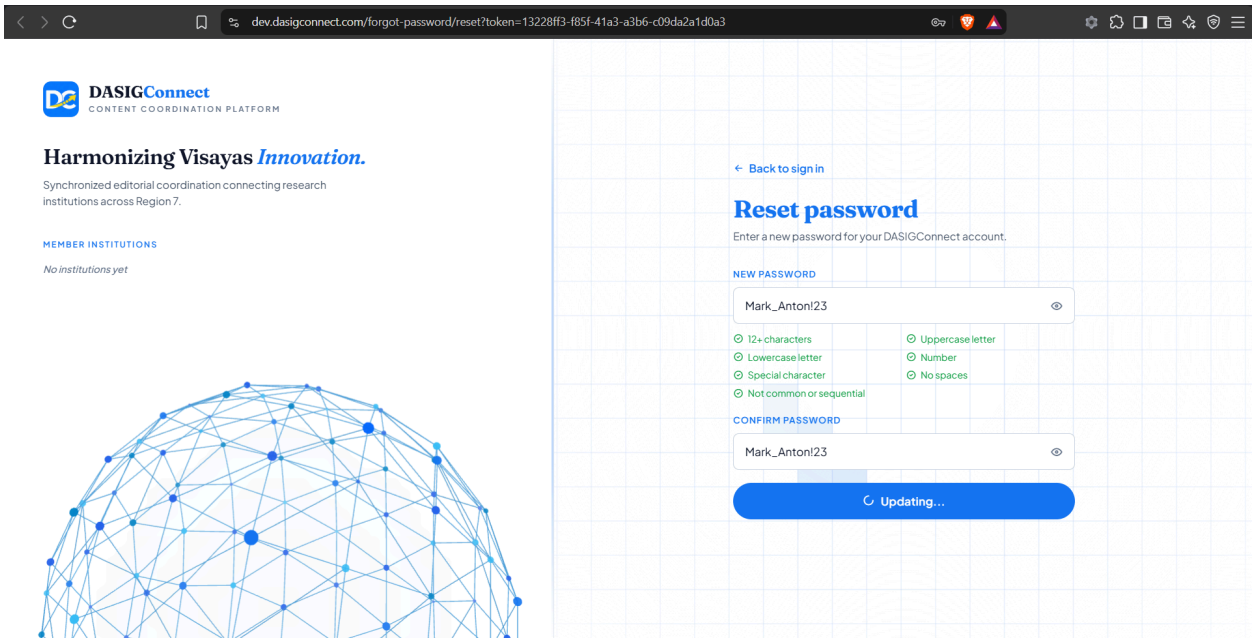

localhost:5173/login
DASIGConnect
CONTENT COORDINATION PLATFORM
Harmonizing Visayas Innovation.
Synchronized editorial coordination connecting research institutions across Region 7.
MEMBER INSTITUTIONS
DASIG-CV
DEV-INST-1
DEV-INST-2
CIT
Back to Overview
Welcome back.
Sign in to your DASIGConnect workspace. Access is by invitation only.
Account has been deactivated. Contact an Moderator.
INSTITUTIONAL EMAIL
markantoncamoro@gmail.com
PASSWORD
Forgot password?
Mark_Anton123
1 of 5 attempts
Sign In
No account? Learn how to get access


localhost:5173/dashboard
DASIGConnect
WORKSPACE
Home
My Submissions
Media Repository
Calendar
Notifications
Analytics
CONTRIBUTOR
DC
Good afternoon, Development.
Contributor · Development Institute One
Guide
Welcome to DASIGConnect! Your account is active and bound to Development Institute One's workspace. Submit photos and videos from your institution's events — your Moderator will review them before they go to the DASIG Facebook page.
MY SUBMISSIONS
0
APPROVED
0
UNDER REVIEW
0
NEEDS REVISION
0
PUBLISHED
0
Quick Actions
Submit Event Content
Upload photos, videos & captions
Recent Activity
EVENT / POST
INSTITUTION
SUBMITTED
STATUS

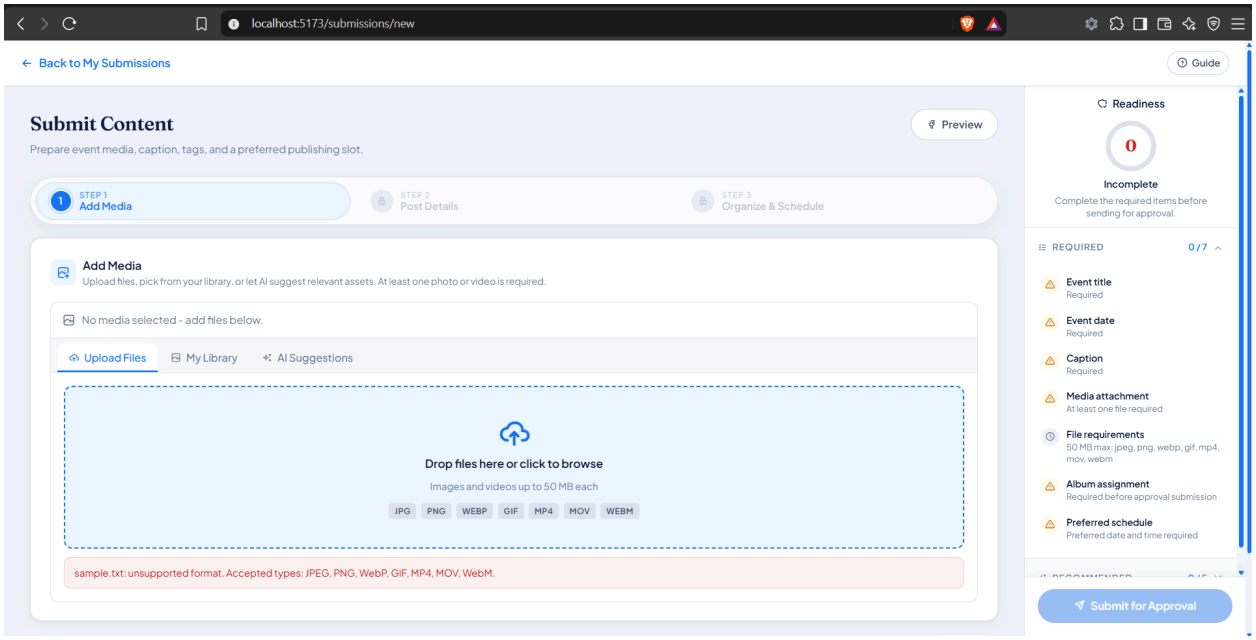

localhost:5173/dashboard
DASIGConnect
WORKSPACE
Home
My Submissions
Media Repository
Calendar
Notifications
Analytics
CONTRIBUTOR
DC
Good afternoon, Development.
Contributor · Development Institute One
Guide
Welcome to DASIGConnect! Your account is active and bound to Development Institute One's workspace. Submit photos and videos from your institution's events — your Moderator will review them before they go to the DASIG Facebook page.
MY SUBMISSIONS
0
APPROVED
0
UNDER REVIEW
0
NEEDS REVISION
0
PUBLISHED
0
Quick Actions
Submit Event Content
Upload photos, videos & captions
Recent Activity
EVENT / POST
INSTITUTION
SUBMITTED
STATUS

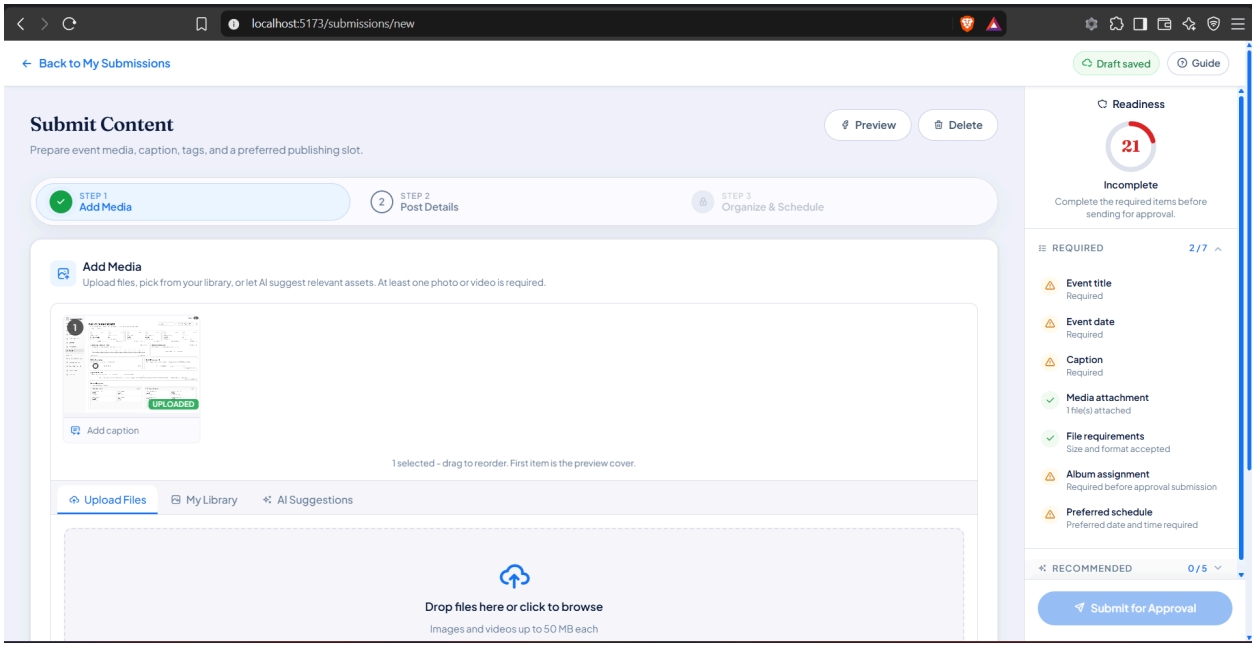

localhost:5173/forgot-password-sent
DASIGConnect
CONTENT COORDINATION PLATFORM
Harmonizing Visayas Innovation.
Synchronized editorial coordination connecting research institutions across Region 7.
MEMBER INSTITUTIONS
DASIG-CV
DEV-INST-1
DEV-INST-2
CIT
Reset email sent.
If markantoncamoro@gmail.com is registered in DASIGConnect, a password reset link has been sent. Check your spam folder if it doesn't arrive shortly.
The reset link is single-use. Setting a new password will invalidate all other active sessions.
Back to Sign In
Reset link expires in 60 minutes

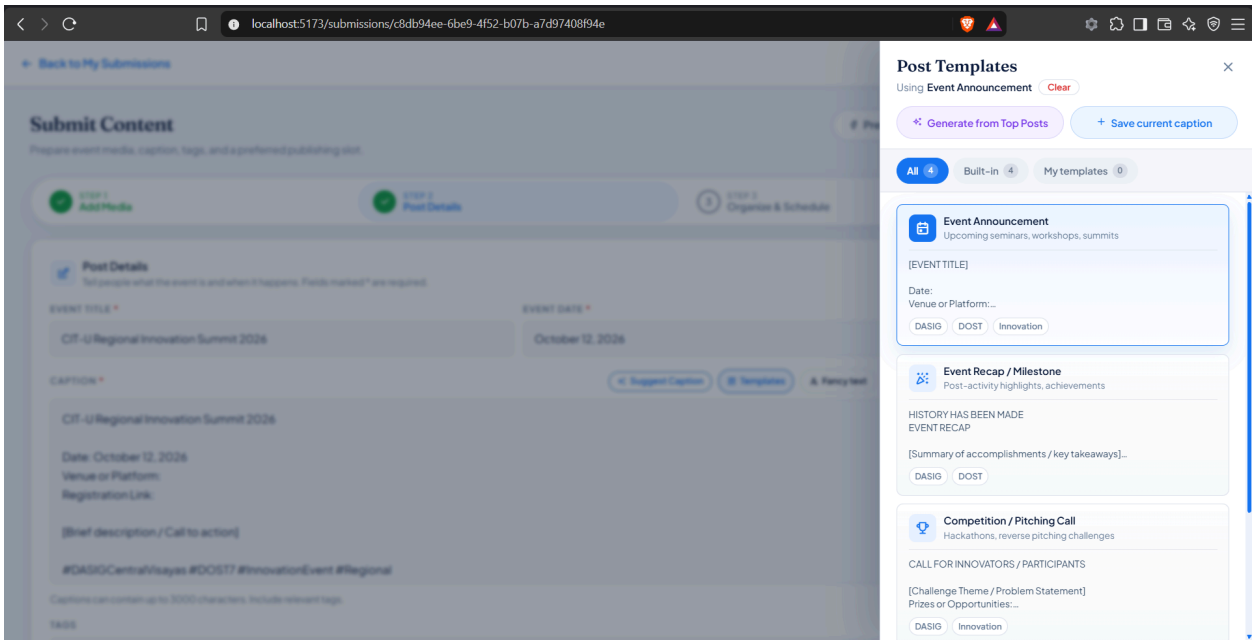

mail.google.com/mail/u/0/#inbox/FMfcgzQhWnztRmhWKKwGRMLPvTXGrnGg
Gmail
Ask Gmail
Compose
Inbox 462
Starred
Snoozed
Sent
Drafts
Purchases 32
More
Labels
1 of 649
DASIGConnect password reset
Inbox
DASIGConnect <no-reply@dasigconnect.com>
to me
3:54 PM (0 minutes ago)
Reset your DASIGConnect password
Use the button below to choose a new password for your DASIGConnect account.
Reset password
If the button does not work, copy and paste this link into your browser:
https://dev.dasigconnect.com/forgot-password/reset?token=13228ff3-f85f-41a3-a3b6-c09da2a1d0a3
This is a transactional message from DASIGConnect. If you did not request this reset, you can ignore this email.
Reply
Forward

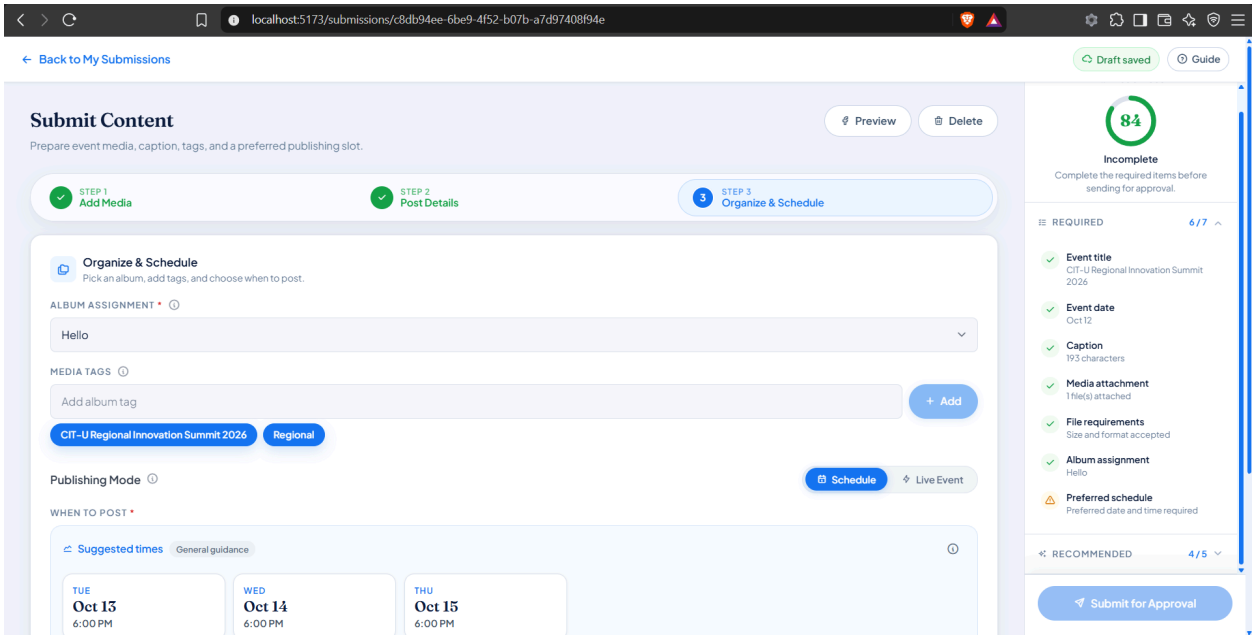

dev.dasigconnect.com/forgot-password/reset?token=13228ff3-f85f-41a3-a3b6-c09da2a1d0a3
DASIGConnect
CONTENT COORDINATION PLATFORM
Harmonizing Visayas Innovation.
Synchronized editorial coordination connecting research institutions across Region 7.
MEMBER INSTITUTIONS
No institutions yet
Back to sign in
Reset password
Enter a new password for your DASIGConnect account.
NEW PASSWORD
Mark_Anton!23
12+ characters
Uppercase letter
Lowercase letter
Number
Special character
No spaces
Not common or sequential
CONFIRM PASSWORD
Mark_Anton!23
Updating...

Screenshots for Test 6 (Case-6 — UC-1.5 Three-Step Post Composer, Readiness & Templates)

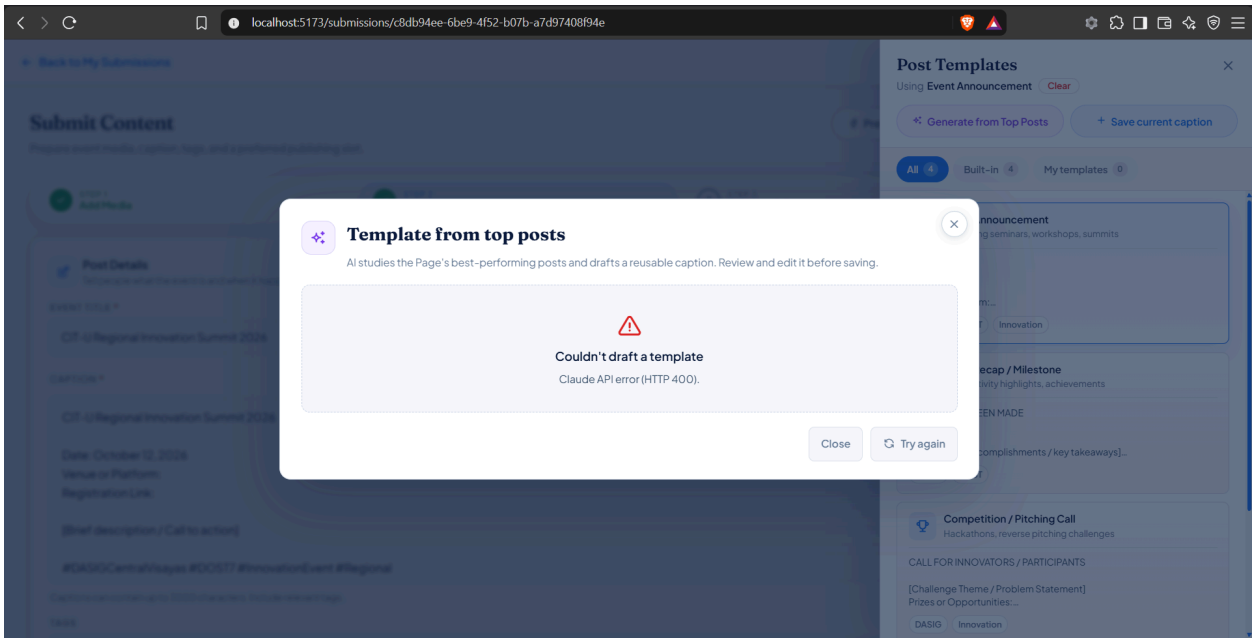




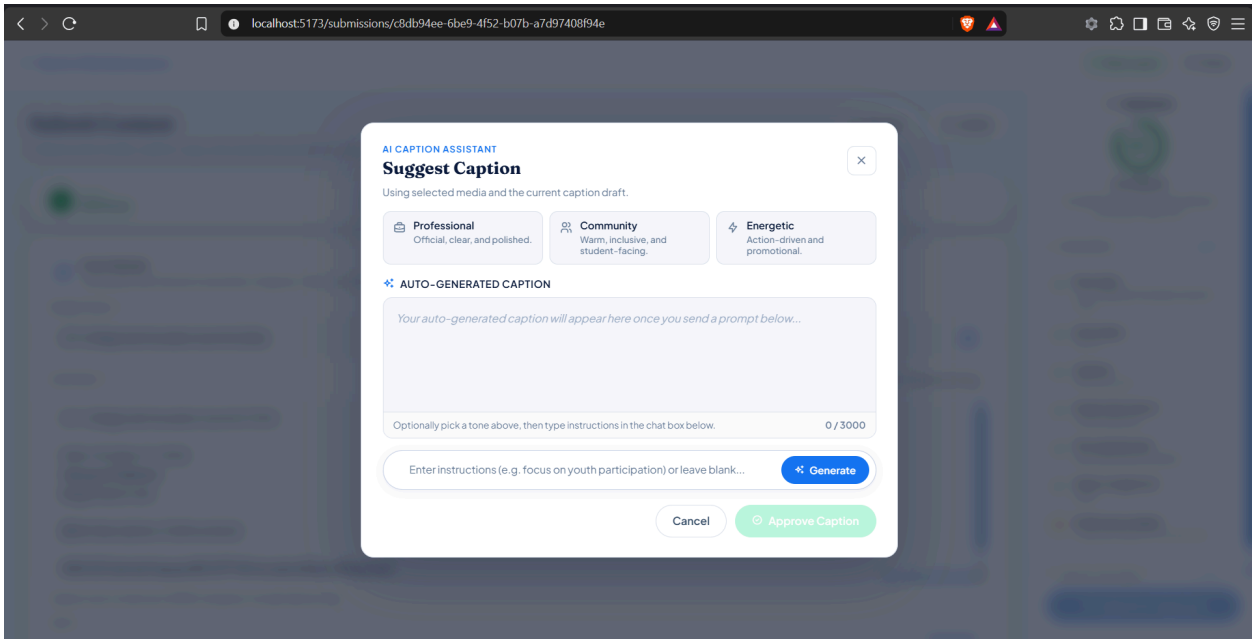

localhost:5173/submissions/c8db94ee-6be9-4f52-b07b-a7d97408f94e
Post Templates
Using Event Announcement
Clear
Generate from Top Posts
Save current caption
All 4
Built-in 4
My templates 0
Event Announcement
Upcoming seminars, workshops, summits
[EVENT TITLE]
Date:
Venue or Platform:...
DASIG
DOST
Innovation
Event Recap / Milestone
Post-activity highlights, achievements
HISTORY HAS BEEN MADE
EVENT RECAP
[Summary of accomplishments / key takeaways]...
DASIG
DOST
Competition / Pitching Call
Hackathons, reverse pitching challenges
CALL FOR INNOVATORS / PARTICIPANTS
[Challenge Theme / Problem Statement]
Prizes or Opportunities:...
DASIG
Innovation

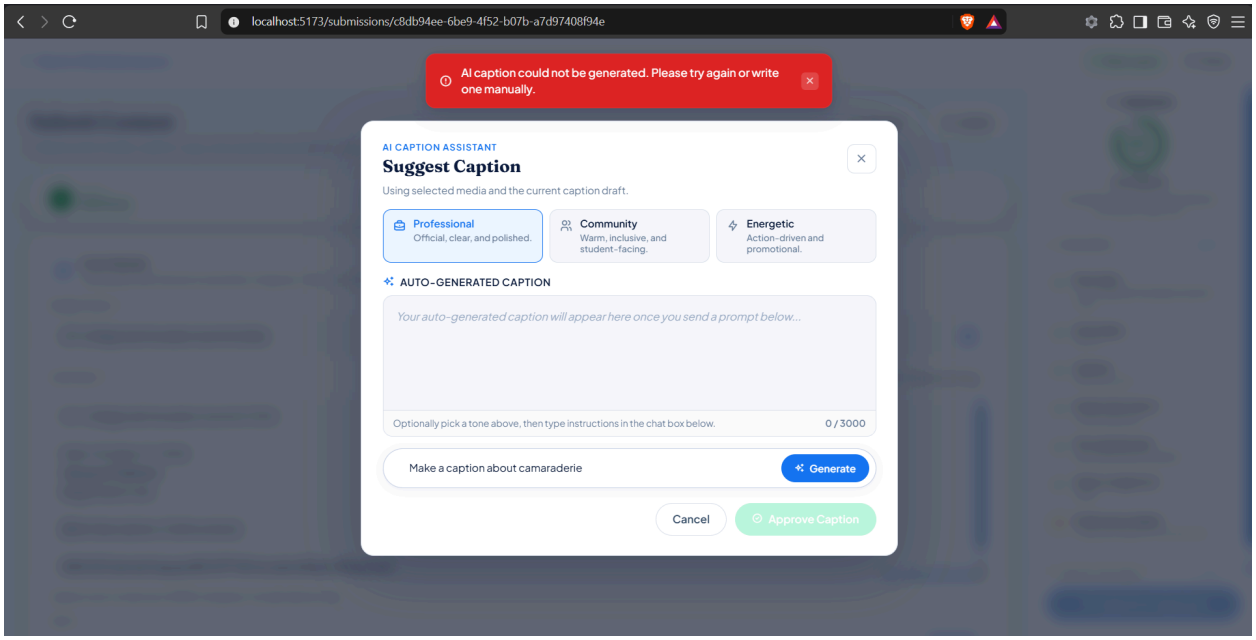

localhost:5173/submissions/c8db94ee-6be9-4f52-b07b-a7d97408f94e
Back to My Submissions
Draft saved
Guide
Submit Content
Prepare event media, caption, tags, and a preferred publishing slot.
Preview
Delete
STEP 1
Add Media
STEP 2
Post Details
STEP 3
Organize & Schedule
Organize & Schedule
Pick an album, add tags, and choose when to post.
ALBUM ASSIGNMENT *
Hello
MEDIA TAGS
Add album tag
Add
CIT-U Regional Innovation Summit 2026
Regional
Publishing Mode
Schedule
Live Event
WHEN TO POST *
Suggested times
General guidance
TUE
Oct 13
6:00 PM
WED
Oct 14
6:00 PM
THU
Oct 15
6:00 PM
84
Incomplete
Complete the required items before sending for approval.
REQUIRED
6 / 7
Event title
CIT-U Regional Innovation Summit 2026
Event date
Oct 12
Caption
193 characters
Media attachment
1 file(s) attached
File requirements
Size and format accepted
Album assignment
Hello
Preferred schedule
Preferred date and time required
RECOMMENDED
4 / 5
Submit for Approval

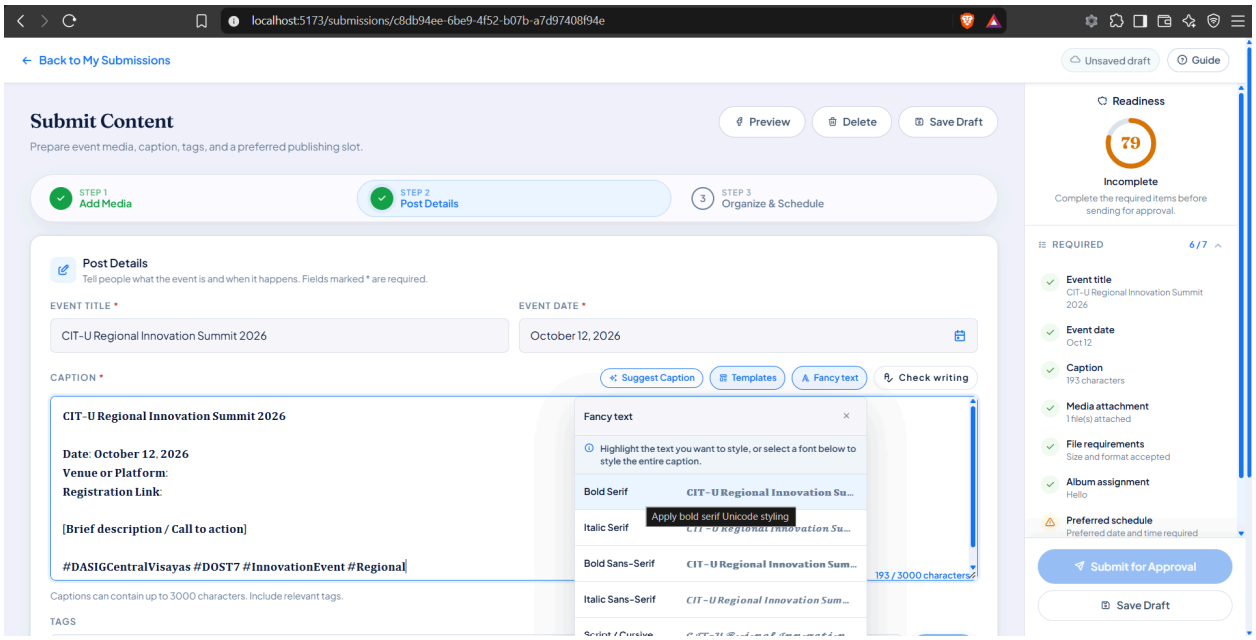
Screenshots for Test 7 (Case-7 — UC-1.5 A10 Generate Template from Top Posts)



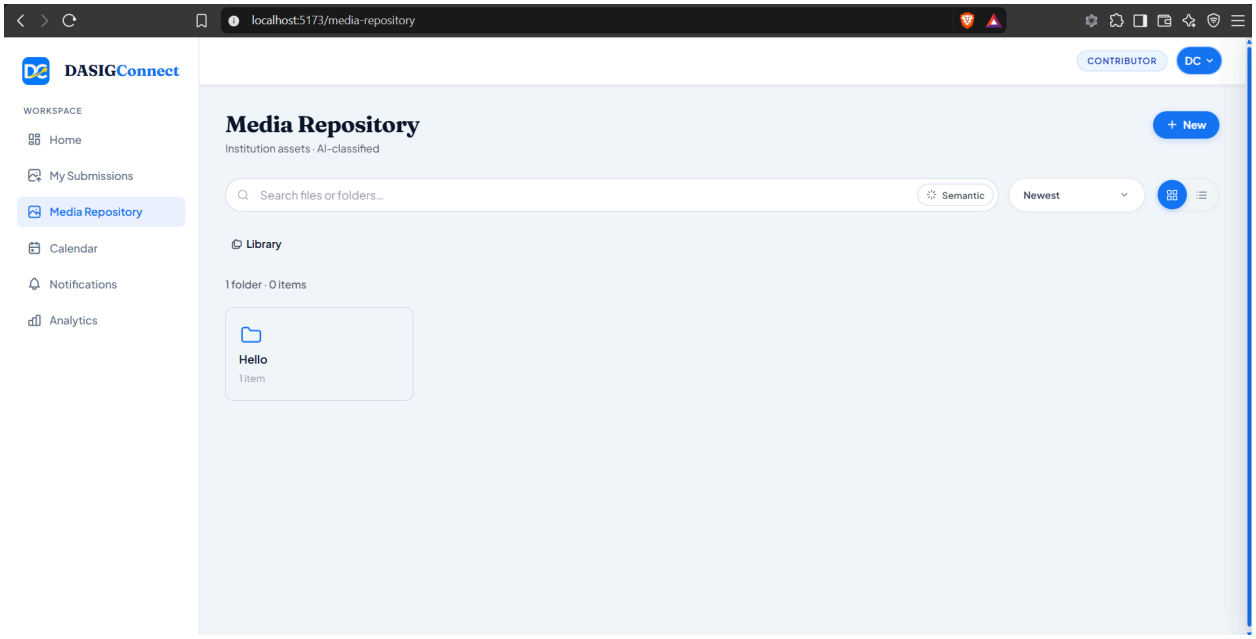
Screenshots for Test 8 (Case-8 — UC-1.6 AI Caption Generation, Writing Check & Fancy Text)

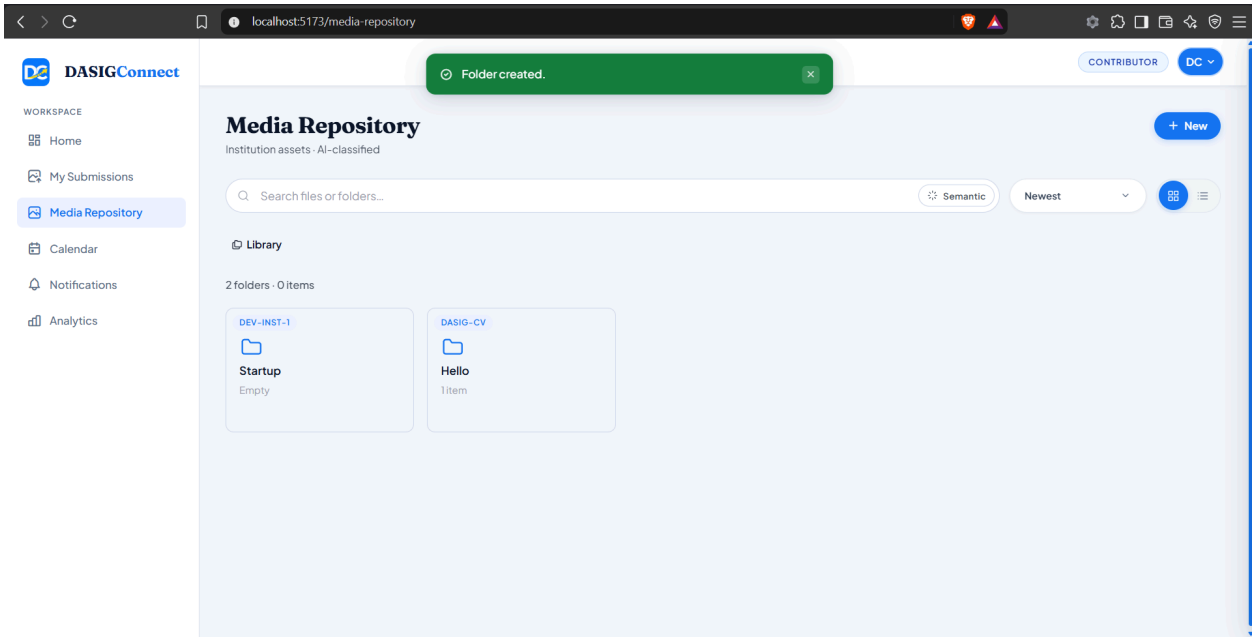


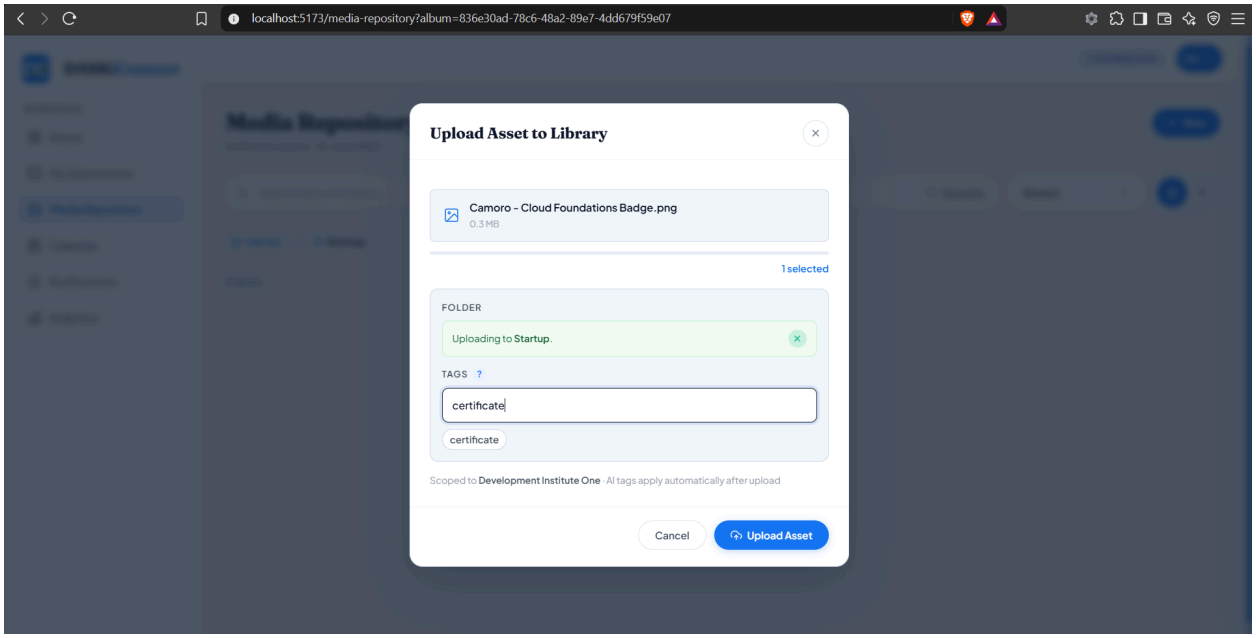

localhost:5173/submissions/c8db94ee-6be9-4f52-b07b-a7d97408f94e
AI caption could not be generated. Please try again or write one manually.
AI CAPTION ASSISTANT
Suggest Caption
Using selected media and the current caption draft.
Professional
Official, clear, and polished.
Community
Warm, inclusive, and student-facing.
Energetic
Action-driven and promotional.
AUTO-GENERATED CAPTION
Your auto-generated caption will appear here once you send a prompt below...
Optionally pick a tone above, then type instructions in the chat box below.
0 / 3000
Make a caption about camaraderie
Generate
Cancel
Approve Caption

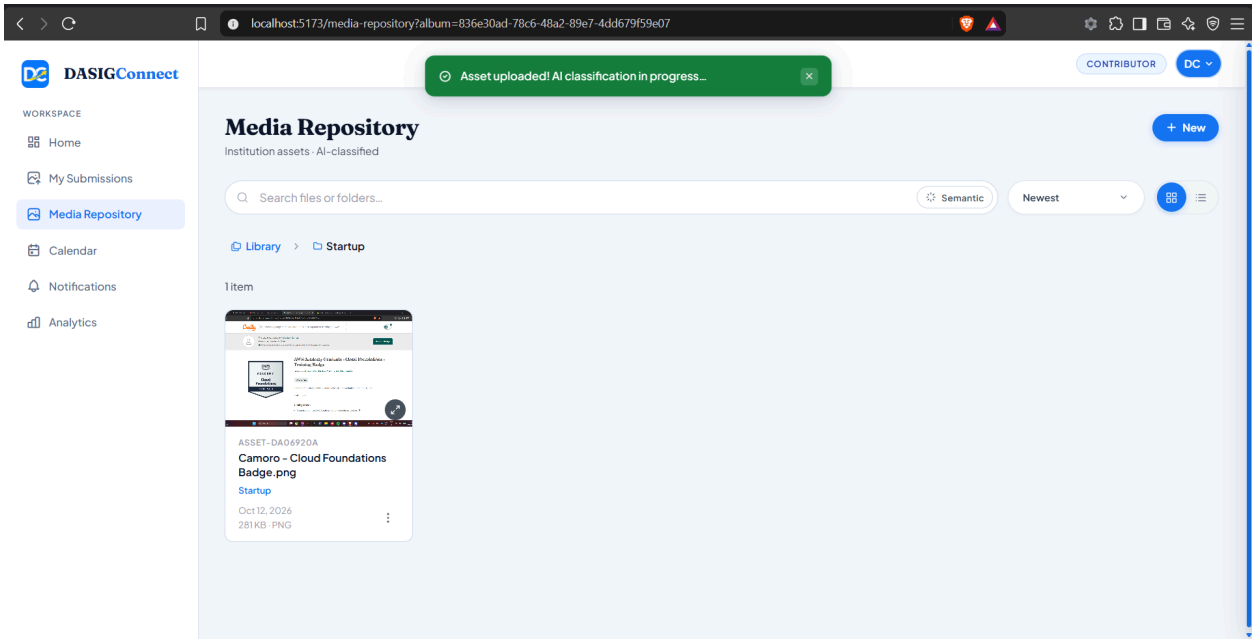

localhost:5173/submissions/c8db94ee-6be9-4f52-b07b-a7d97408f94e
Back to My Submissions
Unsaved draft
Guide
Submit Content
Prepare event media, caption, tags, and a preferred publishing slot.
Preview
Delete
Save Draft
STEP 1
Add Media
STEP 2
Post Details
STEP 3
Organize & Schedule
Post Details
Tell people what the event is and when it happens. Fields marked * are required.
EVENT TITLE *
CIT-U Regional Innovation Summit 2026
EVENT DATE *
October 12, 2026
CAPTION *
Suggest Caption
Templates
Fancy text
Check writing
CIT-U Regional Innovation Summit 2026
Date: October 12, 2026
Venue or Platform:
Registration Link:
[Brief description / Call to action]
#DASIGCentralVisayas #DOST7 #InnovationEvent #Regional
193 / 3000 characters
Captions can contain up to 3000 characters. Include relevant tags.
TAGS
Fancy text
Highlight the text you want to style, or select a font below to style the entire caption.
Bold Serif
CIT-U Regional Innovation Su...
Italic Serif
Apply bold serif Unicode styling
Bold Sans-Serif
CIT-U Regional Innovation Sum...
Italic Sans-Serif
CIT-U Regional Innovation Sum...
Readiness
79
Incomplete
Complete the required items before sending for approval.
REQUIRED
6 / 7
Event title
CIT-U Regional Innovation Summit 2026
Event date
Oct 12
Caption
193 characters
Media attachment
1 file(s) attached
File requirements
Size and format accepted
Album assignment
Hello
Preferred schedule
Preferred date and time required
Submit for Approval
Save Draft

Screenshots for Test 9 (Case-9 — UC-1.7 Media Attachment Rules, AI Suggestions & Album Auto-Match)

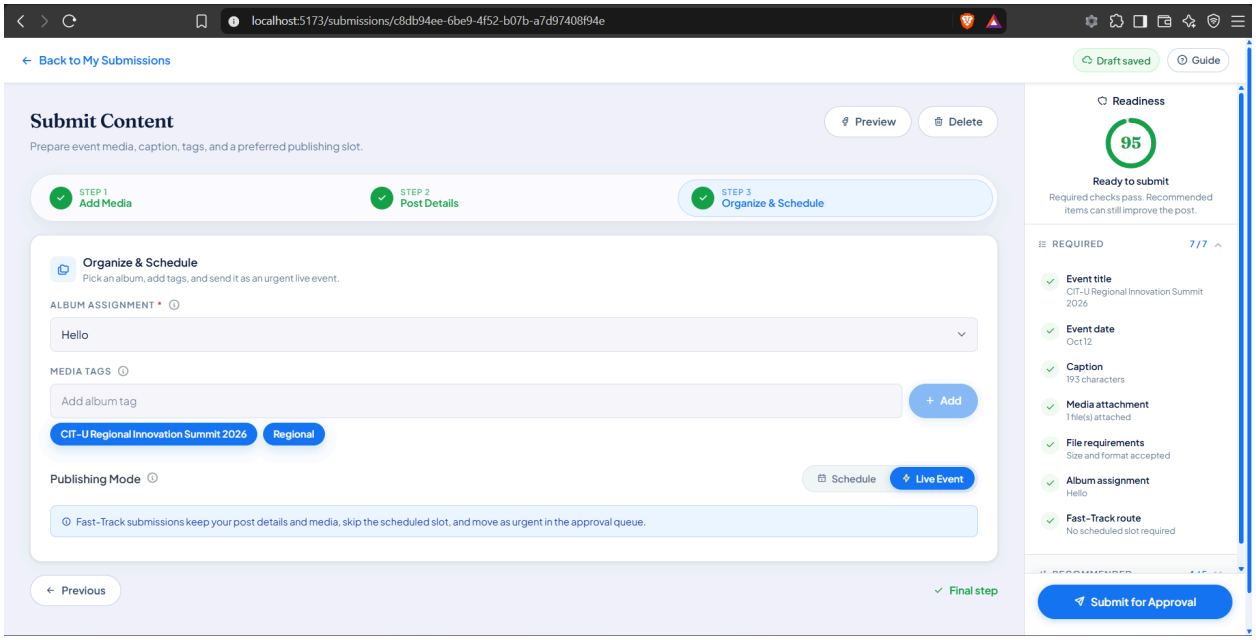


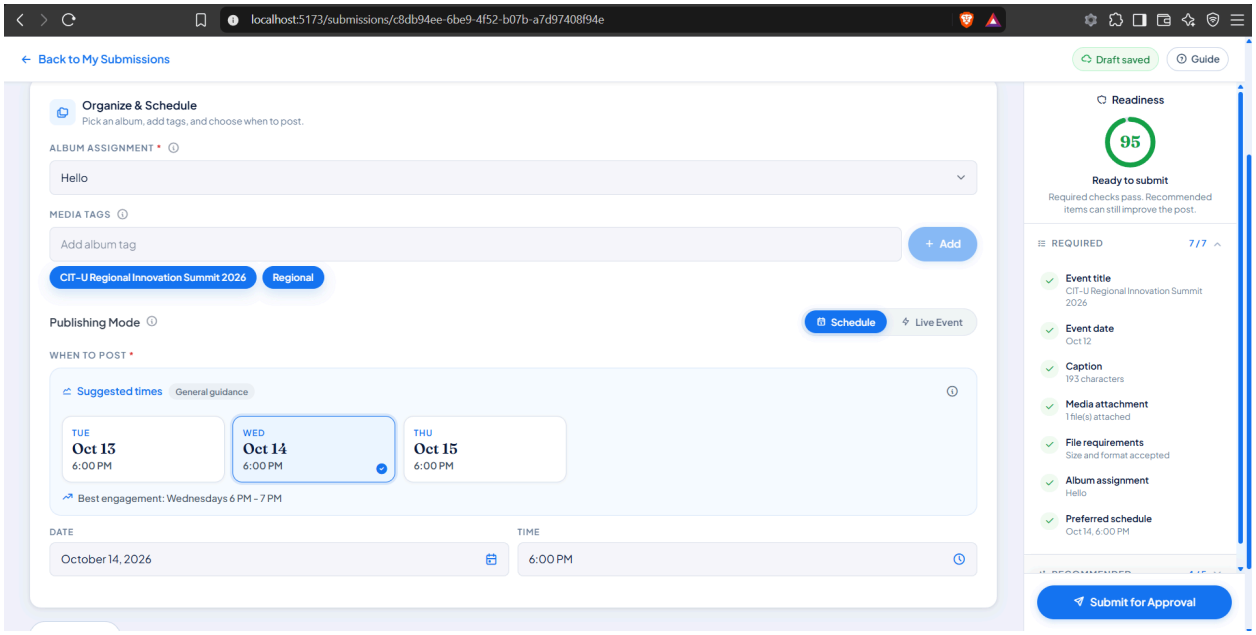



localhost:5173/media-repository?album=836e30ad-78c6-48a2-89e7-4dd679f59e07
Upload Asset to Library
Camoro - Cloud Foundations Badge.png
0.3 MB
1 selected
FOLDER
Uploading to Startup.
TAGS
certificate
certificate
Scoped to Development Institute One · AI tags apply automatically after upload
Cancel
Upload Asset

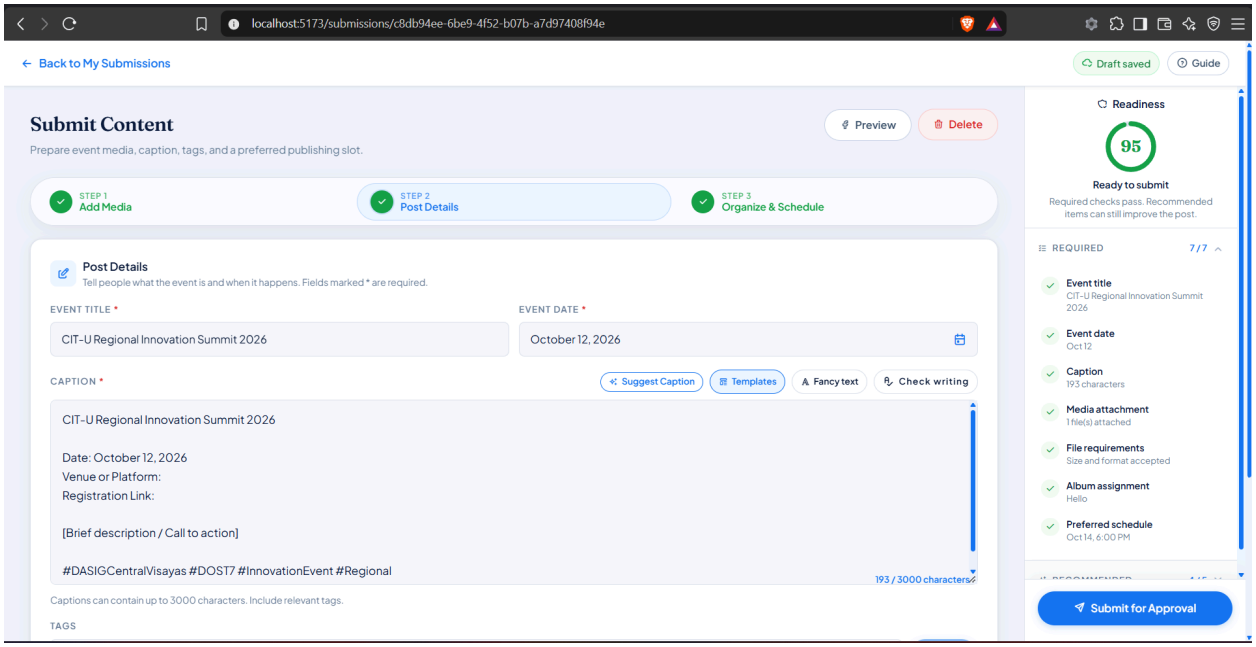

localhost:5173/media-repository?album=836e30ad-78c6-48a2-89e7-4dd679f59e07
DASIGConnect
WORKSPACE
Home
My Submissions
Media Repository
Calendar
Notifications
Analytics
Asset uploaded! AI classification in progress...
CONTRIBUTOR
DC
Media Repository
Institution assets · AI-classified
New
Search files or folders...
Semantic
Newest
Library
Startup
1 item
ASSET-DA06920A
Camoro - Cloud Foundations Badge.png
Startup
Oct 12, 2026
281 KB · PNG

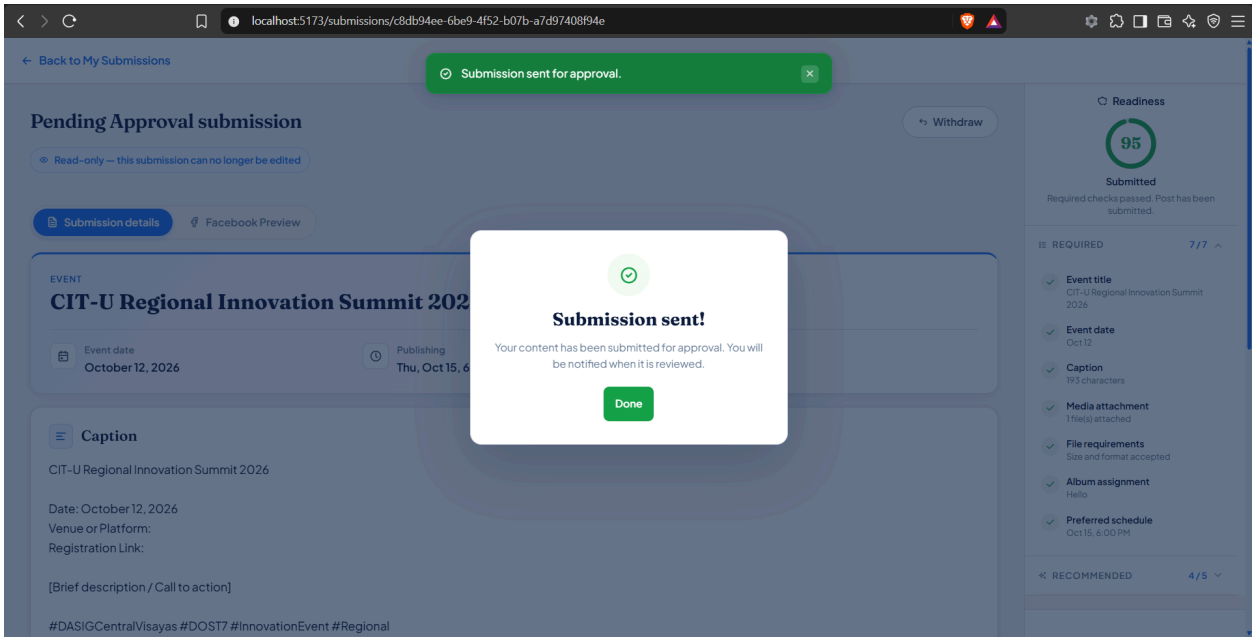
Screenshots for Test 10 (Case-10 — UC-1.8 Peak-Hour Engagement Recommendations)

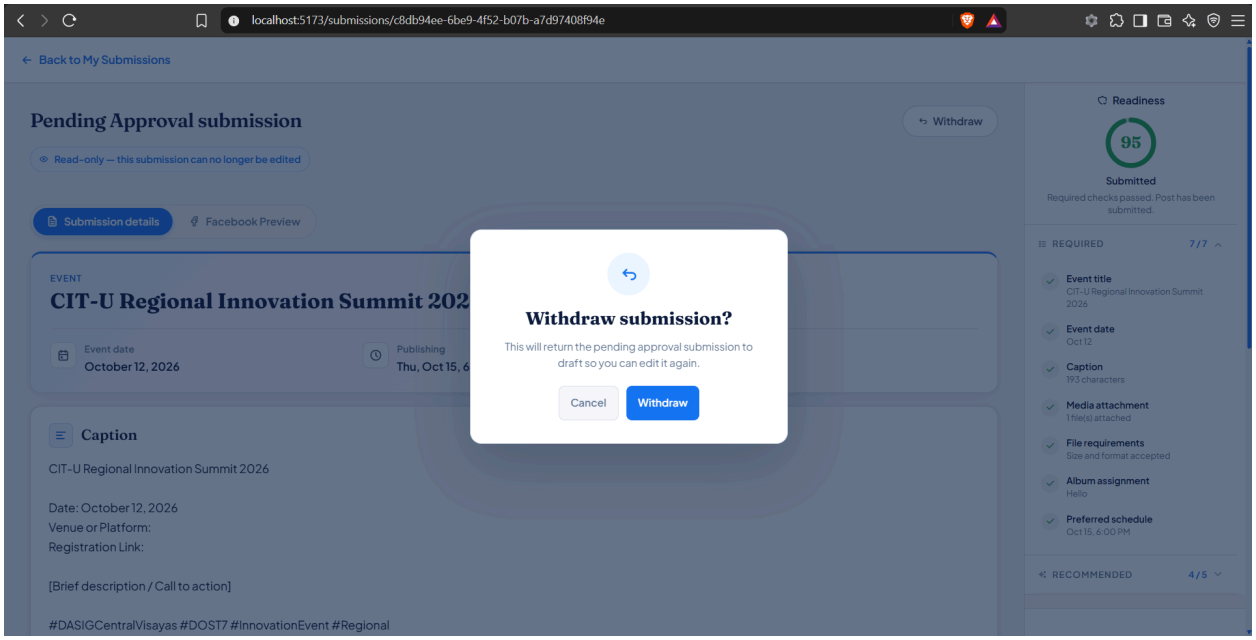


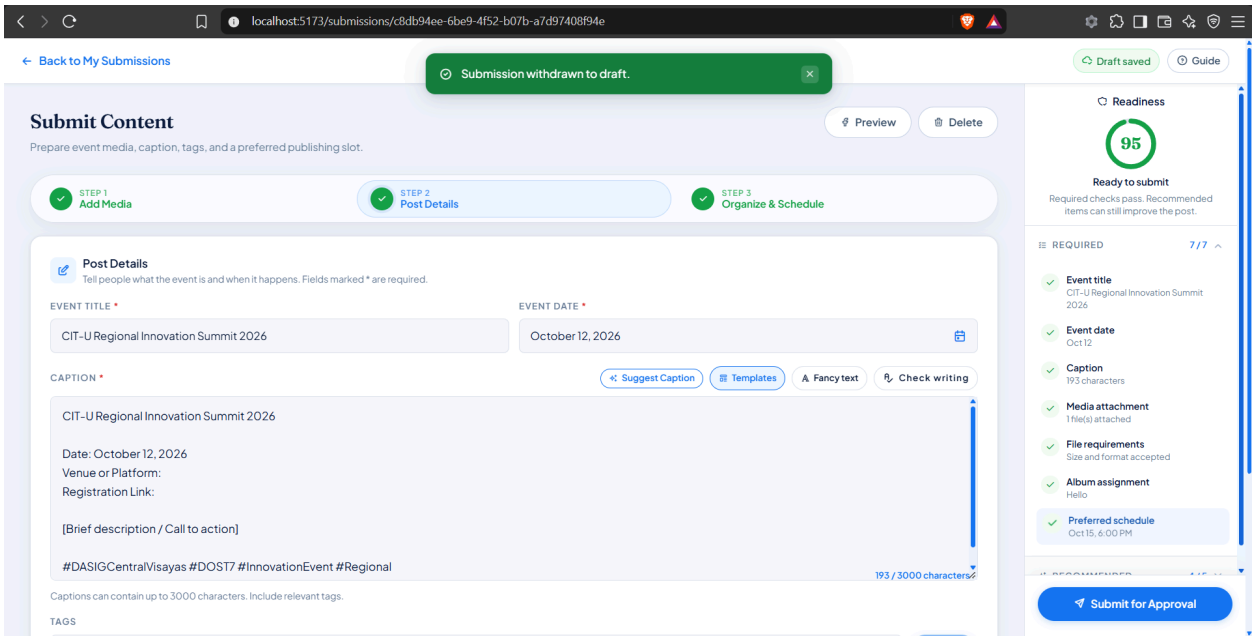


Screenshots for Test 11 (Case-11 — UC-1.9 Draft Submission, Withdrawal & Fast-Track Sorting)

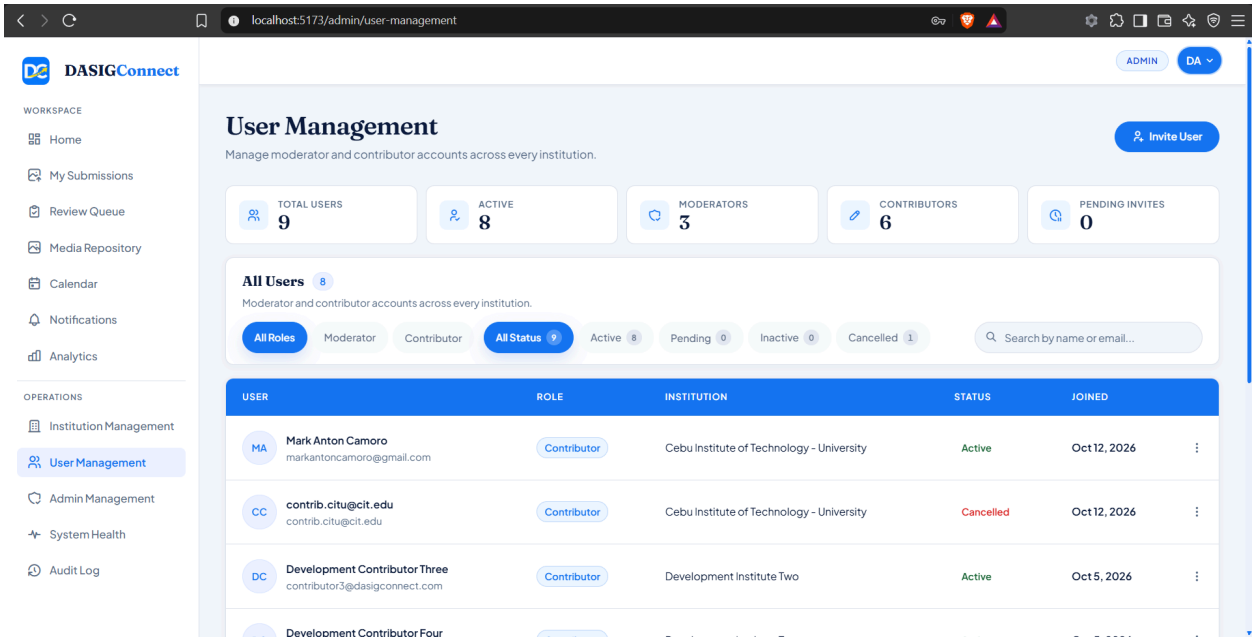


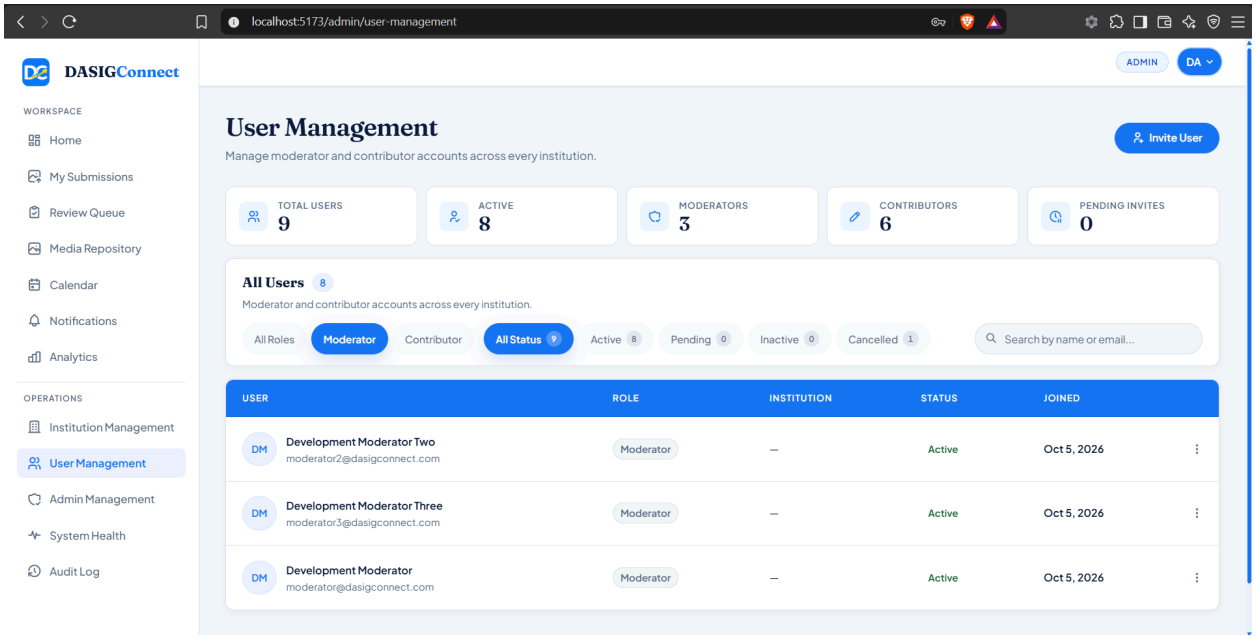


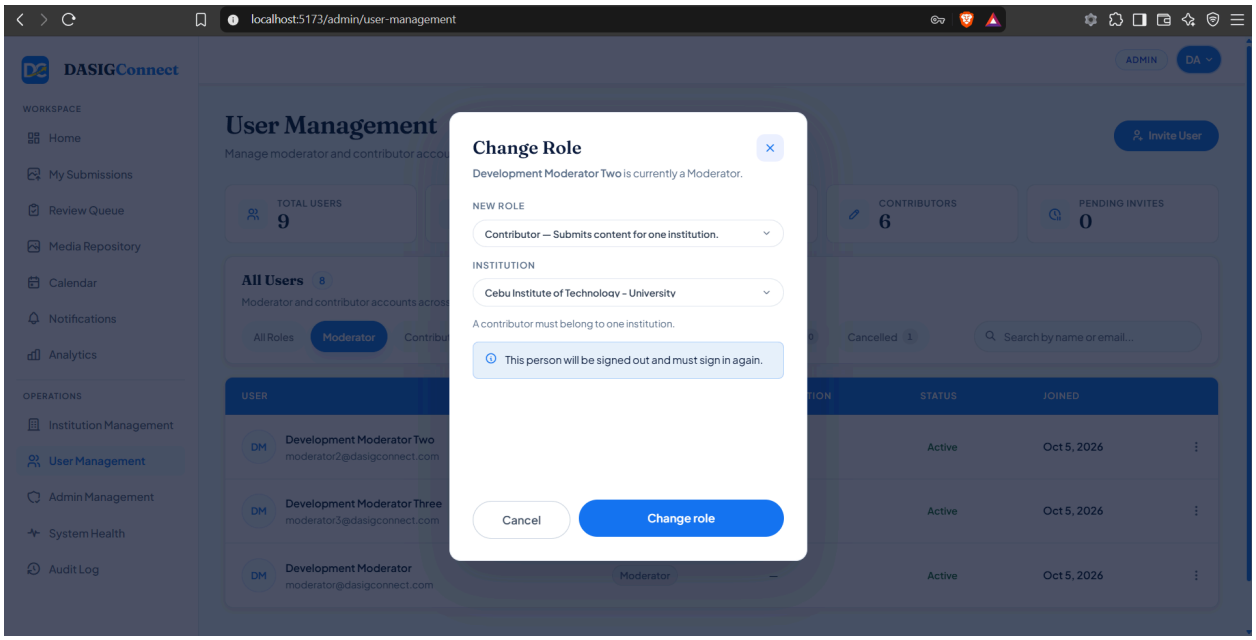

localhost:5173/submissions/c8db94ee-6be9-4f52-b07b-a7d97408f94e
Back to My Submissions
Pending Approval submission
Withdraw
Read-only — this submission can no longer be edited
Submission details
Facebook Preview
EVENT
CIT-U Regional Innovation Summit 202
Event date
October 12, 2026
Publishing
Thu, Oct 15, 6
Caption
CIT-U Regional Innovation Summit 2026
Date: October 12, 2026
Venue or Platform:
Registration Link:
[Brief description / Call to action]
#DASIGCentralVisayas #DOST7 #InnovationEvent #Regional
Withdraw submission?
This will return the pending approval submission to draft so you can edit it again.
Cancel
Withdraw
Readiness
95
Submitted
Required checks passed. Post has been submitted.
REQUIRED
7 / 7
Event title
CIT-U Regional Innovation Summit 2026
Event date
Oct 12
Caption
193 characters
Media attachment
1 file(s) attached
File requirements
Size and format accepted
Album assignment
Hello
Preferred schedule
Oct 15, 6:00 PM
RECOMMENDED
4 / 5

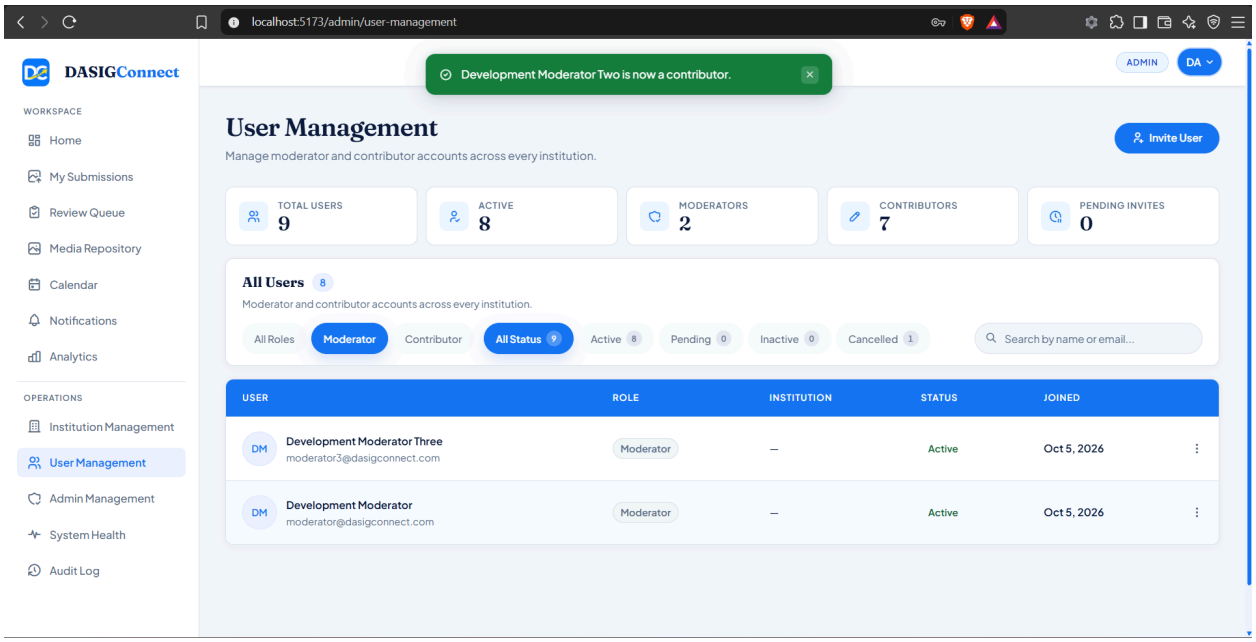

localhost:5173/submissions/c8db94ee-6be9-4f52-b07b-a7d97408f94e
Back to My Submissions
Submission withdrawn to draft.
Draft saved
Guide
Submit Content
Prepare event media, caption, tags, and a preferred publishing slot.
Preview
Delete
STEP 1
Add Media
STEP 2
Post Details
STEP 3
Organize & Schedule
Post Details
Tell people what the event is and when it happens. Fields marked * are required.
EVENT TITLE *
CIT-U Regional Innovation Summit 2026
EVENT DATE *
October 12, 2026
CAPTION *
Suggest Caption
Templates
Fancy text
Check writing
CIT-U Regional Innovation Summit 2026
Date: October 12, 2026
Venue or Platform:
Registration Link:
[Brief description / Call to action]
#DASIGCentralVisayas #DOST7 #InnovationEvent #Regional
193 / 3000 characters
Captions can contain up to 3000 characters. Include relevant tags.
TAGS
Readiness
95
Ready to submit
Required checks pass. Recommended items can still improve the post.
REQUIRED
7 / 7
Event title
CIT-U Regional Innovation Summit 2026
Event date
Oct 12
Caption
193 characters
Media attachment
1 file(s) attached
File requirements
Size and format accepted
Album assignment
Hello
Preferred schedule
Oct 15, 6:00 PM
Submit for Approval

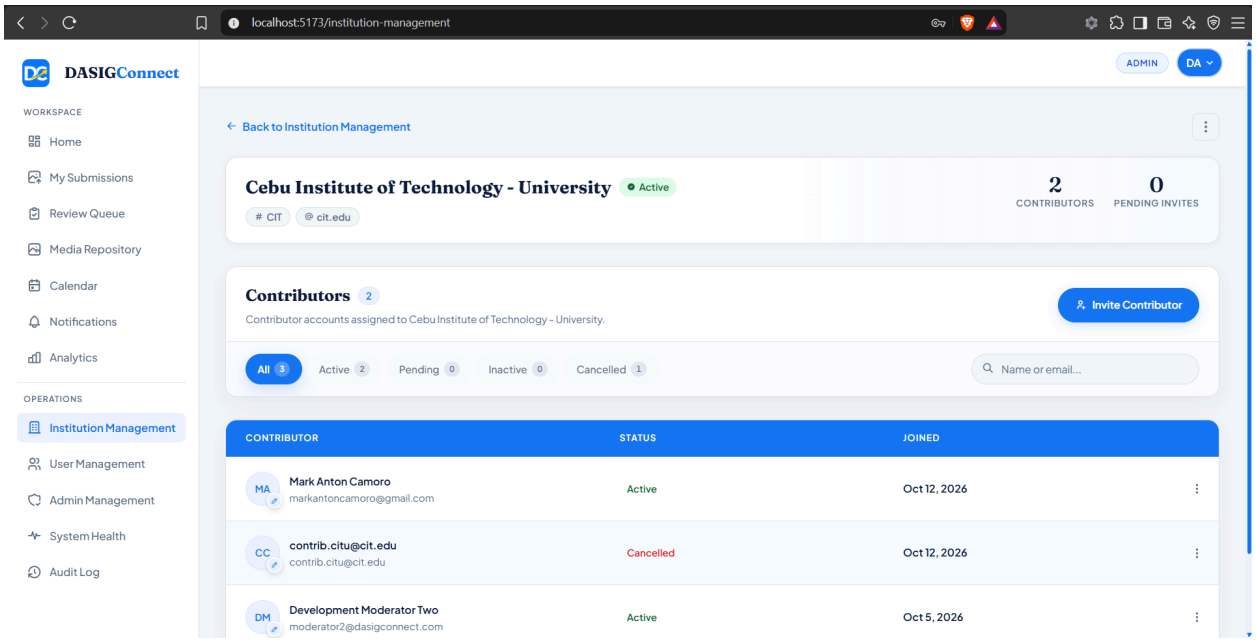
Screenshots for Test 12 (Case-12 — UC-1.10 Moderator Account Management & Lateral Role Change)



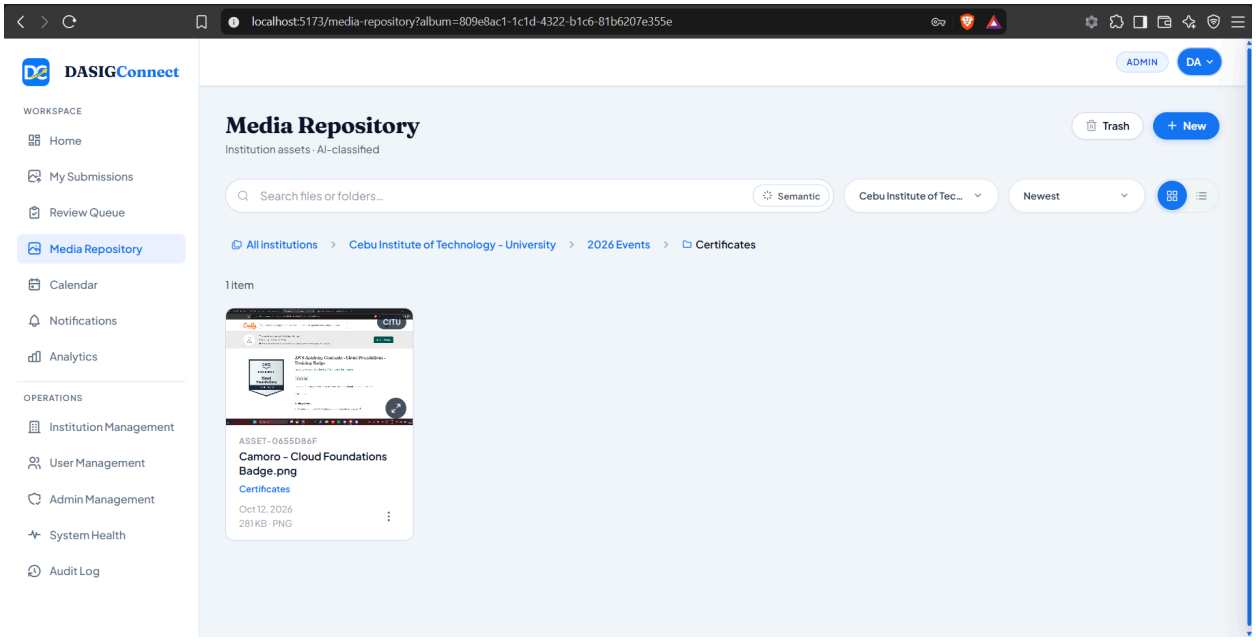



localhost:5173/admin/user-management
DASIGConnect
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
ADMIN
DA
User Management
Manage moderator and contributor accou
Invite User
TOTAL USERS 9
CONTRIBUTORS 6
PENDING INVITES 0
All Users 8
Moderator and contributor accounts acros
All Roles
Moderator
Contribu
Cancelled 1
Search by name or email...
USER
STATUS
JOINED
DM Development Moderator Two moderator2@dasigconnect.com Active Oct 5, 2026
DM Development Moderator Three moderator3@dasigconnect.com Active Oct 5, 2026
DM Development Moderator moderator@dasigconnect.com Moderator – Active Oct 5, 2026
Change Role
Development Moderator Two is currently a Moderator.
NEW ROLE
Contributor — Submits content for one institution.
INSTITUTION
Cebu Institute of Technology - University
A contributor must belong to one institution.
This person will be signed out and must sign in again.
Cancel
Change role

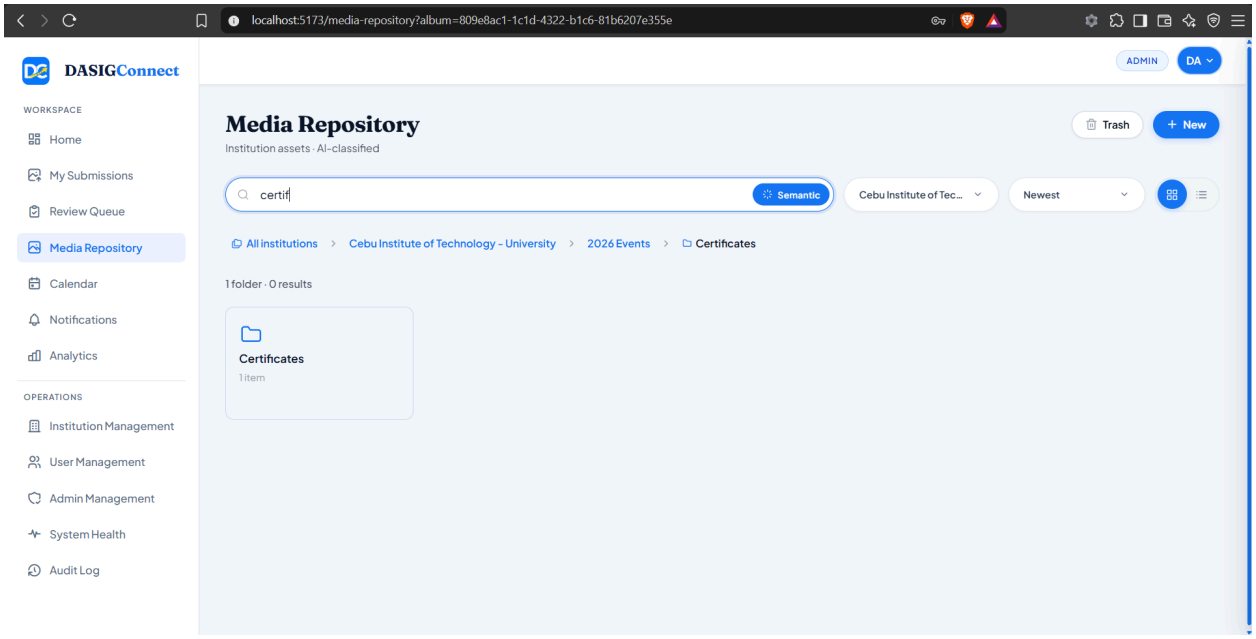

localhost:5173/admin/user-management
DASIGConnect
Development Moderator Two is now a contributor.
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
User Management
Manage moderator and contributor accounts across every institution.
Invite User
TOTAL USERS 9
ACTIVE 8
MODERATORS 2
CONTRIBUTORS 7
PENDING INVITES 0
All Users 8
Moderator and contributor accounts across every institution.
All Roles
Moderator
Contributor
All Status 9
Active 8
Pending 0
Inactive 0
Cancelled 1
Search by name or email...
USER
ROLE
INSTITUTION
STATUS
JOINED
DM Development Moderator Three moderator3@dasigconnect.com Moderator – Active Oct 5, 2026
DM Development Moderator moderator@dasigconnect.com Moderator – Active Oct 5, 2026

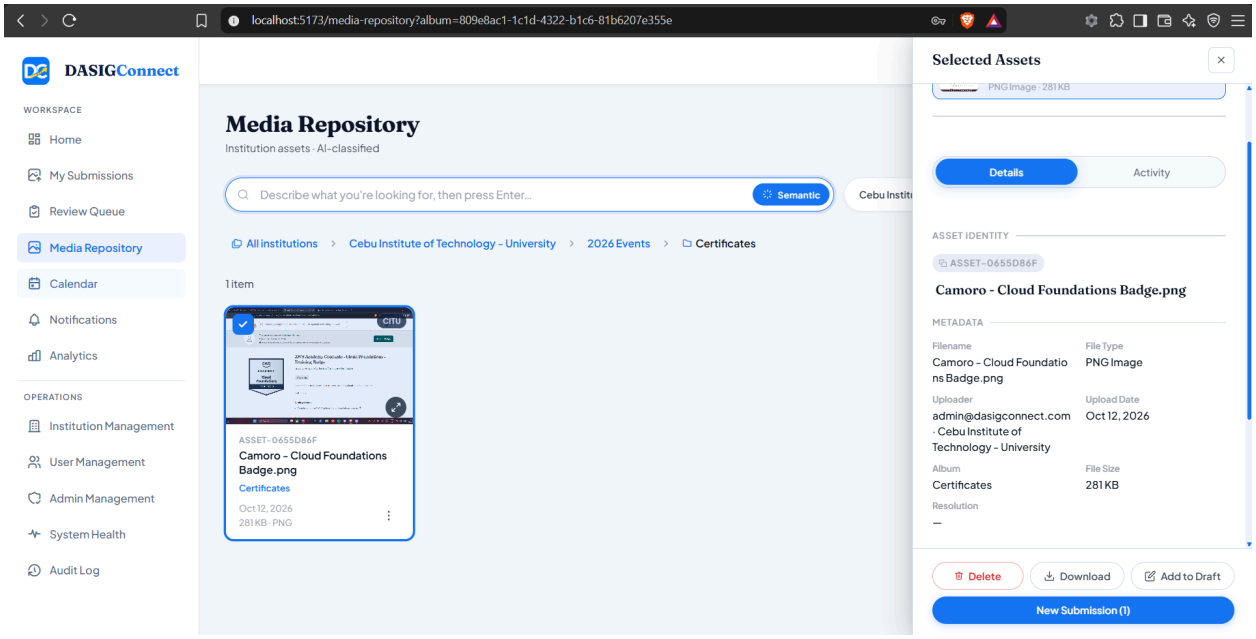


Screenshots for Test 13 (Case-13 — UC-2.1 Standalone Library Uploads, Nested Albums & Tag Floor)

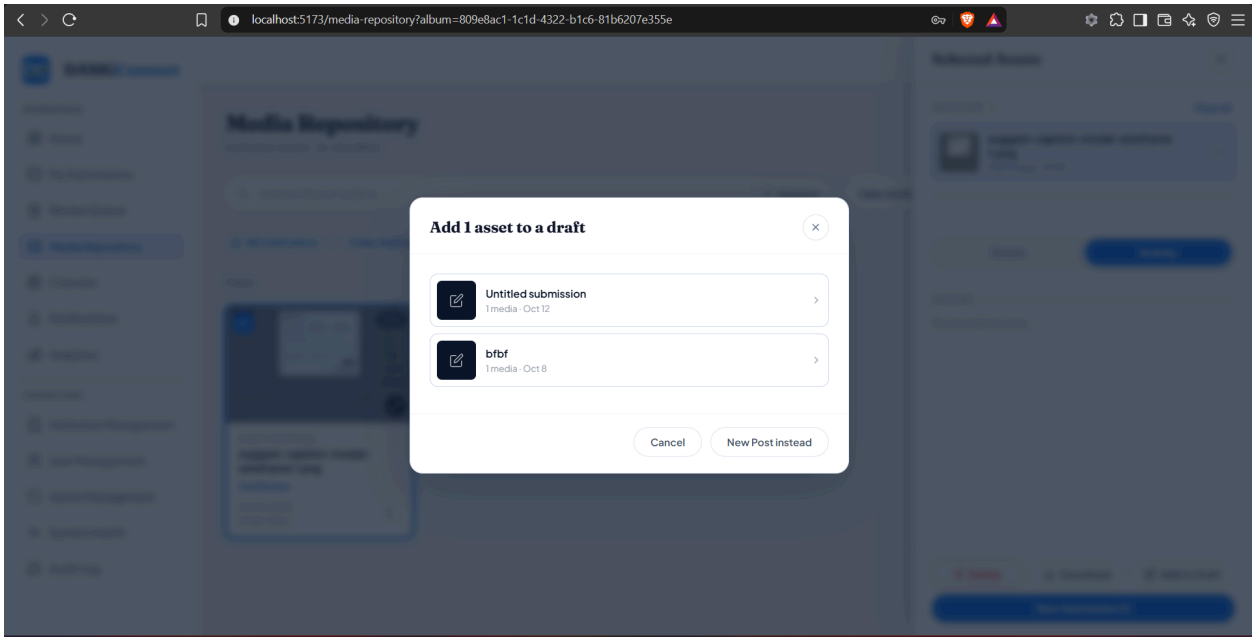


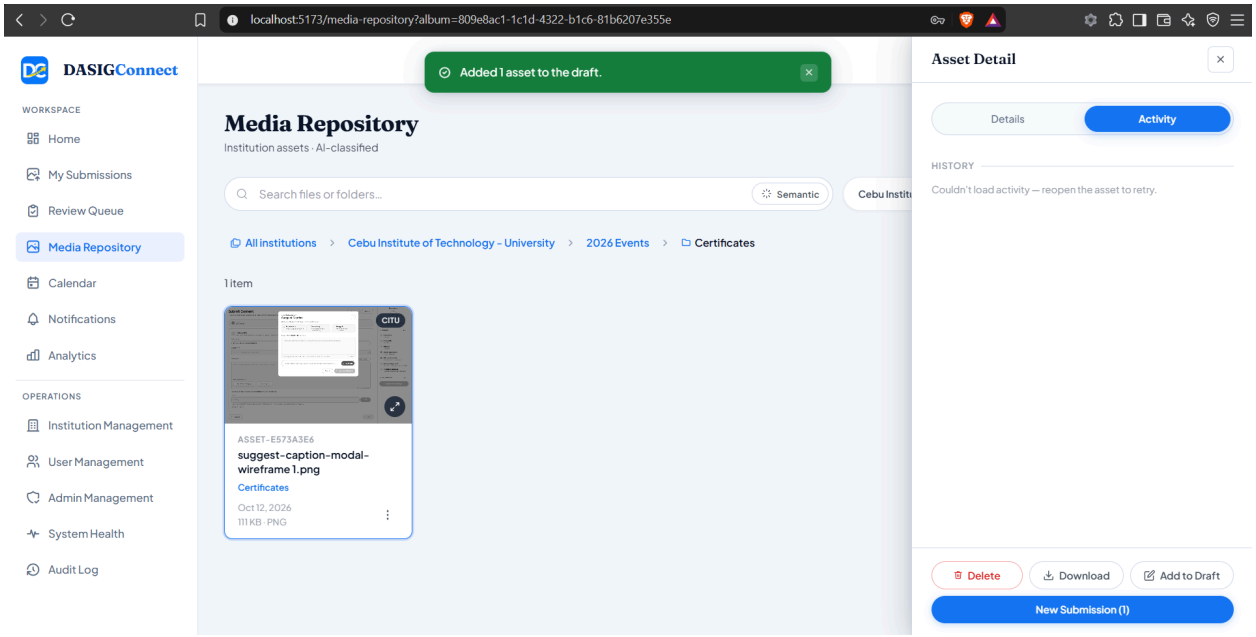
Screenshots for Test 14 (Case-14 — UC-2.2 Semantic Search, Keyword Fallback & Asset Detail Panel)

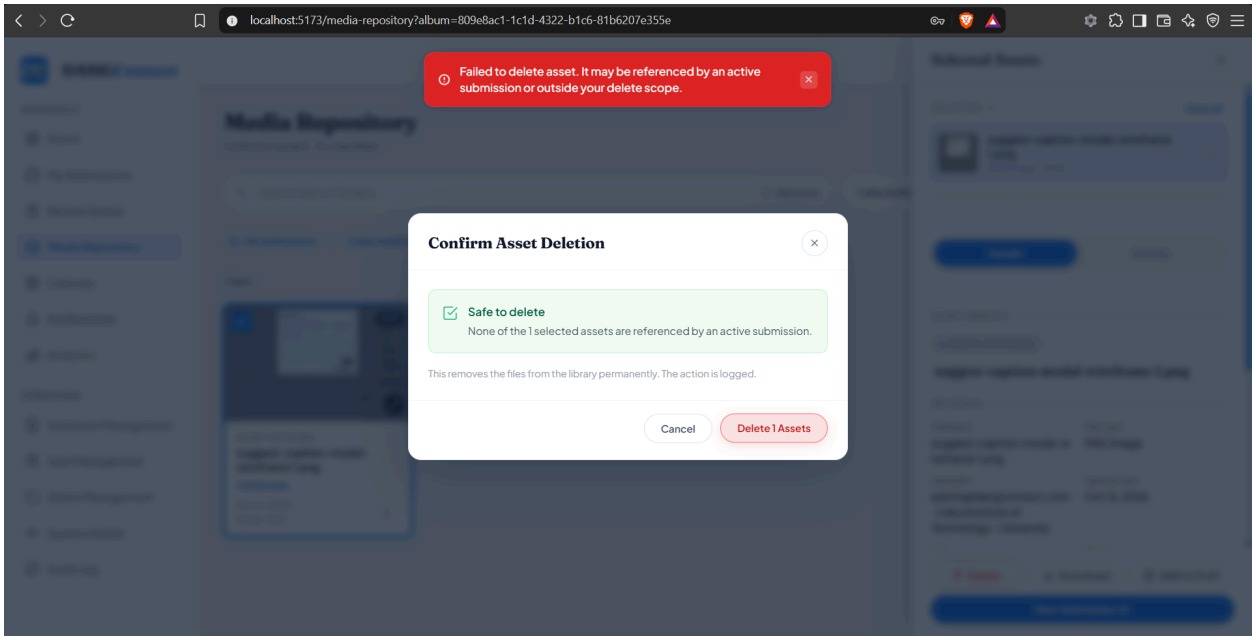


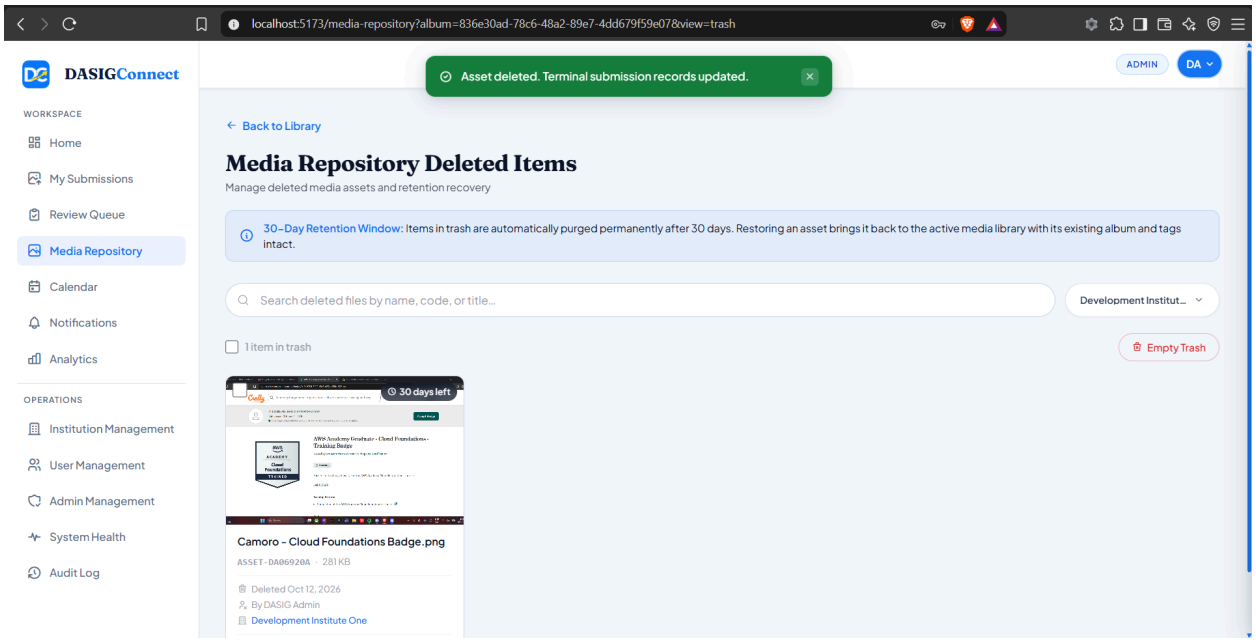


Screenshots for Test 15 (Case-15 — UC-2.3 Media History, Tiered Deletion & Duplicate Detection)

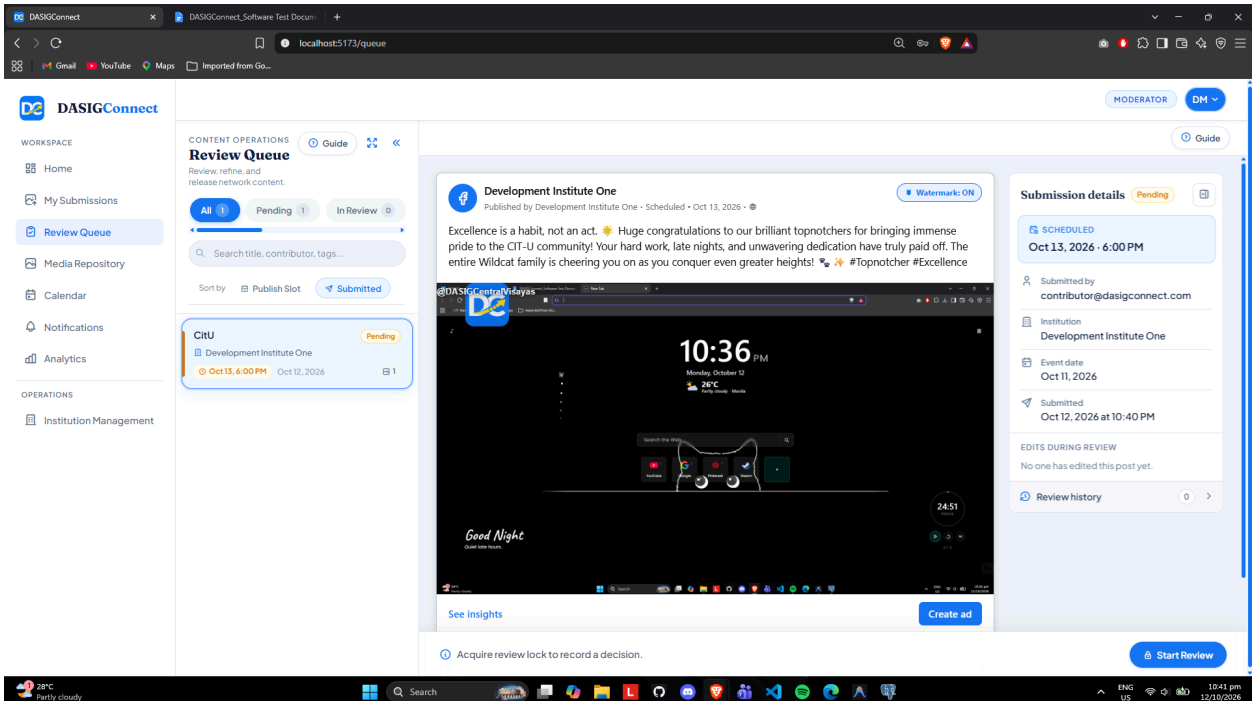


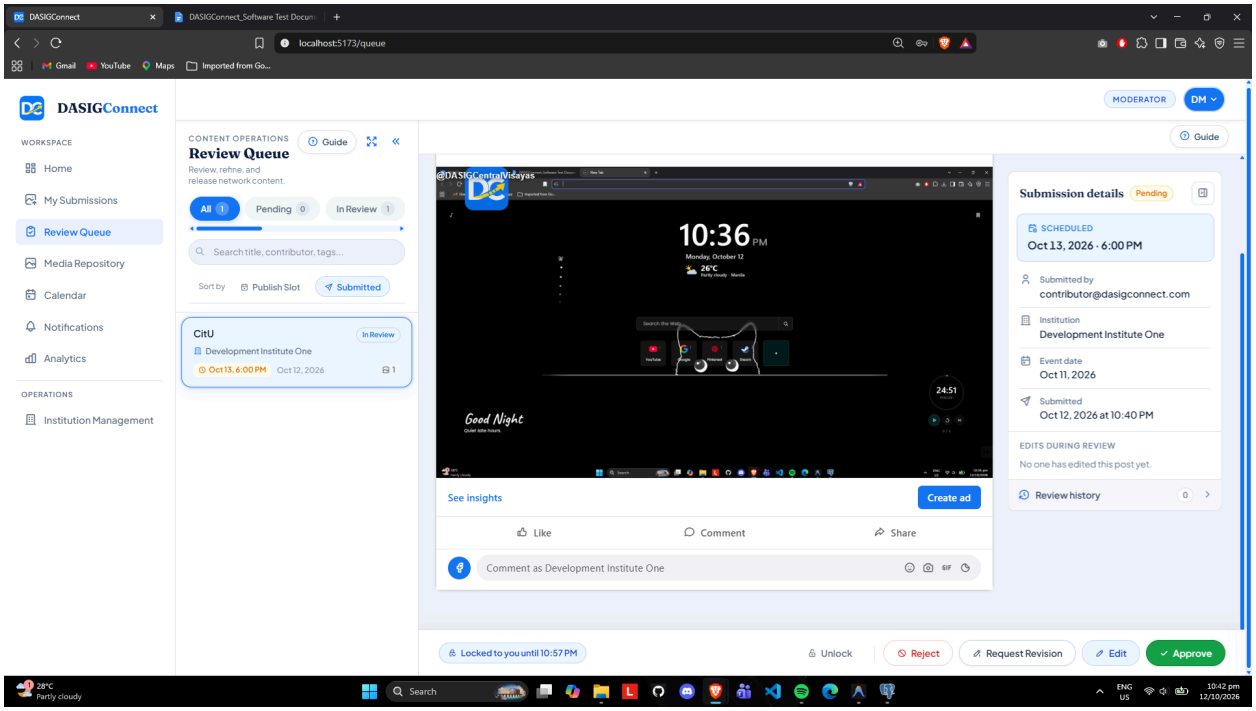



localhost:5173/media-repository?album=809e8ac1-1c1d-4322-b1c6-81b6207e355e
Failed to delete asset. It may be referenced by an active submission or outside your delete scope.
Confirm Asset Deletion
Safe to delete
None of the 1 selected assets are referenced by an active submission.
This removes the files from the library permanently. The action is logged.
Cancel
Delete 1 Assets

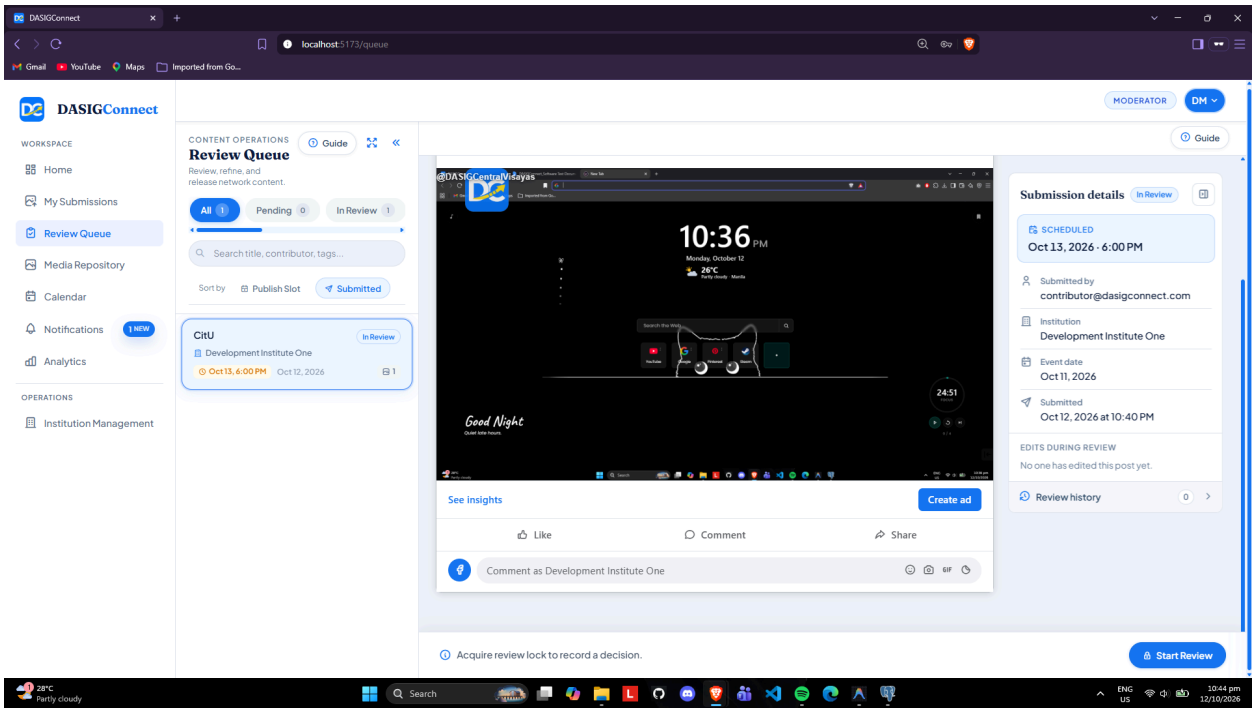

localhost:5173/media-repository?album=836e30ad-78c6-48a2-89e7-4dd679f59e07&view=trash
DASIGConnect
Asset deleted. Terminal submission records updated.
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Back to Library
Media Repository Deleted Items
Manage deleted media assets and retention recovery
30-Day Retention Window: Items in trash are automatically purged permanently after 30 days. Restoring an asset brings it back to the active media library with its existing album and tags intact.
Search deleted files by name, code, or title...
Development Institut...
1 item in trash
Empty Trash
30 days left
Camoro - Cloud Foundations Badge.png
ASSET-DA06920A · 281 KB
Deleted Oct 12, 2026
By DASIG Admin
Development Institute One

Screenshots for Test 16 (Case-16 — UC-2.4 Approval Workflow, 15-Min Lock & Self-Review Block)

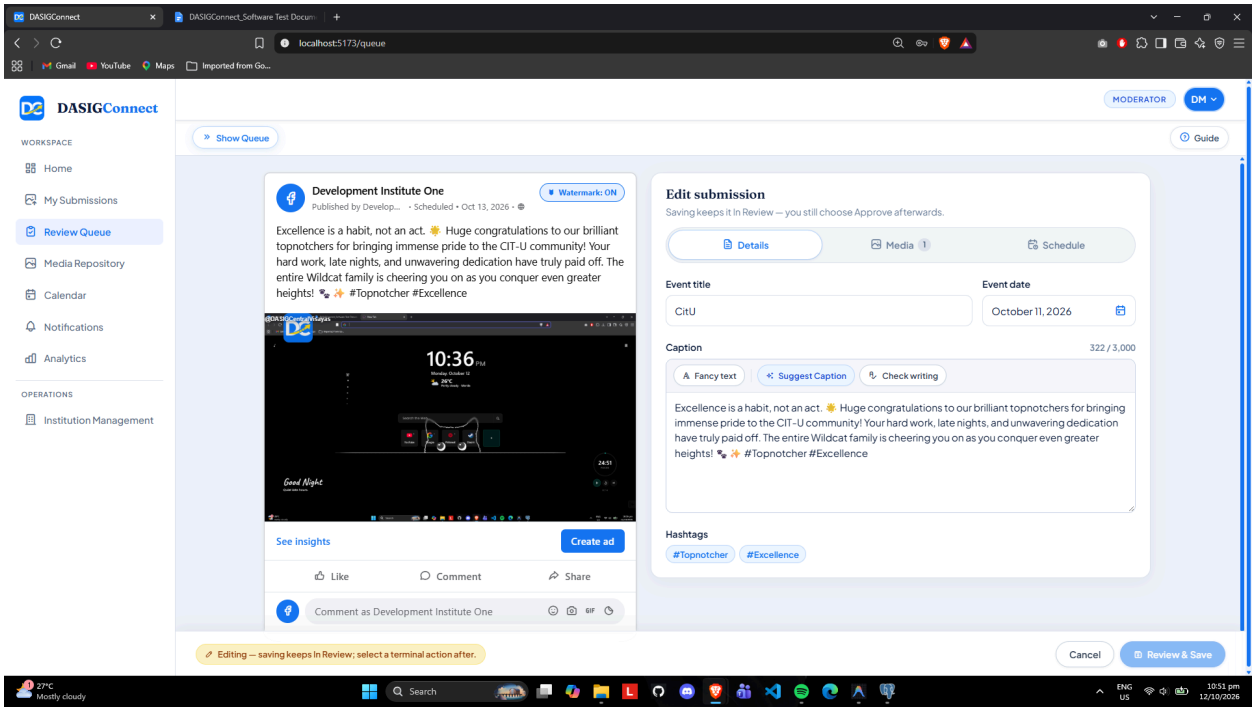


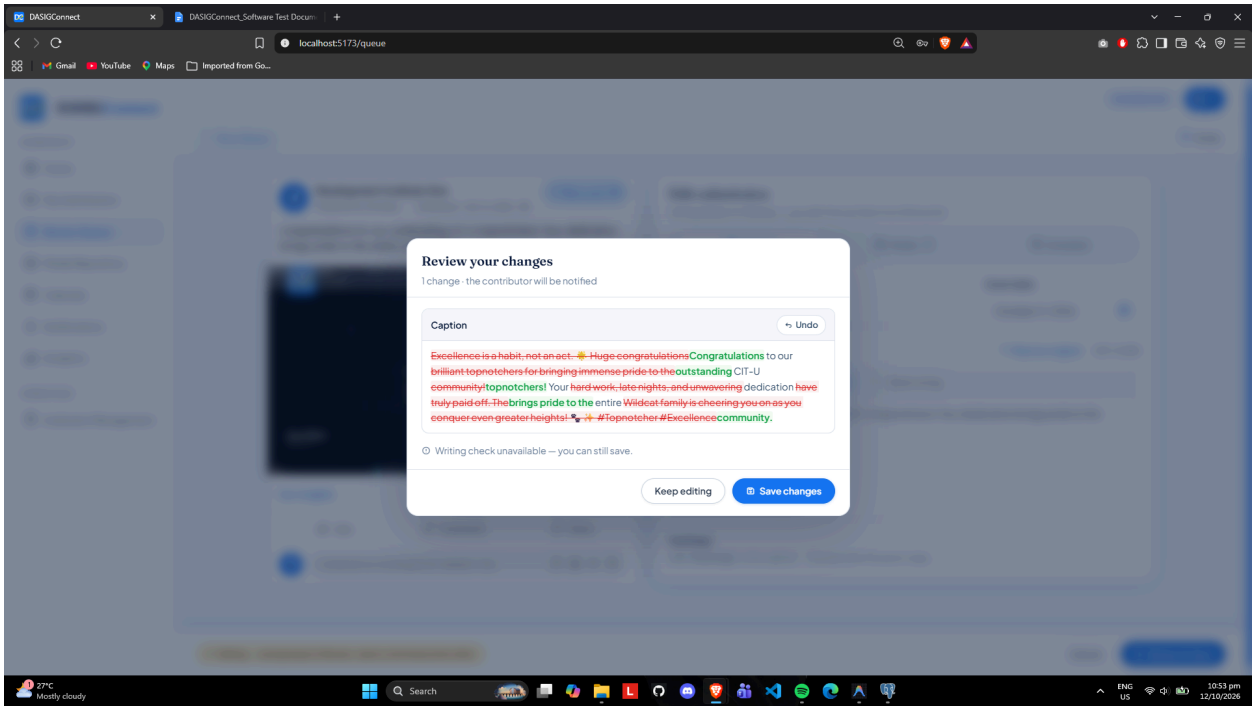


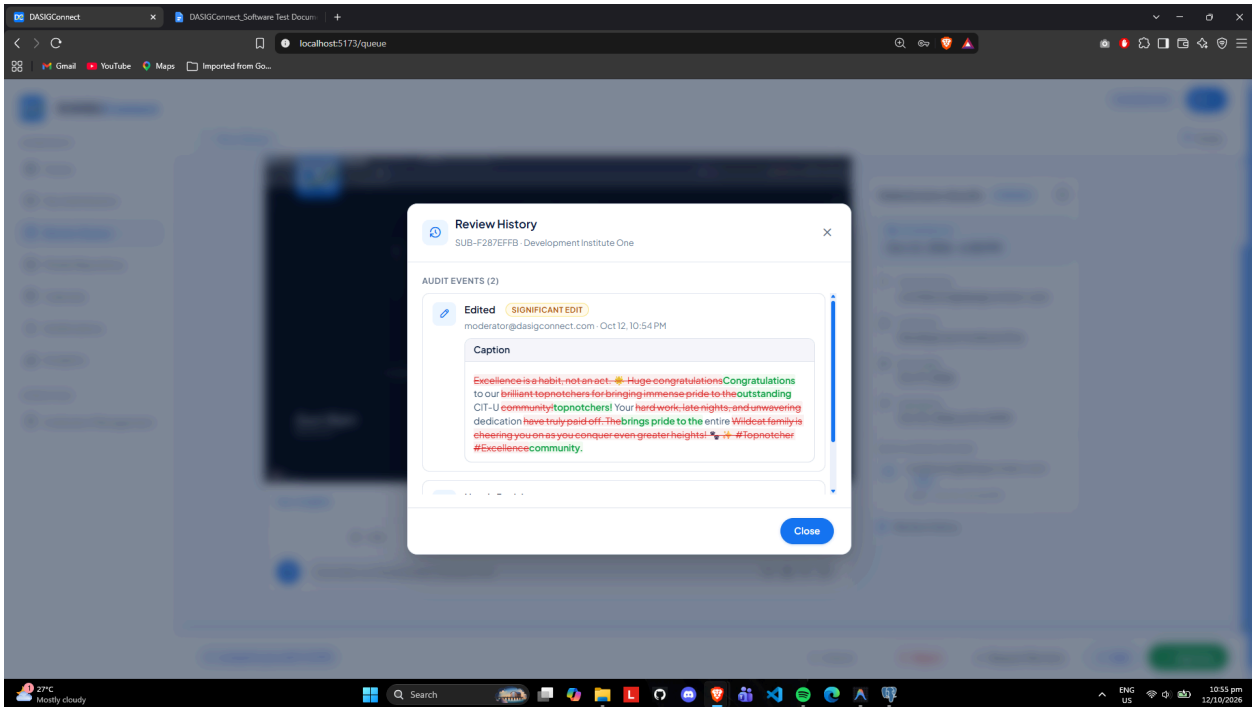
Moderator 2 Pov and the submission is lock.



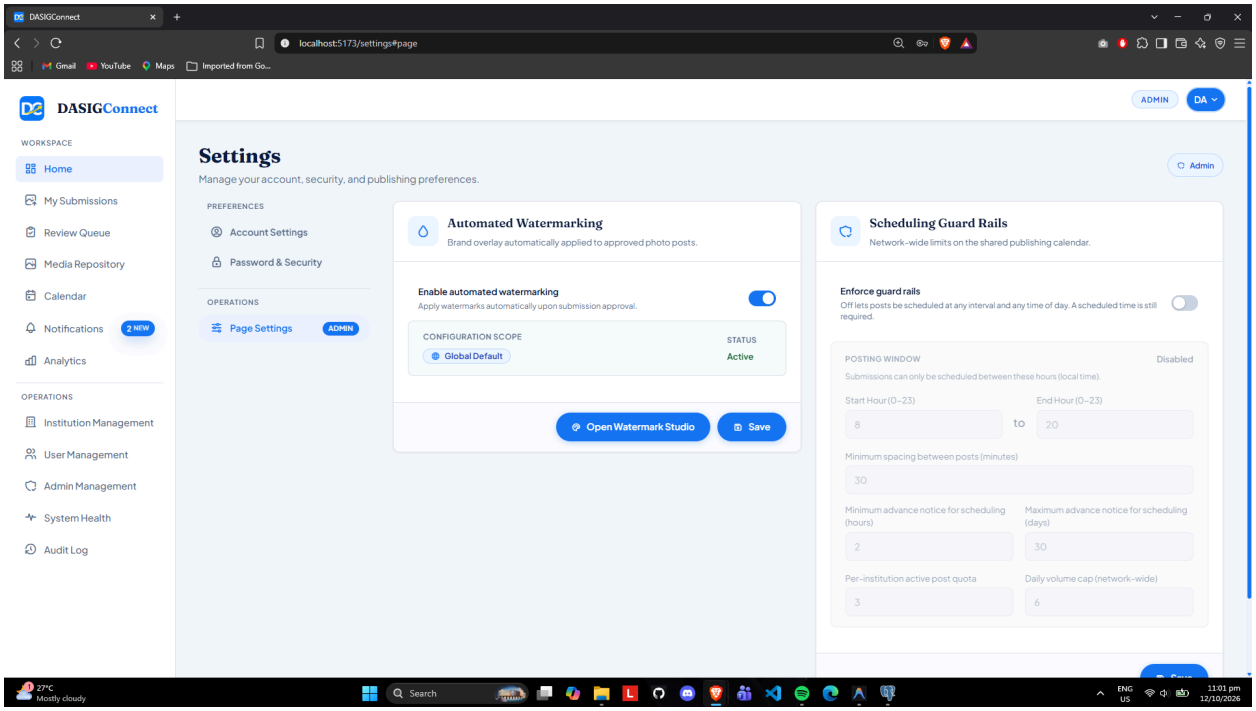
Screenshots for Test 17 (Case-17 — UC-2.4 A9–A15 Edit-During-Review Diffs & Late Publish Override)

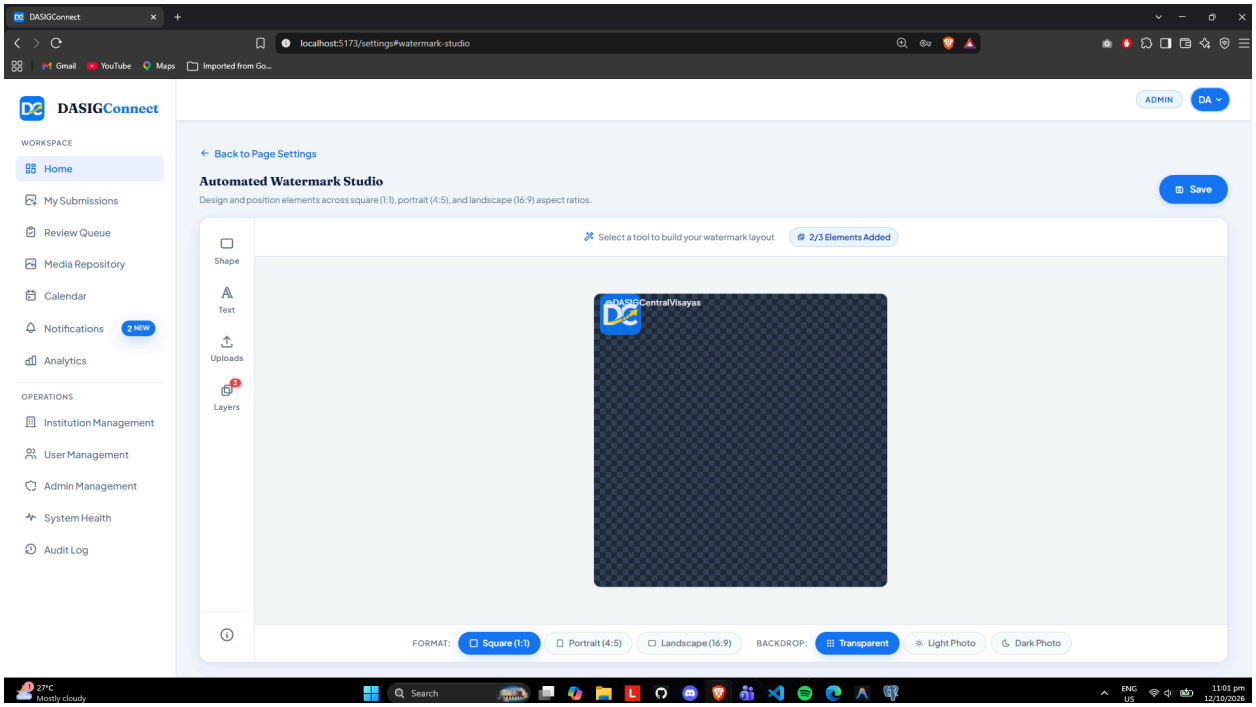


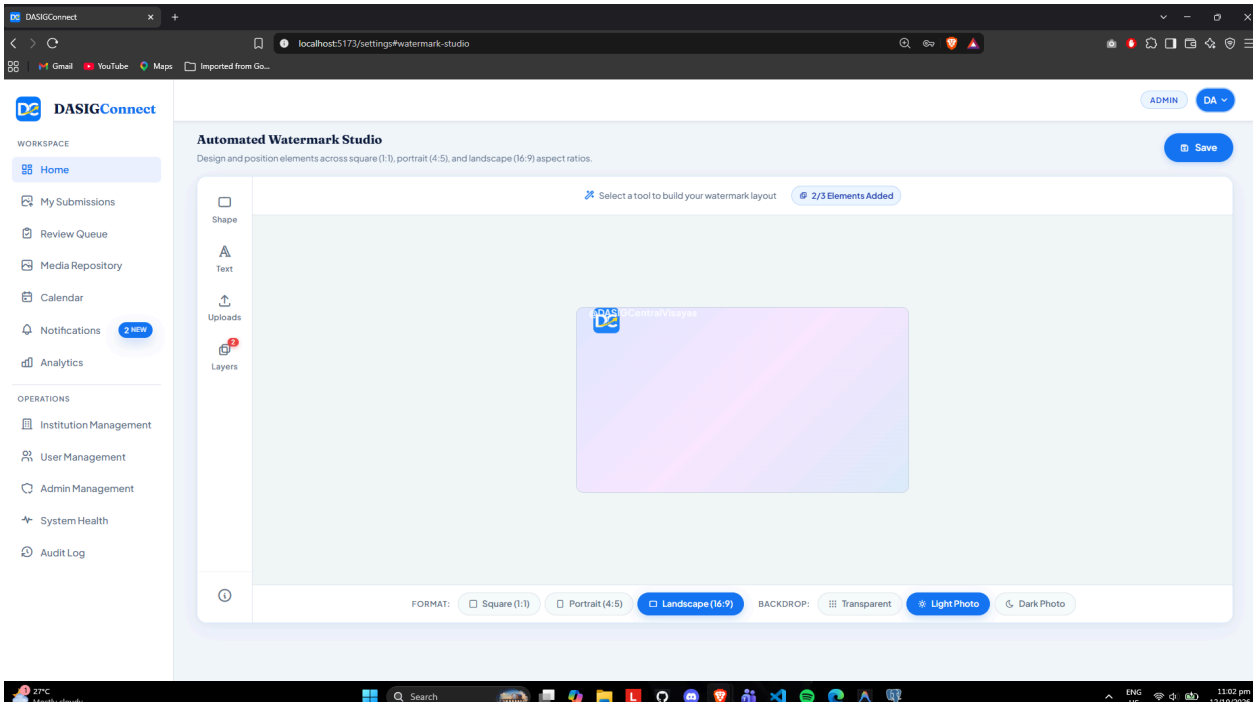

DASIGConnect
DASIGConnect_Software Test Docum
localhost:5173/queue
Gmail
YouTube
Maps
Imported from Go...
Review your changes
1 change · the contributor will be notified
Caption
Undo
~~Excellence is a habit, not an act. 🌟 Huge congratulations~~Congratulations to our ~~brilliant topnotchers for bringing immense pride to the~~outstanding CIT-U ~~community~~topnotchers! Your ~~hard work, late nights, and unwavering~~ dedication ~~have truly paid off. The~~brings pride to the entire ~~Wildcat family is cheering you on as you conquer even greater heights! 🦅✨ #Topnotcher #Excellence~~community.
Writing check unavailable — you can still save.
Keep editing
Save changes
27°C
Mostly cloudy
Search
ENG
US
10:53 pm
12/10/2026


DASIGConnect
DASIGConnect_Software Test Docum
localhost:5173/queue
Gmail
YouTube
Maps
Imported from Go...
Review History
SUB-F287EFFB · Development Institute One
AUDIT EVENTS (2)
Edited
SIGNIFICANT EDIT
moderator@dasigconnect.com · Oct 12, 10:54 PM
Caption
~~Excellence is a habit, not an act. 🌟 Huge congratulations~~Congratulations to our ~~brilliant topnotchers for bringing immense pride to the~~outstanding CIT-U ~~community~~topnotchers! Your ~~hard work, late nights, and unwavering~~ dedication ~~have truly paid off. The~~brings pride to the entire ~~Wildcat family is cheering you on as you conquer even greater heights! 🦅✨ #Topnotcher #Excellence~~community.
Close
27°C
Mostly cloudy
Search
ENG
US
10:55 pm
12/10/2026

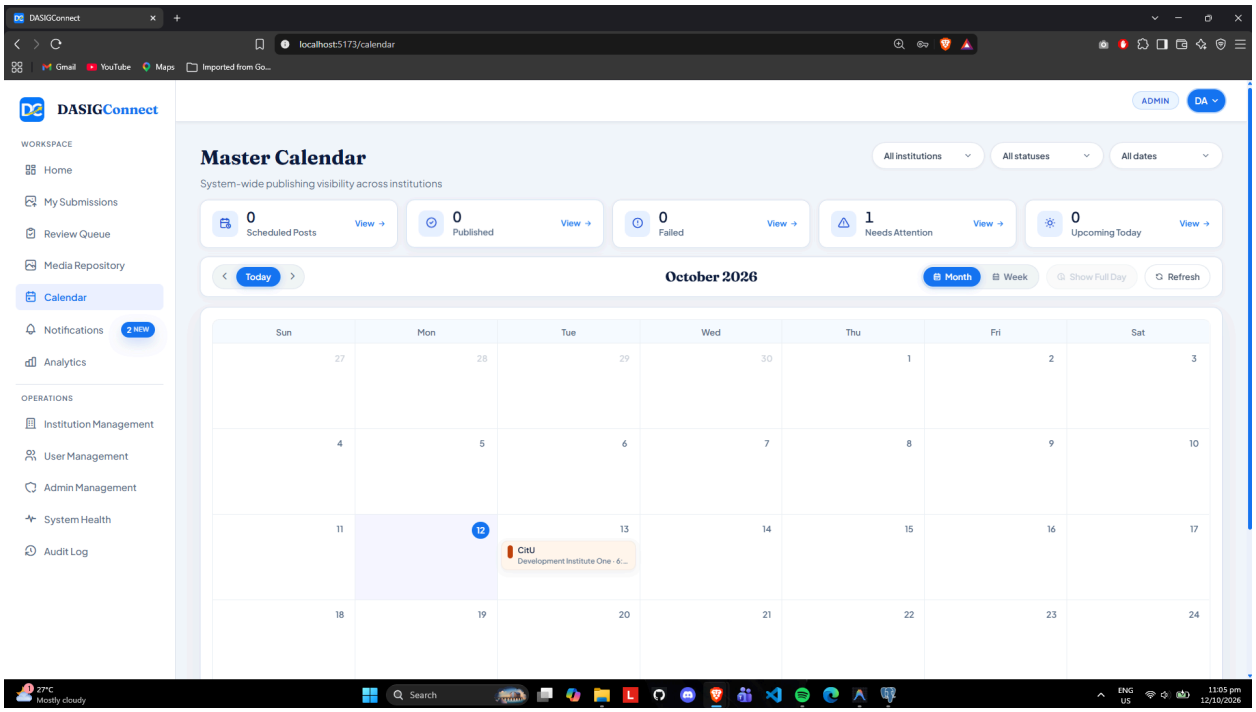
Screenshots for Test 18 (Case-18 — UC-2.5 Automated Brand Watermarking & Derivative Purge)

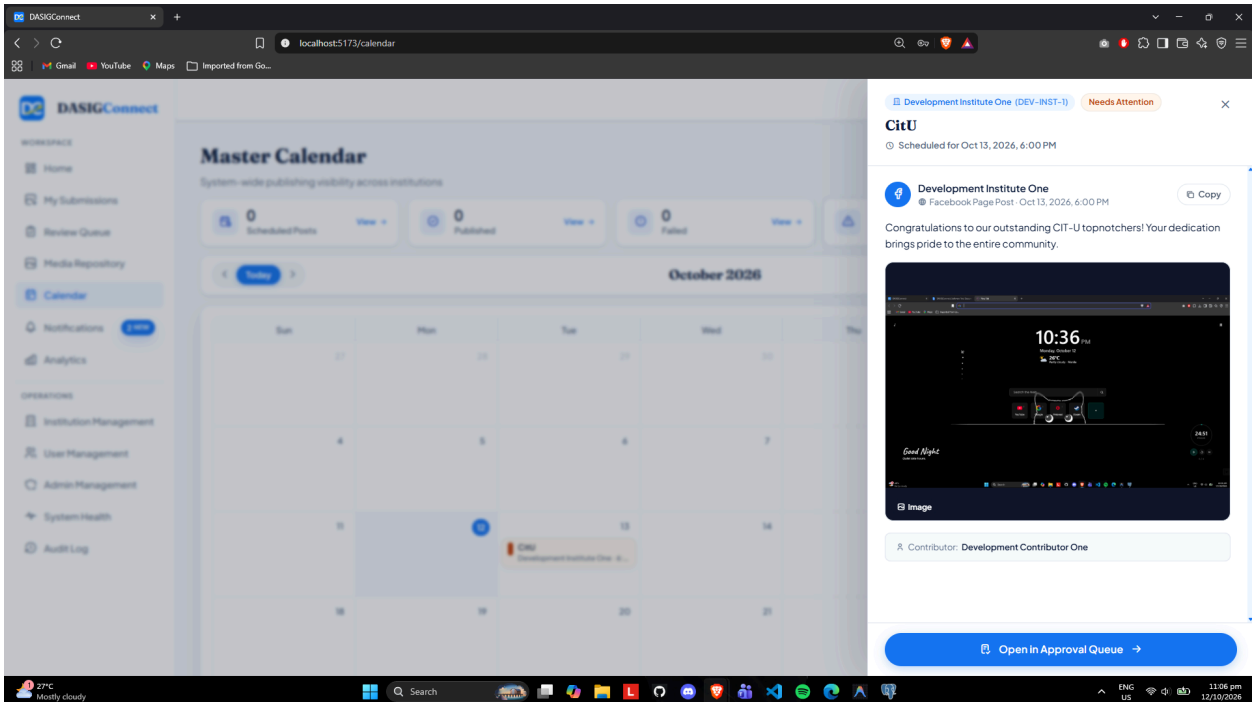




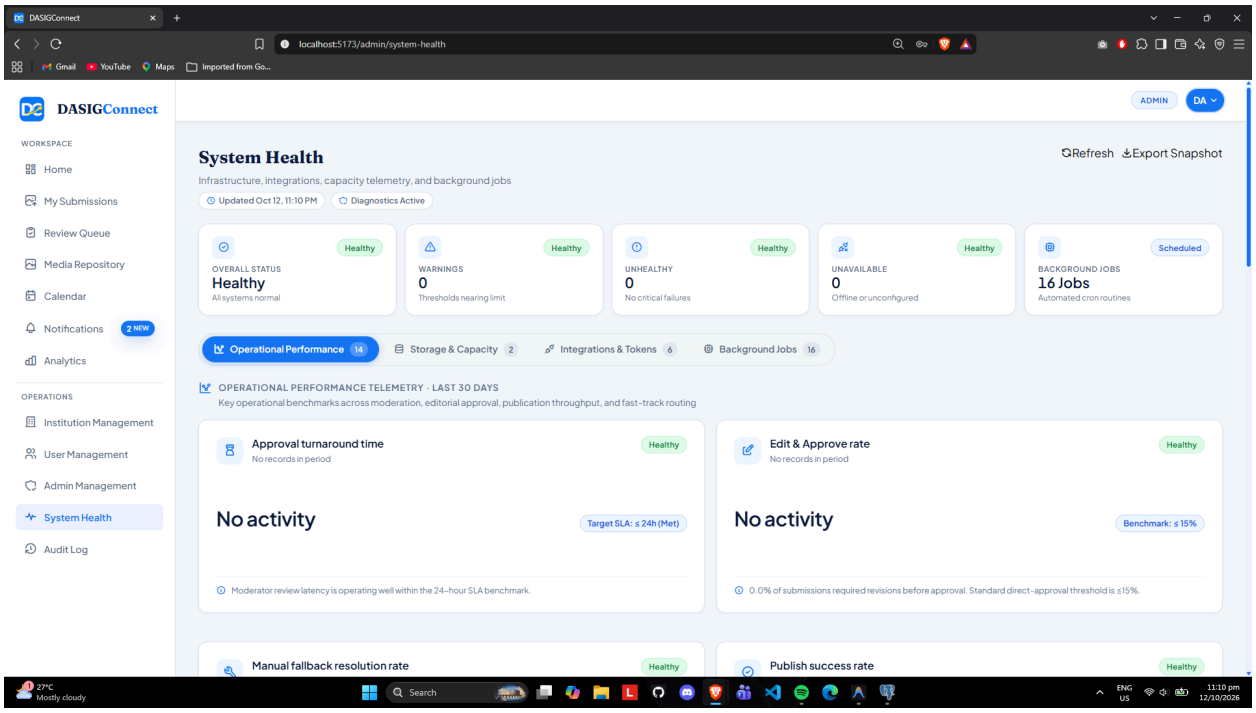


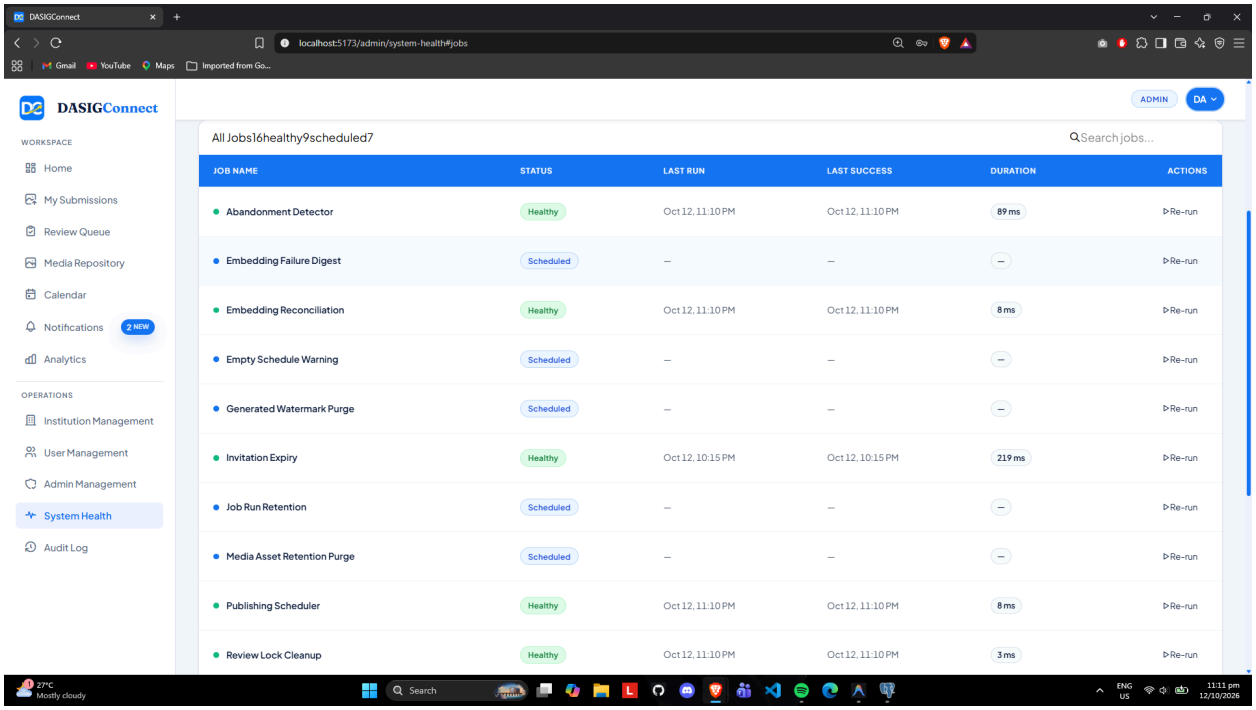
Screenshots for Test 19 (Case-19 — UC-3.1 Master Calendar, Moderator Caps & GiST Conflict Constraint)

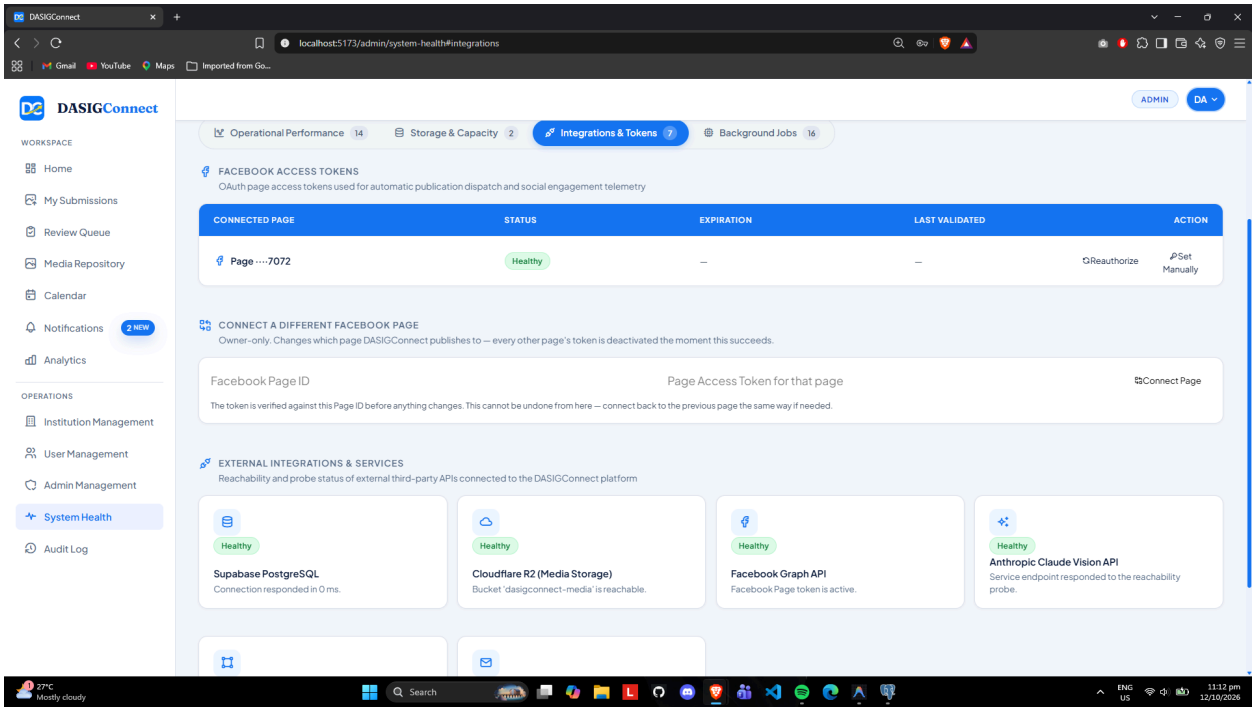




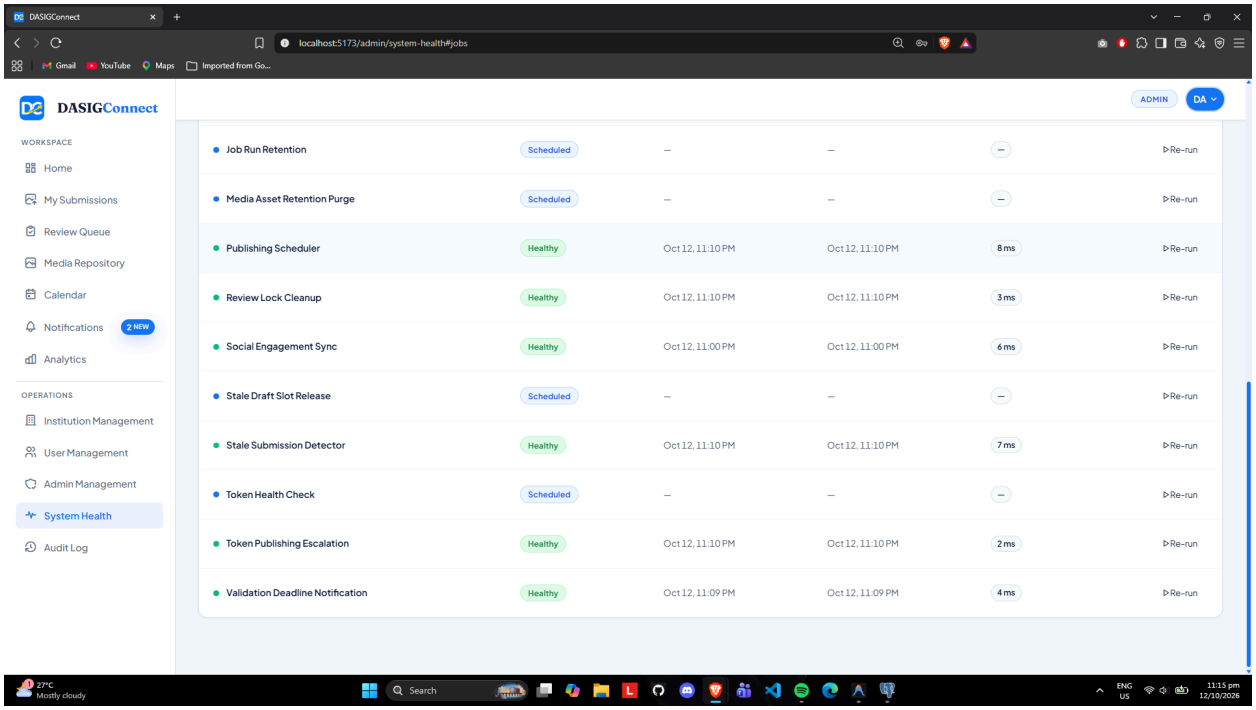
Screenshots for Test 20 (Case-20 — UC-3.2 Automated Facebook Publishing & Atomic Claim)

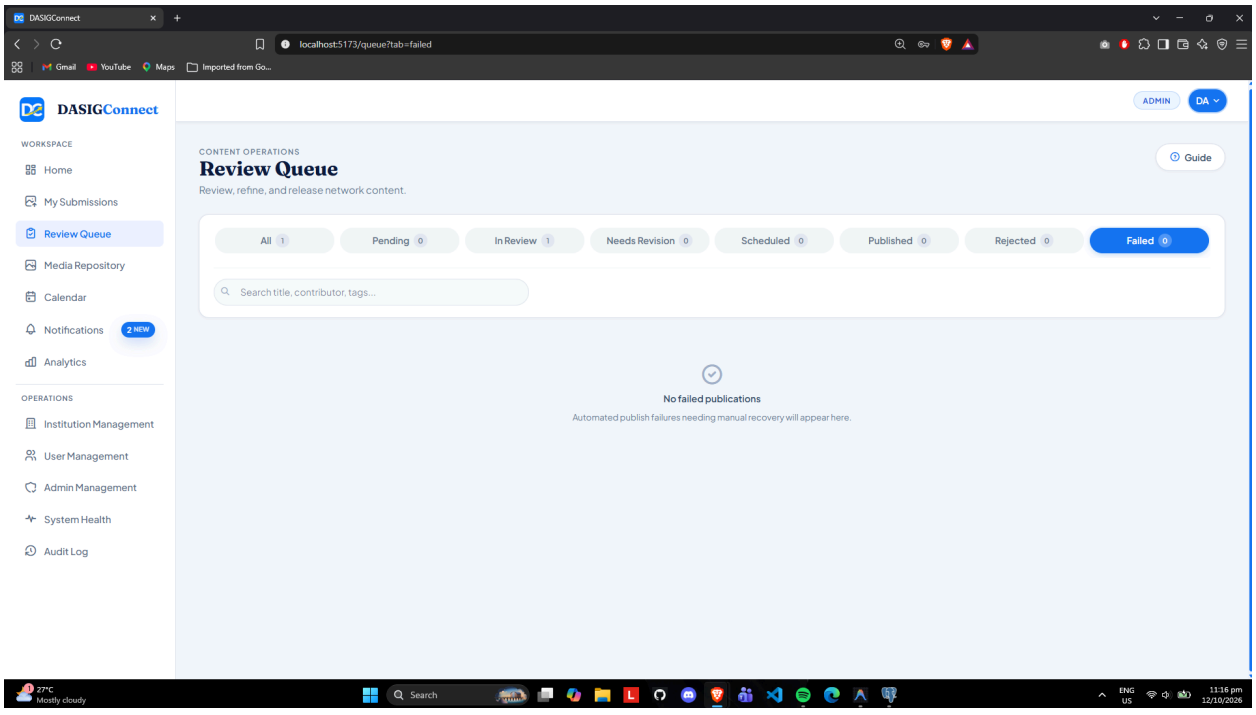



DASIGConnect
localhost:5173/admin/system-health#jobs
Gmail
YouTube
Maps
Imported from Go...
DASIGConnect
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
2 NEW
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
All Jobs16healthy9scheduled7
Search jobs...
JOB NAME
STATUS
LAST RUN
LAST SUCCESS
DURATION
ACTIONS
Abandonment Detector
Healthy
Oct 12, 11:10 PM
Oct 12, 11:10 PM
89 ms
Re-run
Embedding Failure Digest
Scheduled
Embedding Reconciliation
Healthy
Oct 12, 11:10 PM
Oct 12, 11:10 PM
8 ms
Re-run
Empty Schedule Warning
Scheduled
Generated Watermark Purge
Scheduled
Invitation Expiry
Healthy
Oct 12, 10:15 PM
Oct 12, 10:15 PM
219 ms
Re-run
Job Run Retention
Scheduled
Media Asset Retention Purge
Scheduled
Publishing Scheduler
Healthy
Oct 12, 11:10 PM
Oct 12, 11:10 PM
8 ms
Re-run
Review Lock Cleanup
Healthy
Oct 12, 11:10 PM
Oct 12, 11:10 PM
3 ms
Re-run
27°C
Mostly cloudy
Search
ENG
US
11:11 pm
12/10/2026

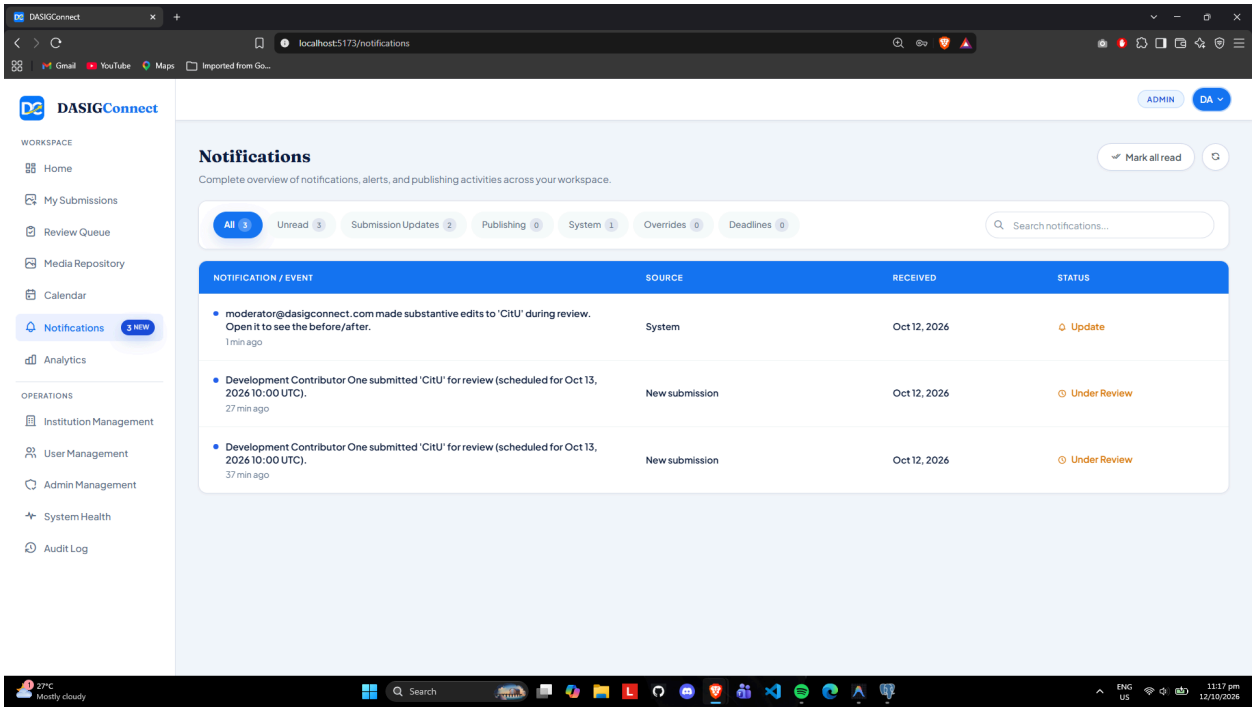

DASIGConnect
localhost:5173/admin/system-health#integrations
Gmail
YouTube
Maps
Imported from Go...
DASIGConnect
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
2 NEW
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Operational Performance 14
Storage & Capacity 2
Integrations & Tokens 7
Background Jobs 16
FACEBOOK ACCESS TOKENS
OAuth page access tokens used for automatic publication dispatch and social engagement telemetry
CONNECTED PAGE
STATUS
EXPIRATION
LAST VALIDATED
ACTION
Page ····7072
Healthy
Reauthorize
Set Manually
CONNECT A DIFFERENT FACEBOOK PAGE
Owner-only. Changes which page DASIGConnect publishes to — every other page's token is deactivated the moment this succeeds.
Facebook Page ID
Page Access Token for that page
Connect Page
The token is verified against this Page ID before anything changes. This cannot be undone from here — connect back to the previous page the same way if needed.
EXTERNAL INTEGRATIONS & SERVICES
Reachability and probe status of external third-party APIs connected to the DASIGConnect platform
Healthy
Supabase PostgreSQL
Connection responded in 0 ms.
Healthy
Cloudflare R2 (Media Storage)
Bucket 'dasigconnect-media' is reachable.
Healthy
Facebook Graph API
Facebook Page token is active.
Healthy
Anthropic Claude Vision API
Service endpoint responded to the reachability probe.
27°C
Mostly cloudy
Search
ENG
US
11:12 pm
12/10/2026

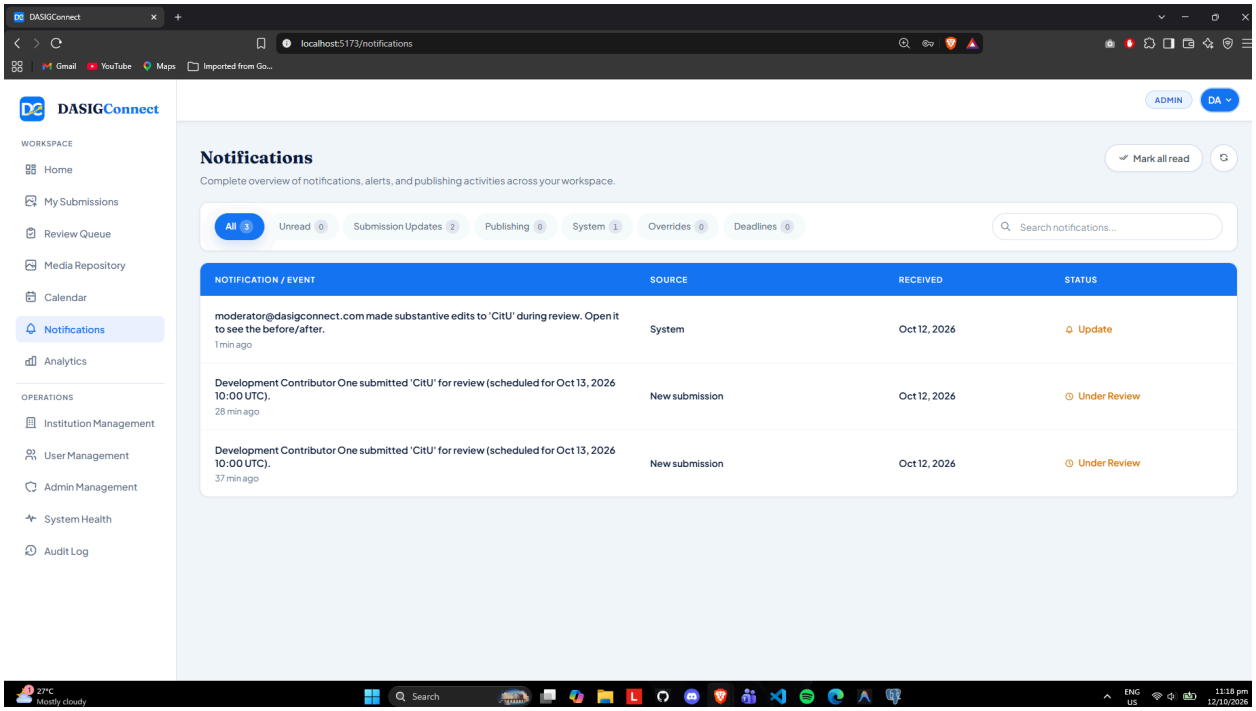
Screenshots for Test 21 (Case-21 — UC-3.2 A2 Stale Sweep & Manual Publishing Fallback)



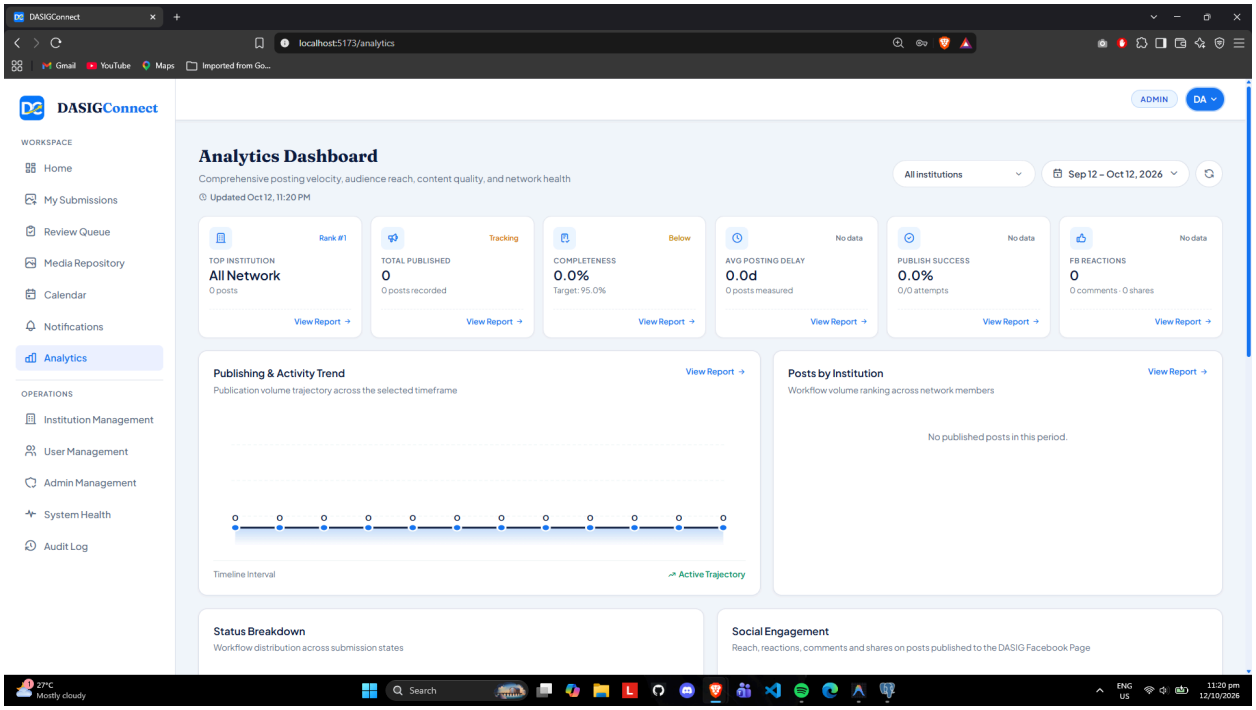


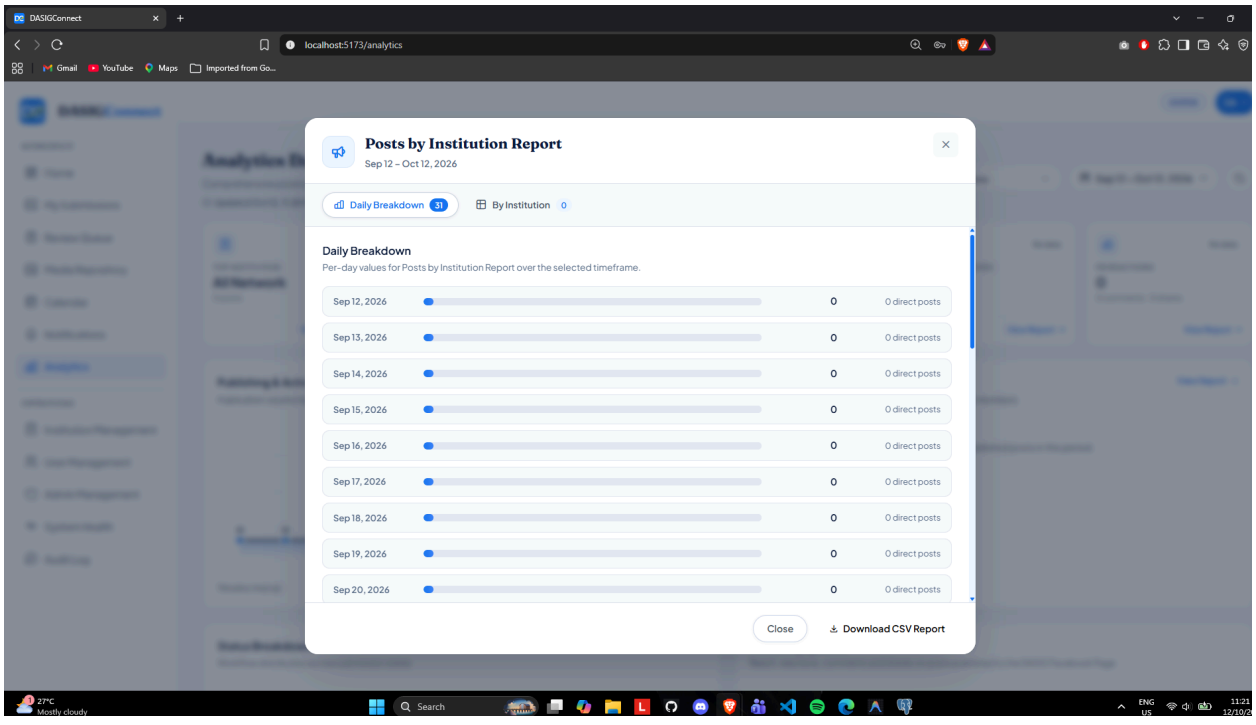
Screenshots for Test 22 (Case-22 — UC-3.3 Real-Time SSE Notifications & Cross-Session Sync)

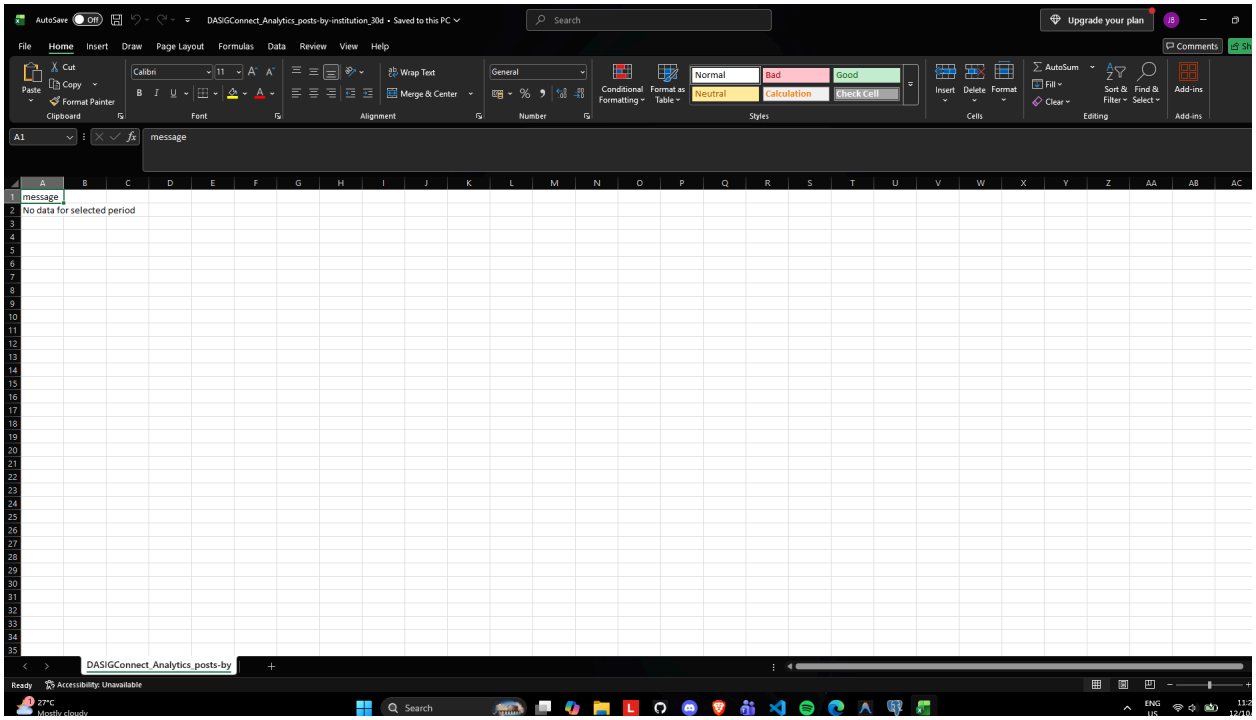




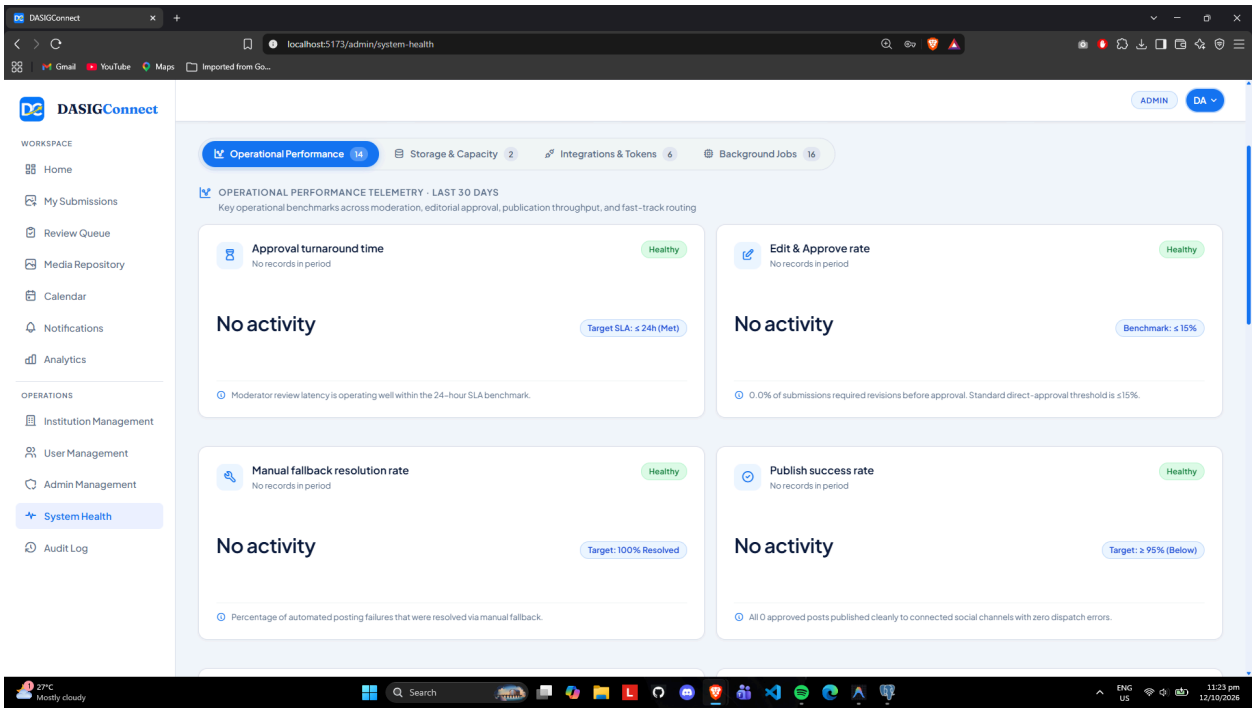
Screenshots for Test 23 (Case-23 — UC-3.4 Social Engagement Analytics & CSV Sanitization)

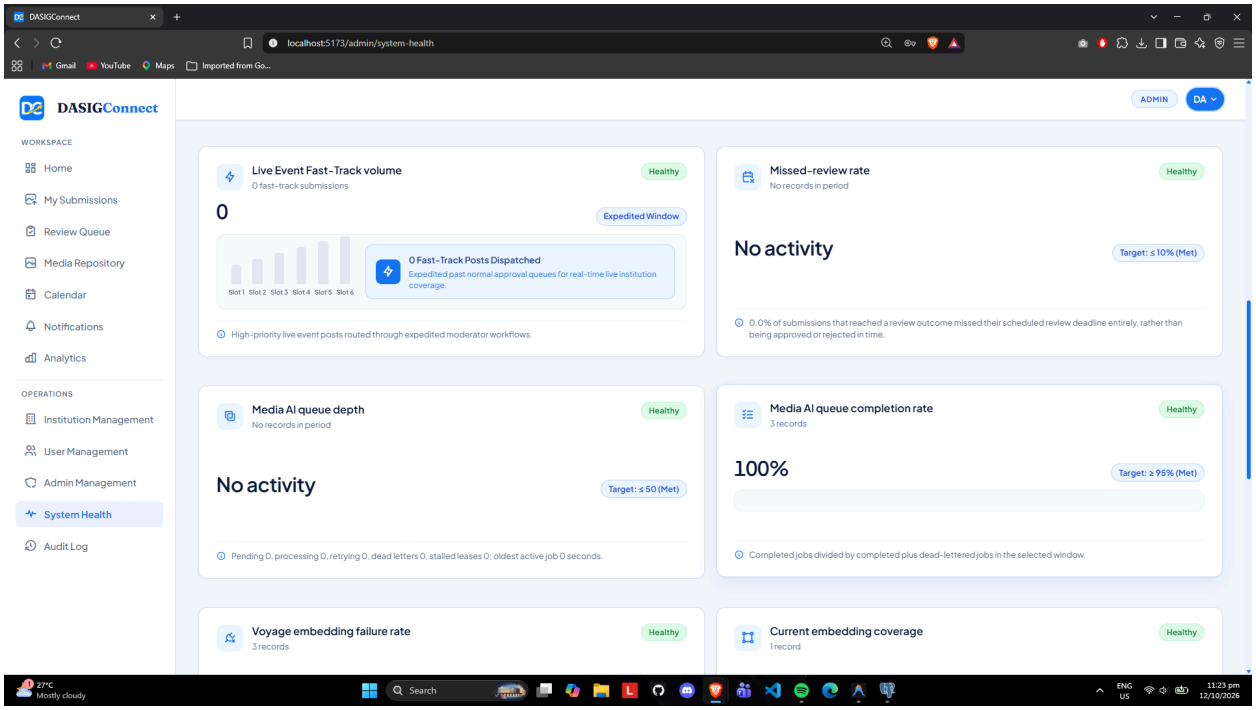


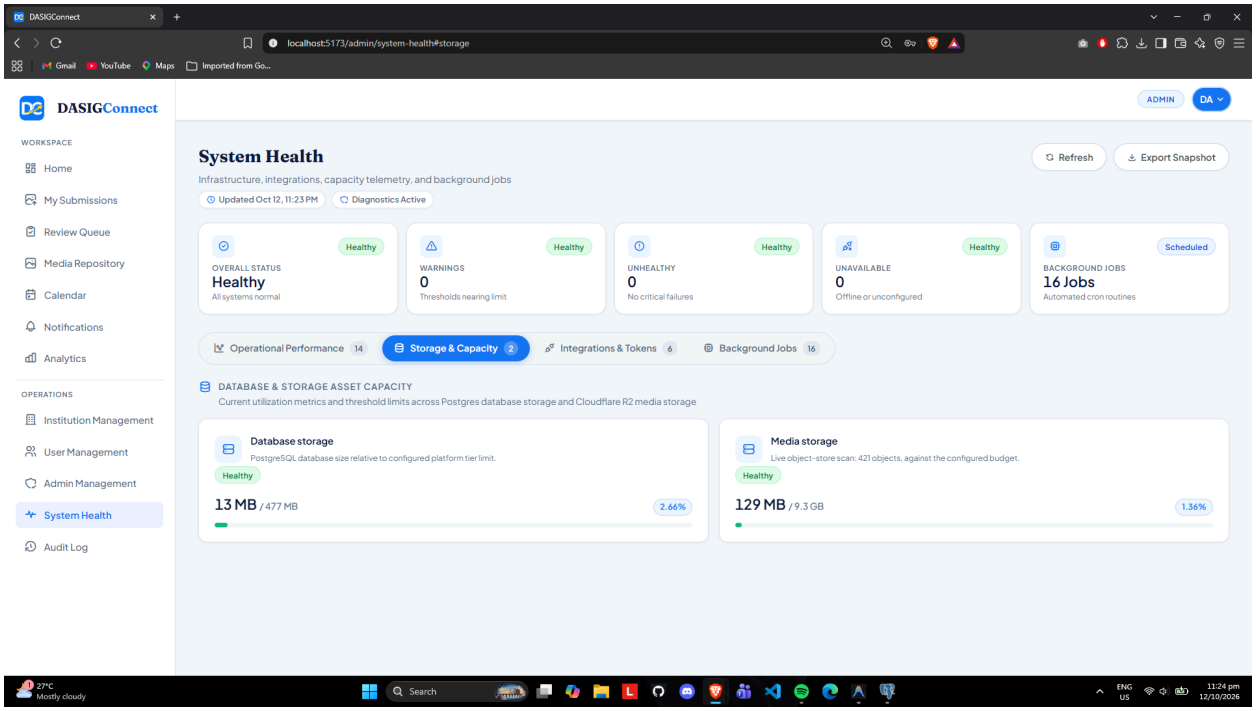




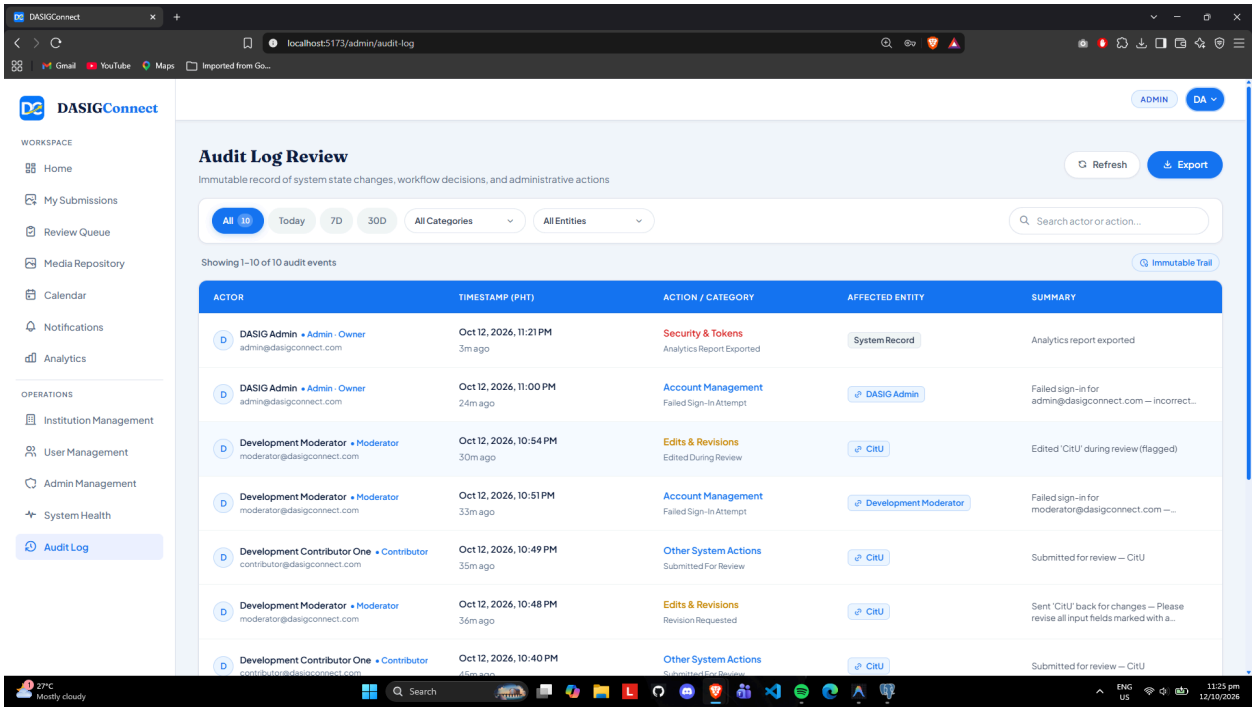
Screenshots for Test 24 (Case-24 — UC-3.5 System Health Analytics & 16 Tracked Jobs)

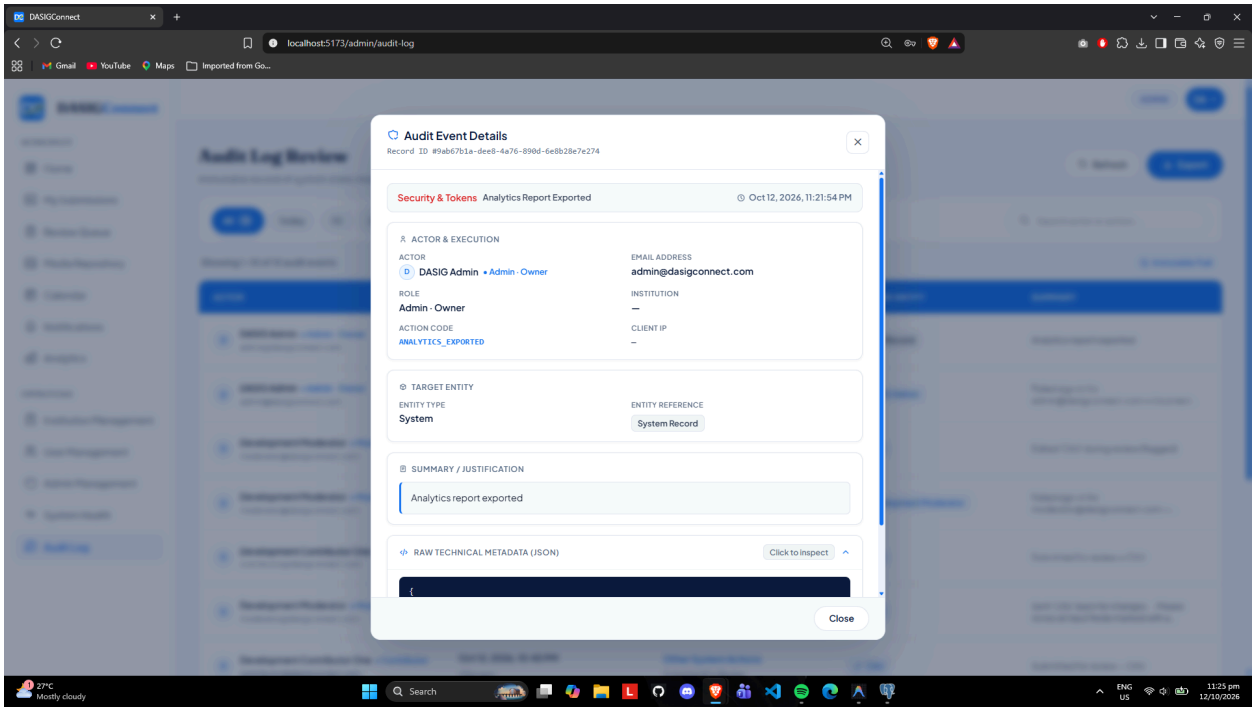


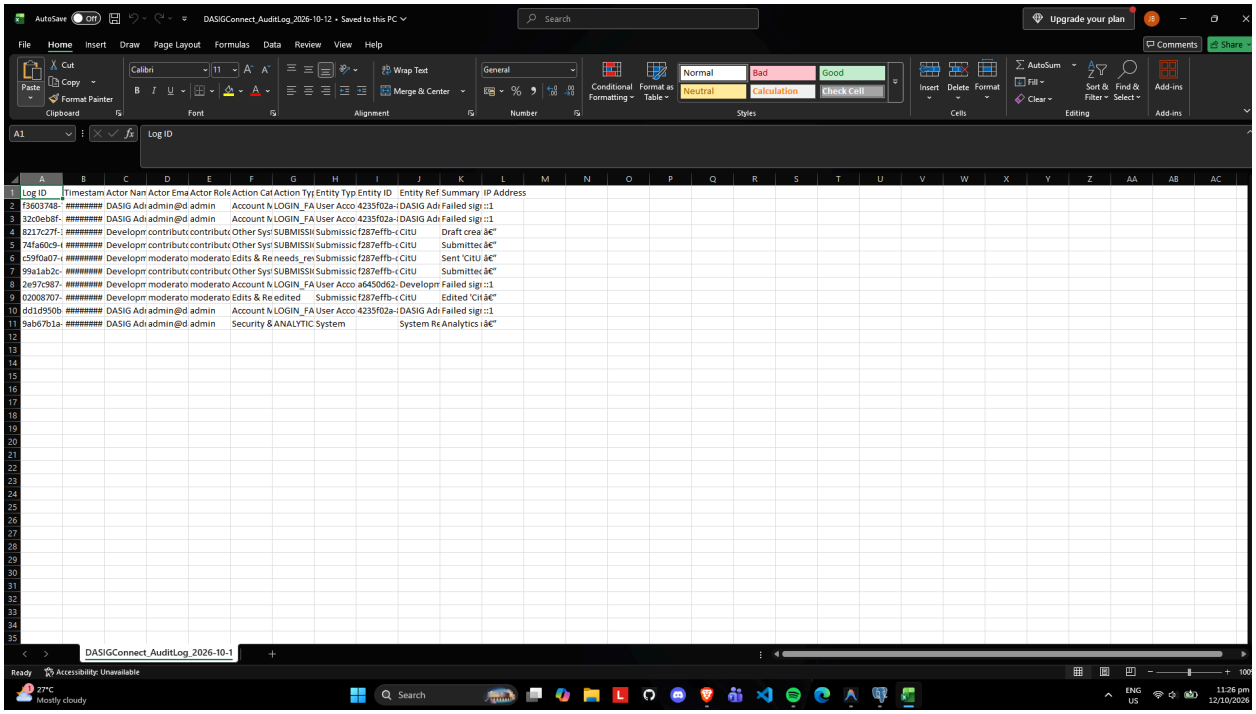

DASIGConnect
localhost:5173/admin/system-health
Gmail
YouTube
Maps
Imported from Go...
DASIGConnect
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
Live Event Fast-Track volume
0 fast-track submissions
Healthy
0
Expedited Window
Slot 1 Slot 2 Slot 3 Slot 4 Slot 5 Slot 6
0 Fast-Track Posts Dispatched
Expedited past normal approval queues for real-time live institution coverage.
High-priority live event posts routed through expedited moderator workflows.
Missed-review rate
No records in period
Healthy
No activity
Target: ≤ 10% (Met)
0.0% of submissions that reached a review outcome missed their scheduled review deadline entirely, rather than being approved or rejected in time.
Media AI queue depth
No records in period
Healthy
No activity
Target: ≤ 50 (Met)
Pending 0, processing 0, retrying 0, dead letters 0, stalled leases 0; oldest active job 0 seconds.
Media AI queue completion rate
3 records
Healthy
100%
Target: ≥ 95% (Met)
Completed jobs divided by completed plus dead-lettered jobs in the selected window.
Voyage embedding failure rate
3 records
Healthy
Current embedding coverage
1 record
Healthy
27°C
Mostly cloudy
Search
ENG
US
11:23 pm
12/10/2026


DASIGConnect
localhost:5173/admin/system-health#storage
Gmail
YouTube
Maps
Imported from Go...
DASIGConnect
ADMIN
DA
WORKSPACE
Home
My Submissions
Review Queue
Media Repository
Calendar
Notifications
Analytics
OPERATIONS
Institution Management
User Management
Admin Management
System Health
Audit Log
System Health
Infrastructure, integrations, capacity telemetry, and background jobs
Updated Oct 12, 11:23 PM
Diagnostics Active
Refresh
Export Snapshot
Healthy
OVERALL STATUS
Healthy
All systems normal
Healthy
WARNINGS
0
Thresholds nearing limit
Healthy
UNHEALTHY
0
No critical failures
Healthy
UNAVAILABLE
0
Offline or unconfigured
Scheduled
BACKGROUND JOBS
16 Jobs
Automated cron routines
Operational Performance 14
Storage & Capacity 2
Integrations & Tokens 6
Background Jobs 16
DATABASE & STORAGE ASSET CAPACITY
Current utilization metrics and threshold limits across Postgres database storage and Cloudflare R2 media storage
Database storage
PostgreSQL database size relative to configured platform tier limit.
Healthy
13 MB / 477 MB
2.66%
Media storage
Live object-store scan: 421 objects, against the configured budget.
Healthy
129 MB / 9.3 GB
1.36%
27°C
Mostly cloudy
Search
ENG
US
11:24 pm
12/10/2026

Screenshots for Test 25 (Case-25 — UC-3.6 Immutable Audit Log & Multi-Institution Tenant Isolation)






DASIGConnect_AuditLog_2026-10-12 • Saved to this PC
Upgrade your plan
File Home Insert Draw Page Layout Formulas Data Review View Help
Comments Share
Paste Cut Copy Format Painter
Calibri 11
Wrap Text Merge & Center
General
Conditional Formatting Format as Table
Normal Bad Good Neutral Calculation Check Cell
Insert Delete Format
AutoSum Fill Clear
Sort & Filter Find & Select
Add-ins
Clipboard Font Alignment Number Styles Cells Editing Add-ins
A1 Log ID
Log ID | Timestam | Actor Nam | Actor Ema | Actor Role | Action Cat | Action Typ | Entity Typ | Entity ID | Entity Ref | Summary | IP Address
f3603748- | ######## | DASIG Adı | admin@d | admin | Account M | LOGIN_FA | User Acco | 4235f02a-i | DASIG Adı | Failed sigı | ::1
32c0eb8f- | ######## | DASIG Adı | admin@d | admin | Account M | LOGIN_FA | User Acco | 4235f02a-i | DASIG Adı | Failed sigı | ::1
8217c27f-1 | ######## | Developm | contributo | contributo | Other Sys | SUBMISSI | Submissic | f287effb-c | CitU | Draft crea | â€"
74fa60c9-1 | ######## | Developm | contributo | contributo | Other Sys | SUBMISSI | Submissic | f287effb-c | CitU | Submittec | â€"
c59f0a07-1 | ######## | Developm | moderato | moderato | Edits & Re | needs_re | Submissic | f287effb-c | CitU | Sent 'CitU | â€"
99a1ab2c- | ######## | Developm | contributo | contributo | Other Sys | SUBMISSI | Submissic | f287effb-c | CitU | Submittec | â€"
2e97c987- | ######## | Developm | moderato | moderato | Account M | LOGIN_FA | User Acco | a6450d62- | Developm | Failed sigı | ::1
02008707- | ######## | Developm | moderato | moderato | Edits & Re | edited | Submissic | f287effb-c | CitU | Edited 'Cit | â€"
dd1d950b- | ######## | DASIG Adı | admin@d | admin | Account M | LOGIN_FA | User Acco | 4235f02a-i | DASIG Adı | Failed sigı | ::1
9ab67b1a- | ######## | DASIG Adı | admin@d | admin | Security & | ANALYTIC | System | | System Re | Analytics i | â€"
DASIGConnect_AuditLog_2026-10-1
Ready Accessibility: Unavailable
27°C Mostly cloudy
Search
ENG US
11:26 pm 12/10/2026